CAI Services for Developers

Student Guide

Version: CAIS-R42-DEV-202404



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CAI Services for Developers

Version: CAIS-R42-DEV-202404

April 2024

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Document Conventions

This guide uses the following formatting conventions:

If you see…	It means…	Example
>	Indicates a sub menu to navigate to.	Click Repository > Connect. In this example, you should click the Repository menu or button and choose Connect.
boldfaced text	Indicates text you need to type or enter.	Click the Rename button and name the new source definition **S_EMPLOYEE**.
UPPERCASE	Database tables and column names are shown in all UPPERCASE.	T_ITEM_SUMMARY
italicized text	Indicates a variable you must replace with specific information.	Connect to the Repository using the assigned *login_id*.
Note:	The following paragraph provides additional facts.	**Note:** You can select multiple objects to import by using the Ctrl key.
Tip:	The following paragraph provides suggested uses or a Velocity best practice.	**Tip:** The m_ prefix for a mapping name is…

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- Documentation and Knowledge Base
- Global Customer Support
- Professional Certification

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Contacting Global Customer Support

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https://www.informatica.com/services-and-training/support-services/contact-us.html

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Class Introduction

- Instructor
  - Role and Informatica
  - Background
  - Contact Details
- Attendee
  - Name
  - Company
  - Role with respect to the product
  - Expectations from the course

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Modules Overview

- Module 1: Overview of Cloud Application Integration
- Module 2: Understanding the Basics: Process Designer
- Module 3: Create a Basic Process
- Module 4: Process Objects
- Module 5: Service Connectors
- Module 6: Data Access Service Connectors
- Module 7: App Connections
- Module 8: Managing and Migrating Assets
- Module 9: Using XQuery in CAI
- Module 10: Adding Web Services to a Process

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Modules Overview (continued)

- Module 11: Fault Handling
- Module 12: Introduction to Guides Designer
- Module 13: Monitoring
- Module 14: API Management
- Module 15: CAI and CDI Integration
- Module 16: Troubleshooting, Tips & Tricks, Best Practices
- Module 17: CAI Live Project Walk-Through

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Module 1

Overview of Cloud Application Integration

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Module Objectives

After completing this module, you will be able to:

- Define Cloud Application Integration (CAI)
- List the features of Cloud Application Integration
- Identify CAI Assets and Components

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IICS Platform

- Supports cloud-to-cloud, cloud-to-on-premise, and on-premise-to-on-premise integration.
- Offers various services such as:
  - Cloud Data Integration
  - Cloud Application and Process Integration
  - API Management
  - Data Masking
  - Data Replication
  - Provides secure access to on-premise applications and middleware platforms

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Informatica Intelligent Cloud Service (IICS) is a next-generation iPaaS solution that provides enterprise data management through a suite of intelligent cloud services.

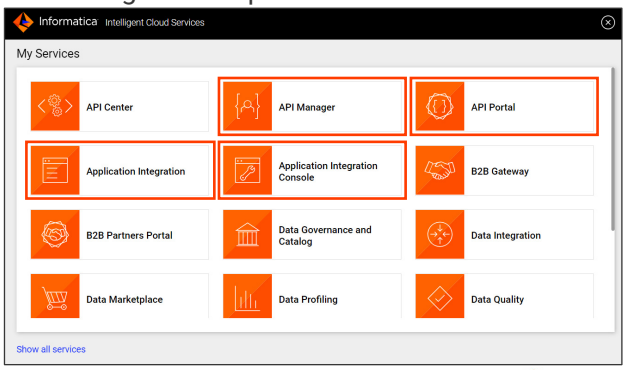
IICS offers:

1. Comprehensive support for cloud-to-cloud, cloud-to-on-premises, and on-premises-to-on-premises integration for easy data, service, and process integration.
2. Various data management and integration services, such as cloud data integration, cloud application and process integration, API management, data masking, and data replication.
3. Secure agent technology for secure access to on-premise applications and middleware platforms for cloud-to-on-premises integrations.
4. Support for Software Development Kit (SDK) and API development.

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Cloud Application Integration

- Perform real-time web service-based integration of processes, applications, and systems.
- Used to design, integrate, and implement business processes spanning different cloud and on-premise applications.
- Four IICS services are available for web-based integration of processes:
  - Application Integration
  - Application Integration Console
  - API Portal
  - API Manager



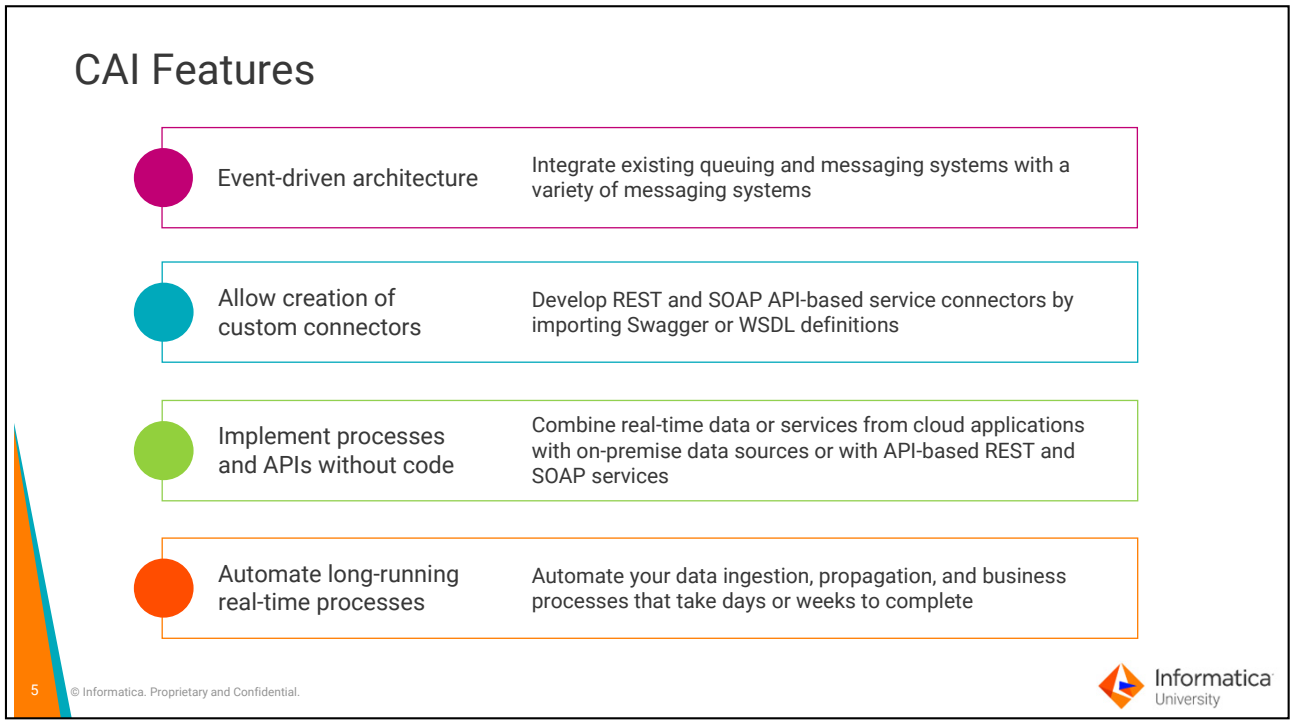
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Informatica Application Integration and Application integration console services allow you to perform real-time web service-based integration of processes, applications, and systems. These services allow you to design, integrate, and implement business processes spanning different cloud and on-premise applications. It also allows you to monitor your Informatica products on-premises and in Informatica Intelligent Cloud Services.

IICS offers the following four services for efficient web-based process integration:
1. Application Integration Console
2. Application Integration
3. API Portal
4. API Manager

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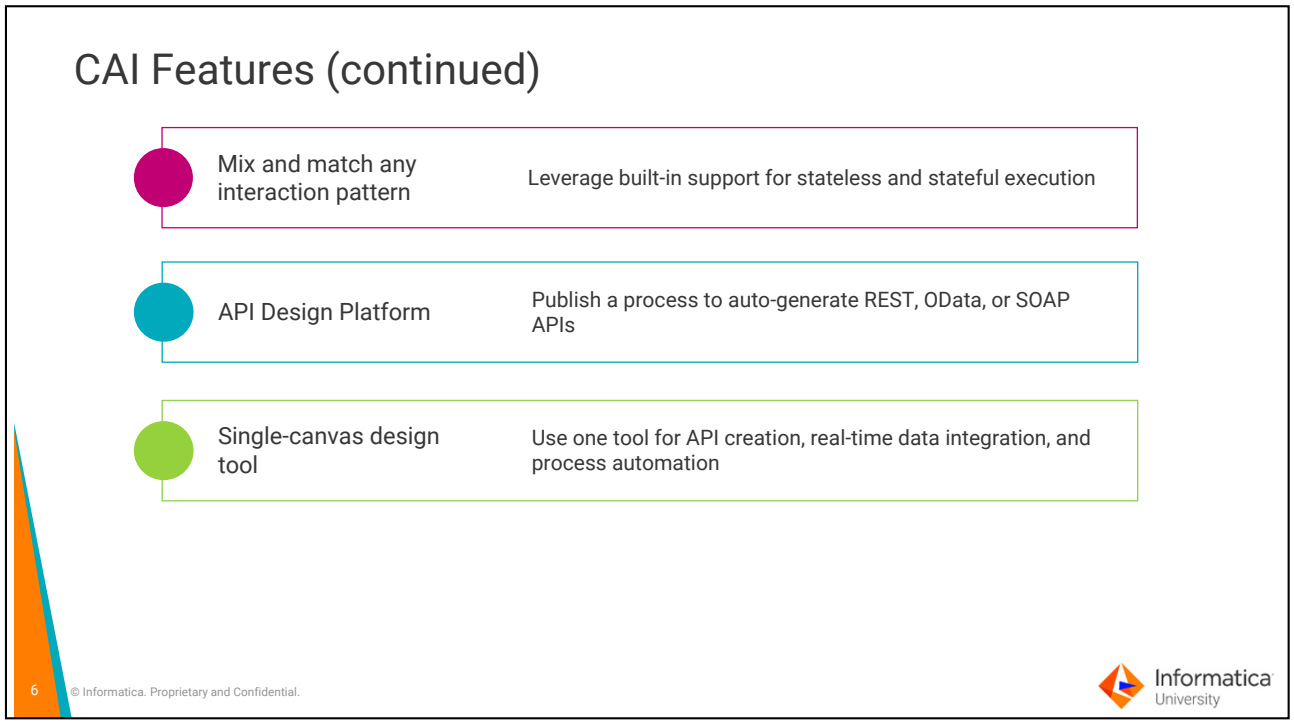
Event-driven architecture:
CAI supports integration of existing queuing and messaging systems with a variety of messaging systems such as JMS, AMQP AWS SNS/SQS, Microsoft Azure Messaging, and so on.

Allow creation of custom connectors:
CAI offers the creation of custom connectors by either importing a Web Service Definition Language (WSDL)/Swagger document or by creating a connector manually.

Implement processes and APIs without code:
With CAI, you can easily combine real-time data or services from cloud applications with on-premise data sources or with API-based REST (XML or JSON), and SOAP services.

Automate long-running real-time processes:
You can automate your data ingestion, propagation, and business processes that take days or weeks to complete, using Informatica CAI's sophisticated orchestration capabilities.

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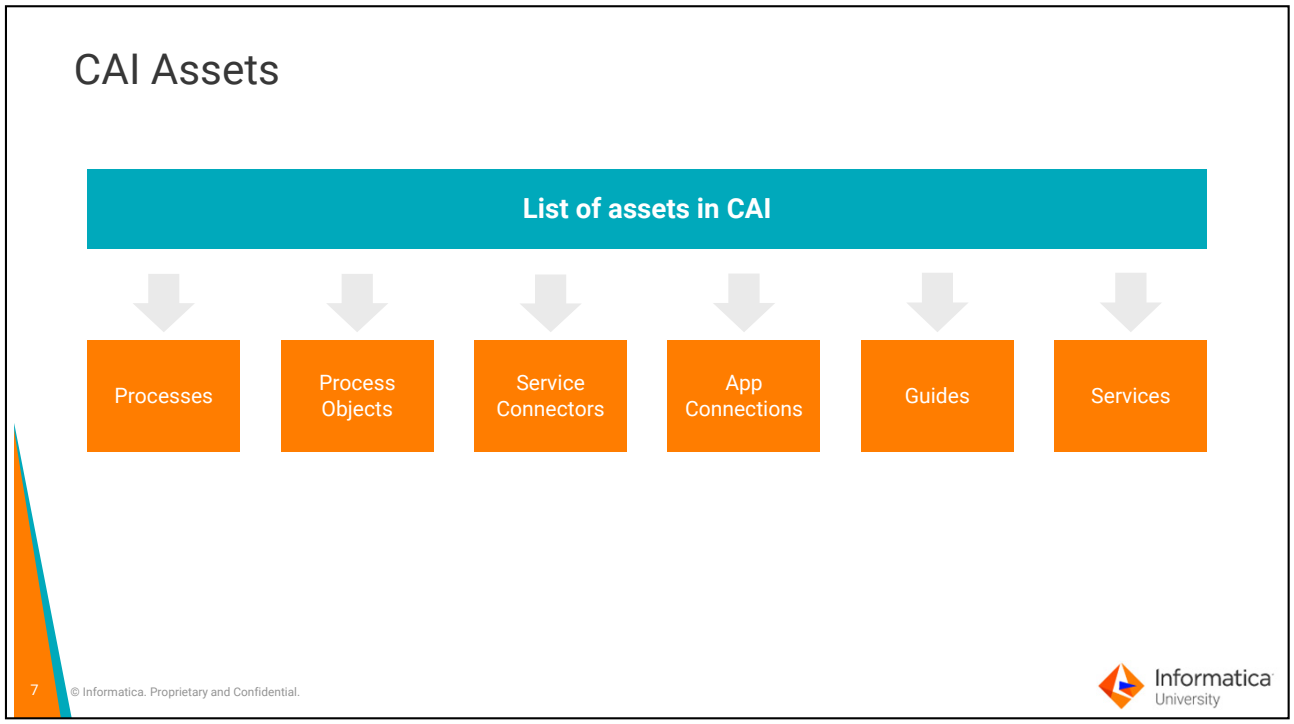


Mix and match any interaction pattern:
With CAI you can leverage built-in support for stateless and stateful execution and one-way, request-response, or correlated asynchronous responses.

API Design Platform:
In CAI, when you publish a process it auto-generates REST, OData, or SOAP APIs.

Single-canvas design tool:
CAI offers a single tool for API creation, real-time data integration, and process automation.

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Processes: In CAI, processes are used to orchestrate and define control-flow logic for business processes that span cloud and on-premise applications. A process can be HTTPS-based or event-based. If you create an HTTPS-based process, you get REST and SOAP APIs that you can invoke, embed, or share. If you create an event-based process, the process executes when the event happens.

Process Objects: A Process Object is used to group and structure data for use in a process. You can also generate Process Objects while creating a service connector.

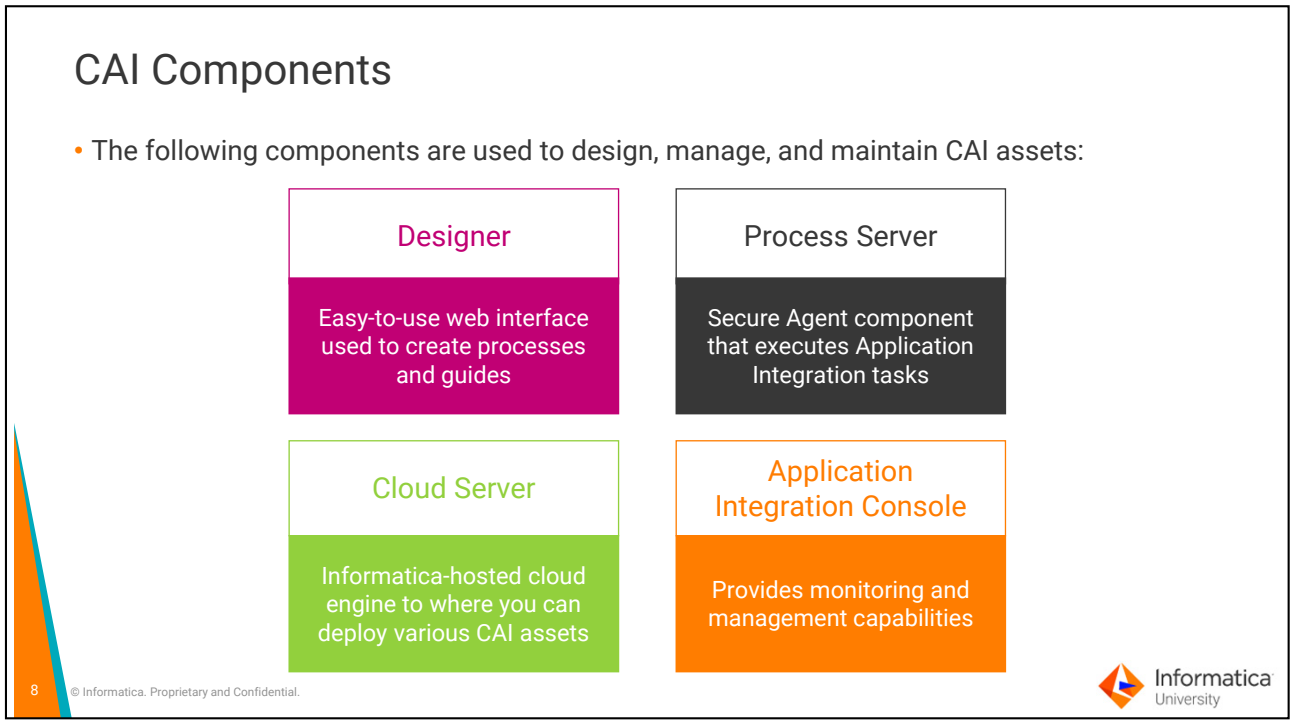
Service Connectors: A Service Connector is used to interact with the third-party programs and services using REST or SOAP APIs. You can also create a service connector by importing the Web Service Definition Language (WSDL) or Swagger JSON file that contains the definition of Service Connectors.

App Connections: Connections define actions that can be performed using a pre-defined connector or a service connector when used in other CAI assets, such as Processes or Guides.

Guides: In CAI, you can create a guide to automate the flow that a user might need to go through often. A guide contains a set of screens as steps that prompts users to review, enter, or confirm data, prompt users to review, enter, or confirm data. For example, a step might display account details or prompt the user to confirm the status of a sales call.

Services: In CAI, Services are used to create a new API proxy in Apigee.
Note: The 'Services' asset is out of scope for this course.

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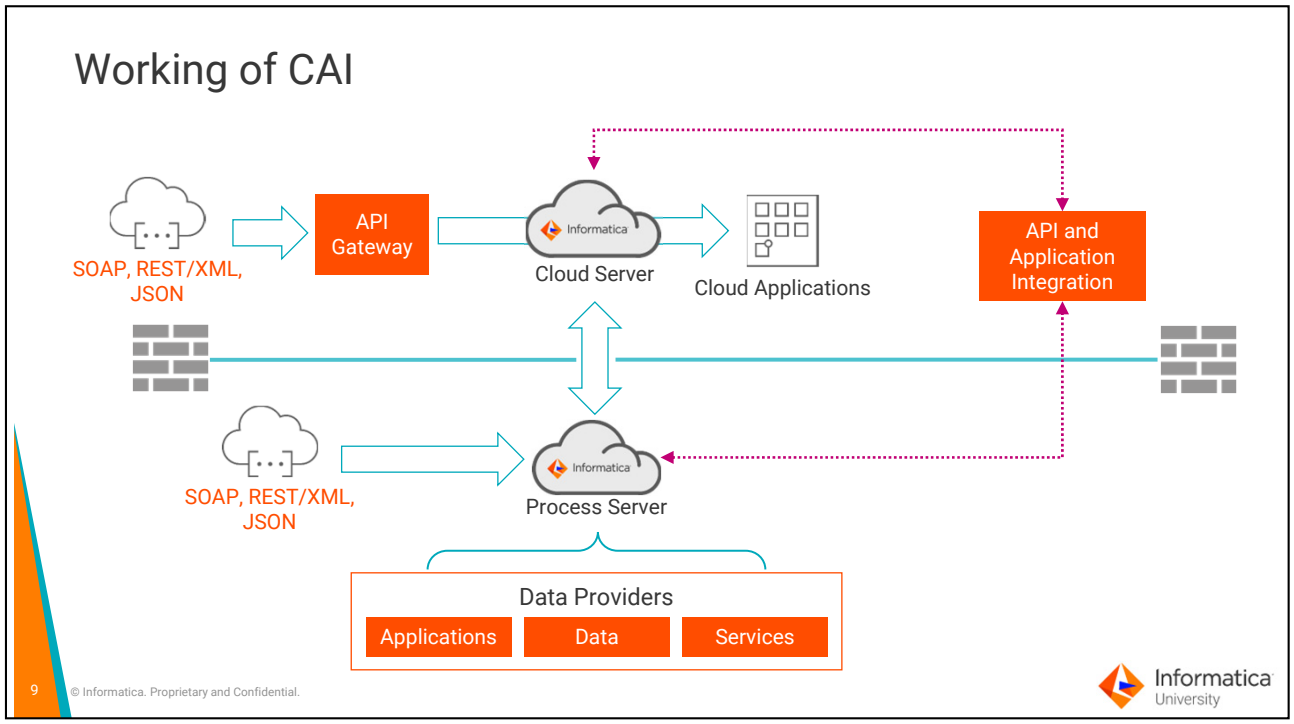
Designer: It is an easy-to-use web interface and can be used to develop Guides and Processes. To create the business logic for your use case, the designer offers various steps that can be used for decision making, parallel processing, and error handling.

Process Server: Process Server is the Secure Agent component that executes the Application Integration tasks. You can deploy the CAI assets to a Secure Agent or to the Cloud Server.

Cloud Server: Cloud Server is the Informatica-hosted cloud engine where you can deploy various CAI assets. As a best practice, it is recommended to deploy assets that contain cloud-based applications to Cloud Server. For an on-premise database, you must deploy the asset to a Secure Agent. For example, you must deploy a JBDC connection to a Secure Agent.

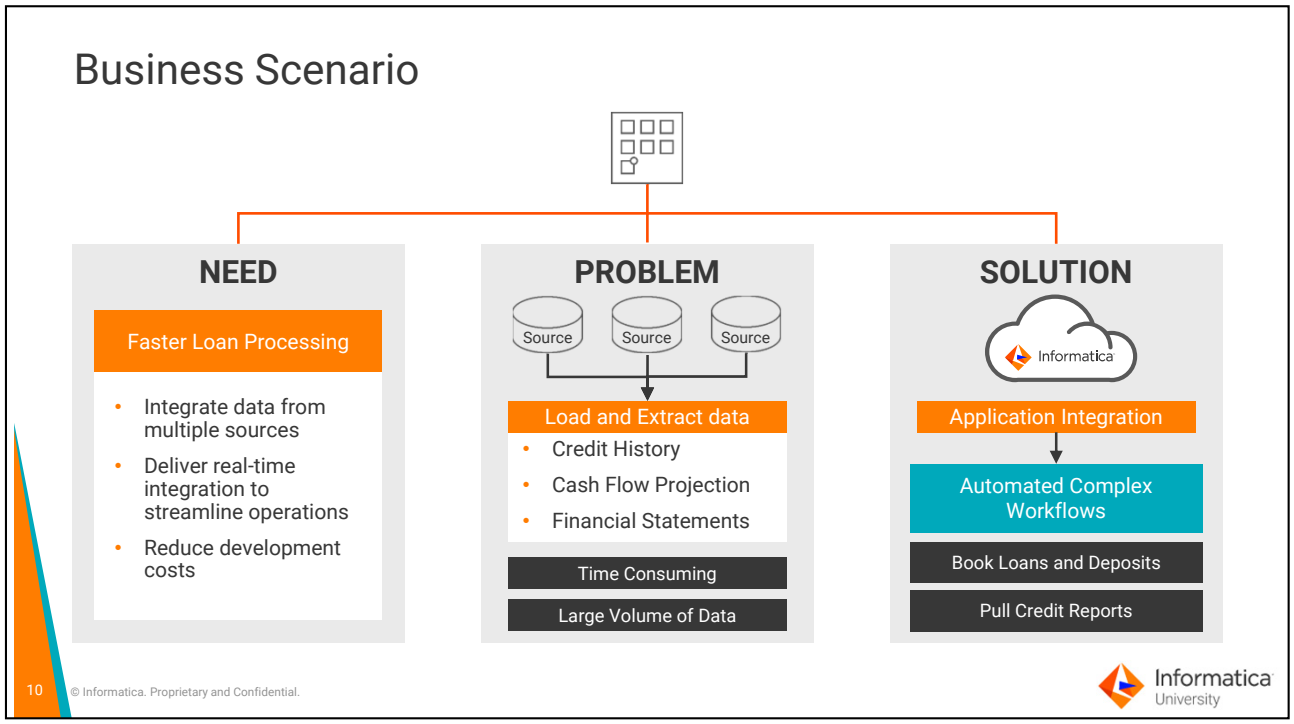
Application Integration Console: The Application Integration Console service provides monitoring and management capabilities for Application Integration.

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Informatica Cloud Application Integration is built for hybrid and multi-cloud environments. It allows users to connect to the cloud, and on-premise applications and processes in real-time through APIs. When an incoming service request originates from a cloud or on-premise consumer over JSON RPC, SOAP and REST (XML/JSON), the APIs send a service request. The service request can be for a new process, for a callback, or some event that the process is waiting to receive. For cloud-based integrations, the API Gateway receives the service request and directs the service request to the cloud server. The API gateway also applies access policies that help in establishing a secure communication channel to the Data Providers via API and Application Integration service. For on-premise integrations, a customer must install a secure agent, on which the CAI processes will run. The process server thus helps in to build on-premise applications and establish API communication.

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A worldwide leader in cloud banking has its Bank Operating System, built on the Salesforce platform. The company aims to makes the loan processing faster and efficient to deliver a high-quality digital experience to its customers. So, it is required to integrate loan data from a group of cloud and on-premise banking systems and deliver real-time integrations to streamline bank operations and reduce development costs.

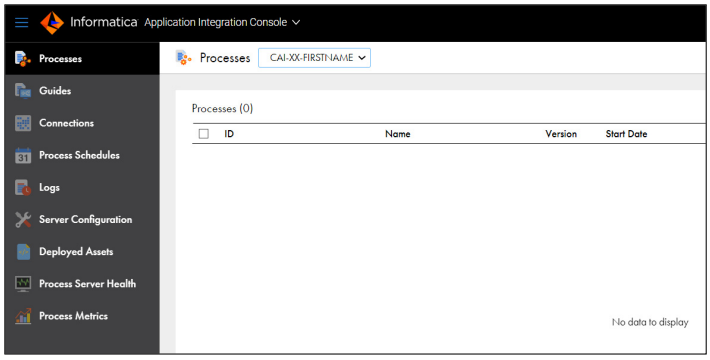
The organization had to accept customer banking data from multiple sources. To decide on a suitable lender, the executives had to first load and map crucial financial data from the borrower including credit history, cash flow projections, business, and personal financial statements, income tax returns, and so on into the lenders' core banking systems. The entire process was time-consuming and took weeks to complete. With so much information to gather, process, and track for each loan applicant, lenders are challenged to keep pace with the volume of data.

The organization used Informatica Cloud as a solution to all their business needs. With Informatica Cloud Data Integration service, the bank was able to integrate nightly batch data from more than 120 banking systems. The company leveraged the Informatica Cloud Application Integration service and designed multiple processes to automate complex workflows. Using CAI real-time integration, the company was able to push data back to core banking systems to automatically book loans and deposits, pull credit bureau reports, and other account documents. .

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Application Integration Console

- Provides options to monitor and manage processes
- Tasks performed in Application Integration Console:
  - Configure Processes Server and Cloud Server
  - Create and view process schedules
  - View server and deployment logs
  - View XML details of all resources
  - Monitor Process Server performance
  - View alarm and receive queue metrics
  - View deployed assets and contributions
  - View invoked processes and guides



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The Application Integration Console service provides monitoring and management capabilities for Application Integration. Depending on user roles and permissions, you can view and configure settings for Process Server or Cloud Server.

The tabs in Application Integration Console are as mentioned below:

Processes: Shows a list of processes and process versions.

Guides: Shows a list of guides that run in the Org. The list includes guides that completed successfully, completed with a fault, and guides that are still running.

Process Schedules: Allows you to configure and view process schedules to perform routine maintenance or recurring tasks.

Connections: Allows you to to view, start, or stop event sources in listener-based connections.

Logs: Displays deployment and server logs.

Server Configuration: Lists Monitor settings, Queue settings, Storage and Scheduled Maintenance settings, System Services settings**,** and Tenant or Process Server properties.

Deployed Assets: View and manage contributions, deployed processes, indexed properties, and mappings.

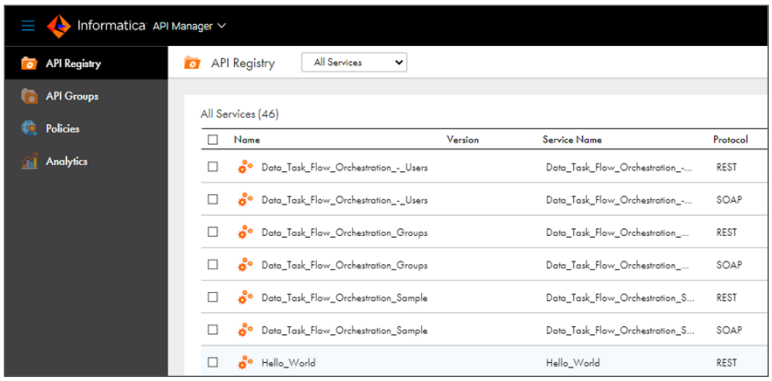
Process Server Health: Displays the overall status of the Process Server, which includes performance, alerts, and Server Statistics. You must select a Secure Agent to see Process Server health details.

Process Metrics: Allows you to filter and view metrics for alarm queues and receive queues.

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API Manager

- API Manager is a cloud-based service used by an organization to manage the APIs
- It deploys, manages, and controls the use of APIs
- Ensures secure use of organizational APIs by internal or external users



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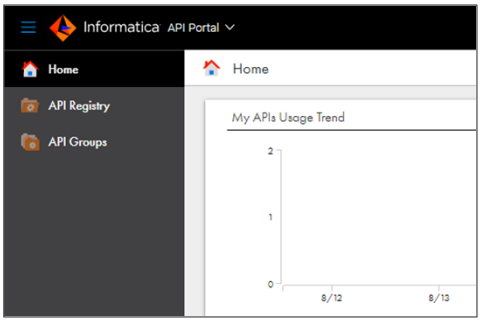
API Manager is a cloud-based service that an organization uses to manage the APIs for services and processes built in Informatica CAI.

API Manager provides the following functionalities:
- Seamless integration with Informatica CAI
- Access to API description and metadata
- Policy management
- API analytics

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API Portal

- Provides API users with secure access to managed APIs.
- Used to view details of API such as status, authentication type, applicable access control policy, and so on.
- Offers **Software Development Kit (SDK)** packages to integrate APIs into applications.



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API Portal provides users with secure access to the created APIs.

The API Portal enables users to interact with managed APIs and view API usage analytics.

The API Registry page shows details of the organization's managed APIs.

You can copy the URL of a managed API and then use it in your applications.

Topic

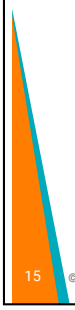
Knowledge Check

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IICS Cloud Application Integration, allows you to perform the web service-based integration of processes, applications, and systems in real time.

- True
- False



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Answer: True

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_____ is used to group and structure data for use in a process.

- Process
- Process Object
- Guide
- Service Connector

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Answer: Process Object

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Module Summary

This module showed you how to:

- Define Cloud Application Integration (CAI)
- List the features of Cloud Application Integration
- Identify CAI Assets and Components

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Module 2

Understanding the Basics: Process Designer

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Module Objectives

After completing this module, you will be able to:

- Describe a process
- Discuss Process Designer
- Identify process steps in Process Designer

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What is a Process?

A **Process** is a graphically represented series of steps executed to perform desired actions.

When do you need an automated process?

Your organization has customers across 240 countries and territories. Regardless of your customer's locations, you need accurate contact data. You need to check the validity of phone numbers to ensure that you have clean data driving of your business.

Process allows you to:

- Send and receive messages to coordinate services
- Write results into fields that are accessible downstream
- Invoke the process from within a process, or externally via SOAP or REST

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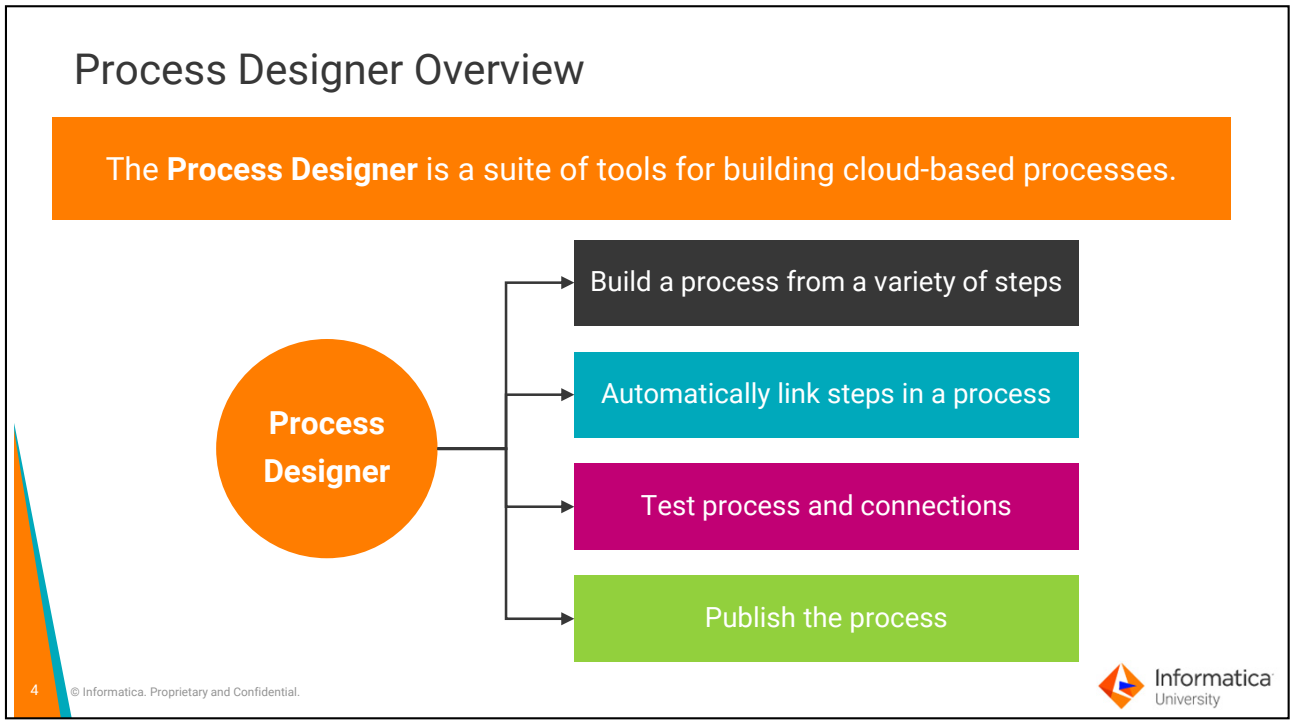
Process is a graphically represented series of steps executed to perform desired actions.

For example, you use Application Integration to create a process that fetches inventory details and sends margin and commission details via email to vendors and consumers.

Process Features:

1. Contains Start and End steps to mark the starting and ending of Process.
2. A Process automatically sends and receives messages to interact with other services configured in your Org.
3. During Process design, you can configure fields that are accessible downstream to write the result of the process.
4. You can invoke a Process from within a process known as a Subprocess, or externally via SOAP or REST.

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A Process Designer is a suite of tools for building cloud-based processes.
With Process Designer you can perform the following activities:

1. Build a Process from a variety of steps such as Assignment, Service, Subprocess, Receive, Wait, and so on.
2. The Process Designer automatically links steps in a Process based on the defined Process properties.
3. Test the Process and connections during design
4. Publish the designed Process

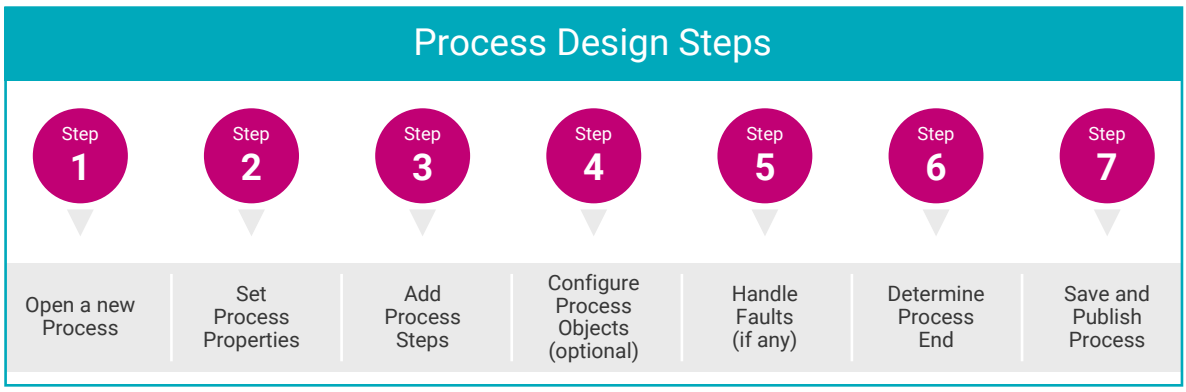
Note: A published process is deployed automatically by Informatica Cloud and is available to be invoked as a REST/XML or JSON service.

Example: To define a Process in CAI to streamline Salesforce order processing, you must perform the following steps:

1. Invoke a Service to obtain shipping information. The Service can be a Connection, Process, or System Service.
2. Perform lookup on inventory database to fetch current pricing information.
3. Submit order details to SAP ERP system.

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Steps in Process Design



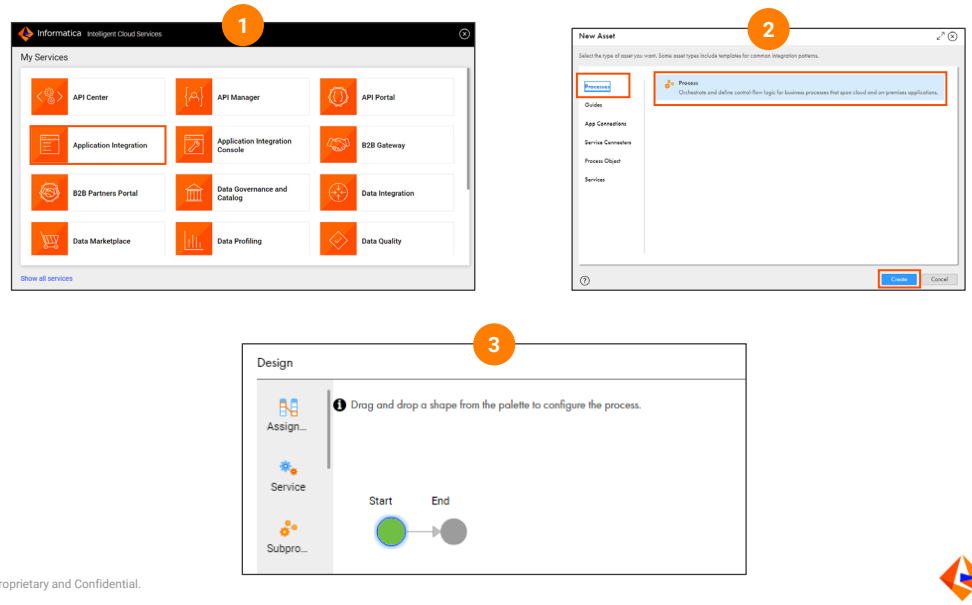
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Create a Process



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To create a Process, you must perform the following three steps:

1. Log in to your IICS Org and select the Application Integration Service.
2. From the Application Integration homepage, click New to open the New Asset dialog box.
3. From the list of Assets, select Processes, and click Create.

A new Process Designer appears with Start and End steps that mark the start and end of the process respectively.

You will then add and configure any steps necessary to complete the process.

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Process Steps

Step Icon	Step Name	Function
	Start Step	Marks the starting of the Process
	Assignment Step	Used to set value for a field
	Service Step	Used to call a service in a Process
	Subprocess Step	Used to call one Process within another Process
	Create Step	Used to create a new instance of an object
	Receive Step	Used to make a Service Call and waits for an event data before the Process can continue
	Human Task Step	Used to include a human decision within an Application Integration process

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Process Steps (continued)

Step Icon	Step Name	Function
	Wait Step	Used to specify the Pause time for Process
	Milestone Step	Used to specify the action to take when the Process ends
	Throw Step	Used for constructing an error payload
	Decision Step	Allows the Process to take one of the defined paths
	Parallel Paths Step	Used to define paths that can run in parallel
	Jump Step	Used to define the target a step can jump to
	End Step	Marks the ending of the Process

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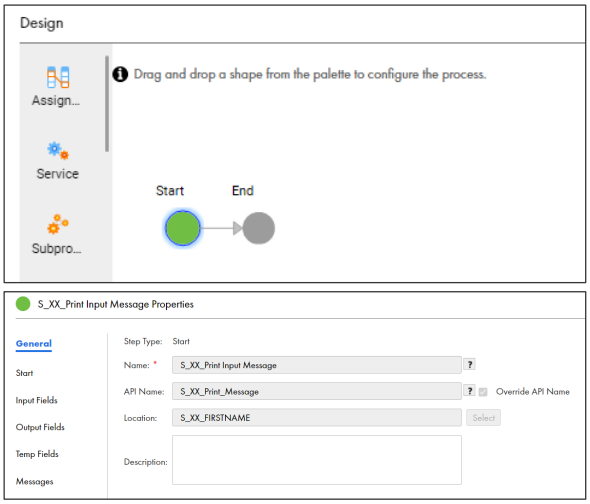
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Start Step

General Properties

- To view or edit process properties, select the Start step from the Process Designer.
- Specify the following General properties for Process:
  - Name
  - Override API Name
  - API Name
  - Location
  - Description



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To view or edit process properties, select the **Start** step from the Process Designer.

You can specify the following General properties for Processes:

Name: A descriptive name to identify the Process. The Process name appears when you want to use it in other objects such as subprocess step.

Override API Name: This is an optional field. If you select this option, it will override the API name that is auto generated when you publish the process with a name that you specify. When you select this option, the API Name field becomes available.

API Name: A unique API name to override the auto-generated API name for the process. The name you specify in this field is used in the generated service URLs.

Location: The location of the project or folder in which you want to save the Process. To select a location for the process, use the Browse option to navigate to the desired project or folder.

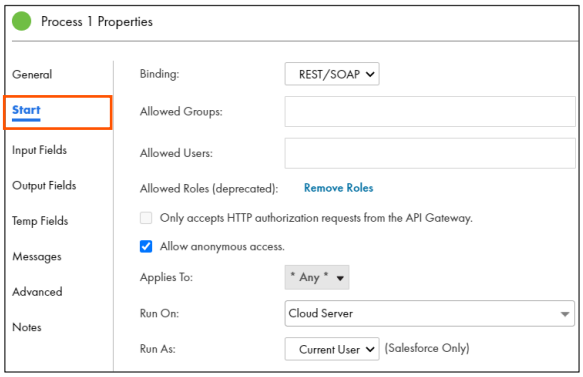
Description: This is an optional field. You can add the process description here.

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Start Step (continued)

Start Properties

- Specify the following Start properties for Process:
  - Binding
  - Allowed Groups
  - Allowed Users
  - Allowed Roles
  - Only accepts HTTP authorization requests from the API Gateway
  - Allow anonymous access
  - Applies To
  - Run On
  - Run As



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1. **Binding**: Defines the method to call the Process. You can select the binding as REST/SOAP or Event. For example, if you want to call the Process after a defined event, select the binding option as Event.
2. **Allowed Groups**: Specify the IICS user group that has access to the Process service URL. Select this option when you want the defined group of users to have access to the service URL. For example, you have a group called 'Approvers'. If you enter Approvers in the Allowed Groups field, all users within the group have access to the service URL.

Note: If you enter a role in the Allowed Groups or the Allowed Roles field, you cannot select Allow anonymous access and vice-versa.

3. **Allowed Users**: Specify the name of IICS user that has access to Process service URL.
4. **Allowed Roles**: Use the Allowed Roles field when you want a specific user to have access to the service URL.
5. **Only accepts HTTP authorization requests from the API Gateway**: Used to enable OAuth2 authentication for the process and to configure the API to accept only HTTP authorization requests from the API Gateway.
6. **Allow anonymous access**: Indicates that access to the Process is not tied to a specific role or user, which means that anyone can access the process.
7. **Applies To**: Used to select the object to which the process can be applied. The object can be Process Object, Connections, and so on.
8. **Run On**: Determines if the Process will run on Cloud Server or Process Server.
9. **Run As**: Used for Salesforce only. You can choose whether to run the process as the Current User or System.

Start Step (continued)

Input Fields Properties

- Fields present in the object to which the Process applies
- Input field values are accessible throughout the Process
- To use an input field in Process, you must define the following fields:
  - Input Format
  - Name
  - Type
  - Description
  - Required



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The Input Fields are the fields present in the object selected in the Applies To field of Start tab of Process properties. The Input field values are accessible throughout the Process.

1. **Input Format**: Determines whether the input field can represent one or more fields or the entire contents of the request. When you select Whole Payload, the input field represents the entire content of the request.

Note: You can configure only one input field if the Input Format is Whole payload.

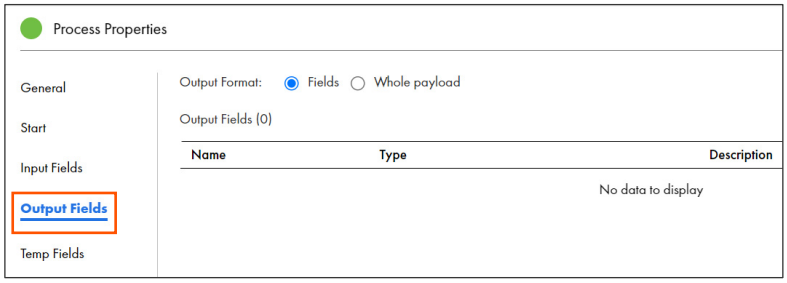
2. **Name**: Defines the name of the input field.
3. **Type**: Determines the data type of the defined input field. The data type can be simple, custom, or connection defined types.
4. **Description**: Description of the input field.
5. **Required**: You must check this option if the field is required for a process to execute, check Required.

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Start Step (continued)

Output Fields Properties

- Output field values are set when the process executes.
- After the value is set for the output field, the field is available for the rest of the Process.
- To use an output field in Process, you must define the following fields:
  - Output Format
  - Name
  - Type
  - Description
  - Initial Value



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The Output Fields are the fields present in the object selected in the Applies To field of Start tab of Process properties. The output field values are set when the process executes. After the value is set for the output field, the field is available for the rest of the Process.

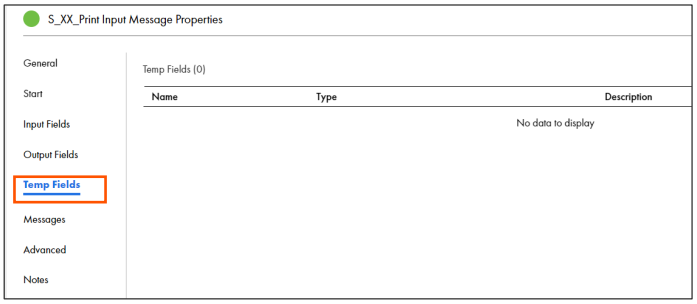
1. **Output Format**: Determines whether the output field can represent one or more fields or the entire contents of the request. When you select Whole Payload, the output field represents the entire content of the response.
2. **Name**: The name of the output field.
3. **Type**: Defines the data type of output field.
4. **Description**: Description of the output field.
5. **Initial Value**: The initial value for the output field.

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Start Step (continued)

Temp Fields Properties

- Temporary field values are available for use in all the steps of the current Process
- To use a temp field in Process, you must define the following fields:
  - Name
  - Type
  - Description
  - Initial Value



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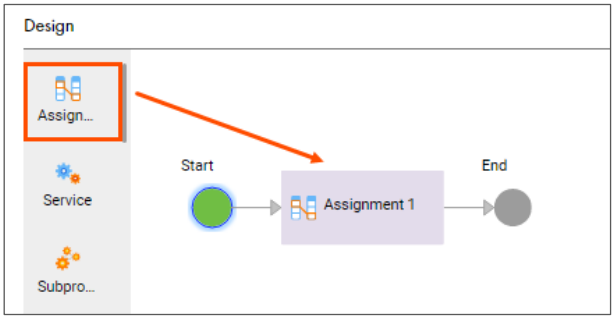
The Temp Fields values are available for use in all of the current Process's steps. However, a temporary field's value is not available to a Subprocess.

1. **Name**: Name of the temporary field.
2. **Type**: Data type of temporary field.
3. **Description**: Description of temporary field.
4. **Initial Value**: Initial value for the temporary field.

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Adding Step to Process

- Drag a step from the design palette onto the canvas



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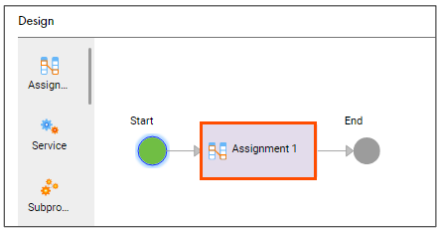
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You can add steps to a Process by dragging them from the palette on the process designer canvas. If you drop the step on the link between two steps, the connection between existing steps and new step is established automatically.

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Assignment Step

- Assignment step sets value for a field
- To assign value for a field, provide the following details:
  - Field
  - Assigned Using
  - From





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Use the Assignment step to set a value to an Input, Output, or Temporary field.

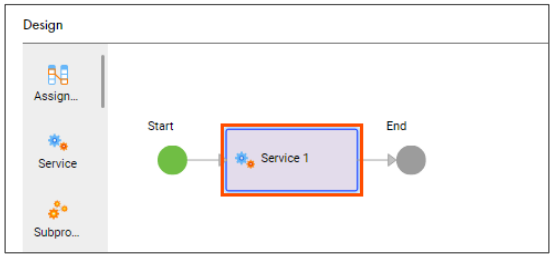
You must configure the following assignment properties:

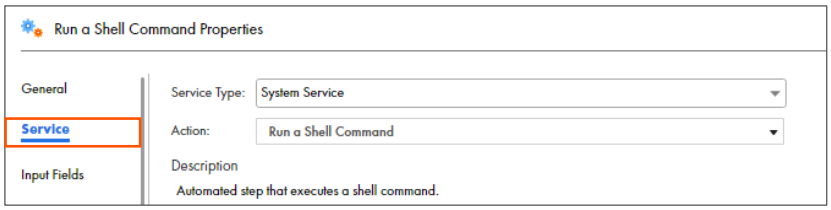
1. **Field**: Name of the field you want to configure. To select the field, use the Add Field option. The fields listed are the fields created in the Start step.
2. **Assigned Using**: Method using which the field is assigned value. The options in Assigned Using drop-down change according to the data type of the field you selected to configure. For example, if the data type of the field is configured as Custom Type, then you see Content, Field, or formula in the Assigned Using drop-down.
3. **From:** Defines the value or expression used to calculate the value for the field. For example, if the Assigned Using option is selected as Formula then, the From field contains the formula used to calculate the value for the field selected for assignment.

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Service Step

- Service step calls a service in a process
- Service type can be
  - Connection
  - Process
  - System Service





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The Service step is used to call a service in the process.

To call service in the current process, you must have an existing service such as connection, process, or system service configured.

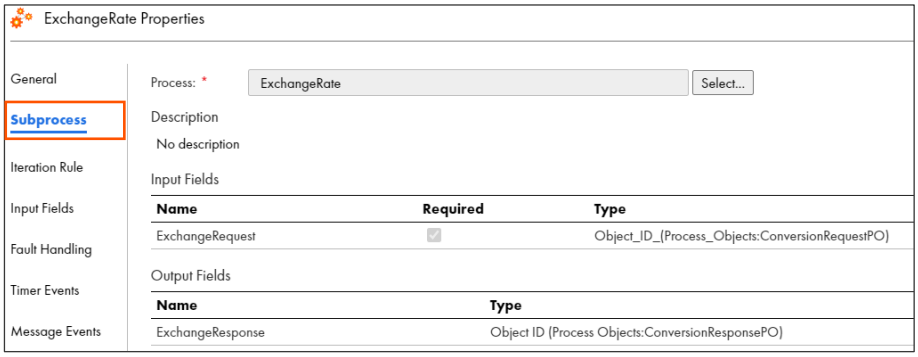
Note: You cannot create a service while designing the process.

The Process automatically creates new input fields for the service that is called by the Service step.

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Subprocess Step

- Subprocess step embeds one process within another process
- You can define the Input and Output associated fields for the Subprocess



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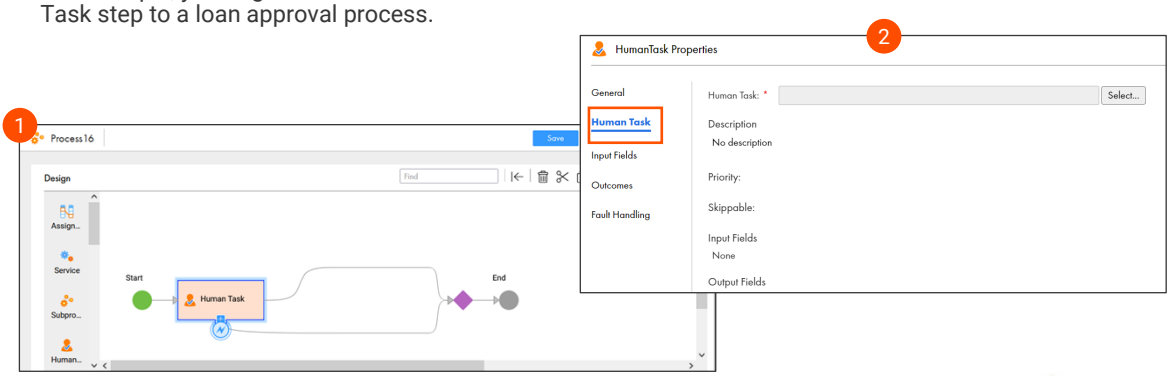
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A Subprocess step embeds one process within another process. When the Subprocess step executes, the embedded process executes instead of the current process. You can define the associated Input and Output fields for the Subprocess step, based on the embedded process.

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Human Task Step

- Use the Human Task step to include a human decision within an Application Integration process.
- A human action is required for approvals and exception management.
  - For example, you might want to add a Human Task step to a loan approval process.



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You need to define the following properties to configure a Human Task step:

General: The name of the Human Task step.

Human Task: Select the human task that you want to include within the Application Integration process.

Input Fields: Enter the values in the input fields for the human task.

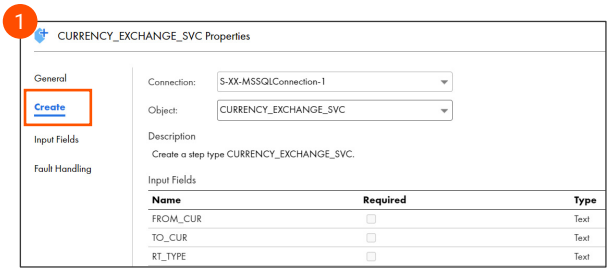
Outcomes: Displays a list containing the outcomes that you defined in the human task asset.

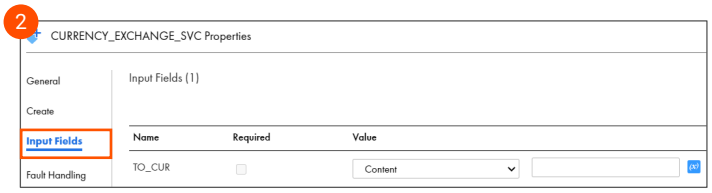
Fault Handling: Select the Catch fields option to enable fault handling for the Human Task step. You can provide a name for the fault handler in the **Fault Field Name** field.

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Create Step

- Create step creates a new object instance.
- To create a new object:
  - From the Create tab, select the **Connection** and **Object** for which you want to create the new object.
  - From the Input Fields tab, select fields to add to a new object.





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The Create step enables the user to create a new instance of the selected object for the selected connection.

To create new objects:

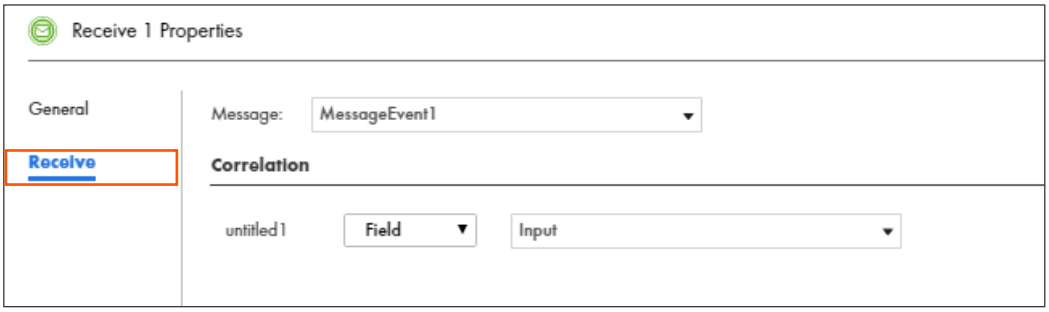
1. Choose the Input Fields tab and select all the fields that you will be adding to the object.
2. For each field name, set its value to one of the following values: Content, Field, or Formula.

For example, if you want to create a new object for Salesforce Account object, then from the Create tab, select Connection as Salesforce. Then, you must select the object as Account. Next, from the Input field tab, configure the fields that you want to keep in the newly created object.

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Receive Step

- Makes a Service Call and waits for an event data before the Process can continue
- Output of a Receive step is available to be used later in the Process



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A Receive step is used in a Process to make a Service call and wait for some event data before the Process can continue.

For example, if the order process includes a step to request more credit information when the order is above a certain threshold, then a Receive step allows you to handle the blocking receive call.

The output of a Receive step is available to be used later in the Process.

When you design a process, you can use message events in connection with the Receive step to interact with a process after the process has started execution. For example, you might want to:

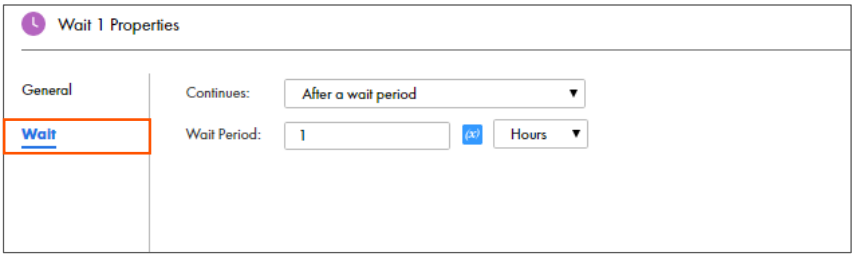
- Query the status of a process
- Update data or provide control to the process

The Correlation field of Receive step enables you to select the fields you are correlating with the incoming message event. You can use these assignments to cancel further execution of a request (interrupting), or to provide the status (non-interrupting).

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Wait Step

- Specifies the amount of time for which the Process will be in a pause state
- You can set the **Continues** field for Process as:
  - At a specific time
  - After a wait period



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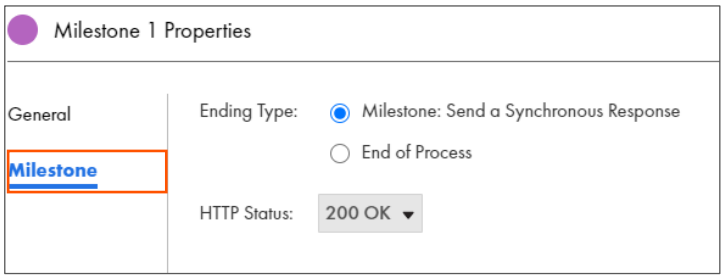
The Wait tab of a Wait Step property determines when and for how long the process pauses. You can choose from the following options:

1. **At a specific time**: Use this option to pause the Process at a particular time. Specify the time you want the process to pause. The delay field is optional. For example, you can set the process to pause at 1:00 am after three days. Here, 1:00 am is the time and three days is the delay.
2. **After a wait period**: Use this option to pause the process after a period. The period begins when the process reaches the Wait step. Enter the wait period that you want the process to pause for. For example, set the process to pause for one hour from the time that the process reaches the Wait step.

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Milestone Step

- Milestone step allows you to specify one of the two actions when the Process ends:
  - Send Synchronous Response
  - End of Process



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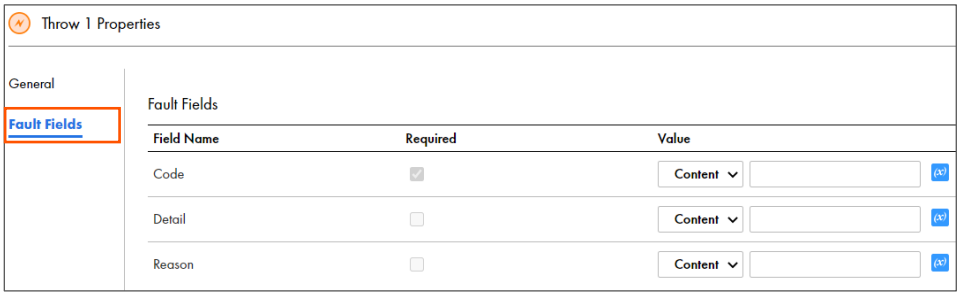
Milestone step allows you to specify one of the two actions when the Process ends.

- **Send Synchronous Response**: Sends a synchronous reply to the original request. This reply has all of the output fields defined in the Process. This is used to send a synchronous message, and then continue execution. If a synchronous reply was already sent, no action occurs. You can have additional steps after a milestone.
  Note: If you insert a Milestone step into a Parallel Paths step, the Ending Type must be set to Milestone.
- **End of Process**: The process ends without any further action.

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Throw Step

- Throw step helps in constructing an error payload
- You can set the following details for fault handling purposes:
  - Code
  - Detail
  - Reason



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A Throw step helps in constructing an error payload.

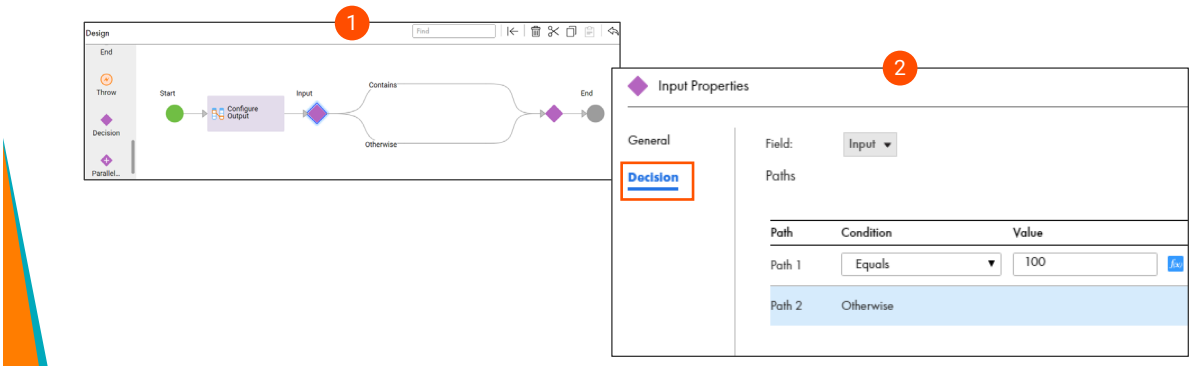
You can set the following details for fault handling purposes:

1. **Code**: A string value that contains the name of the fault info payload. This is a mandatory field.
2. **Detail**: A string value that contains the fault details. This is an optional field.
3. **Reason**: This field specifies the reason for the fault. This is also an optional field.

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Decision Step

- Decision step allows the process to take one of the defined paths
- The path selection is based on the value specified for a field
- You can add multiple conditions to a Decision step



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Decision step allows the process to take one of the defined paths. The process takes the path based on the condition defined. So, it is important to ensure that you construct paths with non-intersecting conditions. The conditions available depend on the field that you select.
For example, if you select a field of type Simple > Text, the following conditions are available:

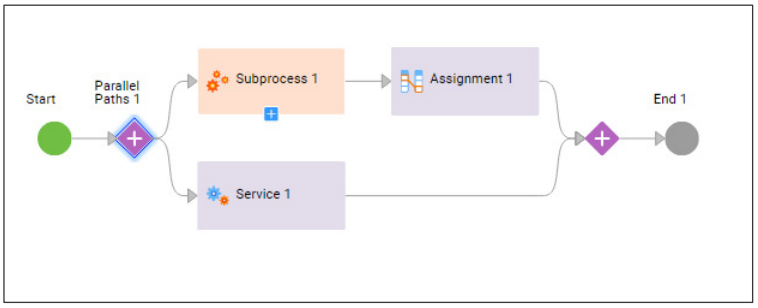
- Equals
- Starts With
- Ends With
- Starts with any of
- Contains.

You can add multiple conditions to a Decision step. For each path that you add, a corresponding branch appears on the UI. Drag branches to rearrange the order in which the branches appear on the UI. Most Decision steps have an Otherwise path. This path handles execution if no data meets the conditions in your tests.
Note: The process evaluates conditions in a top-down manner. Ensure that the Otherwise branch is the last path.

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Parallel Paths Step

- Every branch defined within the step executes at the same time
- There is no evaluation criteria specified
- You can add multiple steps to each branch



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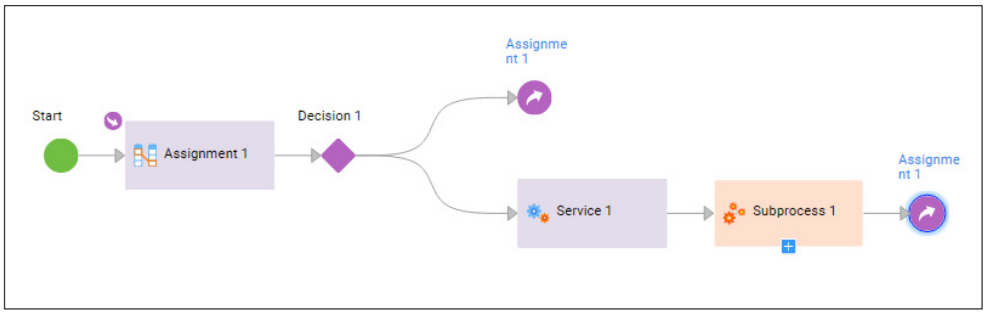
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In the Parallel Paths step, every branch defined within the step executes at the same time. All branches of a parallel path can execute simultaneously, as there is no evaluation criteria specified. You can add multiple steps to each branch. To add steps to a branch, drag and drop a step from the palette onto the canvas.

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Jump Step

- Jump step defines the target, where a step can jump to
- Multiple steps can jump to the same target step



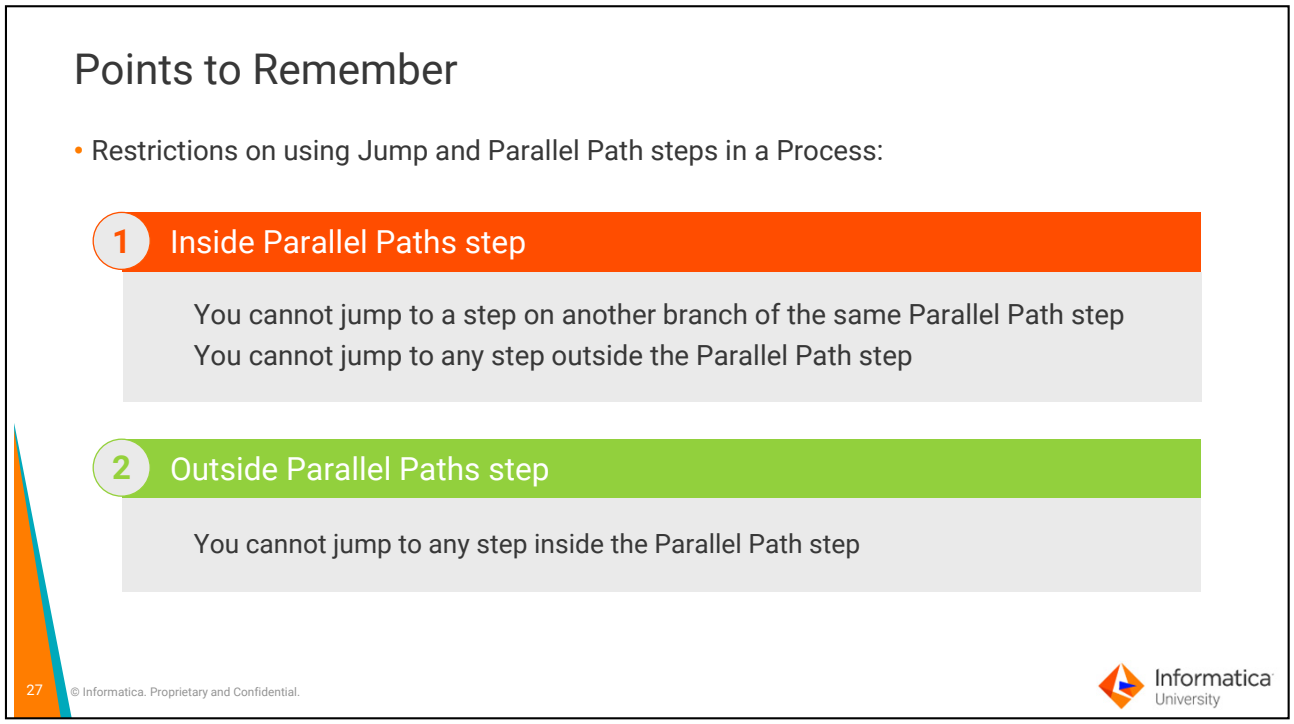
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A Jump step jumps to another step or target step. Multiple steps can jump to the same target step. To see how many Jump steps have a particular step as their target, place the cursor over the arrow next to the target step.

In the image shown above, both the jump steps have target step as Assignment.

Using a jump step prevents the duplication of the same steps for separate branches.

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When you use the Jump step in conjunction with the Parallel Path step, you can only jump to another step on the same Parallel Path branch.

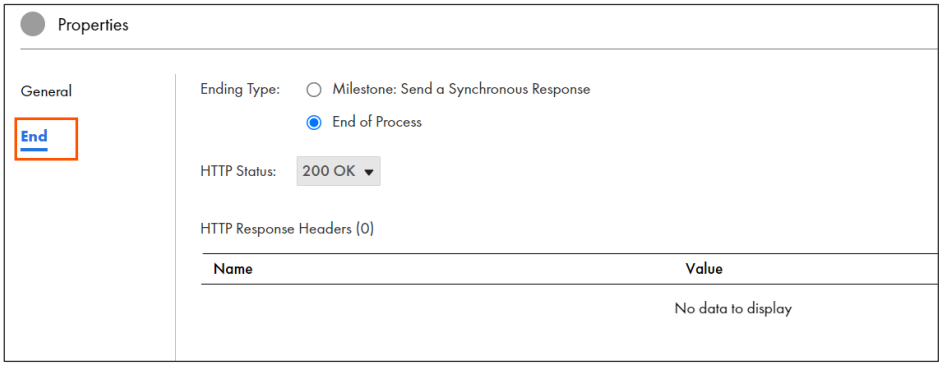
Following restrictions apply when you use the Jump step and the Parallel Path step together in a Process:

1. If you are in a Parallel Path step, you cannot jump to a step on another branch of the same Parallel Path step.
2. If you are in a Parallel Path step, you cannot jump to any step outside the Parallel Path step.
3. If you are outside a Parallel Path step, you cannot jump to any step inside the Parallel Path step.

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End Step

- Indicates the end of the Process
- When execution reaches this step, the Process completes

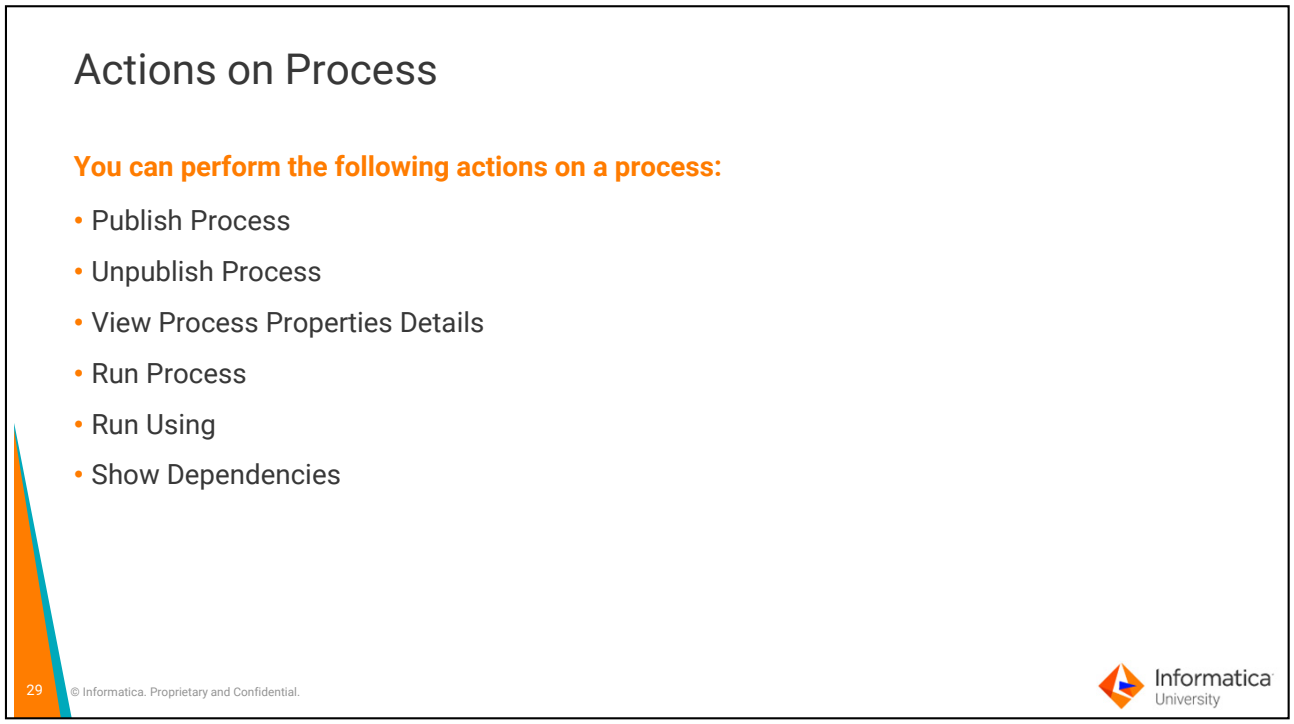


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An End step indicates the end of the process. When execution reaches this step, the process completes. For the End step, the ending type is set to End of Process by default. You cannot edit this value. The HTTP Status response status code is set to 200 OK by default. You can edit this value.

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After you design and save a process, you can perform any one of the following actions on the process:

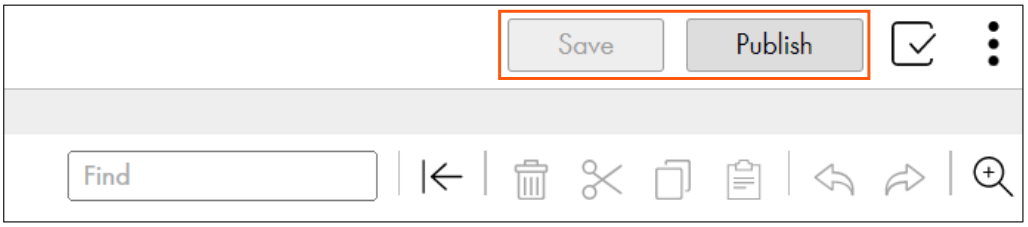
1. Publish Process
2. Unpublish Process
3. View Process Properties Details
4. Run Process
5. Run Using
6. Show Dependencies

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Publish Process : Method 1

Method 1

- In the Process Designer page, click **Save** to save the process, and then click **Publish** to publish the process.



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To generate the process endpoints, you must publish the Process.

In CAI, you can publish a process in two ways:

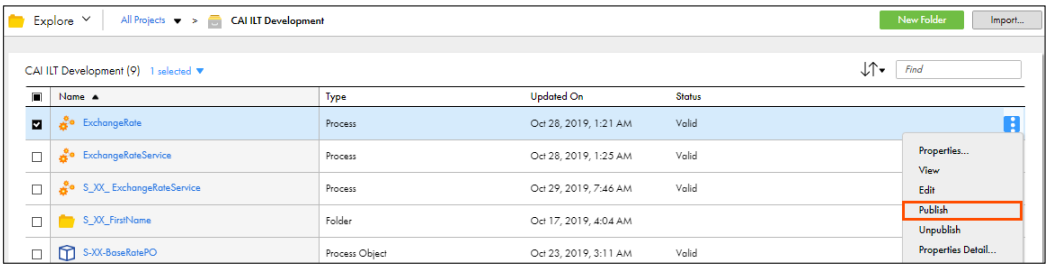
Method 1: In the **Process Designer** page, click Save and then click **Publish**.

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Publish Process: Method 2

Method 2

- From the **Explore** page, select the Process, click **Actions**, and select **Publish**.



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Method 2: You can also publish processes and view service URLs by clicking the Actions icon next to a process on the Explore page.

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Similar to publishing the process, you can unpublish the process in following two ways:

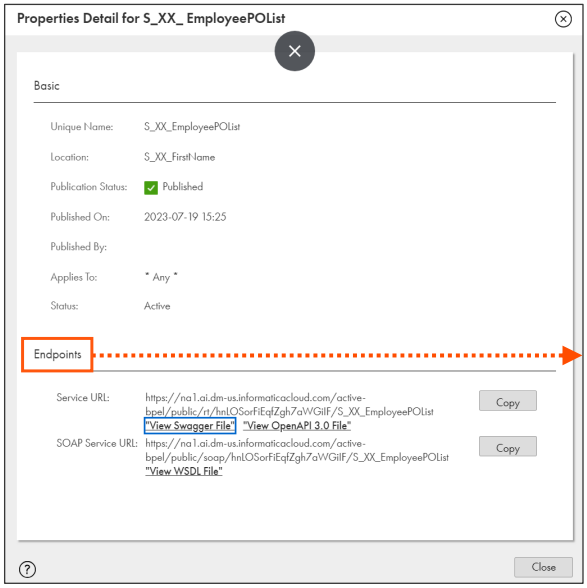
1. From the Process Designer Actions menu, select Unpublish.
2. From Explore page, select the process and from the Actions menu, select Unpublish.

In CAI, you must Unpublish a process when you want to perform the following actions on the process:

1. Rename an already published process.
2. Edit a published process.

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Process Properties Detail



View Process Endpoint

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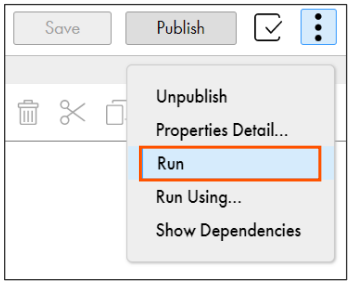
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You can view the SOAP or REST endpoints generated for a process in the Properties Details page of the Actions menu.

Run a Process

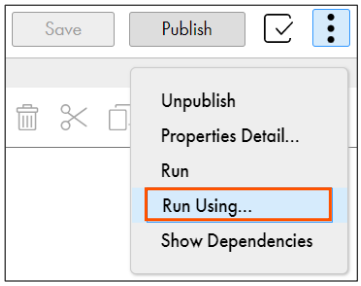
Run Option

- From the **Actions** menu, select **Run**. This option runs the process with the last successfully run process input.



Run Using Option

- From the **Actions** menu, select **Run Using**. This option allows you to run a process with a specific process or all the saved process inputs.



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1. To execute the process, you can use the Run option to run the process with the last successfully run process input. The Process Execution Status page displays the details of the successful and unsuccessful process executions.

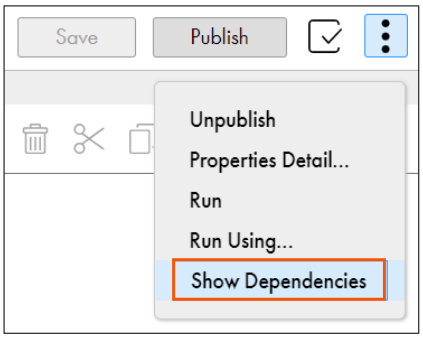
If you have not created process inputs or run the process before, the Test Process Input Collection page opens where you can create process inputs and run the process.

2. To run a process with a specific process input or all the process inputs, select the Run Using option. Specify and save the input and run the process.

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Show Dependencies

- Before you publish, unpublish, or delete an asset, you must check the object dependencies.
- You can also view object dependencies for a process, guide, app connection, service connector, and process object from the corresponding asset designer page.
- Open the asset, click Actions, and click **Show Dependencies**.



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Before you publish, unpublish, or delete an asset, you must check the object dependencies. To view object dependencies for an asset, in the row that contains the asset, click Actions and select Show Dependencies. The Dependencies page opens showing the Uses tab by default. The Uses tab lists the objects that the selected asset uses.

For example, a process might use another process. The Used By tab lists the objects that use the selected asset. For example, a connection might be used by one or more processes.

You can view object dependencies for a process, guide, app connection, service connector, and process object from the corresponding asset designer page.

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Topic

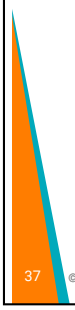
Knowledge Check

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Steps are used to guide a user through a process.

- True
- False

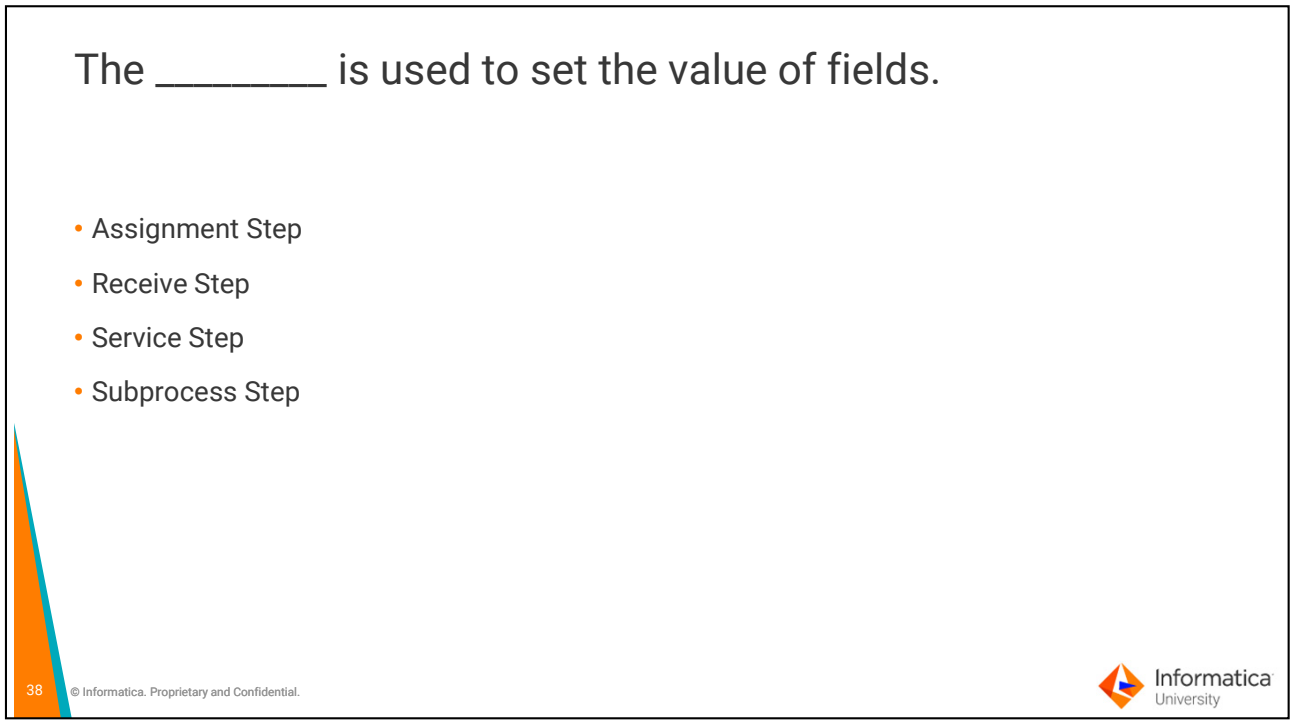


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Answer: True

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Answer: Assignment Step

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Module Summary

This module showed you how to:

- Describe a process
- Discuss Process Designer
- Identify process steps in Process Designer

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Module Objectives

After completing this module, you will be able to:

- Create a basic process to display static message
- Create a basic process to display user input

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Topic

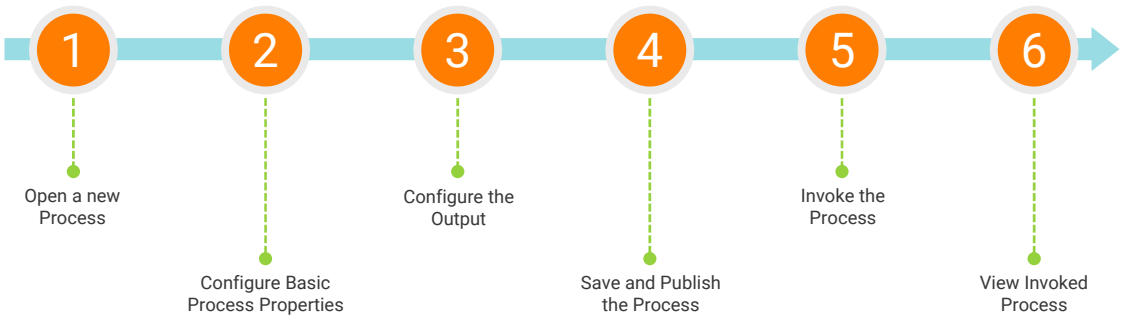
An Example – Display Static Message



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A Simple Process

A simple process that prints “Hello World”



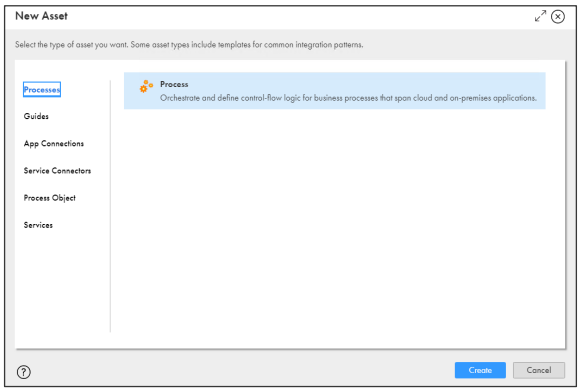
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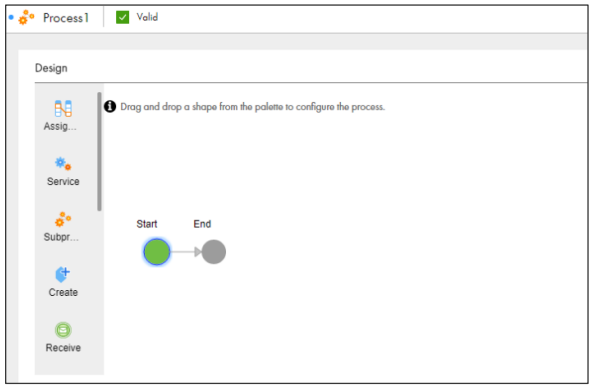
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Example: A Simple Process

Step 1: Create a Process





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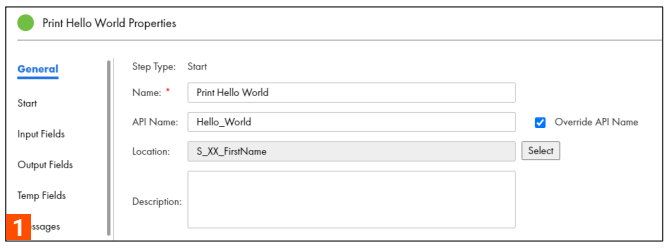
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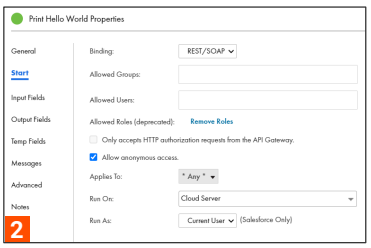
In the Application Integration service, create a new process. The Start and End steps are included by default.

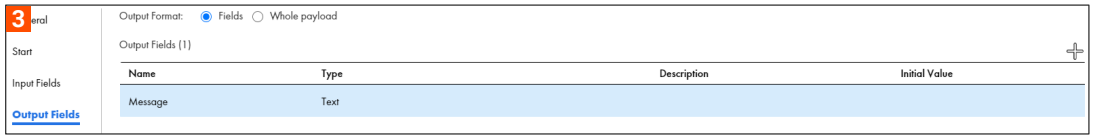
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Example: A Simple Process (continued)

Step 2: Configure the Basic Process Properties







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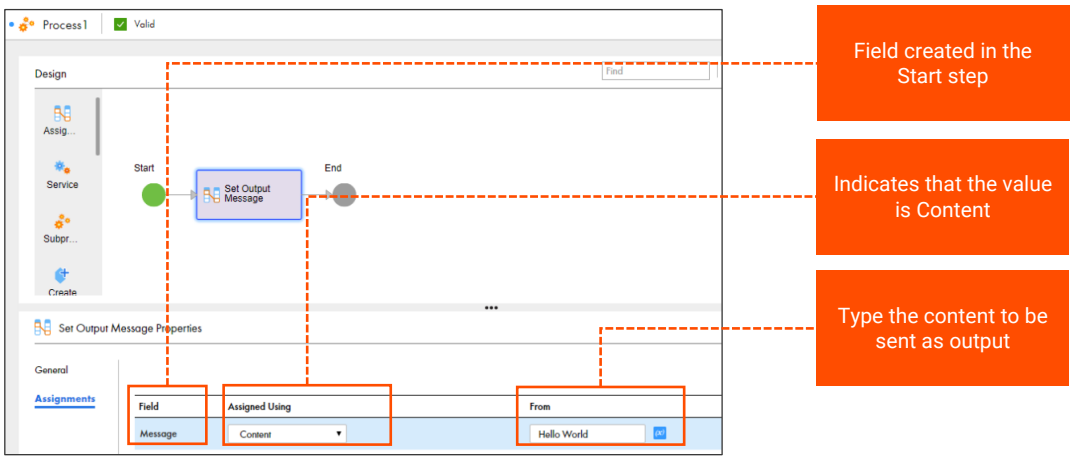
Configure the Start step.

1. Provide the General property details such as Name, API Name, Location, and Description.
2. In the Start tab:
   a. Select the Binding property as REST/SOAP.
   b. Enable Allow Anonymous Access. This ensures that all the users can access the process.
   c. Select Cloud Server as the Run On option
3. Configure an output field such as ‘Message’ of type ‘Text’.

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Example: A Simple Process (continued)

Step 3: Configure the Output



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To configure the output, add an assignment step between the Start and End steps.

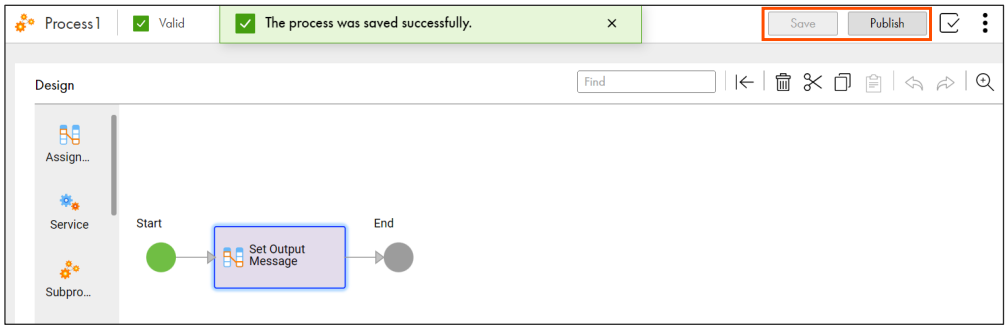
For the assignment step:

1. Provide a name for the step.
2. In the Assignments tab, add a field.
3. Select the field that you created in the Start step.
   For example, the 'Output' field.
4. Select the Assigned Using field as Field, and in the From field, select Input.

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Example: A Simple Process (continued)

Step 4: Save and Publish the Process



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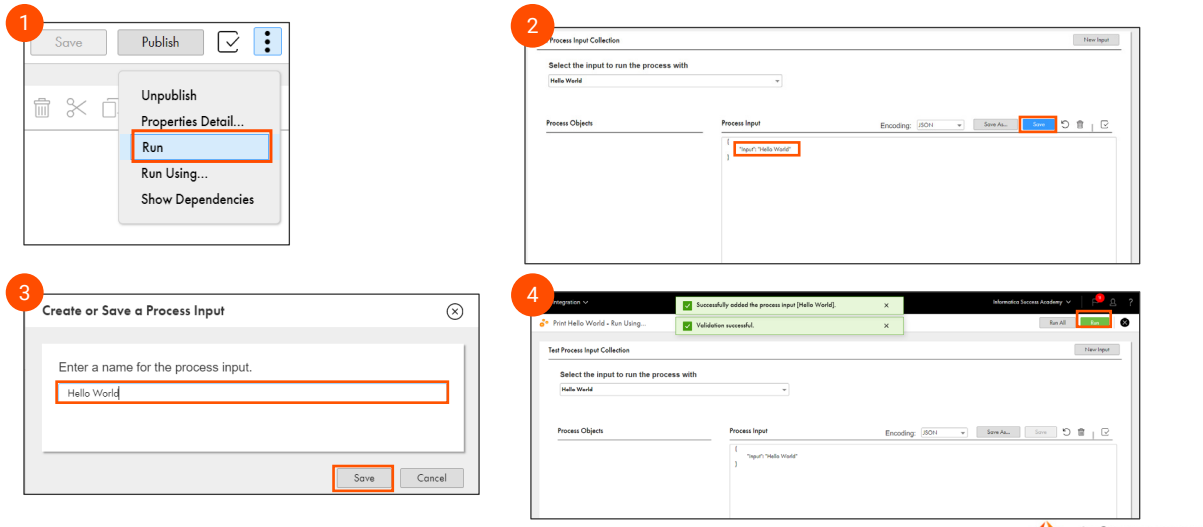
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Save and publish the process.

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Example: A Simple Process (continued)

Step 5: Invoke the Process



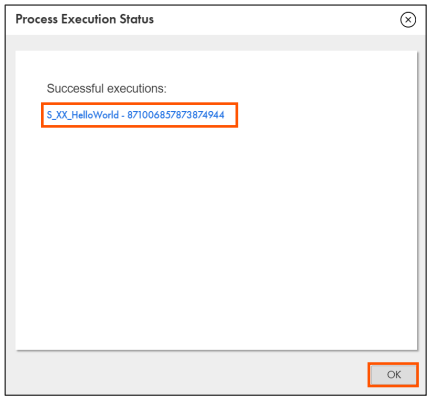
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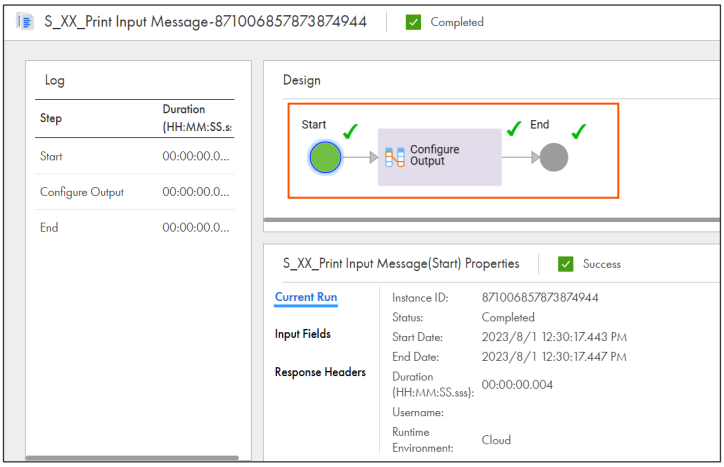
Run the process. In the Process Input section, enter the input as ‘Hello World’. The Input entered here will be available as the output.

Enter the process input name and save it. Click **Run**.

Example: A Simple Process (continued)

Step 5: Invoke the Process (continued)





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To check the Process Execution Status, click on the Process Execution link, and click **OK**.

This directs you to Application Integration Console to view the executed process.

You have now successfully created a basic process and invoked it from the interface.

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Topic

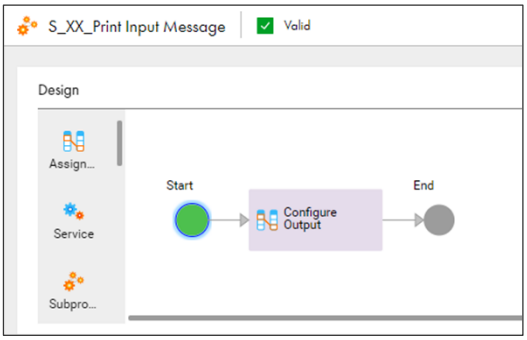
An Example – Display User Input

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Create a Basic Process to Display User Input

1. One Input Field of type Text
2. One Output Field of type Text
3. Use Assignment step to configure the field





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To create a basic Process that accepts text as an input and returns text as an output, follow the steps mentioned below:

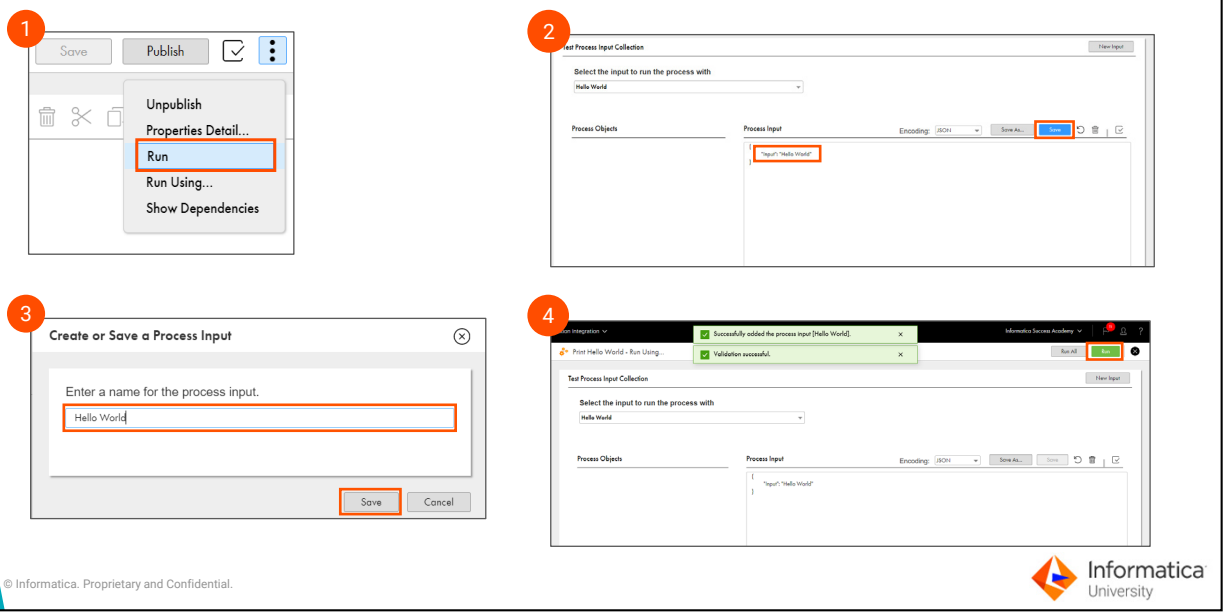
1. To invoke the process using a third-party application, create a Process with binding as REST/SOAP.
2. To accept text input for the service, create an input field with type as Text.
3. As you want to print the accepted input as output, you need to create an output field with type as Text.
4. Now, you will use an Assignment step to set the output field value to input field value.

Note: To provide an input to the process at runtime, you have configured the assigned using as 'field' in assignment step. It takes values from the input field and displays it as an output.

5. To verify the Process configuration, validate and publish the Process.
6. After the process is published, run the process and select the **Advanced View** to check the output.

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Create a Basic Process to Display User Input (continued)



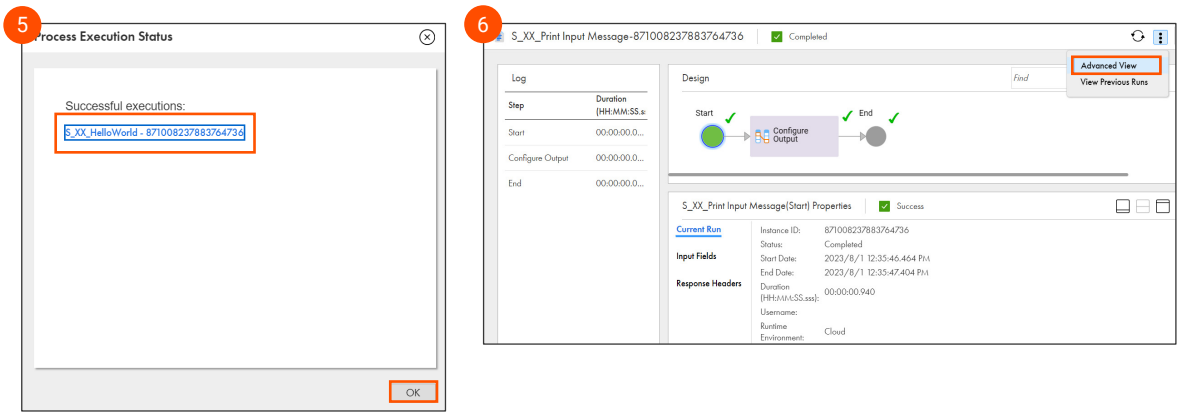
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Run the process. In the Process Input section, enter the input as ‘Hello World’. The Input entered here will be available as the output.

Enter the process input name and save it. Click **Run**.

Create a Basic Process to Display User Input (continued)



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To check the Process Execution Status, click the **Process Execution link** and click OK.

This directs you to Application Integration Console to view the executed process.

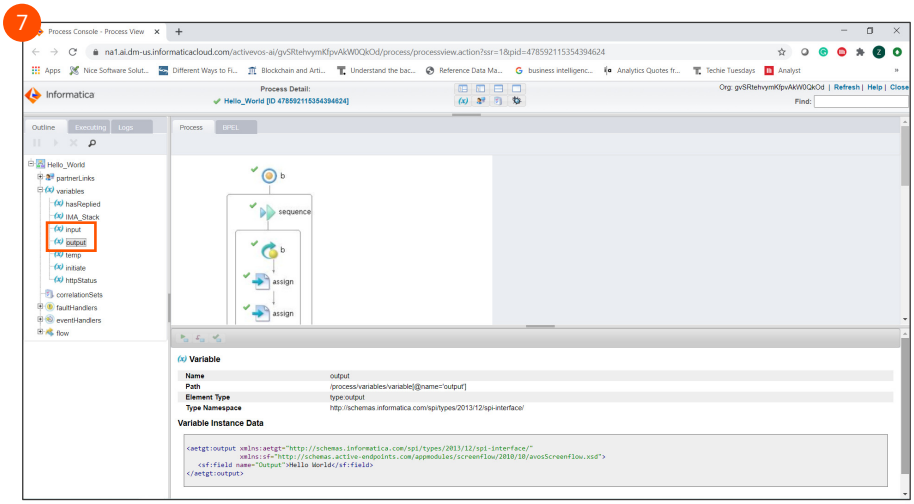
To check the output in detail, from the Actions menu, select **Advanced View**.

You have now successfully created a basic process and invoked it from the interface.

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Create a Basic Process to Display User Input (continued)

- Process Input and Output:



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To view the output, expand variables and click **output**.

Topic

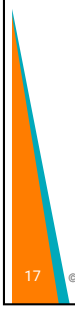
Knowledge Check



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Can you use a Process without publishing it?

- Yes
- No



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Answer: No

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Assignment step takes values from the input field and displays it as an output.

- True
- False

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Answer: True

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Lab Activity

3-1 Create a Basic Process to Display User Input

In this lab, you will perform the following:

- Create a basic Process
- Invoke the Process

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Module Summary

This module showed you how to:

- Create a basic process to display static message
- Create a basic process to display user input

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Module Objectives

After completing this module, you will be able to:

- Describe and Create Process Objects

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Topic

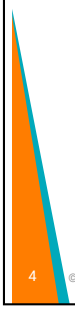
Design Process Objects

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Process Object

- Is used to group data and create a structured object
- Is used to retrieve sets of structured data required to design a business process
- Enforces schema validation



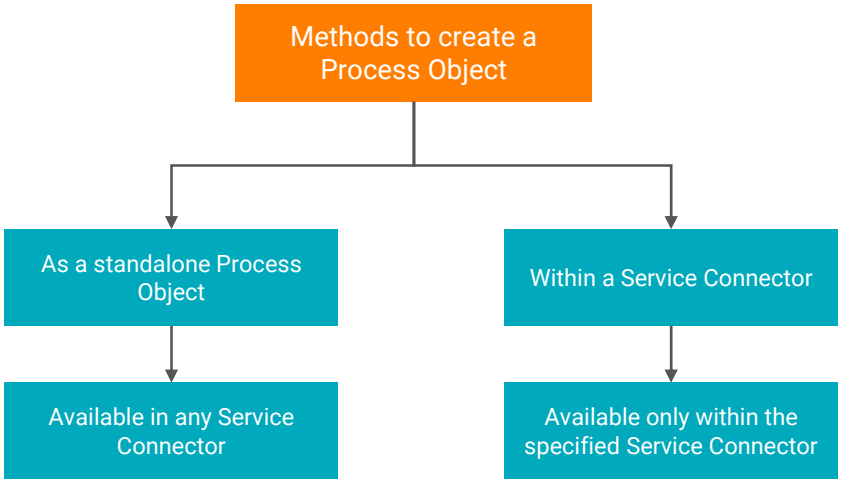
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A Process Object is used to group data and create a structured object. It helps to identify the inputs and outputs for a Process. With Process Objects, you can retrieve sets of structured data required to design a business process. For example, instead of creating separate items for each kind of demographic data, you can create a group that contains the name, address, and phone number. These groups are often called Objects. Process Objects group and structure this data for use in Processes and Guides. One of the advantages of using Process Objects is that it avoids the need of writing XPath to retrieve data from a Process written in XML or JSON format. A Process Object enforces schema validation, which ensures that the input or output data confirms a well-formed XML or JSON Schema.

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Methods to Create a Process Object



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In CAI, you can define Process Objects in two ways:

1. As a standalone Process Object for use in your applications
2. Within a Service Connector to handle data returned from a service

Process Objects generated in a Service Connector are available only within the Service Connector.

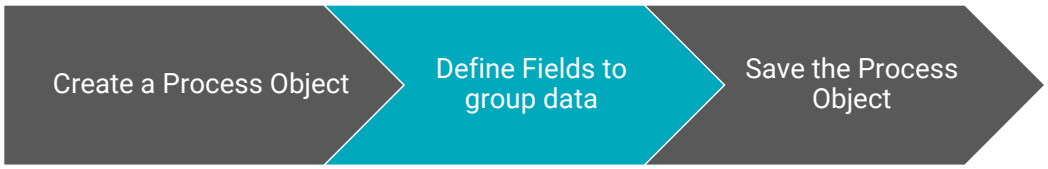
Standalone Process Objects are available in any Service Connector. Both are used in a similar manner in processes. Using the Process Object editor, you can define Standalone Process Objects. They are used in the same way as built-in data types such as integer, text, or phone.

In the Service Connector editor, you can generate Process Objects based on the data available from the service and the actions, bindings, and other properties that you specify.

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Example: Create a Process Object to Group Data

- To group data of Integer and Text data type, perform the following steps:



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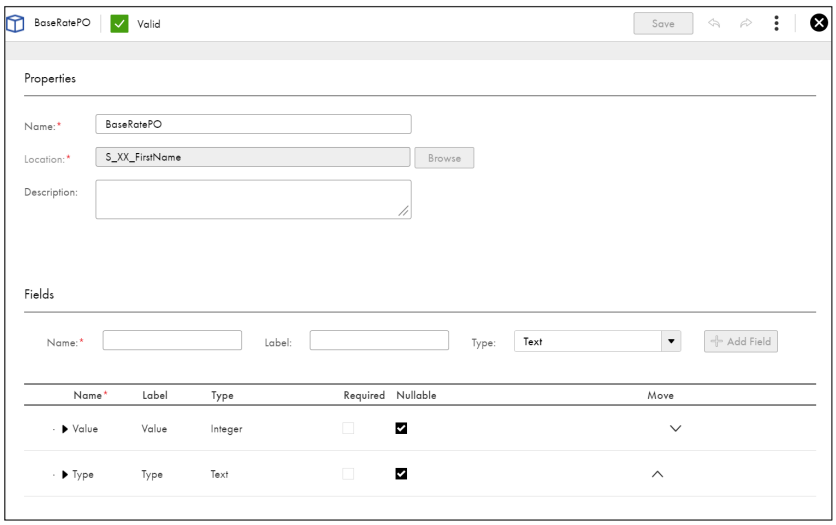
To create a Process Object in CAI, you must perform the following steps:

1. In the Application Integration service, create a new Process Object. In this example, you will create a Process Object named BaseRatePO to group integer and text data.
2. In the Properties section of Process object, provide the name and location to save the Process Object.
3. In the Fields section, provide the name and type of the field you want to group.
4. In this example, you will create 2 fields: Value and Type.
5. The Value field will store data of Integer data type and the Type field will store data of Text data type.
6. Save the Process Object to use it in Processes and Guides.

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Example: Create a Process Object to Group Data (continued)

BaseRatePO Process Object – Configuration



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Knowledge Check

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With Process Objects, you can retrieve sets of structured data required to design a business process.

- True
- False

9

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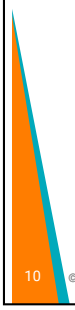
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Answer: True

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Process Objects generated in a Service Connector are available only within the Service Connector.

- True
- False



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Answer: True

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Lab Activity

4-1 Create Process Objects

In this lab, you will perform the following:

- Create a Process Object – BaseRatePO
- Create a Process Object – ConversionRatePO
- Create a Process Object – ConversionRequestPO
- Create a Process Object – ConversionResponsePO

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Module Summary

This module showed you how to:

- Describe and Create Process Objects

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Module Objectives

After completing this module, you will be able to:

- Define the Connectors
- Describe Service Connectors
- Create Service Connectors using different methods

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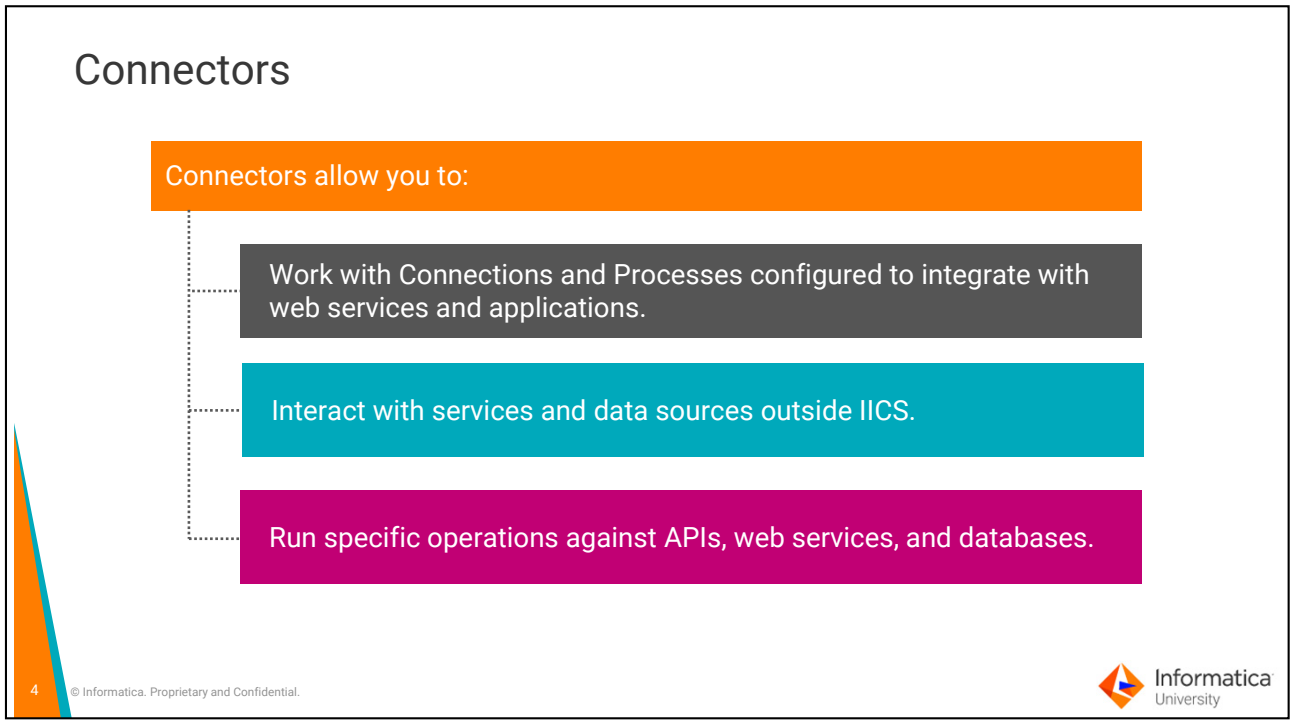
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Topic

Connectors

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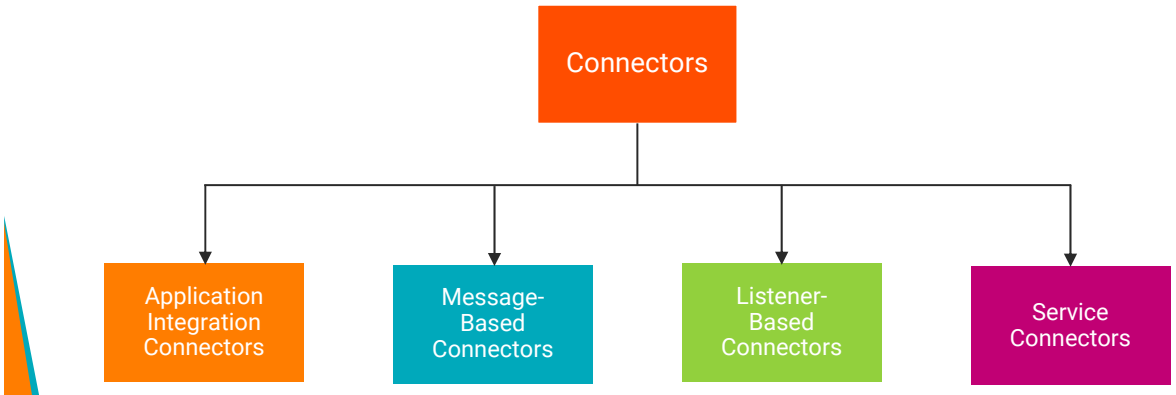


Informatica Application Integration service offers multiple connectors that are designed to work with the App Connections and Processes you configure for integration with web services and applications. All the supported connectors allow you to interact with services or data sources outside IICS. You can also run specific operations against an API, web service, or database using app connectors.

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Types of Connectors

- You can use the following types of connectors:



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In CAI, you can use the following types of connectors:

Application Integration Connectors are designed to access data to and from the cloud for each of your applications. Connectors for JDBC, Workday, SAP, OData, and Salesforce are in this category.

Message-Based Connectors are designed so you can configure queue-based message brokers like ActiveMQ and JMS. The AMQP and Amazon SQS connectors are in this category.

Listener-Based Connectors are designed to perform the following tasks:

- Monitor file-based systems for files or objects on a file system or other type of storage. You can retrieve files and process the contents of the files or perform file operations like moving or reading file metadata.
- Access event services to perform tasks like reading XML from a process object and creating comma-delimited files or reading binary files from a process stream and writing that binary content to the target file system.

Service connectors are designed with Process Designer to specify the parameters and actions that associate with a specific service that you want to make available in a process.

Note: In this course, we will focus on Service Connectors.

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Topic

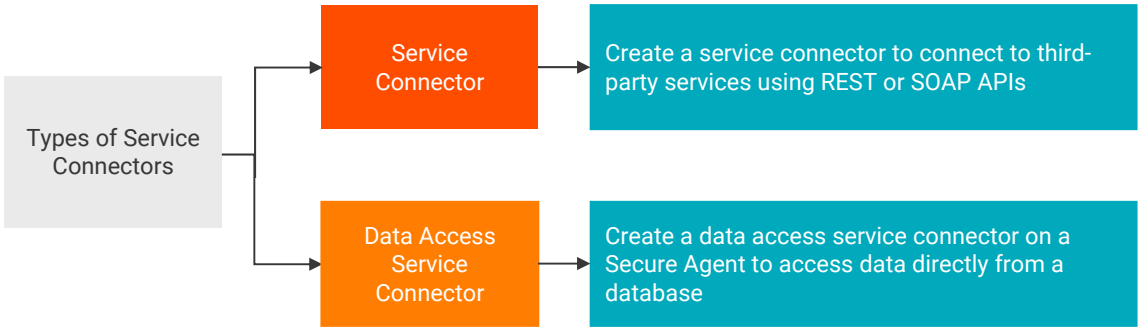
Service Connectors

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Service Connectors

- In CAI, you can create two types of service connectors:



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You can use Process Designer to interact with programs and services that are in Data Integration, inside your firewall, or anywhere in the cloud, as long as they have an API. For example, Google provides many APIs to access the services that Google provides.

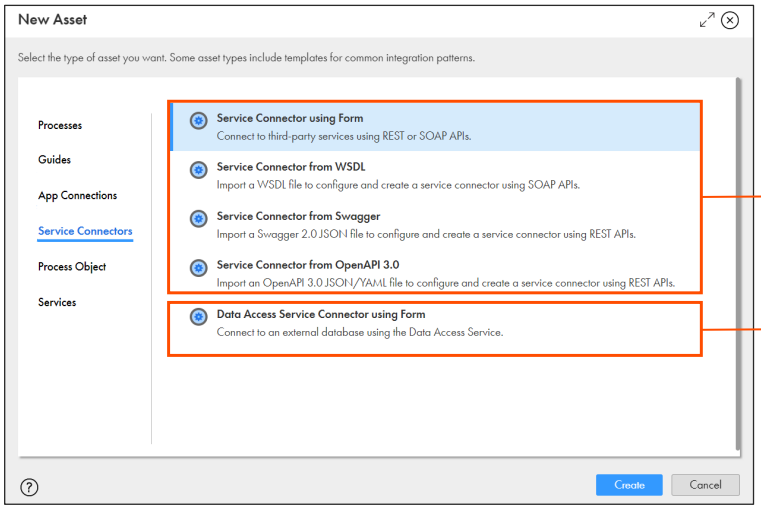
You can create two types of service connectors:

Service Connector: You can create a service connector to connect to third-party services using REST or SOAP APIs.

Data Access Service Connector: You can create a data access service connector on a Secure Agent to access data directly from a database.

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Service Connectors (continued)



These options are used to create Service Connectors

This option is used to create Data Access Service Connectors

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To Create a Service Connector, you can select one of the following options:

- Service Connector using Form
- Service Connector from WSDL
- Service Connector from Swagger
- Service Connector from OpenAPI 3.0

To create a Data Access Service Connector, you can select **the Data Access Service Connector using Form** option.

Note: You will learn more about **Data Access Service Connector**, in the next module (Module 6: Data Access Service Connectors).

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Methods to Create a Service Connector

- Use one of the following methods to create a service connector:

Create a new service connector manually

Create a service connector by importing a File (WSDL, Swagger JSON, or OpenAPI 3.0 JSON/YAML file)

Download a service connector from GitHub

Download a service connector from Informatica Marketplace

Download a service connector from Administrator

9

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In CAI, you can use one of the following ways to create a service connector:

- Manually create a service connector
- Generate a service connector by importing a WSDL, Swagger JSON, or OpenAPI 3.0 JSON or YAML file
- Download a service connector from GitHub
- Download a service connector from Informatica Marketplace
- Download a service connector from Administrator

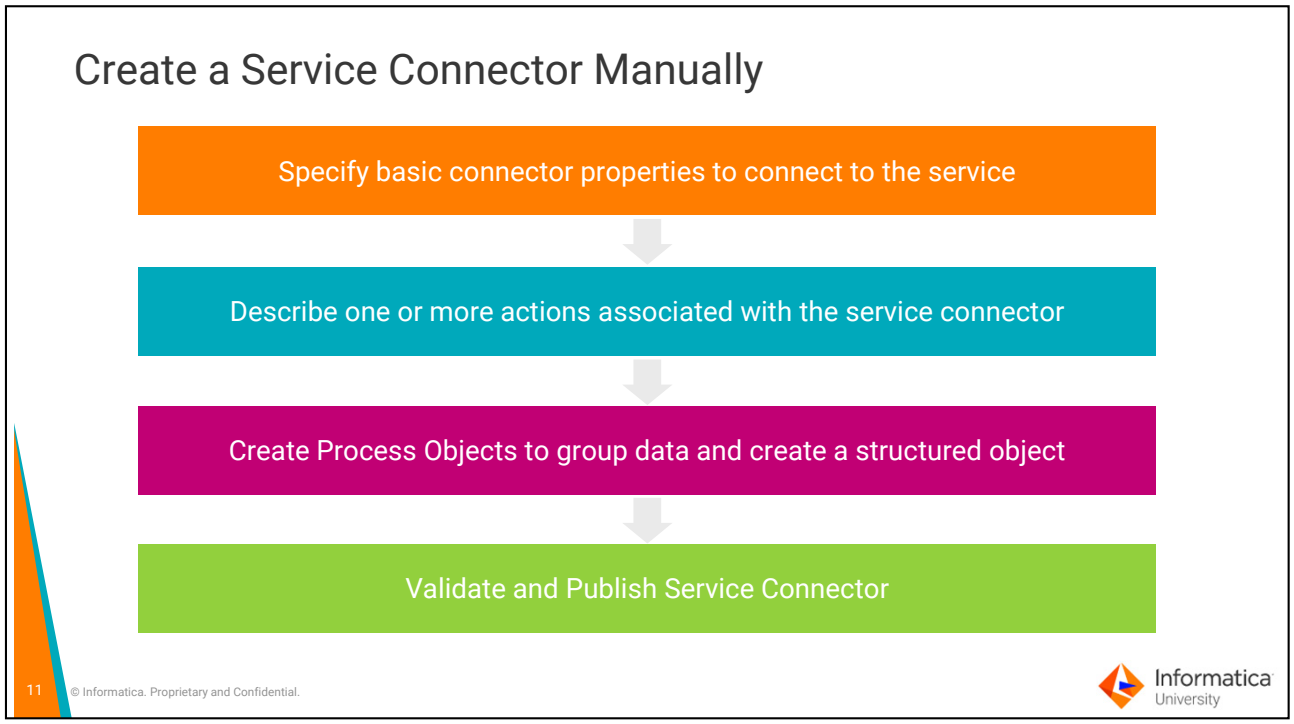
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Topic

Create a Service Connector Manually

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When you create a service connector in CAI, you have an option to specify the connection properties, describe actions associated with a service connector, and create process objects.

To create a service connector, follow these steps:

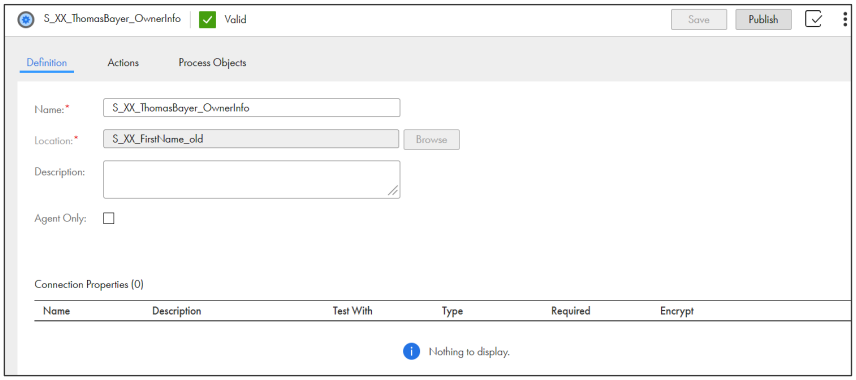
1. Login to your IICS Org and select the **Application Integration** service.
2. From the Application Integration homepage, to open the New Asset dialog box, click **New**.
3. From the list of Assets, select Service Connector, and click **Create**.

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Create a Service Connector Manually (continued)

Definition

- Define the connection properties in the Definition tab of the Service Connector
- Specify the following properties for the Service Connector:
  - Name
  - Location
  - Description
  - Agent Only
  - Connection Properties



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Name: Name of the service connector. It is used to identify the service connector for use in a Process.

Location: Name of project or folder to save the service connector.

Description: Enter a description for the service connector.

Agent Only: Check this if you want the service connector to be available to run on the Secure Agent.

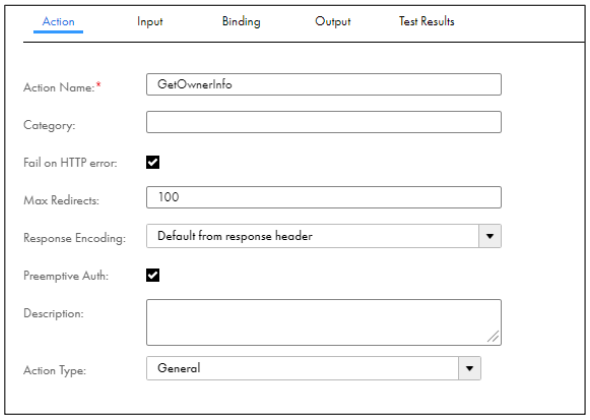
Connection Properties: Enter the properties required to connect to this service. In connection properties, you can define the property name, value to test the defined connection property, make the field mandatory, and so on.

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Create a Service Connector Manually (continued)

Actions

- **Action tab:** Used to define actions associated with service connector
- **Input tab:** Used to define input data items for service connector
- **Binding tab:** Used to specify interface and parameters required for communication
- **Output tab:** Used to define the way service connector will parse data and place it in variables
- **Test Results tab:** Used to test the service connector



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Action tab: You can define one or more actions associated with a service connector in this tab.

For each action, you can specify additional details in the following tabs:

Input tab: Use the Input tab to define input data items that are unique to the service to which you are sending data.

Binding tab: For each action in a service connector, the Binding options enable you to specify the interface and parameters required to communicate with a service.

Output tab: Use the Output tab to define how the service connector will parse data returned from service and place it into variables. Typically, the returned data is XML or JSON converted to XML. You can map this XML to the output fields in two ways:

1. Directly map the data to a property based on XML tag names or JSON properties.
2. Use Expression to extract elements.

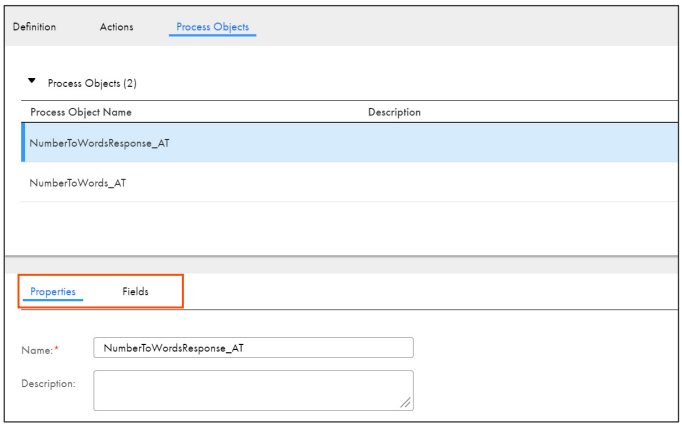
Test Results tab: Use the Test Results tab to test the service connector.

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Create a Service Connector Manually (continued)

Process Objects

- Define the process objects to group data in the Process Objects tab of Service Connector
- Specify the following properties for the Service Connector:
  - Properties
  - Fields



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In the Process Objects tab, you can define one or more process objects for a service connector, to group data and create a structured object.

You can associate each process object with one or more data elements returned by the service.

In the Properties tab, specify the Name and Description for each process object.

In the Fields tab, specify the Field Name and Type to select the data type of the value being written to the variable.

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Validating and Publishing Service Connector

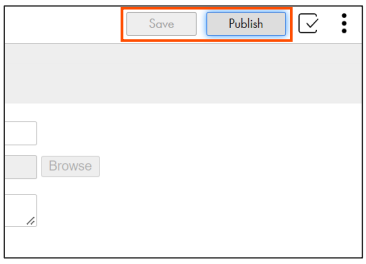
Validating

- Validation panel lists the error details
- Errors are grouped based on the tabs
- You cannot publish a service connector that contains errors



Saving and Publishing

- You must save a service connector before you can publish it
- You must publish the service connector to use it in a Process



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When you create a service connector, you can use the Validation panel to verify if there are validation errors and view the error details. Errors are grouped based on the tabs where they occur. The error list in the Validation panel is refreshed after the error is resolved.

To use a service connector in a Process, you must save and publish the connector. Use the Save option to save the service connector configuration. After the connector is saved, the option to Publish the connector is enabled under the Actions icon. To disable a published service connector, you must unpublish the service connector. After you make the necessary changes, publish the service connector again for the changes to get reflected.

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Topic

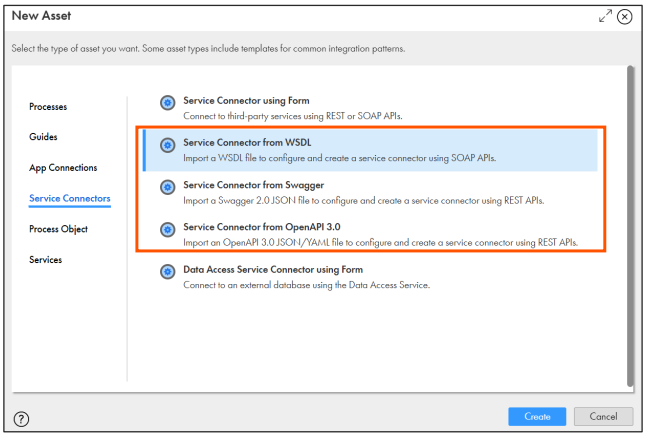
Create a Service Connector by Importing a File

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Create a Service Connector by Importing a File

- To create a Service Connector, you can import:
  - Web Service Definition Language (WSDL) file
  - Swagger 2.0 JSON file
  - OpenAPI 3.0 JSON or YAML file



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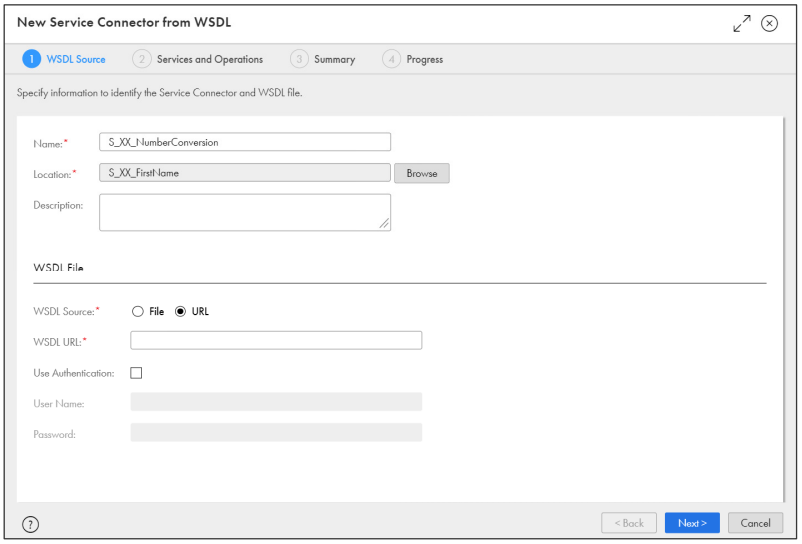
To create a Service Connector, you can import a Web Service Definition Language (WSDL), a Swagger JSON file, or an or OpenAPI 3.0 JSON/YAML file.

To create a service connector by importing a file, you must perform the following steps:

1. In Application Integration, click **New**.
2. In the New Asset dialog box, select one of the following options based on the type of file you want to import :
   - Service Connector from WSDL
   - Service Connector from Swagger
   - Service Connector from OpenAPI 3.0
3. Click **Create**.

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Create a Service Connector from WSDL



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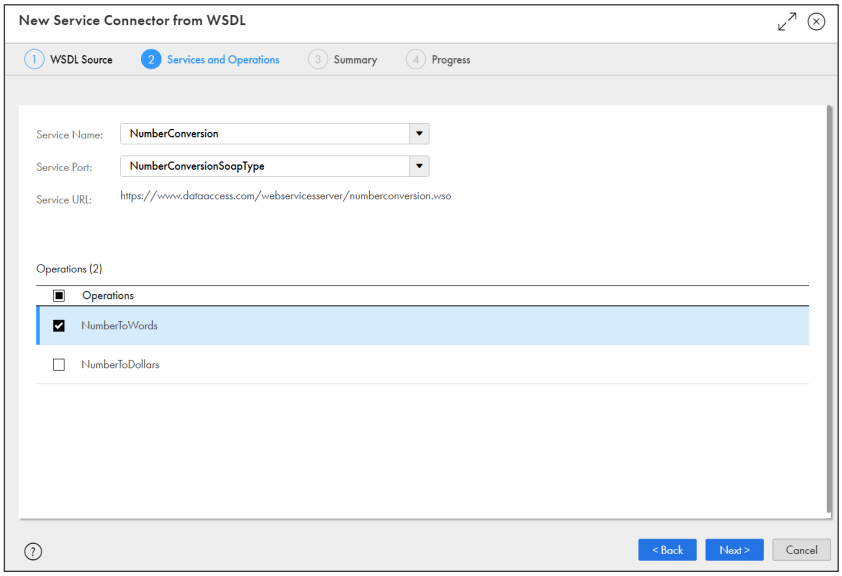
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In the WSDL Source step, provide a name and location to save the connector. You can import a WSDL file in one of the following ways:

1. To use a WSDL file on your local system, select the File option and select the file. You can also select a .zip file that contains a WSDL file along with an XML schema file.
2. To use a URL that contains a WSDL file, select the **URL** option.
3. To proceed to the Service and Operations steps, click **Next**.

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Create a Service Connector from WSDL (continued)



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The **Services and Operations** tab appears.

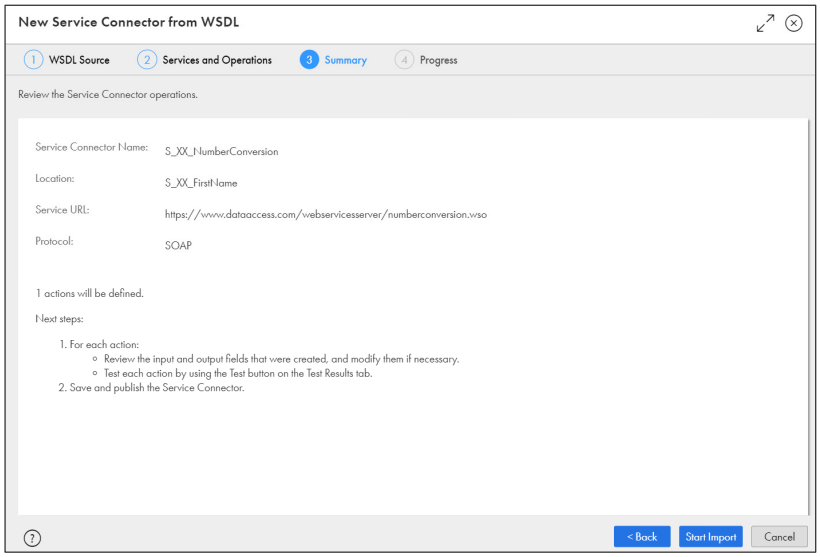
1. In the Service Name field, select the service name of the imported WSDL file.
2. The Service URL field displays the URL where you want the service connector to send requests.
3. In the Service Port field, select the **port** that you want the service connector to use.
4. In the Operations section, select the operations that you want the service connector to provide.

   Note: The operations that you select will be available to you as actions when you create a process and use this service connector.

5. To proceed to the Summary steps, click **Next**.

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Create a Service Connector from WSDL (continued)



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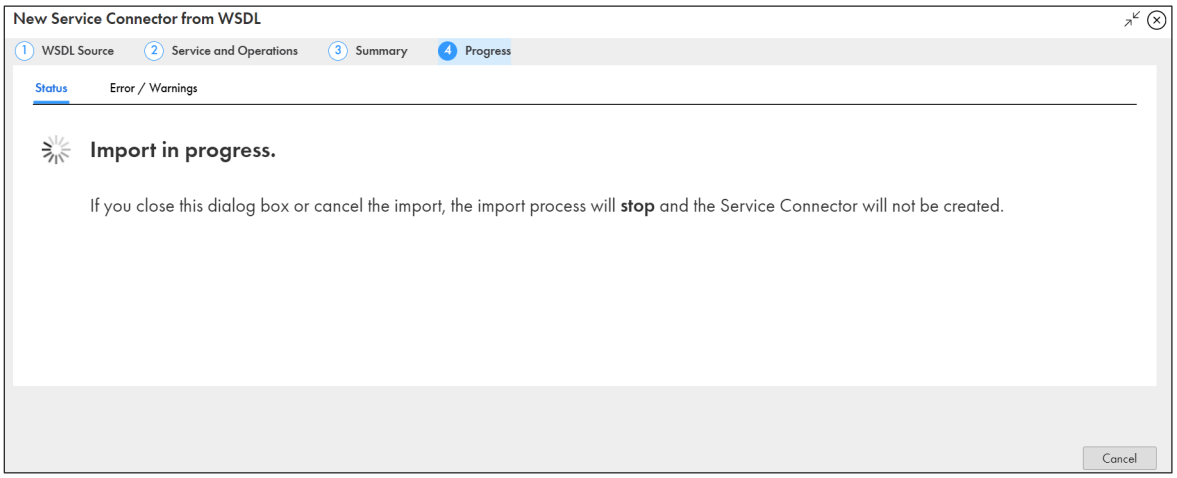
The **Summary** tab appears.

Review the service connector configuration and click **Start Import**.

Application Integration starts importing the WSDL file and displays the status on the **Progress** tab.

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Create a Service Connector from WSDL (continued)



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The **Progress** tab appears.

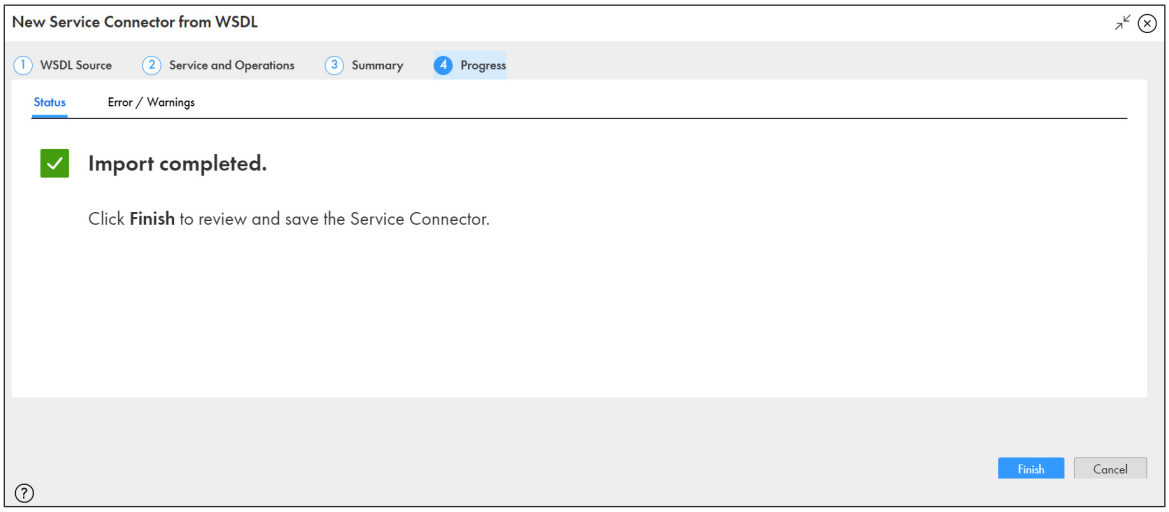
The **Progress** tab indicates whether the import completed successfully, failed, or completed with warnings.

If the import fails or completes with warnings, click the **Error/Warnings** tab to understand why the import did not complete successfully. You can fix the issues and import the WSDL file again.

Note: If you close the dialog box or cancel the import, the import process stops and the service connector is not created.

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Create a Service Connector from WSDL (continued)



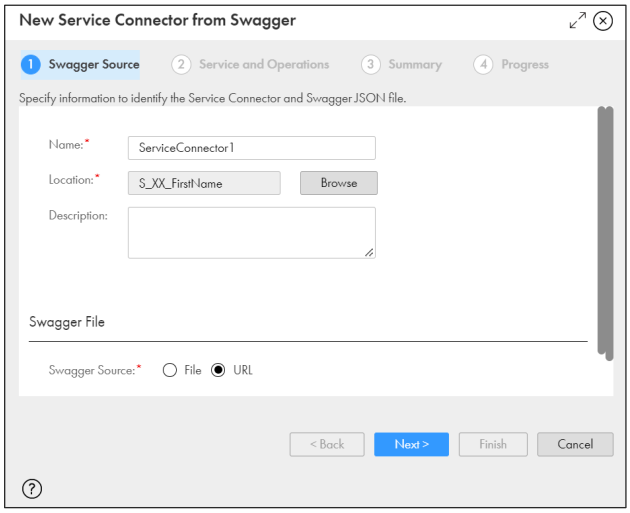
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Click **Finish** to review and save the service connector.

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Create a Service Connector from Swagger JSON



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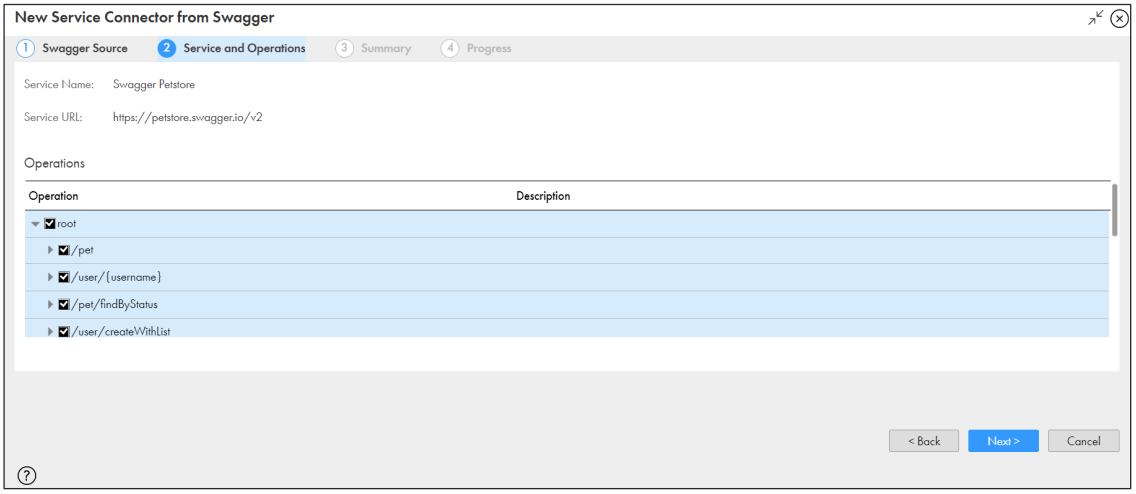
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In the Swagger Source step, provide a name and location to save the connector. You can import a Swagger JSON file in one of the following ways:

1. To use a Swagger JSON file on your local system, select the **File option** and select the file.
2. To use a URL that contains a Swagger JSON file, select the **URL** option.
3. To go to Service and Operations steps, click **Next**.

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Create a Service Connector from Swagger JSON (continued)



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The **Service and Operations** tab appears.

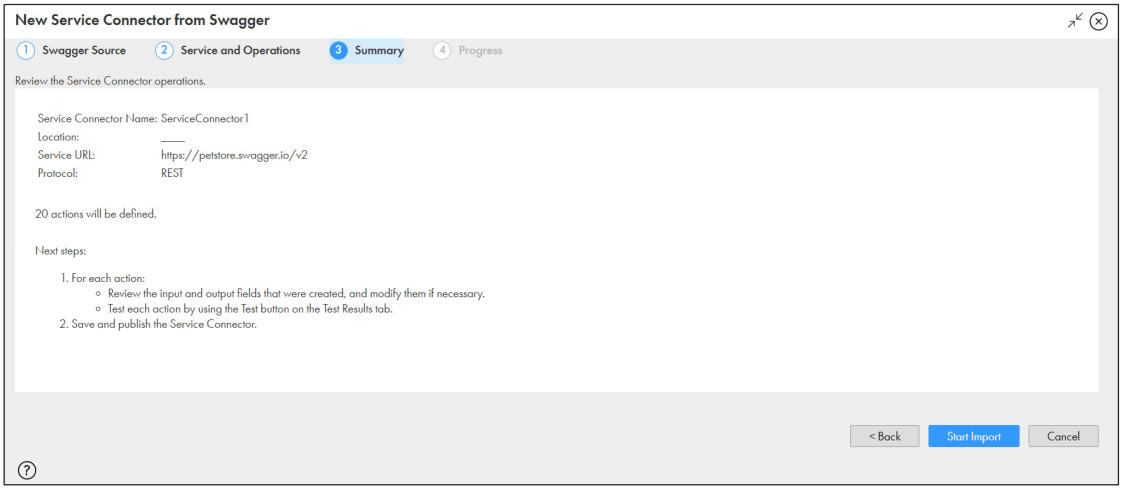
1. The Service URL field displays the URL where you want the service connector to send requests. The Service Name and Service URL fields are displayed automatically.
2. In the Operations section, select the operations that you want the service connector to provide. To remove an operation from the service connector, unselect that operation.

   Note: The operations that you select will be available to you as actions when you create a process and use this service connector.

   You can import a Swagger file that contains the DELETE, PUT, POST, PATCH, and GET HTTP operations. The generated service connector has actions corresponding to the HTTP operation is the Swagger file. For example, if you use a swagger file with a POST HTTP operation, you get a service connector with a POST action.
3. To go to Summary steps, click **Next**.

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Create a Service Connector from Swagger JSON (continued)



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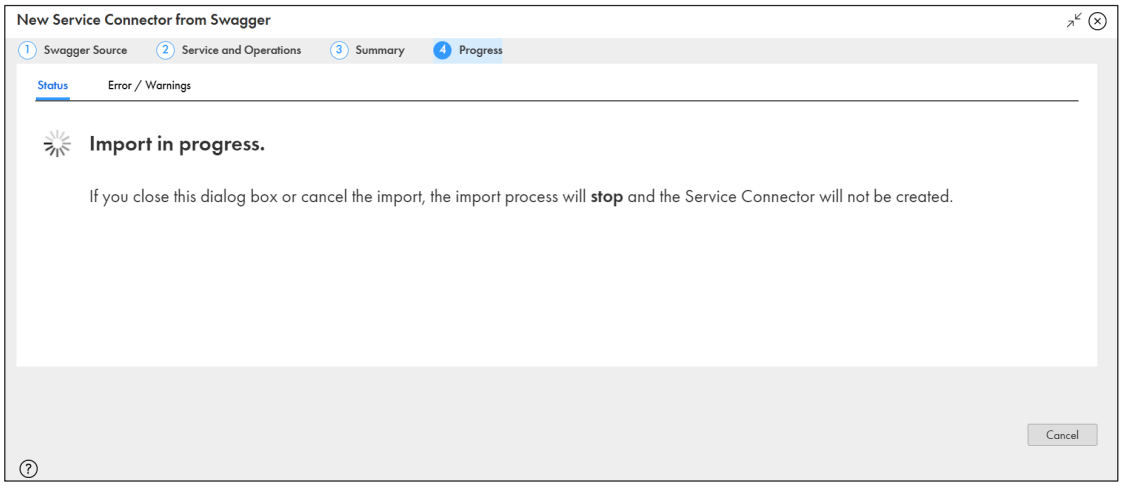
The **Summary** tab appears.

Review the service connector configuration and click **Start Import**.

Application Integration starts importing the Swagger JSON file and displays the status on the **Progress** tab.

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Create a Service Connector from Swagger JSON (continued)



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The **Progress** tab appears.

The Progress tab indicates whether the import completed successfully, failed, or completed with warnings.

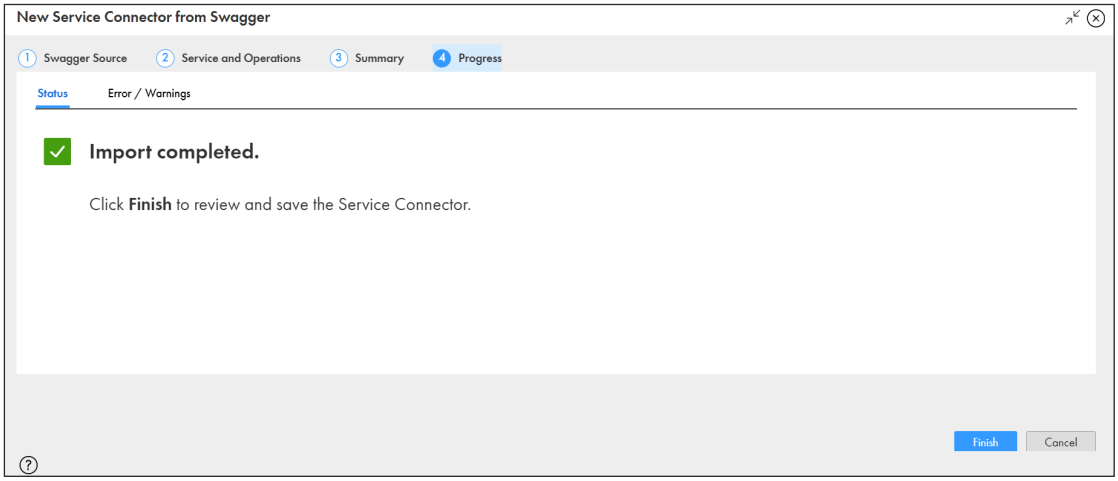
If the import fails or completes with warnings, click the **Error/Warnings**

tab to understand why the import did not complete successfully. You can fix the issues and import the Swagger JSON file again.

Note: If you close the dialog box or cancel the import, the import process stops and the service connector is not created.

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Create a Service Connector from Swagger JSON (continued)



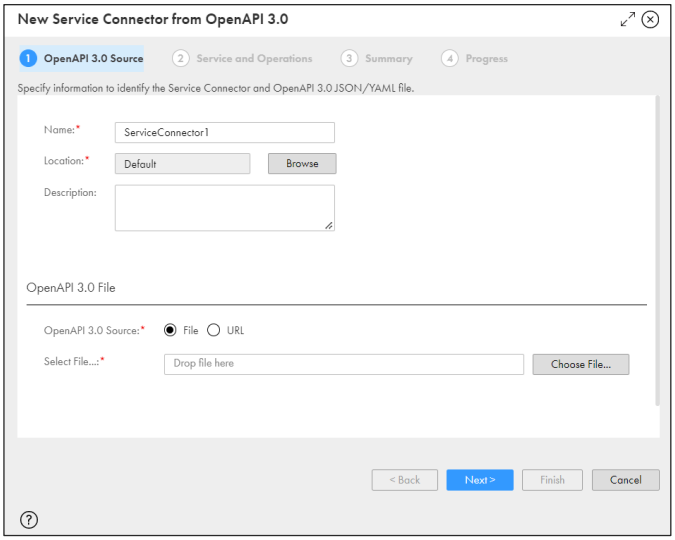
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Click **Finish** to review and save the service connector.

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Create a Service Connector from OpenAPI 3.0



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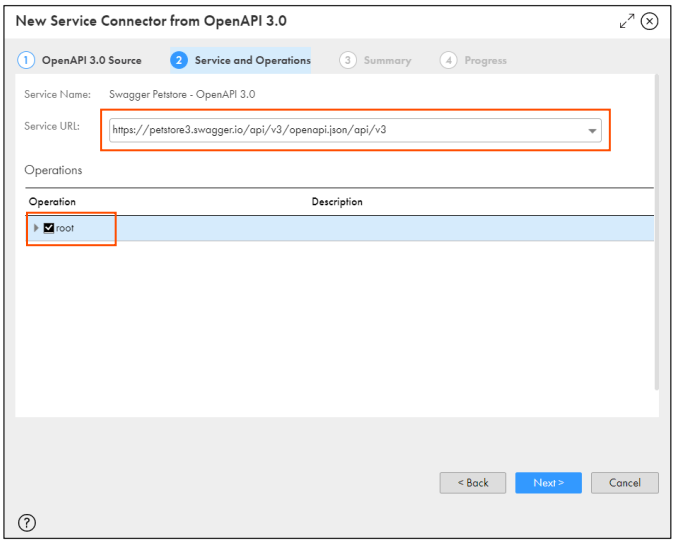
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In the **OpenAPI 3.0 File** tab, use one of the following options to specify the OpenAPI 3.0 file that you want to import:

1. To use an OpenAPI 3.0 file from your local system, select **File** and then select the OpenAPI 3.0 JSON or YAML file. You can either select a single .json or .yaml file, or a single .zip file that contains self-contained .json or .yaml files.
2. To use a URL that contains an OpenAPI 3.0 file, select **URL**. The **OpenAPI 3.0 URL** and **Use authentication** fields appear. In the **OpenAPI 3.0 URL** field, enter the URL that contains the OpenAPI 3.0 JSON or YAML file. Select the **Use authentication** option to specify the user name and password to access the OpenAPI 3.0 URL.
3. To go to Service and Operations steps, click **Next**.

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Create a Service Connector from OpenAPI 3.0 (continued)



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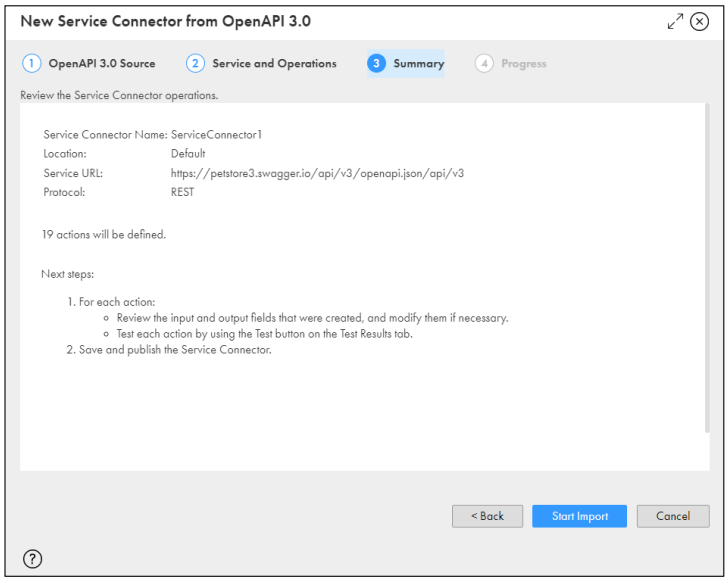
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The **Services and Operations** tab appears.

1. From the **Service URL** list, select the endpoint that you want to use from the imported OpenAPI 3.0 file to create a service connector.
2. In the **Operations** section, select the operations that you want the service connector to provide.

   Note: The operations that you select will be available as actions when you create a process and use the service connector.
3. To go to Service and Operations steps, click **Next**.

Create a Service Connector from OpenAPI 3.0 (continued)



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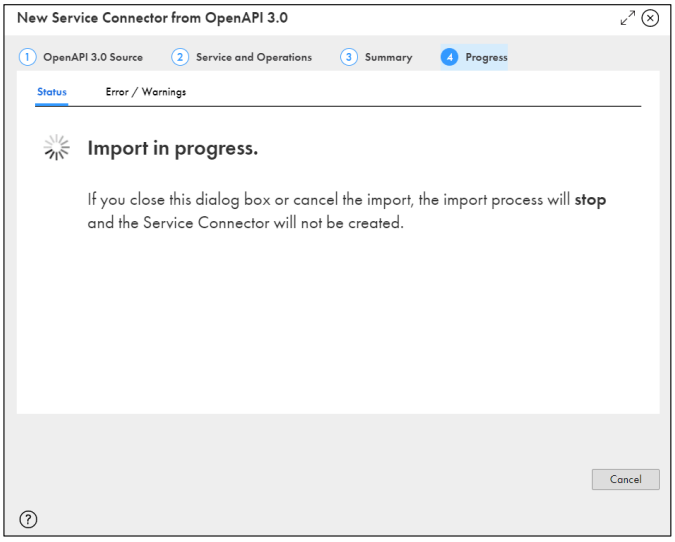
The **Summary** tab appears.

Review the service connector configuration and click **Start Import**.

Application Integration starts importing the OpenAPI 3.0 file and displays the status on the **Progress** tab.

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Create a Service Connector from OpenAPI 3.0 (continued)



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The **Progress** tab appears.

The Progress tab indicates whether the import completed successfully, failed, or completed with warnings.

If the import fails or completes with warnings, click the **Error/Warnings**

tab to understand why the import did not complete successfully. You can fix the issues and import the OpenAPI 3.0 file again.

Note: If you close the dialog box or cancel the import, the import process stops and the service connector is not created.

Click **Finish** to save the service connector.

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Topic

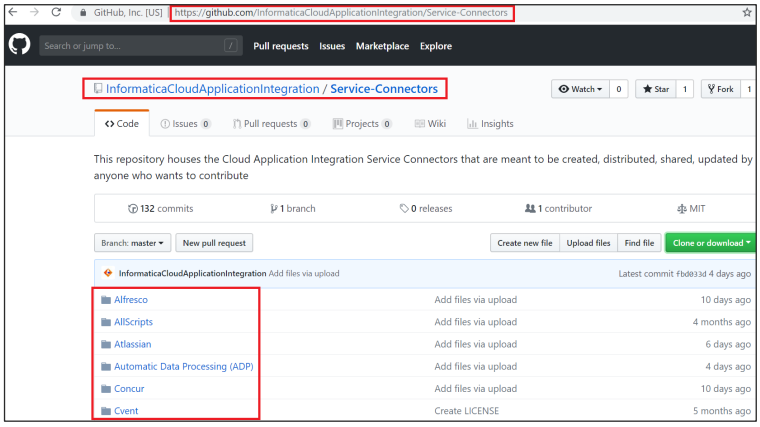
Download a Service Connector from GitHub

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Download a Service Connector from GitHub

- Informatica provides ready-to-use REST and SOAP API-based service connectors on GitHub
- To access GitHub, use the URL: https://github.com/InformaticaCloudApplicationIntegration/Service-Connectors



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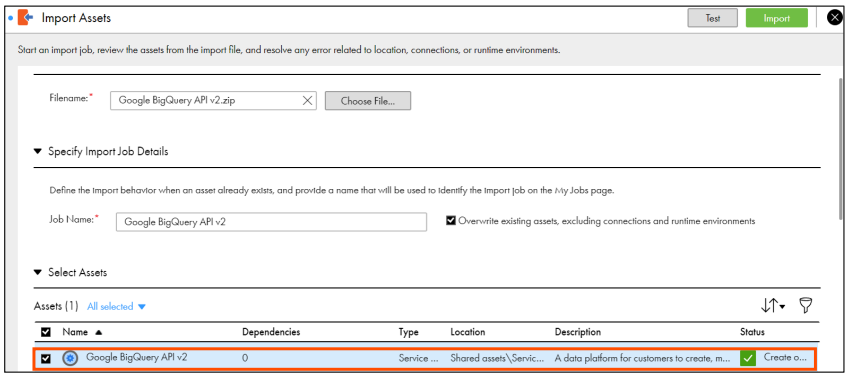
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Informatica provides several REST and SOAP API-based service connectors on GitHub. You can download the service connectors for free and import them into your organization.

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Download a Service Connector from GitHub (continued)

- Download the Service Connector .zip file
- Import Service Connector .zip file in Application Integration
- After the import completes, the service connector appears in the Select Assets section



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1. Download the service connector .**zip** file that you want to use.
2. Log in to Informatica Intelligent Cloud Services.
3. In the Application Integration Explore tab, click **Import** and select the service connector .zip file downloaded from GitHub.
4. Open the My Import/Export Logs page to check the import status.
5. After the import completes, the service connector is available in your organization.

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Topic

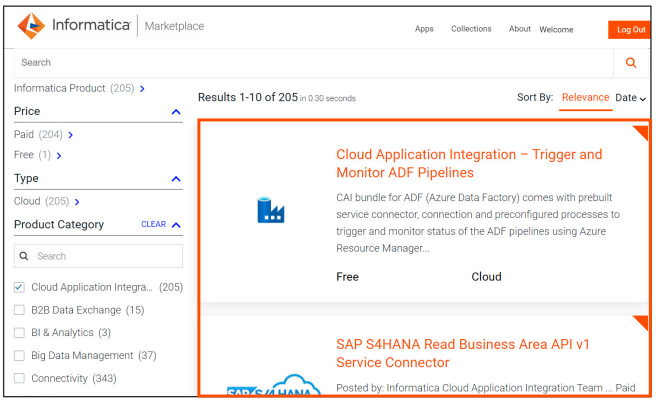
Download a Service Connector from Informatica Marketplace

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Download a Service Connector from Informatica Marketplace

- Access the https://marketplace.informatica.com/ URL
- Create an account and log in
- From the **Product Categories** list on the left pane, click **Cloud Application Integration**.
- A page appears displaying the available service connectors.



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Informatica has uploaded several REST and SOAP API-based service connectors on Informatica Marketplace. After you create an account in Informatica Marketplace, you can download the Informatica service connectors and import them into your organization.

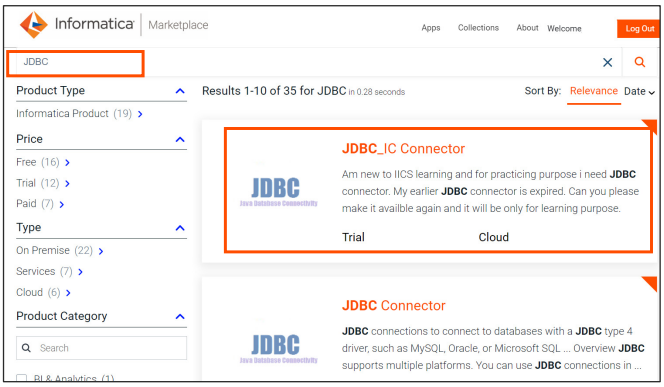
1. Access the following URL: https://marketplace.informatica.com/
2. Create an account and log in.
3. From the Product Categories list on the left pane, click **Cloud Application Integration**.

A page appears displaying the available service connectors.

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Download a Service Connector from Informatica Marketplace (continued)

- In the Search box, enter a name for the service connector that you want to download
  - You can also use the **Price** list and **Type** list to filter the service connectors.
- Click the name of the service connector that you want to download



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1. In the Search box, enter a name for the service connector that you want to download. You can also use the **Price** list and **Type** list to filter the service connectors.
2. Click the name of the service connector that you want to download.

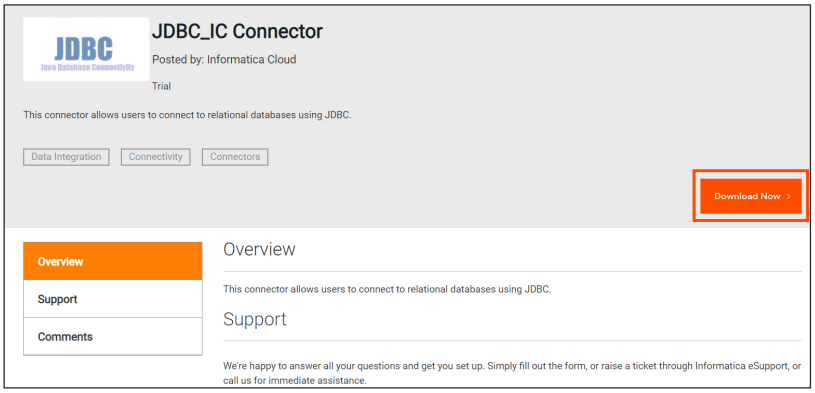
The service connector details appear.

Notes: If the service connector contains specific API-based service connectors, you see all the connector details along with links to access the API documentation and API specification.

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Download a Service Connector from Informatica Marketplace (continued)

- Click **Download Now**
- A form appears prompting you to enter the user details
- Fill in the form, accept the terms and conditions, and submit the form



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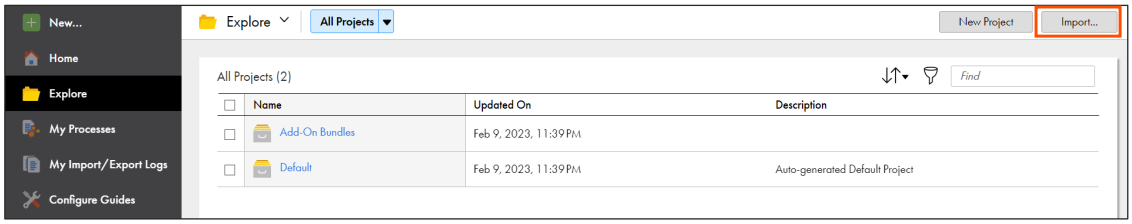
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1. Click **Download Now**. A form appears prompting you to enter the user details.
2. Fill in the form, accept the terms and conditions, and submit the form. The service connector is downloaded.

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Download a Service Connector from Informatica Marketplace (continued)

- Log in to **Informatica Intelligent Cloud Services** and click **Application Integration**
- Click **Import** and select the service connector .zip file that you downloaded from Informatica Marketplace
- The service connector appears in the **Select Assets** section.
- Click **Import** and open the **My Import/Export** Logs page to check the import status
- The service connector is available in your organization in the following locations: **Shared Assets\Service Connectors**



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1. Log in to **Informatica Intelligent Cloud Services** and click **Application Integration**.
2. On the **Explore** page, click **Import** and select the service connector .zip file that you downloaded from Informatica Marketplace. The service connector appears in the **Select Assets** section
3. Click **Import** and open the **My Import/Export Logs** page to check the import status.
   After the import completes, the service connector is available in your organization in the following location:

   Shared assets\Service Connectors

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Topic

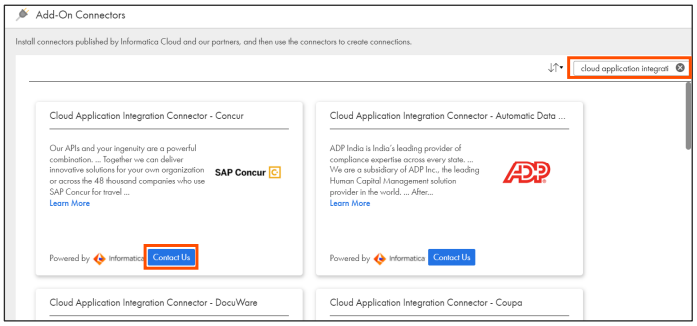
Download a Service Connector from Informatica Administrator

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Download a Service Connector from Informatica Administrator

- Log in to **Informatica Intelligent Cloud Services** and click **Administrator**
- On the left pane, click **Add-On Connectors**
- On the right pane, enter the search keyword **cloud application integration connector** in the Search box
- Click **Contact Us** against the service connector that you want to download
- The **Informatica Marketplace** page for the service connector opens



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You can download REST and SOAP API-based service connectors from Informatica Intelligent Cloud Services Administrator and import them into your organization.

1. Log in to Informatica Intelligent Cloud Services and click **Administrator**.
2. On the left pane, click **Add-On Connectors**.
3. On the right pane, enter the search keyword as **cloud application integration connector** in the **Search** box. The Cloud Application Integration service connectors that you can download are displayed.
4. Click **Contact Us** against the service connector that you want to download.

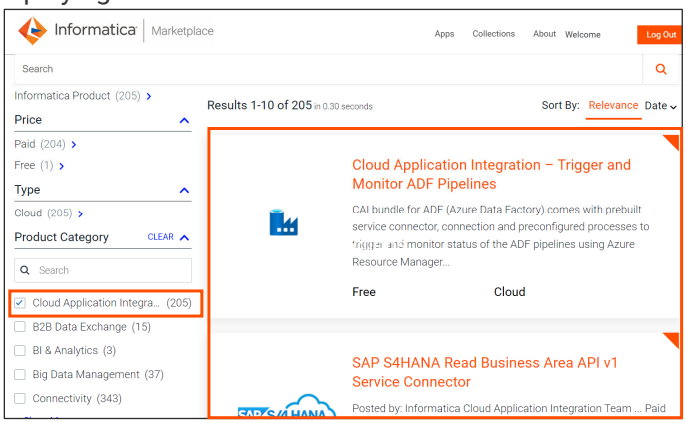
   Note: The Cloud Application Integration service connectors are prefixed with the term **Cloud Application Integration Connector**.

5. The Informatica Marketplace page for the service connector opens.

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Download a Service Connector from Informatica Administrator (continued)

- Create an account and log in to Informatica Marketplace
- From the **Product Categories** list on the left pane, select **Cloud Application Integration**
- A page appears displaying the available service connectors



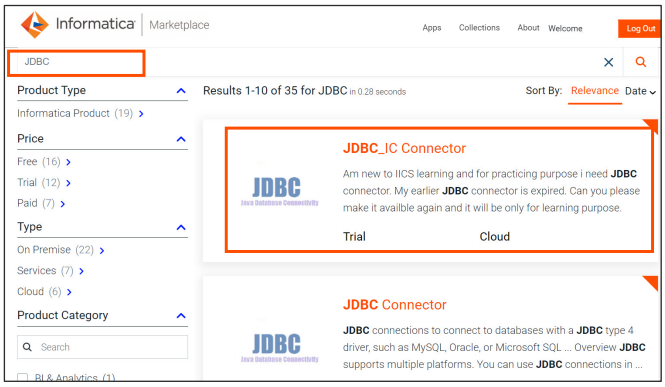
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1. Create an account and log in to **Informatica Marketplace**.
2. From the **Product Categories** list on the left pane, click **Cloud Application Integration**. A page appears displaying the available service connectors.

Download a Service Connector from Informatica Administrator (continued)

- In the **Search** box, enter a name for the service connector that you want to download
  - You can also use the Price list and Type list to filter the service connectors
- Click the name of the service connector that you want to download



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1. In the **Search** box, enter a name for the service connector that you want to download. You can also use the **Price** list and **Type** list to filter the service connectors.
2. Click the name of the service connector that you want to download.

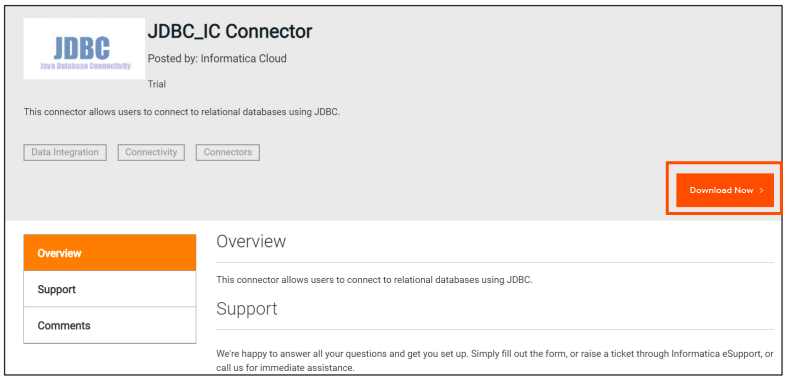
The service connector details appear.

Notes: If the service connector contains specific API-based service connectors, you see all the connector details along with links to access the API documentation and API specification

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Download a Service Connector from Informatica Administrator (continued)

- Click **Download Now**
- A form appears prompting you to enter the user details
- Fill in the form, accept the terms and conditions, and submit the form



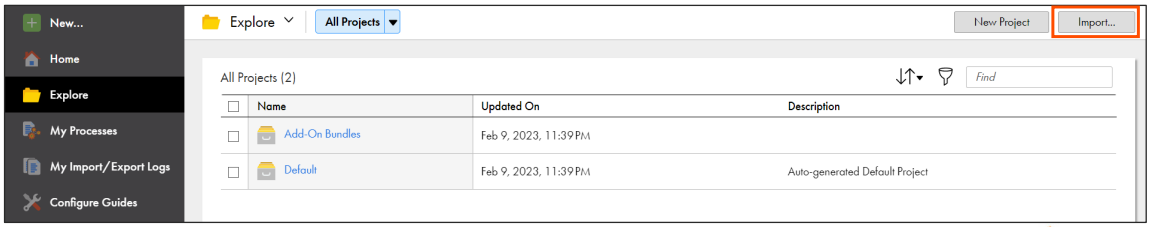
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1. Click **Download Now**. A form appears prompting you to enter the user details.
2. Fill in the form, accept the terms and conditions, and submit the form. The service connector is downloaded.

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Download a Service Connector from Informatica Administrator (continued)

- Log in to **Informatica Intelligent Cloud Services** and click **Application Integration**
- Click **Import** and select the service connector .zip file that you downloaded from Informatica Marketplace
- The service connector appears in the **Select Assets** section
- Click **Import** and open the **My Import/Export Logs** page to check the import status
- The service connector is available in your organization in the following locations: **Shared Assets\Service Connectors**



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1. Log in to **Informatica Intelligent Cloud Services** and click **Application Integration**.
2. On the **Explore** page, click **Import** and select the service connector .zip file that you downloaded from Informatica Marketplace. The service connector appears in the **Select Assets** section
3. Click **Import** and open the **My Import/Export Logs** page to check the import status.
   After the import completes, the service connector is available in your organization in the following location:

   Shared assets\Service Connectors

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Topic

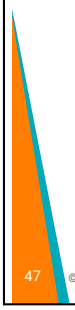
Knowledge Check

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Service Connectors cannot be configured using WSDL or Swagger file.

- True
- False



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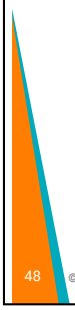
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Answer: False

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Can you use a Service Connector in a Process?

- Yes
- No



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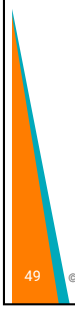
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Answer: Yes

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You can use a service connector in a Process without publishing it?

- True
- False



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Answer: False

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Lab Activity

5-1 Create a Service Connector

In this lab, you will perform the following:

- Create a new Service Connector
- Define Service Connector input parameters
- Define a Service Connector Action and Binding
- Test the Service Connector

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Lab Activity

5-2 Create a Service Connector from WSDL

In this lab, you will perform the following:

- Import a WSDL File
- Modify an existing Action
- Test the Service Connector

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Lab Activity

5-3 Create and Test a Swagger File

In this lab, you will perform the following:

- Sign up to Access OpenWeatherAPI
- Create Swagger files
- Test the Swagger file

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Lab Activity

5-4 Create a Service Connector from a Swagger File

In this lab, you will perform the following:

- Import a Swagger JSON File
- Modify an existing Action
- Test the Service Connector

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Lab Activity

5-5 Create a Service Connector from OpenAPI 3.0

In this lab, you will perform the following:

- Obtain the OpenAPI 3.0 URL
- Create a Service Connector from OpenAPI 3.0
- Add Parameter to an existing Action
- Test the Service Connector

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Module Summary

This module showed you how to:

- Define the Connectors
- Describe Service Connectors
- Create Service Connectors using different methods

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Module 6

Data Access Service Connectors

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Module Objectives

After completing this module, you will be able to:

- Describe and create Data Access Service Connectors

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Topic

Data Access Service Connectors Overview

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Data Access Service Connectors

- You can create a data access service connector on a Secure Agent to access data directly from a database.
- You can use the data access service connector to connect to the following databases:
  - IBM DB2
  - Microsoft SQL Server
  - MySQL
  - Oracle
  - PostgreSQL

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You can create a data access service connector on a Secure Agent to access data directly from a database. You can use the data access service connector to connect to the following databases:

- IBM DB2
- Microsoft SQL Server
- MySQL
- Oracle
- PostgreSQL

Note: Before you create a data access service connector, you must copy the required JDBC JAR files to the following location and restart the Secure Agent:
<Secure Agent installation directory>/apps/process-engine/ext

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Topic

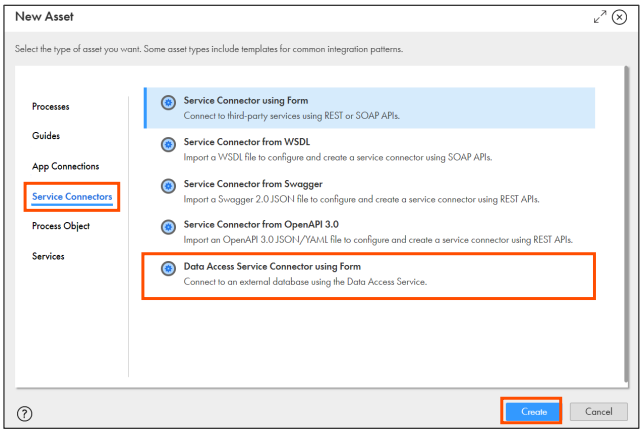
Create Data Access Service Connectors

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Create a Data Access Service Connector

- In Application Integration, click **New** > **Service Connectors**
- Click **Data Access Service Connector using Form**, and then click **Create**



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To create a data access service connector, perform the following steps:

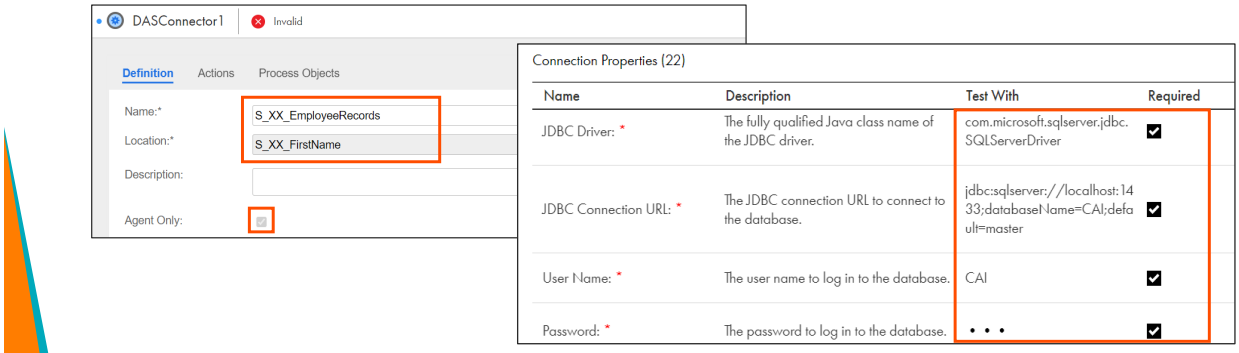
- In Application Integration, click **New** > **Service Connectors**.
- In the New Asset dialog box, click **Data Access Service Connector using Form**, and then click **Create**.

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Create a Data Access Service Connector (continued)

Define Properties

- In the **Definition** tab, define the basic properties for the data access service connector
- In the **Connection Properties** section, define the general properties to connect to an external database



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In the **Definition** tab, define the following basic properties for the data access service connector :

Name: The name by which the data access service connector is made available for processes.
Location: Specify the project or folder to save the data access service connector.
Agent Only: Selected by default because a data access service connector is only available to run on the Secure Agent.

In the **Connection Properties** section, define the following general properties to connect to an external database:

JDBC Driver: The fully qualified Java class name of the JDBC driver.

JDBC URL: The connection URL to connect to an external database.

User Name: The user name to connect to the database. This is a required field.

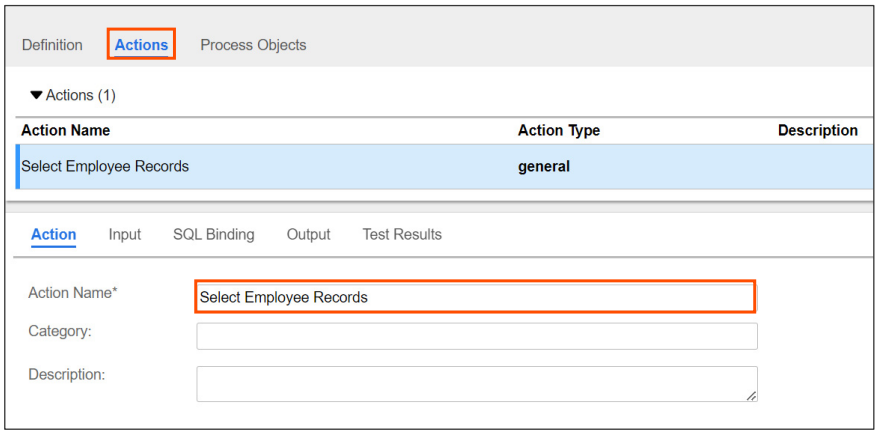
Password: The password to connect to the database. This is a required field.

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Create a Data Access Service Connector (continued)

Define Actions

- On the **Actions** tab, you can create and describe one or more actions associated with a data access service connector



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On the **Actions** tab, you can create and describe one or more actions associated with a data access service connector.

Click the row that you want to edit or click **+** to add a new row, and then enter the following information on the **Action** tab:

Action Name: Enter the name that appears in lists when referencing this action in data access service connectors and connections. Do not use spaces or special characters in the action name. This is a required field.

Category: If you have many data access service connectors, each with multiple actions, you can create categories to help users navigate within processes.

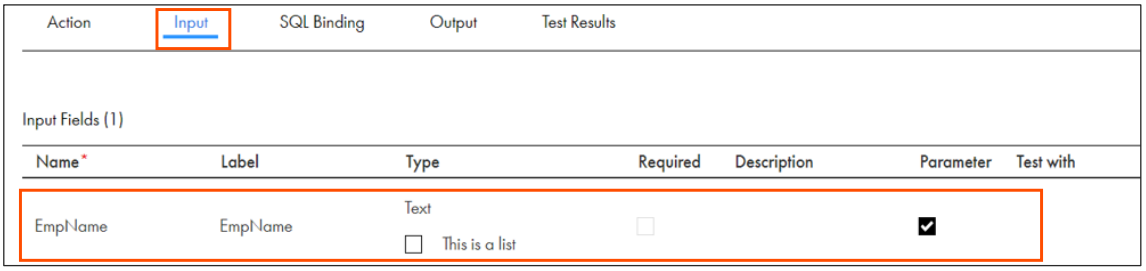
Description: Enter a description or notes about this action.

For each action, specify additional details on the **Input**, **SQL Binding**, **Output**, and **Test Results** tabs.

Create a Data Access Service Connector (continued)

Define Action: Input Tab

- Use the **Input** tab to define input data items that are unique to the service to which you are sending data



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Use the **Input** tab to define input data items that are unique to the service to which you are sending data. For each item, enter the following properties:

Name: The name of the input data item.

Label: The name of the item to display within a process. If not specified, the name is displayed.

Type: Select the data type of the item from the list. If you select the Reference or Object List type, you can also select a Process Object.

Required: Select if a value must be set for this parameter.

Description: Enter a description of the parameter.

Test with: Enter a value that Process Designer will use to test the action. If the input is an attachment, you can also upload a sample attachment file.

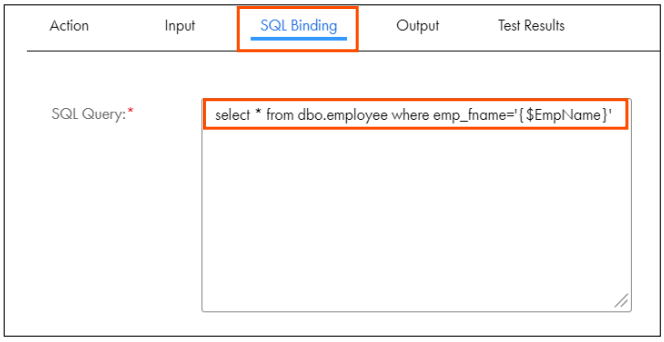
Use the **+** icon to add a new item. Click X to delete the current row.

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Create a Data Access Service Connector (continued)

Define Action: SQL Binding Tab

- Use the SQL Binding tab to specify the SQL query to access the data from the data source
- In the SQL field, enter a valid SQL SELECT statement that returns at least one row
- You can execute any SQL statement for selecting, inserting, updating, or deleting data from a data source



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For each Action in a data access service connector, use the **SQL Binding** tab to specify the SQL query to access the data from the data source.

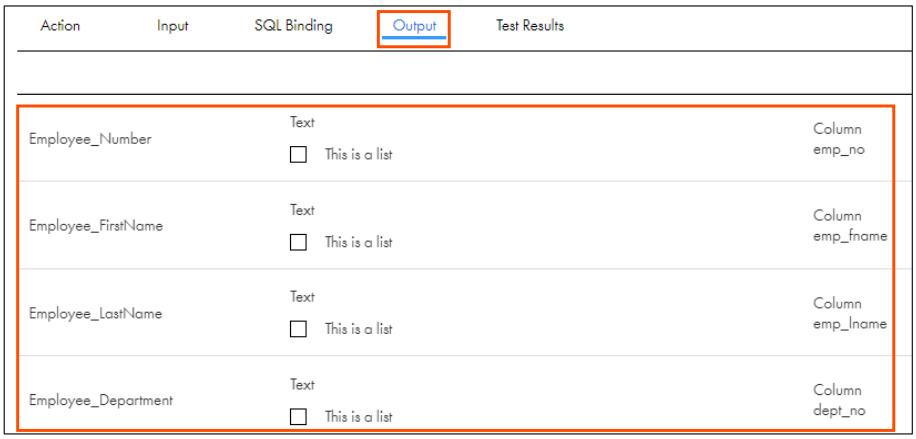
In the **SQL** field enter a valid SQL SELECT statement that returns at least one row. You can execute any SQL statement for selecting, inserting, updating, or deleting data from a data source.

Note: The SQL query must not end with a semicolon (;). Otherwise, an error occurs.

Create a Data Access Service Connector (continued)

Define Action: Output Tab

- Use the **Output** tab to define how the data access service connector must parse the data returned from the database



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Use the **Output** tab to define how the data access service connector must parse data returned from the database. or each output data item, specify the following properties:

Name: The name of a field into which a returned value is placed.

Type: The data type of the value being written to the field.

Description: Enter a description of the output field.

Get From: Select one of the following options:

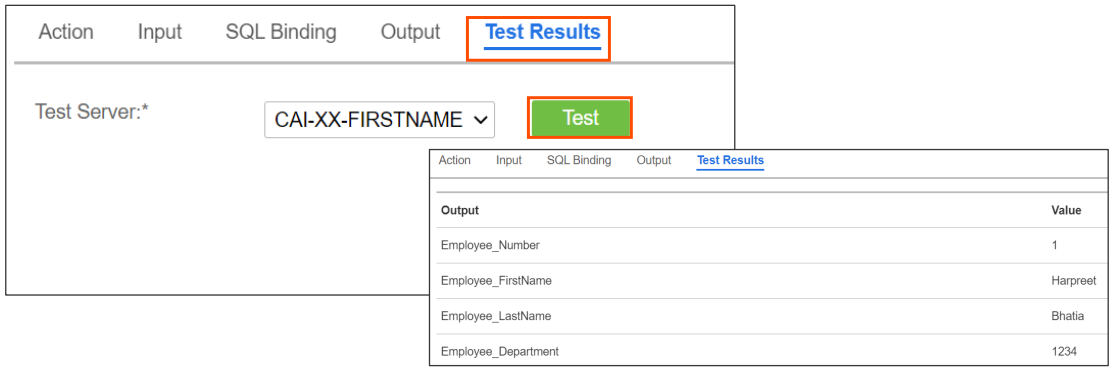
- **Column:** To enter a column name to be placed within the output field.
- **Expression:** To write an expression to parse the output returned by the database.
- **Result Set:** To assign the complete contents of the response from the database to the output field.
- **Entire Response As Attachment:** To handle the entire response as an attachment.
- **Attachments:** To work with multiple attachments and pass the entire list of attachments to the selected variable except the part used as the payload.

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Create a Data Access Service Connector (continued)

Test the Data Access Service Connector

- Click **Test** to send a request to the database and display the response data on the **Test Results** tab



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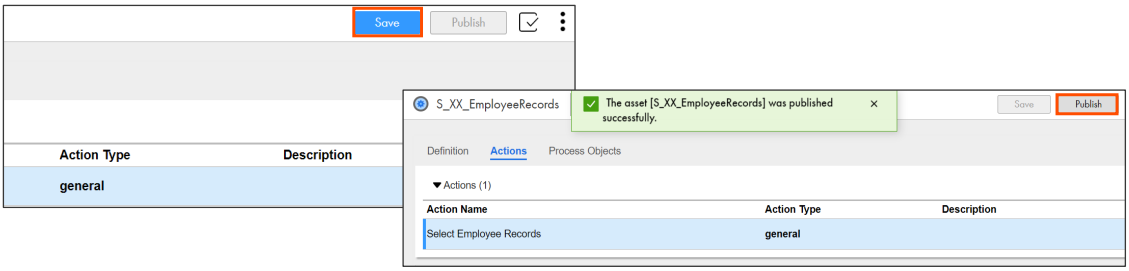
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Create a Data Access Service Connector (continued)

Save and Publish the Data Access Service Connector

- Click **Save** to save the service connector, and then click **Publish** to publish the service connector



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Topic

Knowledge Check

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Which of the following databases can be connected using data access service connector?

- Microsoft SQL Server
- MySQL
- Oracle
- All of the above

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Answer: All of the above

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On the Actions tab, you can create and describe only one action associated with a data access service connector.

- True
- False

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Answer: False

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Lab Activity

6-1 Create a Data Access Service Connector Using Form

In this lab, you will perform the following:

- Create a Data Access Service Connector
- Define properties and actions for the Data Access Service Connector
- Test the Data Access Service Connector

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Lab Activity

6-2 Create a Currency Exchange Data Access Service Connector

In this lab, you will perform the following:

- Create a Data Access Service Connector
- Define properties and actions for the Data Access Service Connector
- Test the Data Access Service Connector

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Lab Activity

6-3 Using Currency Exchange Data Access Service Connector in a Process

In this lab, you will perform the following:

- Create an App Connection
- Create a Process using the Data Access Service Connector
- Publish and execute the Process

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Module Summary

This module showed you how to:

- Describe and create Data Access Service Connectors

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Module 7
App Connections

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Module Objectives

After completing this module, you will be able to:

- Describe the App Connection
- Create App Connections
- Utilize CAI Connectors and Service Connectors in App Connection

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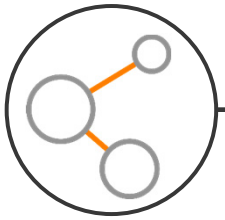
Topic

App Connection Overview



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What is an App Connection?



- Defines a link to a data source, a database, or a Service Connector.
- Can use any out-of-the-box Connectors or Service Connectors.
- Can be used in any Guide or Process.

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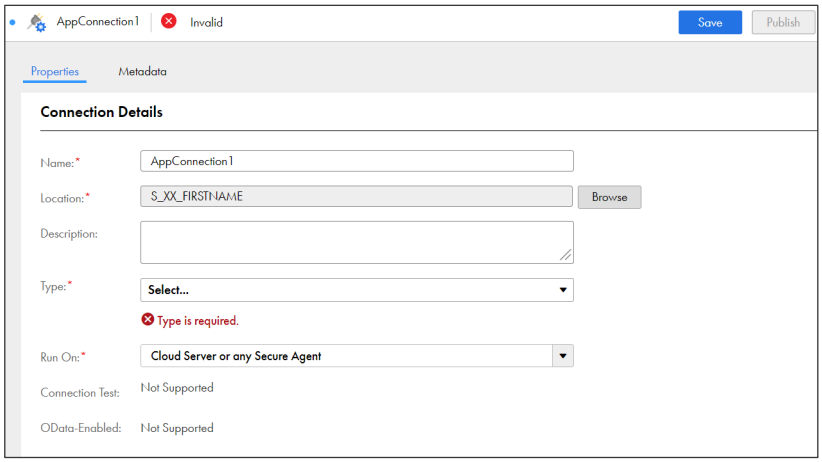
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App Connection defines a link to a data source, a database, or a Service Connector. A CAI App Connection can use any out-of-the-box connector or any Service connector that is published. After that, you can use the published app connection in any Guide or Process in CAI.

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Basic App Connection Properties

- Specify the following properties for Connection:
  - Name
  - Location
  - Description
  - Type
  - Run On
  - Connection Test
  - OData-Enabled



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App Connection defines actions that you can perform using a pre-defined connector or a service connector.

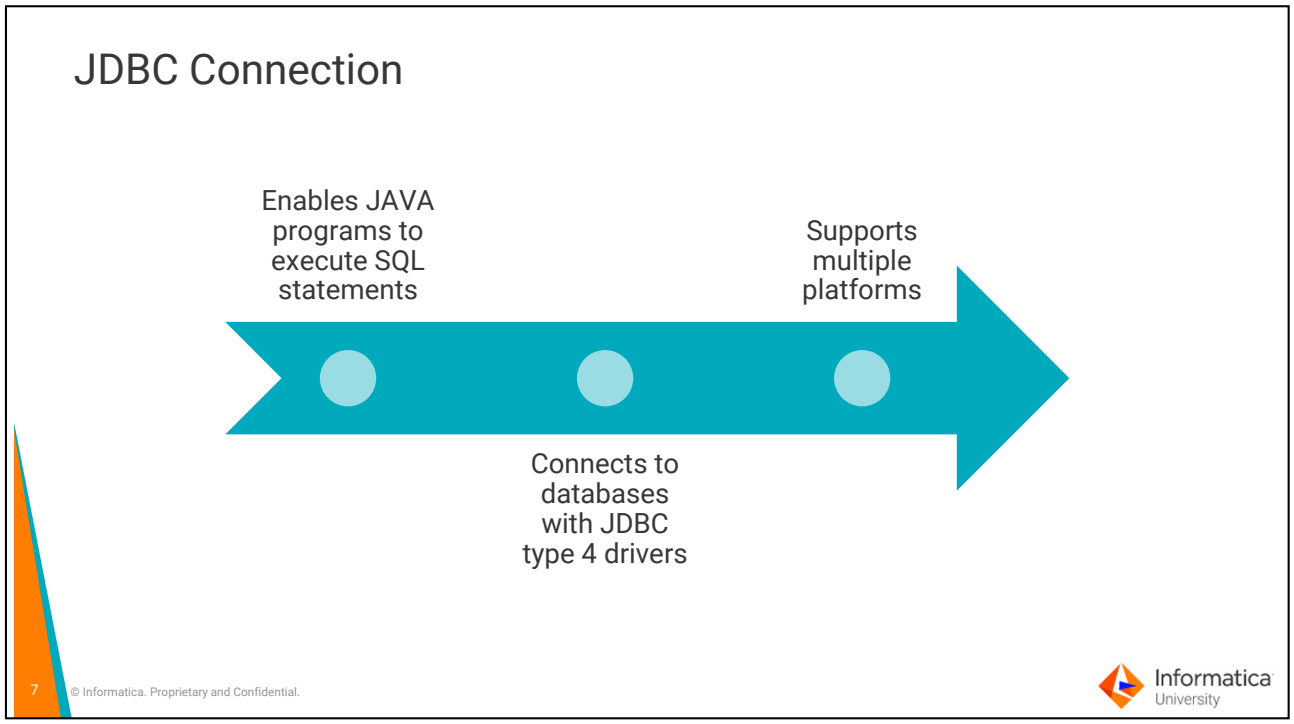
Name and Description: Enter a suitable name and description of the connection. You must enter a unique name for the connection in the Name field to identify it in the Process Designer.
Location: This field specifies the folder where you save the connection.
Type: It is a mandatory property to specify the built-in connector or service connector for this connection.
Run On: This is also a mandatory property. Select the agent where you want to access a service behind your firewall or Cloud Server, or any agent. The Run On drop-down lists the agents available for your organization.
Connection Test: Displays the results of any connection test run on the Cloud Server or the agent named in the Run on field.
OData Enabled: Choose **Yes** to utilize the OData protocol to access data services from the Cloud that exist on-premise such as those available via a JDBC or a SAP Table Reader connection.

Topic

App Connection Examples

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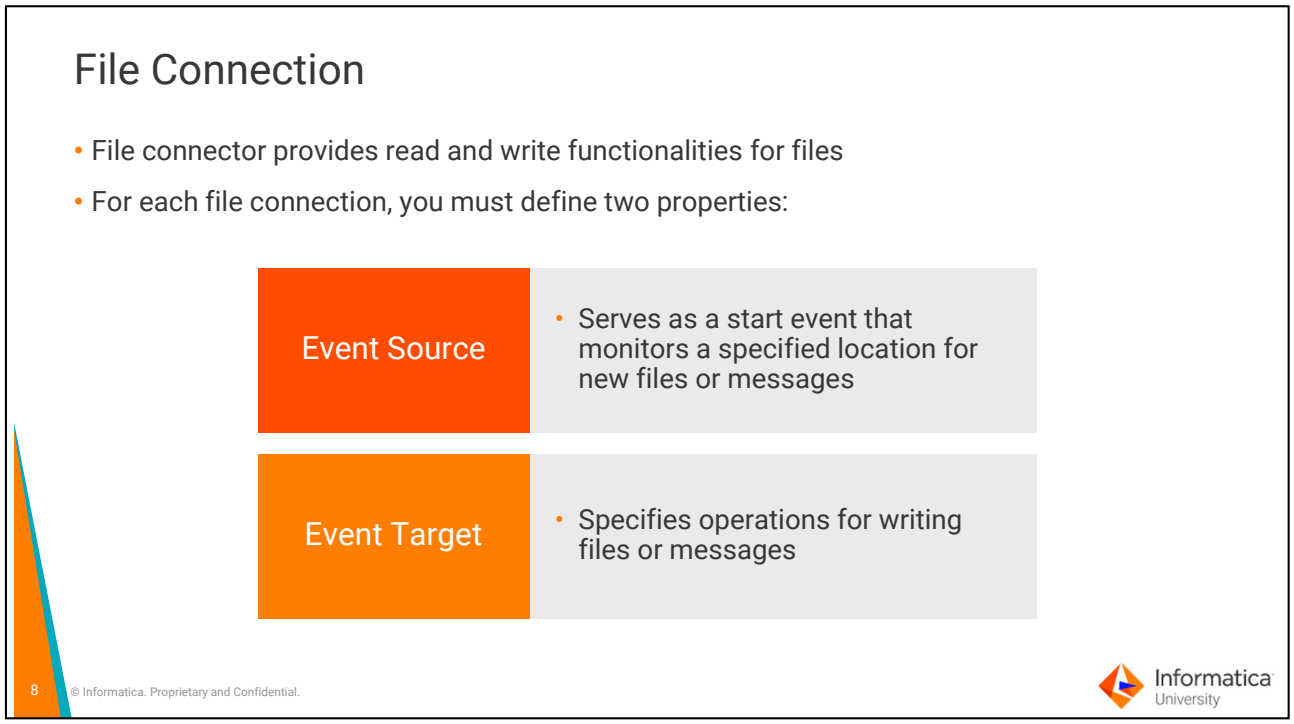
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Java Database Connectivity (JDBC) is a Java API that enables Java programs to execute SQL statements, and interact with any SQL-compliant database. You can use JDBC Connections to connect to databases with a JDBC type 4 driver such as IBM DB2, Oracle, MySQL, PostgreSQL, and Microsoft SQL Server. JDBC supports multiple platforms.

When you create and save a JDBC connection, the Secure Agent validates the app connection properties. After you publish the JDBC connection, you can view the generated process objects for the app connection in the Metadata tab.

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The File connector in Informatica CAI provides reading and writing functionalities for files. For example, you can use the File Connector, to write log files needed as part of a larger integration, or to monitor a file system for new .csv files.
For each file connection, you must define two properties - Event Sources and Event Targets, along with the basic connection properties.

Event Sources: An event source serves as a start event that listens to or monitors a specified location for new files or messages. After you define an event source for a connection, you can publish the connection only on a Secure Agent. You can access the event source in a process and deploy that process on a Secure Agent to consume the Process Objects generated by the event source downstream.

Event Target: For each connection, you can include one or more Event targets to specify operations for writing files or messages, or when the event target is called from a Process. For example, you must define an event target that reads from a process object and writes to comma-delimited files.

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Amazon S3 Connection

- Provides read and write functionalities for files residing in S3 Storage system
- For Amazon S3 connection, you must define following connection properties:
  - Access Key ID
  - Secret Access Key
  - Policy
  - Amazon S3 Endpoint
  - Region
  - User Data Encryption
  - Encryption Algorithm
  - Encryption Key
  - Encryption Password



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Access Key ID: Defines AWS access key ID of the requester. It is a mandatory field.
Secret Access Key: Defines AWS secret access key of the requester. It is a mandatory field.
Policy: Defines bucket policy to be applied to any new buckets created by the process.
Amazon S3 Endpoint: The region-specific website endpoint where the S3 bucket should reside if the process creates a new bucket. For example, s3-website-us-east-1.amazonaws.com
Region: The region where a new bucket should be located. For example, us-east1
User Data Encryption: If Yes, the content of S3 objects will be decrypted and encrypted by event sources and targets.
Encryption Algorithm: Encryption algorithm that should be used to decrypt and encrypt the S3 objects.
Encryption Key: Base64 encoded secret key used for encryption.
Encryption Password: Encryption password that should be used to generate an Encryption Key. You can either enter the Encryption Key or generate the key using this parameter.

Demonstration

Create Amazon S3 Connection

In this demonstration, the instructor will demonstrate the following:

- The steps to create Amazon S3 connection

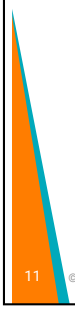
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Kafka Connection

- Allows you to connect to a Kafka stream and read data from or write data to a topic.
- For Kafka connection, you must define the following connection properties:
  - Kafka Host
  - Kafka Port
- Requires a Kafka Consumer, Kafka Producer, and a Kafka Topic.



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Kafka is a distributed publish-subscribe messaging system that is fast, scalable, and durable. It stores messages in the form of topics. Consumers read data from a topic. Producers write data to a topic. Kafka topics are partitioned and replicated across multiple nodes that allow distributed processing.

A Kafka broker works as an intermediary between consumers and producers to pass data. Kafka Connector is implemented by using a Camel event listener.

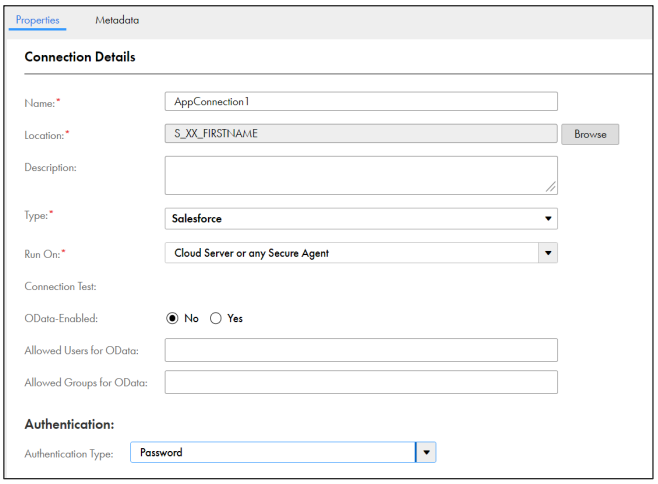
- **Kafka Host**: The Kafka host ensures that the Secure Agent must connect to a TCP connection.
- **Kafka Port**: Port number used for communication.

Kafka uses Apache Zookeeper to manage the Kafka cluster.

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Salesforce Connection

- Allows you to connect to Salesforce objects and services
- Used to create guides and processes that read and write information to and from Salesforce
- Configure the connection using Password, OAuth authentication, or OAuth JWT



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In CAI, you must create a Salesforce connection to connect to Salesforce objects and services. The Salesforce connector lets you create guides and processes that read and write information to and from Salesforce. When you create a Salesforce connection, you can configure the connection using Password or OAuth authentication properties. Default is Password authentication.

After you create a Salesforce connection, save the connection. You can then publish the Salesforce connection and click the Metadata tab to view the generated process objects for the connection.

When you publish the connection, the Secure Agent validates the connection properties.

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Salesforce Connection (continued)

Password Authentication

- For Password authentication, you must define the following connection properties:
  - User Name
  - Password
  - Security Token
  - Service URL



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To authenticate the Salesforce connection using Password authentication, select Password in the Authentication Type list and configure the following properties:

User Name: Enter the user name for Salesforce developer account.

Password: Enter the Salesforce developer account password.

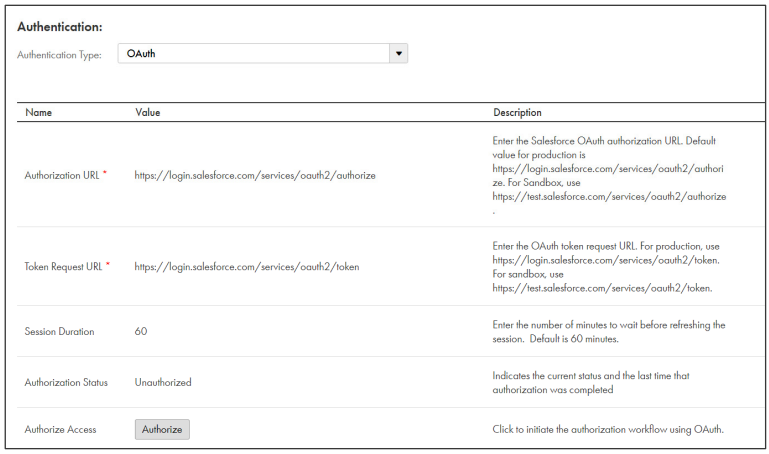
Security Token: Enter the Salesforce security token. This is a case-sensitive alphanumeric code used as a second level of authentication.

Service URL: The SOAP service URL of the Salesforce endpoint. For salesforce sandbox, use the **test.salesforce.com** as a service URL.

Salesforce Connection (continued)

OAuth Authentication

- For OAuth authentication, you must define the following connection properties:
  - Authorization URL
  - Token Request URL
  - Session Duration
  - Authorization Status
  - Authorize Access



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Use OAuth to connect to Salesforce through an API. To authenticate the Salesforce connection using OAuth authentication, select OAuth in the Authentication Type list and configure the following properties:
Authorization URL: Enter the endpoint used to make OAuth authorization requests to Salesforce.

Token Request URL: Endpoint used to make OAuth token requests to Salesforce.

Session Duration: Number of minutes before the OAuth token expires. When this duration ends, the connection makes a token request and performs authorization again.

Authorization Status: The current authorization status. The name of the last user who authorized the connection and the time of authorization appears.

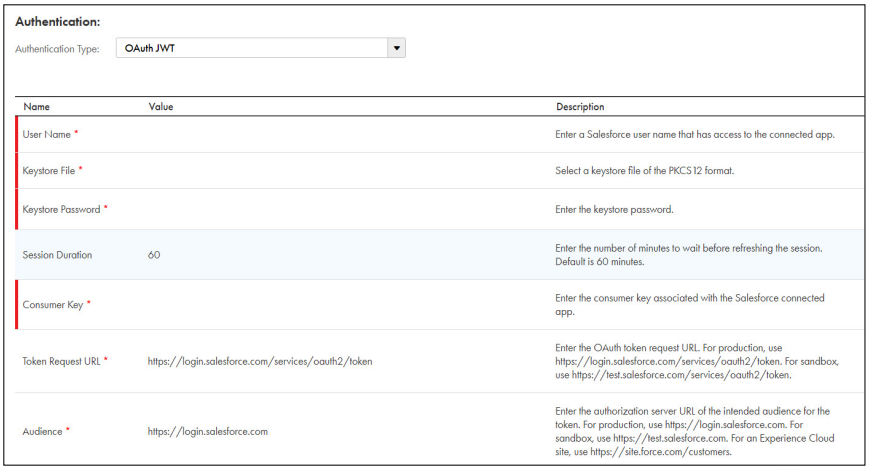
Authorize Access: Starts the OAuth authentication process and perform the following tasks:
- Click Authorize.
- Enter your Salesforce developer account credentials.
- Click Allow next to the message that prompts you to authorize access.

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Salesforce Connection (continued)

OAuth JWT

- For OAuth JWT authentication, you must define the following connection properties:
  - User Name
  - Keystore File
  - Keystore Password
  - Session Duration
  - Consumer Key
  - Token Request URL
  - Audience



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Use OAuth JWT authentication to authorize servers to access data without logging in each time the servers exchange information. The OAuth JWT authentication uses a certificate to sign the JWT request and does not require explicit user interaction.

User Name: Enter Salesforce user name that has access to the connected app.

Keystore File: Select a keystore file of the PKCS12 format.

Keystore Password: Enter the keystore password.

Session Duration: Number of minutes after which the session expires.

Consumer Key: Enter the consumer key associated with the Salesforce connected app..

Token Request URL: Endpoint used to make OAuth token requests to Salesforce.

Audience: Endpoint of the authorized servers of the intended audience for the token.

Salesforce Platform Events Connection

- Connect business processes within the Salesforce and the external apps through exchange of real-time event data
- Platform Events are secure and scalable messages that contain data
- Use the Salesforce Platform Events connector to configure the integration between Salesforce Platform Events and connected business applications
  - Listens for events that are published through Platform Events
  - Listens for events from connected applications to update Platform Events and trigger actions
  - This listening provides real-time connectivity and minimizes resource requirements
- Set up two components to configure the connector:
  - **Salesforce Platform Events connection**: Uses the OAuth 2.0 protocol for authentication and authorization
  - **Salesforce Platform Events operation**: Listen and Send messages

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Salesforce Platform Events connect business processes within the Salesforce and the external apps through exchange of real-time event data. Salesforce Platform Events are secure and scalable messages that contain data.

You can use the Salesforce Platform Events connector to configure the integration between Salesforce Platform Events and connected business applications.

The Salesforce Platform Events connector listens for events that are published through Platform Events. It also listens for events from connected applications to update Platform Events and trigger actions. This listening provides real-time connectivity and minimizes resource requirements.

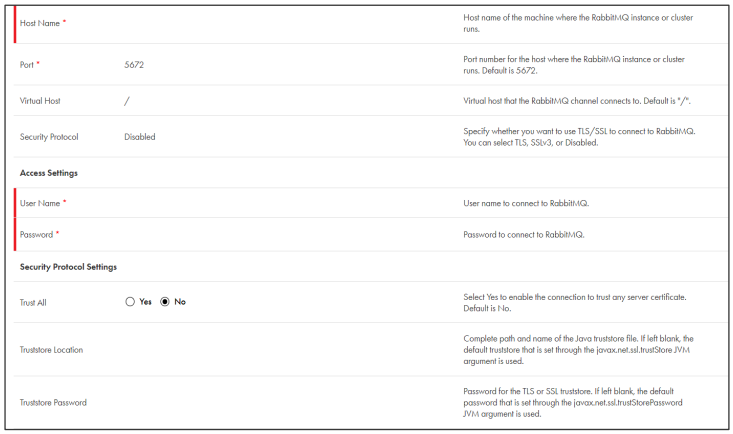
You can set up two components to configure the Salesforce Platform Events connector :

- **Salesforce Platform Events connection:** Uses the OAuth 2.0 protocol for authentication and authorization
- **Salesforce Platform Events operation:** Listen and Send messages

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RabbitMQ Connection

- Used to connect to a RabbitMQ broker and read data from or write data to a queue
- For RabbitMQ connection, you must define following connection properties:
  - Host Name
  - Port
  - Virtual Host
  - Security Protocol
  - User Name and Password
  - Trust All
  - Truststore Location and Password
  - Connection and Heartbeat Timeout
  - Enable Mandatory Message Routing
  - Enable Guaranteed Message Delivery



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RabbitMQ is a distributed message broker system that uses Advanced Messaging Queuing Protocol (AMQP) for secure transfer of messages. Producer publishes messages to an exchange. After the exchange receives the messages, it routes the messages to different queues. Based on the message routing key, the exchange creates a binding with the queue and routes the messages to the queue. The message stays in the queue until a consumer connects to the queue and subscribes to the message.

Host Name: Hostname of the machine where the RabbitMQ instance runs.
Port: Port number for communication. Default is 5762.
Virtual Host: Specify the virtual host that the RabbitMQ channel connects to.
Security Protocol: Specify protocol to use for connection.
User Name and Password: Credentials to connect to RabbitMQ.
Trust All: Select **Yes** to enable the connection to trust any server certificate.
Truststore Location and Password: Location and Password of the truststore file that contains the TLS or SSL certificate.
Connection and Heartbeat Timeout: Number of seconds to connect to RabbitMQ Connection. Default is 60 seconds.
Enable Mandatory Message Routing: Select Yes to throw an exception when a message cannot be routed to a queue.
Enable Guaranteed Message Delivery: Select Yes to throw an exception when the message cannot be delivered and the mandatory message routing is enabled.

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File Transfer Protocol (FTP) Connection

- Provides connectivity between Application Integration and remote FTP servers
- Allows you to monitor an FTP server for new files, move files, read and write content from files, and select options to handle processed files
- Use the FTP connection, for example, to:
  - Write log files needed as part of a larger integration
  - Monitor a file system for new .csv files
  - Read the delimited content from the files

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The FTP connection provides connectivity between Application Integration and remote FTP servers. With the FTP connection, you can monitor an FTP server for new files, move files, read and write content from files, and select options to handle processed files.

You might use the FTP Connector, for example, to write log files needed as part of a larger integration, or to monitor a file system for new .csv files, and to read the delimited content from the files.

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Topic

Salesforce Managed Package

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Salesforce Managed Package Overview

You can use Salesforce Managed package to:

- Integrate Application Integration with Salesforce
- Test and publish a Salesforce connection to fetch the metadata from Salesforce
- Work with Salesforce objects when running a process
- Work with Salesforce guides
  - Use a Salesforce Outbound Message (OBM) to trigger a process

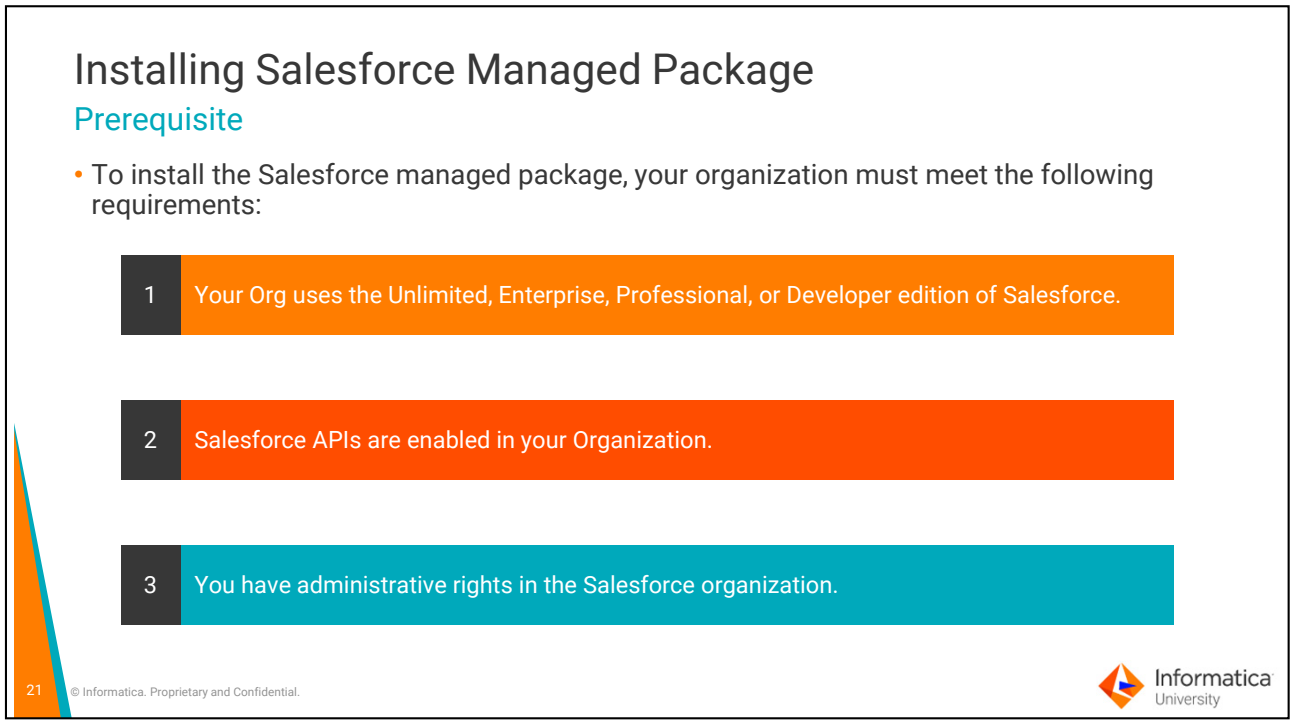
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You can use Salesforce Managed package to:

1. Integrate Application Integration with Salesforce.
2. Test and publish a Salesforce connection in Application Integration to fetch the metadata from Salesforce.
3. Work with Salesforce objects when you run a process.
4. Work with Salesforce guides.
5. Access Application Integration and Salesforce resources with a single sign-on.
6. Use a Salesforce outbound message (OBM) that triggers an Application Integration process.

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To install Salesforce managed package in your Organization, ensure that you meet the following requirements:

1. Your organization uses the Unlimited, Enterprise, Professional, or Developer edition of Salesforce.

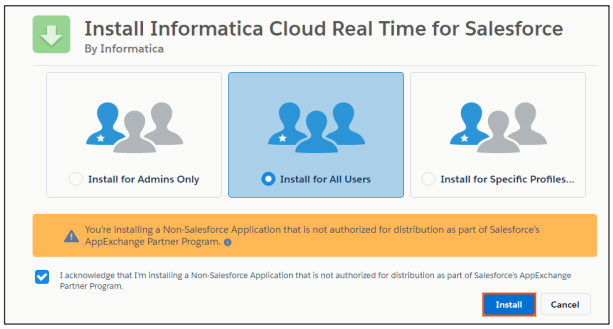
Note: When you use the trial version, which is the Professional edition of Salesforce, you might not be able to configure the managed package or connect to Salesforce. This happens because the API Enabled option is not available for the trial organizations. Informatica recommends that you use the Unlimited, Enterprise, or Developer edition of Salesforce.

2. Salesforce APIs are enabled.
3. You have administrative rights in the Salesforce organization where you want to install the managed package.

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Steps to Install Salesforce Managed Package

- Log in to your production or sandbox Salesforce developer account using any of the following URLs:
  - For Production: https://login.salesforce.com/?ec=302&startURL=%2Fpackaging%2FinstallPackage.apexp%3Fp0%3D04t1J0000007uro
  - For Sandbox: http://test.salesforce.com/?ec=302&startURL=%2Fpackaging%2FinstallPackage.apexp%3Fp0%3D04t1J0000007uro
- Select the user type and click Install



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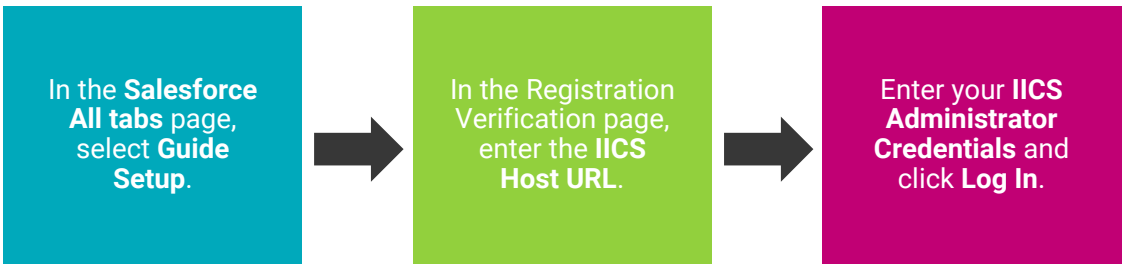
1. To install Salesforce Managed package, you must login to your Salesforce account using the following URL:

For Production:
https://login.salesforce.com/?ec=302&startURL=%2Fpackaging%2FinstallPackage.apexp%3Fp0%3D04t1J0000007uro
For Sandbox:
http://test.salesforce.com/?ec=302&startURL=%2Fpackaging%2FinstallPackage.apexp%3Fp0%3D04t1J0000007uro

2. On the Install Informatica Cloud Real Time for Salesforce page, select one of the following user profiles:
   - Install for Admins Only. Select this option to allow only admin users in Salesforce to run guides.
   - Install for All Users. Select this option to allow all users in Salesforce to run guides on the objects they work with. Most organizations grant access to all users.
   - Install for Specific Profiles. Select this option to configure custom security settings.
3. Click Install.

Note: If the installation takes a long time, a message appears stating that you will receive an email when the installation is complete. Click Done.

Steps to Install Salesforce Managed Package (continued)



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After the installation is complete, you must perform the following steps in Salesforce:

1. From the Salesforce home page, select **All Tabs** > **Guide Setup**.
2. In the Registration Verification page, enter the Host URL.

Note: The Host URL consist of POD-specific Cloud Application URL and IICS org ID. You must construct the Host URL in the following format:

<POD-specific Cloud Application URL for Salesforce>,<Informatica Organization ID>

3. After you verify the Host URL, enter your IICS credentials and click Log In.
4. If the authentication is successful, the following confirmation message appears:

"Your Salesforce Org has been configured for Informatica CAI. Click on Advanced settings to change the configuration".

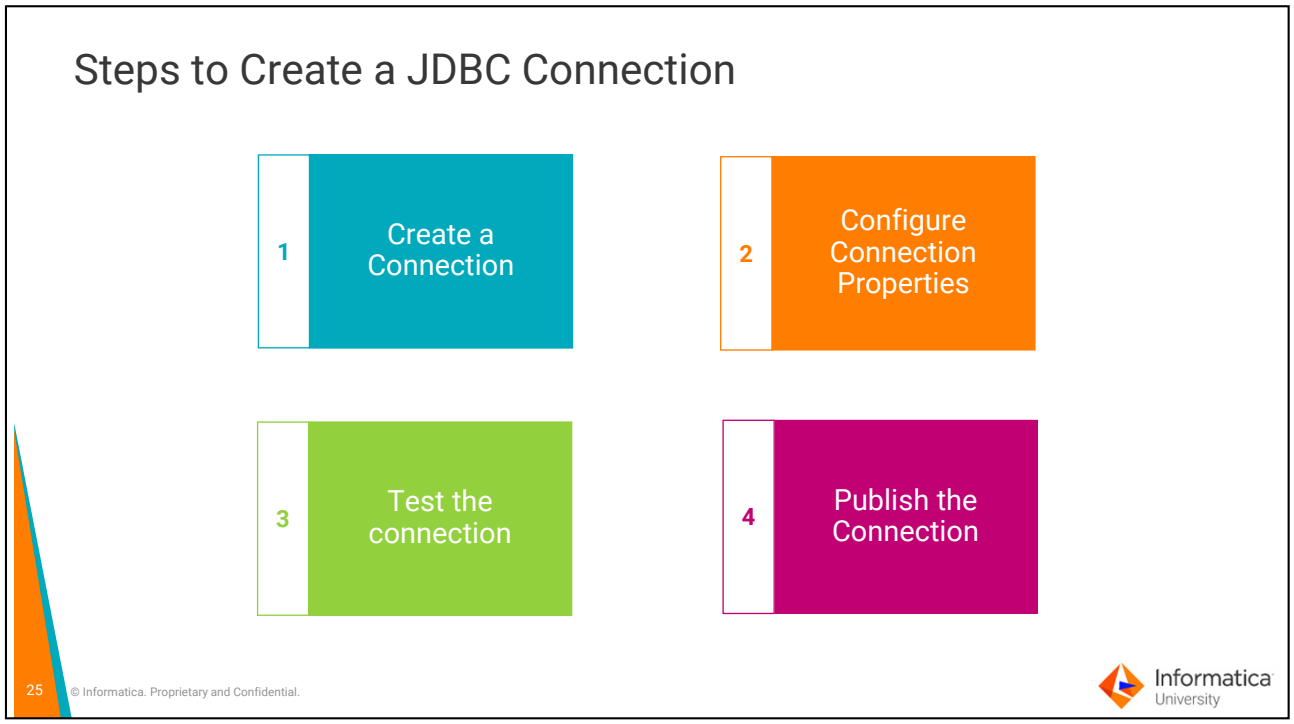
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Topic

Creating a JDBC Connection

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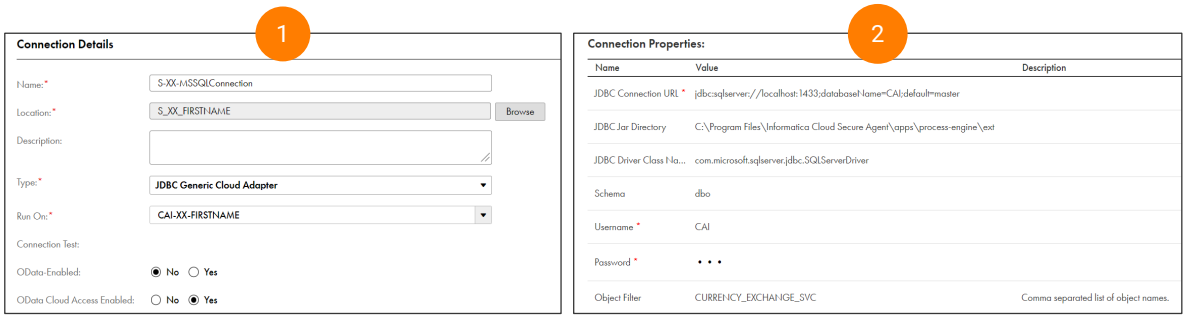
To create a JDBC connection in CAI, you must perform the following steps:

1. In the Application Integration service, create a new **Connection**.
2. In the connection wizard, to configure the connector, in the basic connection properties enter the details such as Name, Location, Type, and Run On. As you are creating a JDBC connection, select connection type as JDBC Generic Cloud Adapter and Run on as your local secure agent to communicate with the MS SQL server.
3. In the Connection Properties section, specify the Connection URL, location on your secure agent machine where the jdbc.jar file is located. You also need to provide the credentials to access the MS SQL server.
4. To validate the connection configuration, use the Test option to test the connection.
5. If the status of the connection is Passed, you can save and publish the connection to use it in other CAI assets.

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Configure Connection Properties

- Creating a JDBC Connection



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To configure the JDBC connection, in the basic connection properties, enter the details such as Name, Location, Type, and Run On.

In the Connection Properties section, specify the connection-specific details to allow the cloud server or secure agent to communicate with the connector.

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JDBC Connection Properties

- Specify the following properties for the JDBC Connection:
  - JDBC Connection URL
  - JDBC JAR Directory
  - JDBC Driver Class Name
  - Schema
  - Username
  - Password
  - Object Filter
  - Time Zone

Connection Properties:

Name	Value	Description
JDBC Connection URL *	jdbc:sqlserver://localhost:1433;databaseName=CAI;default=master	
JDBC Jar Directory	C:\Program Files\Informatica Cloud Secure Agent\apps\process-engine\ext	
JDBC Driver Class Na...	com.microsoft.sqlserver.jdbc.SQLServerDriver	
Schema	dbo	
Username *	CAI	
Password *		
Object Filter	CURRENCY_EXCHANGE_SVC	Comma separated list of object names.
TimeZone	Default	Database timezone. Use this property if DB and Agent are in different timezones

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JDBC Connection URL: This is a mandatory field. It defines the URL schema for the database. The format varies from database to database.
JDBC JAR Directory: The path to the JDBC driver .jar file. For example, you can enter the path: C:/jdbclf
If you do not specify a directory path, the Secure Agent obtains the .jar file from the process-engine/ext directory.
JDBC Driver Class Name: The name of the JDBC driver class. The driver class name varies from database to database.
Schema: Schema name to fetch data. It varies from one database to another database.
Username: Username required to connect to database.
Password: Password required to connect to database.
Object Filter: A comma-separated list of object names.
Time Zone: The time zone of the database. You must specify the value of the database time zone when the machine hosting the database and the machine hosting the Secure Agent are in different time zones.

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Lab Activity

7-1 Create an App Connection using Service Connector

In this lab, you will perform the following:

- Create an App Connection using Service Connector
- Publish the App Connection

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Lab Activity

7-2 Create a JDBC Connection

In this lab, you will perform the following:

- Create a JDBC connection
- Publish the connection

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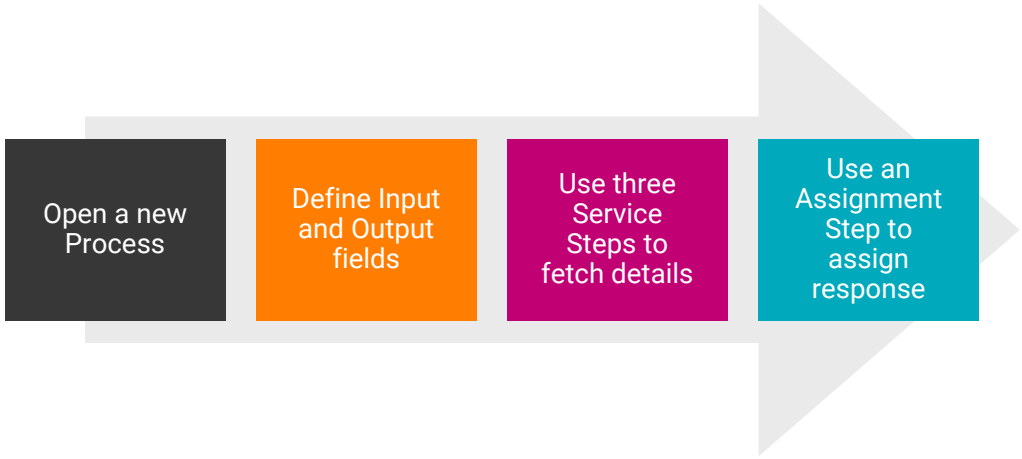
An Example – Customer and Order Details Process

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Create a Customer-and-Order-Details Process

- Create a Process to obtain the Customer and Order details.



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In this Process, you want to fetch customer order details by calling a service in the Process. To create this process, you must perform the following steps:

1. In the Application Integration service, create a new Process. In the Process Designer, provide name to the process and configure an input field CustomerId in of type Text.
2. To save the response that will be received by the Process, define four output fields, all of type Text.
3. Then you will call 3 different services to get Customer details, Order details, and item details. You must make the appropriate field assignment to map the input fields defined in the process to that of field defined in the service connector.

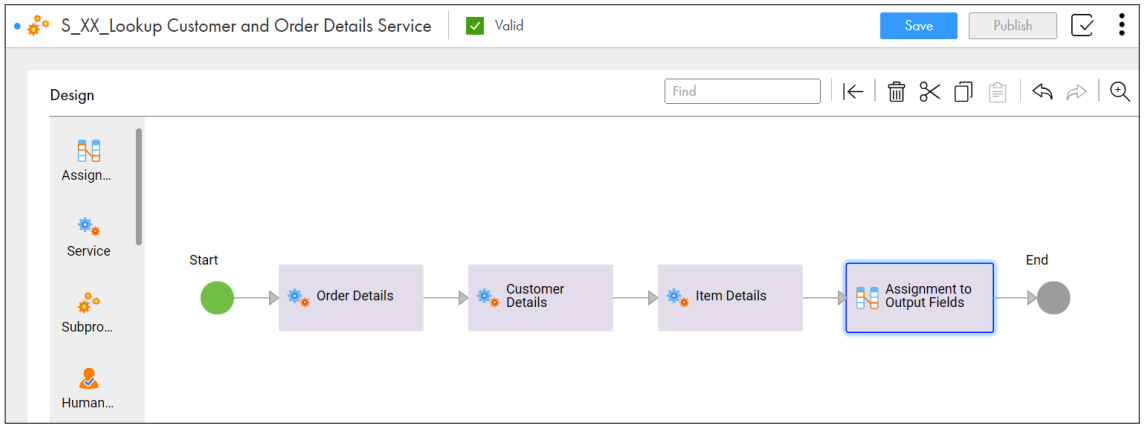
Note: Here, the service type is connection. This connection is using the service connector. You have defined 3 actions in the service connector, where each action is used to fetch Customer details, Order details, and item details respectively.

4. In the Assignment step, you have to assign the response received from service to the output fields defined in the Process.
5. Once this configuration is complete, save and publish the process to get Service URL.
6. You will use this Service URL to call the service using any third-party application.

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Example: Create a Process

Process Configuration



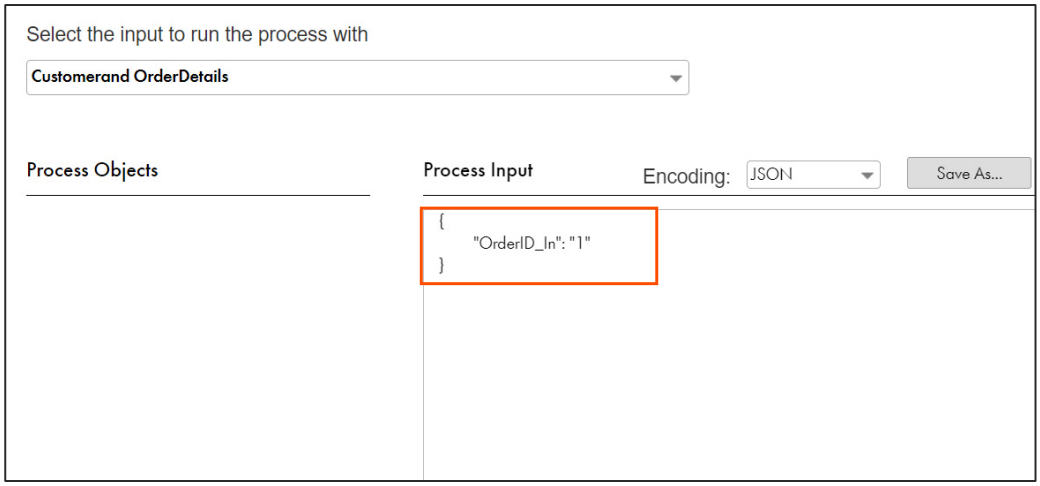
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Example: Create a Process (continued)

Process Input



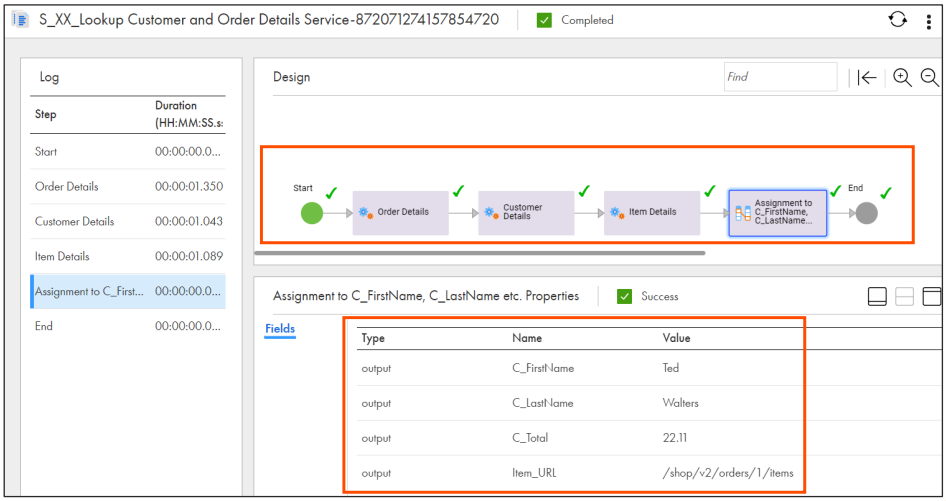
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As an input to the process, you are providing the Order id for which you want to get the details.

Example: Create a Process (continued)

Execution Details



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Note: Click the Assignment step to check the output.

Topic

Knowledge Check



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Which of the following databases can be connected using JDBC connector?

- PostgreSQL
- Microsoft SQL Server
- MySQL
- Oracle
- All of the above

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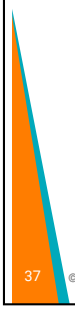
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Answer: All of the above

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When you create and publish a Salesforce connection in CAI, a list of Objects and Actions appear in the Metadata tab.

- True
- False



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Answer: True

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_____ enables interoperability between disparate data sources.

- FTP
- HTTP
- IP
- Odata

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Answer: Odata

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Lab Activity

7-3 Create a Customer and Order Details Process

In this lab, you will perform the following:

- Create a new Process
- Add and configure service steps
- Publish and execute the Process

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Lab Activity

7-4 Create a File Connection

In this lab, you will perform the following:

- Create a File connection
- Publish the connection

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Lab Activity

7-5 Create a Salesforce Connection

In this lab, you will perform the following:

- Install Managed Package
- Create Salesforce Connection

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Lab Activity

7-6 Create a Kafka Connection

In this lab, you will perform the following:

- Start Apache ZooKeeper and Kafka server
- Create a topic
- Create a Kafka connection
- Publish the connection
- Create and publish a process to write data to a topic

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Lab Activity

7-7 Create a Flat File Application Connection

In this lab, you will perform the following:

- Create a Flat File connection

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Lab Activity

7-8 Create a Process with File Listener using a Flat File connection

In this lab, you will perform the following:

- Create a Process
- Configure a File Listener
- Run the mapping and view results

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Module Summary

This module showed you how to:

- Describe the App Connection
- Create App Connections
- Utilize CAI Connectors and Service Connectors in App Connection

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Module 8

Managing and Migrating Assets

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Module Objectives

After completing this module, you will be able to:

- Manage assets in CAI
- Import and export assets in CAI

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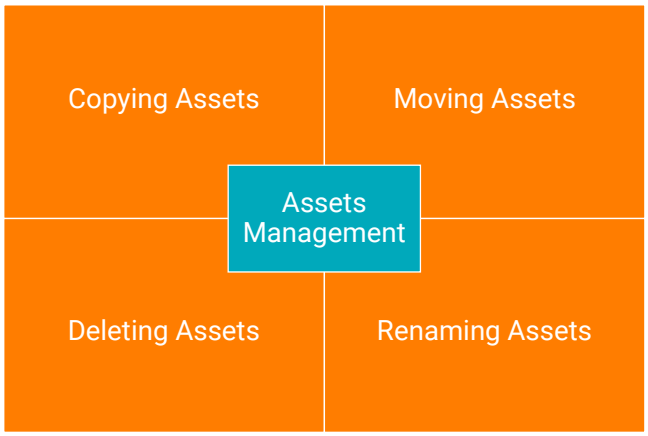
Topic

Assets Management

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Assets Management



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In CAI, you can perform the following Asset Management activities:

1. Copying Assets
2. Deleting Assets
3. Moving Assets
4. Renaming Assets

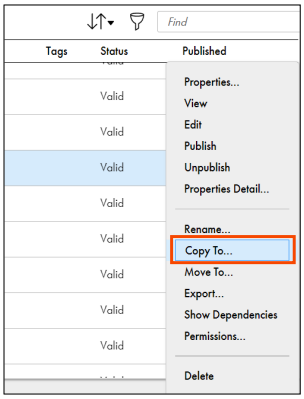
In CAI, you can create the assets in different Projects and folders. The Explore tab of Application Integration service lists all the Projects in your IICS Org. You can create a new folder inside a Project to manage the assets.

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Copying Assets

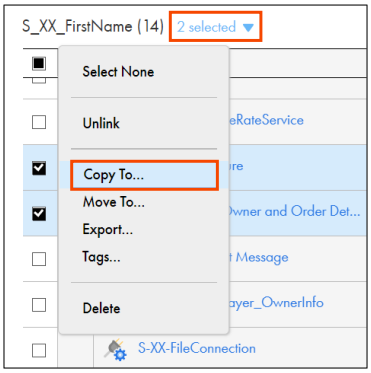
METHOD 1

- Navigate to the asset you want to copy.
- Select the asset and use the 'Copy To' option.



METHOD 2

- Navigate to the asset you want to copy.
- Select the asset and from the 'selected' drop-down, select the 'Copy To' option.



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To copy an asset, perform the following steps:
1. On the Explore page, navigate to the asset that you want to copy
2. In the row that contains the object, click Actions, and select Copy To.
3. You can also use the selected drop-down to copy the asset.

In IICS, you can copy an asset in the same folder where the asset already exists or to a different folder.

If you want to copy an asset in the same folder, IICS provides you two options: To keep both assets or to cancel the operation.

If you select the keep both option, IICS appends the asset name with 'Copy x' where x is the sequential copy number.

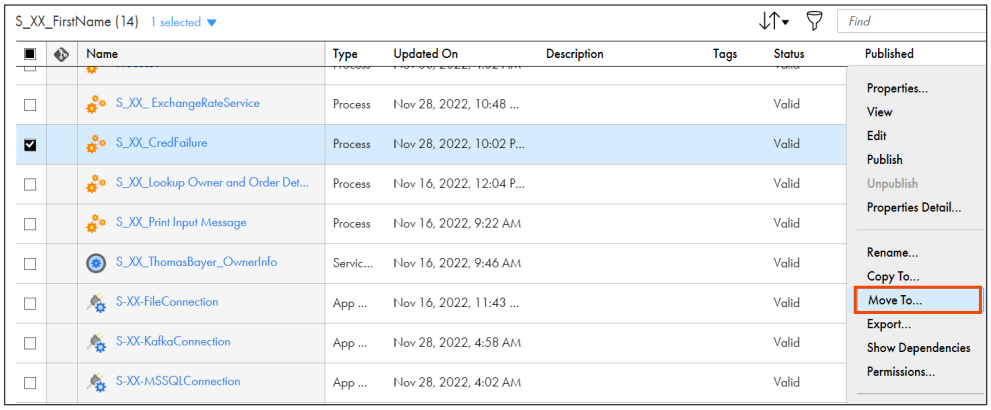
For example, you have an asset named SQLServerConnection. If you copy this asset in the same location, IICS will append Copy 1 to the asset name. So, the copied asset name will be SQLServerConnection-Copy 1.

When you copy an asset into a different folder that already contains an asset with the same name, IICS provides you with the following 3 options:
1. To keep both assets
2. To replace the existing asset with the copied asset
3. To cancel the operation

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Moving Assets

- To move an asset:
  - Navigate to the asset you want to move
  - Select the asset and use the 'Move To' option



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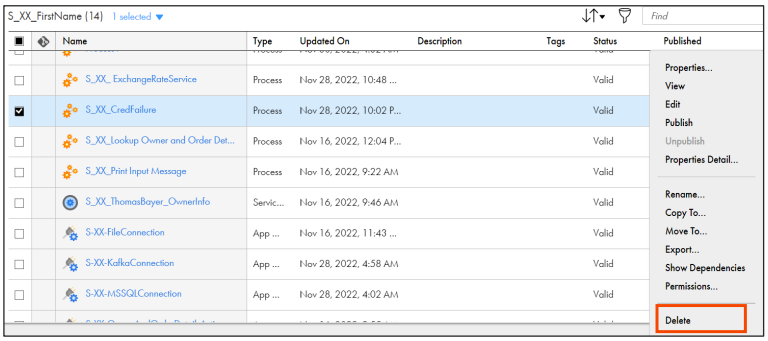
To move an asset from one folder to another project, perform the following steps:

1. On the Explore page, navigate to the asset that you want to copy
2. In the row that contains the object, click Actions and select **Move To**.
3. From the opened Move To window, select the location where you want to move the asset.

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Deleting Assets

- To delete an asset:
  - Navigate to the asset you want to delete
  - Select the asset and use the 'Delete' option
- You cannot delete an asset in the following situations:
  - Asset has cyclic dependency on other assets
  - Asset is a currently running task
  - Asset is used by another asset
  - Asset is published



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You can delete an asset if you no longer need it. To delete an asset, perform the following steps:

1. On the Explore page, navigate to the asset that you want to delete.
2. In the row that contains the object, click Actions and select Delete.

Note: You cannot retrieve projects, folders, or assets after deletion.

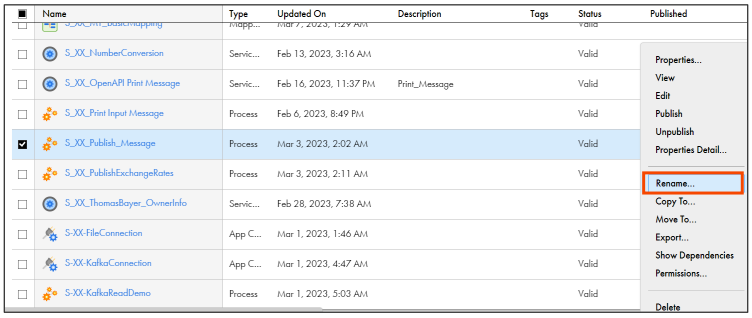
You cannot delete an asset in the following situations:

1. The asset is a task that is currently running.
2. The asset is used by another asset. You must first delete the dependencies of the asset, and then delete the asset. For example, if a connection is used by a process, you cannot delete the connection. You must either delete the process or update the process to use a different connection, and then delete the connection.
3. The asset has a cyclic dependency on other assets. You must delete all the assets simultaneously. For example, if Process A is used by Process B, Process B is used by Process C, and Process C is used by Process A, you cannot delete any of these processes individually. You must delete all of them simultaneously. To do so, select all the processes, click Actions, and then select Delete.
4. The asset is published. You need to first unpublish the asset and then delete the asset.

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Renaming Assets

- To rename an asset:
  - Navigate to the asset that you want to rename
  - Select the assets that you want to rename
  - Click **Action** and use the 'Rename' option
  - Enter the new name and click **Save**



Note: You cannot use special characters in a name or use the same name as another object that is in the same folder.

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You can rename an asset without losing references to objects that they use.

For example, if an asset named S_XX_CustomerOrderList uses an asset named S_XX_ProductList, you can rename the S_XX_CustomerOrderList asset to S_XX_CustomerOrder and it will still reference the S_XX_ProductList asset.

To rename an asset:

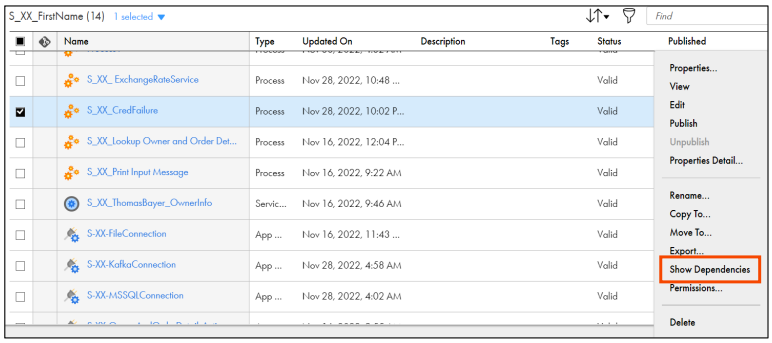
- Navigate to the asset that you want to rename
- Select the assets that you want to rename
- Click **Action** and use the 'Rename' option
- Enter the new name and click **Save**

Note: You cannot use special characters in a name or use the same name as another object that is in the same folder

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Asset Dependencies

- To view an asset dependency:
  - Navigate to the asset of interest
  - Select the asset and use the 'Show Dependencies' option
- The Dependencies page shows the following tabs:
  - Uses
  - Use By



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You can view object dependencies for an asset on the Explore page. To view object dependencies for an asset, in the row that contains the asset, click Actions, and select Show Dependencies.

The Dependencies page shows the Uses tab by default.

The **Uses** tab lists the objects that the asset depends on.
For example, you have a process and a connection. The process uses the connection. Then, the dependencies page for process will list the Connection in Uses tab.

The **Used By** tab lists the objects that the asset is dependent on.

For example, the dependencies page for a connection will list the process in the Used By tab.

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Topic

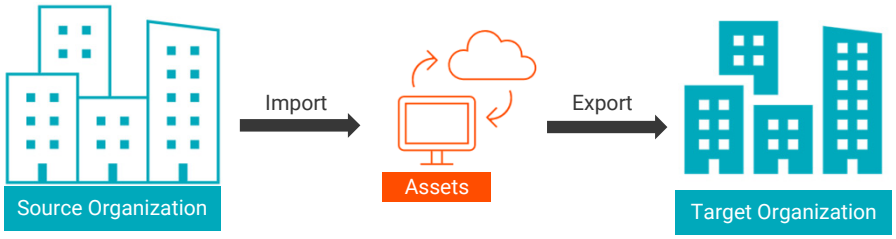
Assets Migration

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Migrating Assets

- You can migrate assets from one Org to another Org
- To migrate assets, you export the assets from the source organization and then import the assets into the target organization
- To export an asset, use the 'Export' option.
- To import an asset, use the 'Import' option.



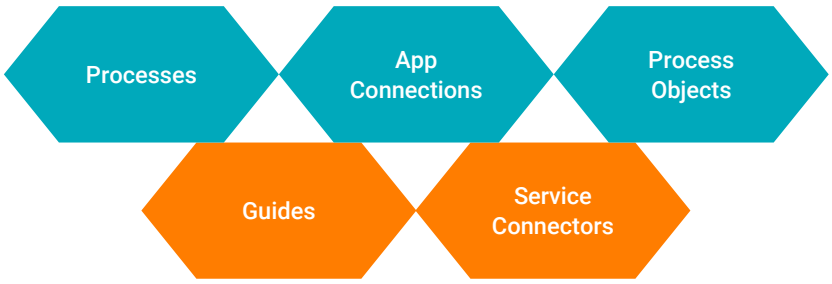
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In IICS, you can migrate assets from one org to another org. To migrate assets, you export the assets from the source org and then import the assets into the target org. You can export single assets, groups of assets, or export all of the assets in a project.

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Migrating Assets (continued)

- You can import and export the following types of assets:



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You can import and export the following types of assets:

1. Processes
2. Guides
3. App Connections
4. Service Connectors
5. Process Objects

If you export a project or folder, the file structure remains intact so that when you perform the import in the target organization, you can duplicate the original structure.

Note: You can view the export/import status in the My Import/Export Logs tab.

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Conditions for Assets Migration

- Requirements for assets migration:
  - The source and target Orgs must have the Import/Export license
  - The target Org must have the licenses required for the imported assets
  - You must have a user account with import and export privileges

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To migrate an asset, ensure that you meet the following requirements:

1. The source and target orgs must have the Import/Export license.
2. The target Org must have the required licenses for the assets that you want to import.
3. You must have a user account in the source and target Orgs with a role that has import and export privileges, such as the Admin or Designer role.

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Export Assets

- When you export assets, Informatica Intelligent Cloud Services creates an export ZIP file that contains the assets that you selected for export
- To export an asset, you need the following privileges and permissions:
  - Your user role must have privileges to export assets
  - You must have read permission on the asset
- You can export a single asset, multiple assets, or a project
- When you export a project or folder, the export file includes all the assets in the project or folder

Note: Informatica recommends that you include no more than 1000 objects in an export file.

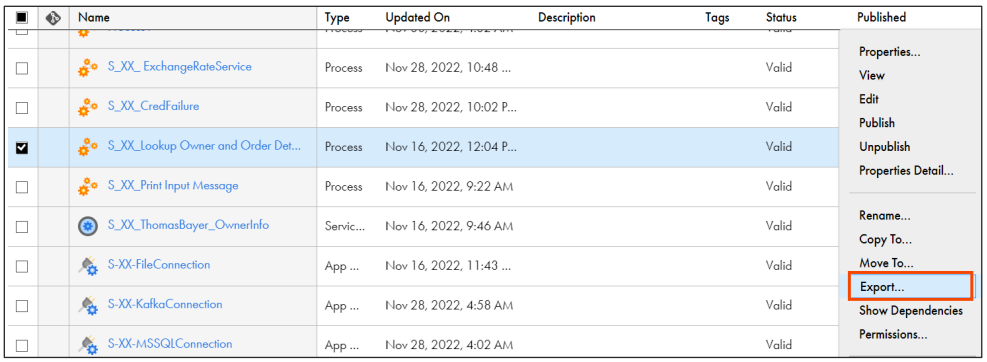
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Export Assets (continued)

- To export an asset:
  - Navigate to the asset you want to move
  - Select the asset and use the 'Export' option
  - From the Import/Export Logs tab, download the exported file



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To export an asset, perform the following steps:

1. On the Explore page, navigate to the asset that you want to export.
2. In the row that contains the object, click Actions, and select **Export**.
3. Download the exported file from the Import/Export Logs tab and save it on your local machine.

Import Assets

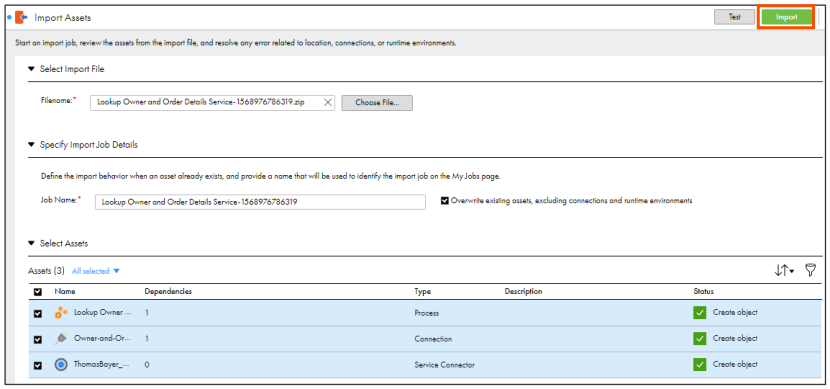
- You can import all the assets in an export file or select the assets that you want to import
- To import an asset, you need the following privileges and permissions:
  - Your user role must have privileges to import assets
  - If you import an asset into the target project as a new asset, you must have create, update, and read permissions on the asset
  - If you overwrite an asset in the target project, you must have update and read permissions on the asset
- The target organization must have all the required licenses for the assets being imported

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Import Assets (continued)

- To import an asset:
  - Login to the target Org
  - On the Explore page, click the 'Import' option



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To import an asset, perform the following steps:

1. Login to the target Org, and from the Explore page, select the **Import** option.
2. On the Import Assets page, select the file you want to import.
3. While importing, you can choose if you want to overwrite existing assets with the assets in the import.

If you choose to overwrite existing assets, when an asset has the same name as an asset in the target project, the asset replaces the existing asset in the target project. If you do not choose this option, if an asset with the same name exists in the target project, the asset is not imported.

4. If you want to publish assets when you import them, select **Publish Application Integration assets after import**.
5. You can use the **Test** option to see the potential results of the import.

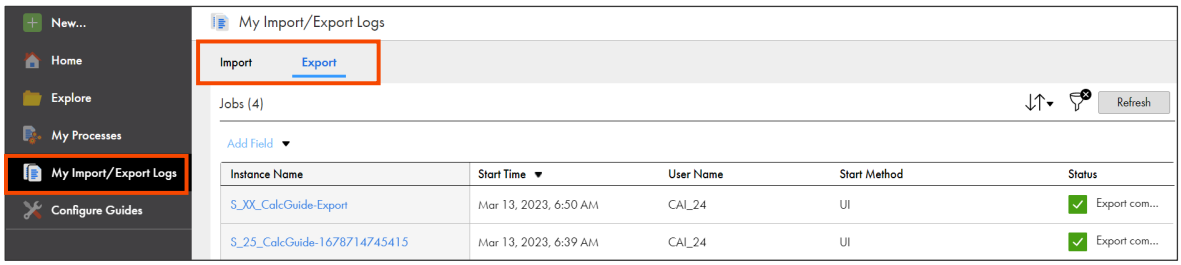
For example, the exported file contains three assets that are dependent on each other, and while importing you choose only two assets to import. So, when you click Test, the test result may show that the test failed as one or more objects that are dependent on it are missing for import.

4. Click the **Import** option to import the assets.

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Viewing Import and Export Logs

- To view properties for an import or export instance, click the **Import** or **Export** tab at the top of the My Import/Export Logs page.
- By default, the My Import/Export Logs page lists the import and export instances that were started within the last 30 days.



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Topic

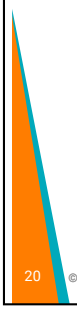
Knowledge Check

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Can you delete an asset if it is used by another asset?

- Yes
- No



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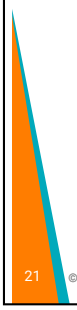
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Answer: No

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You can rename an asset without losing references to objects that they use.

- True
- False



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Answer: True

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Lab Activity

8-1 Export an Asset

In this lab, you will perform the following:

- Export an asset
- View the log details

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Module Summary

This module showed you how to:

- Manage assets in CAI
- Import and export assets in CAI

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Module 9
Using XQuery in CAI

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Module Objectives

After completing this module, you will be able to:

- Describe XQuery and its functions

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Topic

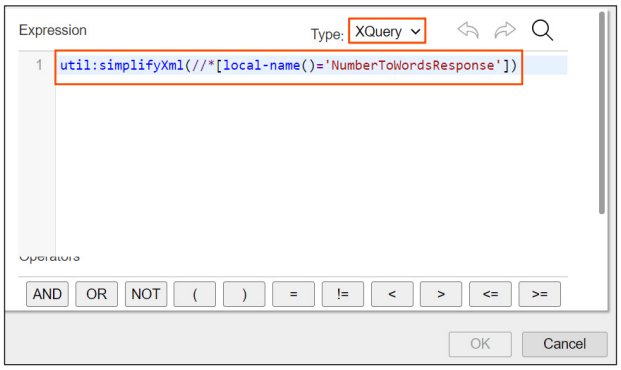
XQuery

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What is XQuery?

- XQuery is a language for finding and extracting elements and attributes from XML documents
- XQuery for XML is like SQL for databases
- XQuery can be used to:
  - Extract information to use in a Web Service
  - Generate summary reports
  - Transform XML data to XHTML
  - Search Web documents for relevant information



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XQuery Basic Syntax Rules

- XQuery is case-sensitive
- XQuery elements, attributes, and variables must have valid XML names
- XQuery string value can be in single or double quotes
- XQuery variable is defined with a $ followed by a name
  - **For example, $employee**
- XQuery comments are delimited by (: and :)
  - **For example, (: XQuery Comment :)**

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XQuery FLWOR Expression

- FLWOR (pronounced as flower) is an acronym for **For**, **Let**, **Where**, **Order by**, **Return** clauses which are used in XQuery
- A FLWOR expression is used to select and filter elements from the database
- The following table describes the function of each clause:

Clause	Function
For	Selects a sequence of nodes
Let	Binds a sequence to a variable
Where	Filters the nodes
Order by	Sorts the nodes
Return	What to return in the result

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The for Clause

- The for clause binds a variable to each item returned by the expression
- The for clause results in iteration
- There can be multiple for clauses in the same FLWOR expression

Expression

```
1  for $i in 1 to 10
2  return
3  <Employee>
4     <EMP_ID>{$i}</EMP_ID>
5     <FIRST_NAME>{fn:concat('Harpreet',$i)}</FIRST_NAME>
6     <LAST_NAME>{fn:concat('Bhatia',$i)}</LAST_NAME>
7     <CREATED_DATE>{xs:string(fn:current-date())}</CREATED_DATE>
8     <CLOBFIELD></CLOBFIELD>
9     <TEST2>Test</TEST2>
10 </Employee>
```

7

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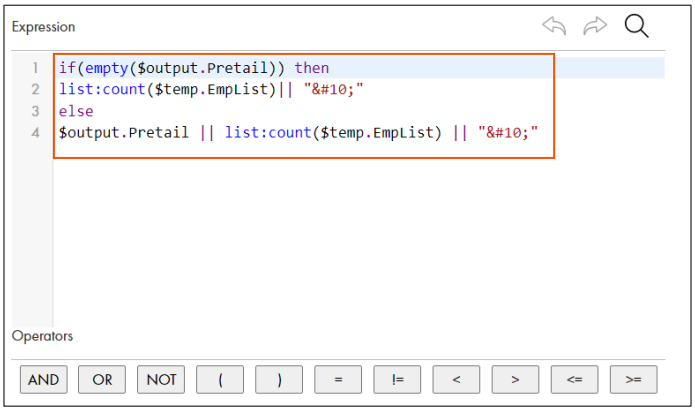
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if-then-else Expression

- XQuery allows conditional expressions using the keywords if, then, and else
- if-then-else checks the validity of the input values passed
- The syntax of a conditional expression is:

if (<expr>) then <expr> else <expr>



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XQuery Functions

- XQuery includes over 100 built-in functions
- XQuery includes functions for string values, numeric values, date and time comparison, sequence manipulation, boolean values, and more
- Functions are specified with the fn: prefix
  - **For example: fn:string()**

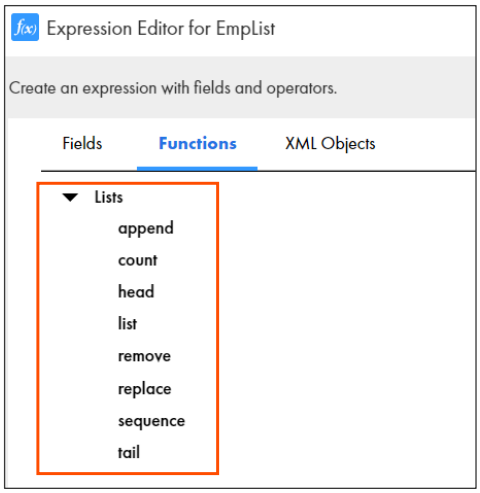
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Lists Functions in XQuery

- You can use the following Lists functions in XQuery:
  - append
  - count
  - head
  - list
  - remove
  - replace
  - sequence
  - tail



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Lists Functions in XQuery (continued)

- The following table describes the usage of each Lists functions in XQuery:

Function	Usage
append	Returns an array containing all the elements of a supplied array, plus one additional member at the end
count	Counts and returns the total elements present in a sequence
head	Returns the first element in a sequence
list	Returns the list of elements present in a sequence
remove	Removes an item in a given sequence from any position
replace	Replaces the matched input string with a given string
sequence	Returns an ordered collection of items
tail	Returns an array containing all members except the first from a supplied array

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Commonly Used Functions in XQuery

- The following table describes some commonly used functions of XQuery:

Function	Usage
string-length()	Returns the length of the string
concat()	Returns the concatenated string as output
string-join()	Returns the combination of items in a sequence separated by a delimiter
current-date()	Returns the current date
current-time()	Returns the current time
current-datetime()	Returns both the current date and the current time

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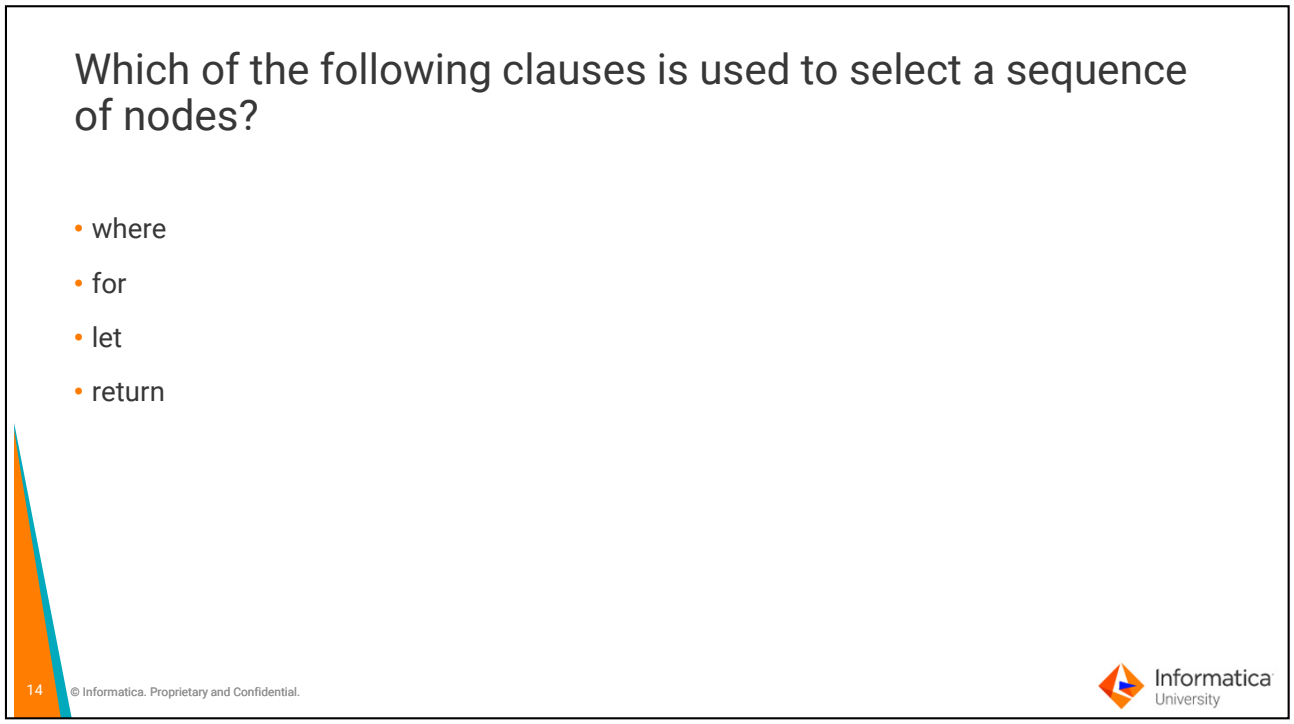
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Knowledge Check

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Answer: for

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Which of the following Lists functions is used to return an array containing all members except the first from a supplied array?

- count
- head
- append
- tail

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Answer: tail

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Lab Activity

9-1 Create a Process Using Lists Function of XQuery

In this lab, you will perform the following:

- Create a Process Object – Employee
- Create a Process using Lists functions

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Module Summary

This module showed you how to:

- Describe XQuery and its functions

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Module 10

Adding Web Services to a Process

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Module Objectives

After completing this module, you will be able to:

- List types of web services
- Add the Synchronous and Asynchronous web service calls to processes

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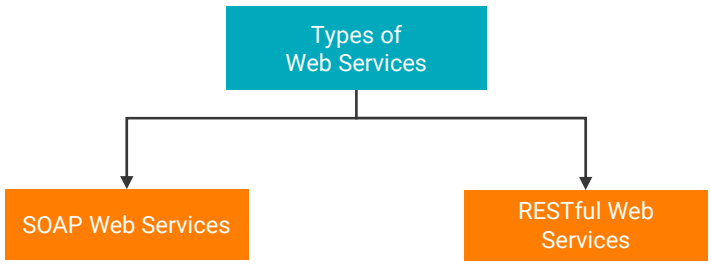
Web Services Overview

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Web Services

- Collection of open protocols and standards for exchanging data between applications or systems



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A web service is a collection of open protocols and standards used for exchanging data between applications or systems. Software applications written in various programming languages and running on various platforms can use web services to exchange data over the Internet.
Web service operations include requests for information, requests to update data, or requests to perform tasks. For example, a web service client sends a SOAP request to run a web service operation called getCustomerOrders. The web service client passes a customer ID in the request. It retrieves the customer and the order information and returns the information to the client in response. The components of web services define its purpose and how the client can communicate with the web service.

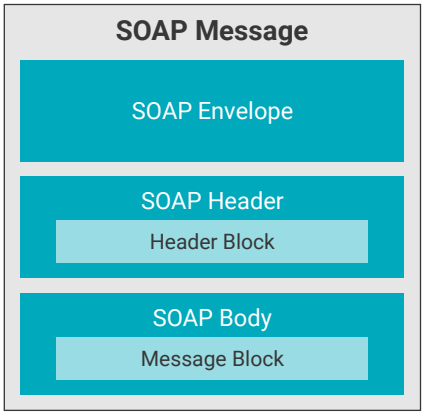
Service Operations component defines one or more operations a web service can perform. Each operation corresponds to an action in the web service.

There are mainly two types of web services supported by IICS:
1. SOAP web services
2. RESTful web services

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Simple Object Access Protocol (SOAP)

- SOAP is the communication protocol for web services
- A SOAP message contains the following sections:
  - SOAP Envelope
  - SOAP Header
  - SOAP Body



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SOAP is the communications protocol for web services. It defines the format for web service requests, responses, and fault messages. The SOAP web service contains a WSDL component, which is an XML document that describes the protocols, formats, and signatures of the web service operations.

A SOAP message contains the following sections:

SOAP envelope: It encapsulates all of the necessary details of the SOAP messages, which are exchanged between the web service and the client application. The SOAP envelope element is used to indicate the beginning and end of a SOAP message.

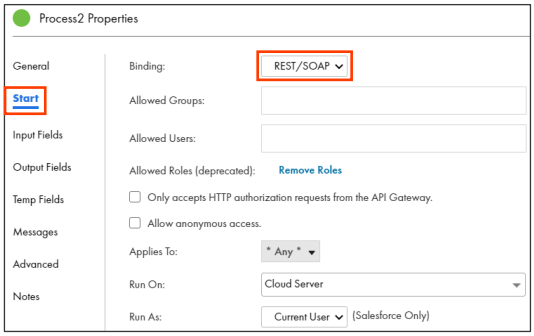
SOAP header: It includes authentication information and details about how to process the SOAP message.

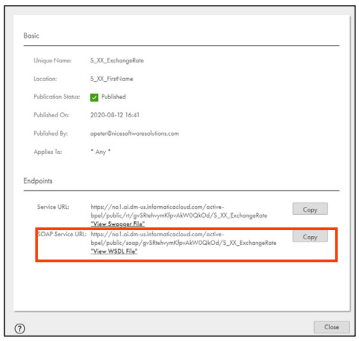
SOAP body: It contains the data that the client and web service provider pass between each other. SOAP messages are in the XML format.

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SOAP Endpoint in Process Designer

- To define a SOAP **Endpoint** in Process Designer:
  - In the process properties, for the Start step, select Binding as **REST/SOAP**.
  - To access the SOAP URL, from the Actions menu, select Properties Details.
  - Use the SOAP Service URL generated after publishing the Process.





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Process Designer exposes Web Services through an automatically generated WSDL interface that you can access and use as a SOAP endpoint. Each Web Service you interact with contains a set of methods, known as operations, hosted on an application server. You can invoke these operations remotely in the cloud or over a network. When you invoke an operation, you create a SOAP message and send it to the Web Service, typically over HTTP/ HTTPS.

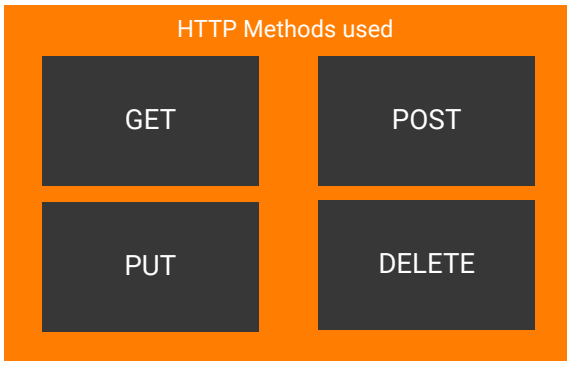
To define a SOAP endpoint in Process Designer, follow the given steps:

1. In the process properties, for the Start step, select **Binding** as REST/SOAP.
2. After you define and publish the process, CAI automatically generates a SOAP Service URL for the Process.
3. To access the SOAP service URL, from the process designer, select **Actions** > **Properties** details.
4. You can use the 'View WSDL File' option to open the complete WSDL file and view the SOAP message generated for the process.
5. You can copy the SOAP Service URL to use it as a SOAP endpoint.

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Representational State Transfer (REST)

- Stands for Representational State Transfer
- A web standards-based architecture that uses HTTP Protocol



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REST stands for Representational State Transfer. It is a web standards based architecture and uses HTTP Protocol.

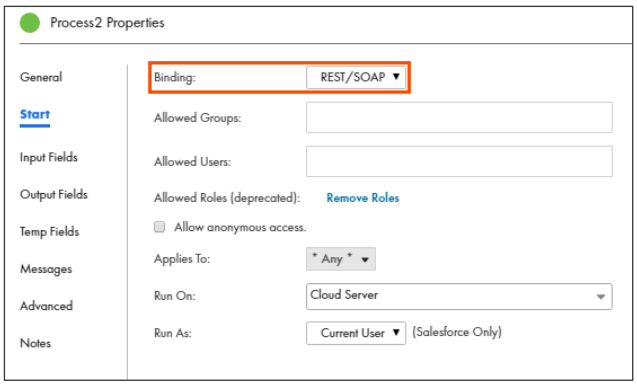
In REST, every component is a resource and a resource is accessed by a common interface using HTTP standard methods. A client can access the resource using the unique Uniform Resource Locator (URL) and a representation of the resource is returned. With each new resource representation, the client is said to transfer the state. While accessing RESTful resources with HTTP protocol, the following four HTTP methods are commonly used:

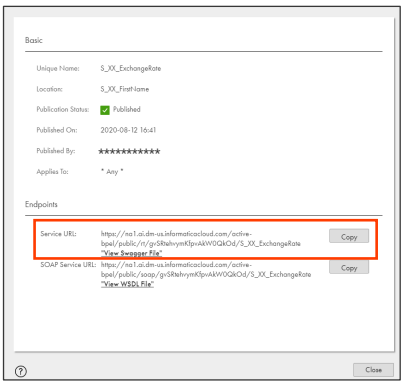
1. **GET**: Provides read-only access to a resource.
2. **POST**: Used to create a new resource.
3. **DELETE**: Used to remove a resource.
4. **PUT**: Used to update an existing resource or create a new resource.

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REST Endpoint in Process Designer

- To define a REST Endpoint in Process Designer:
  - In the process properties, select **Start**, and from the Binding drop-down, select 'REST/SOAP'.
  - Use the Service URL generated after publishing the Process.





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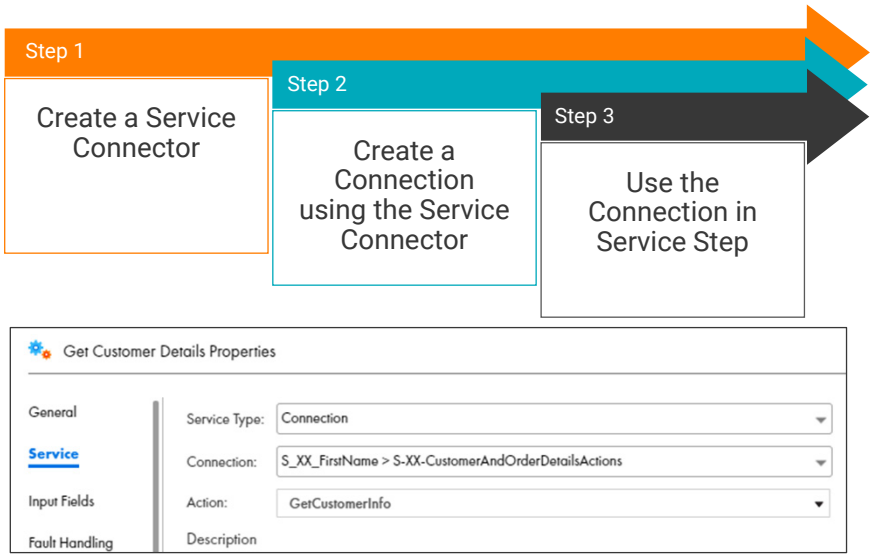
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Similar to SOAP endpoint, to define a REST endpoint in Process Designer, follow the given steps:

1. In the process properties, for the Start step, select Binding as REST/SOAP.
2. After you define and publish the process, CAI automatically generates a REST Service URL for that Process.
3. You can access the Service URL for a Process from Actions > Properties Details option in process designer.
4. You can use the 'View Swagger File' option to view the Swagger JSON specification of the payload for a process that you create in the Informatica Process Designer.
5. You can copy the Service URL to use it as a REST endpoint.

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Adding Web Service in a Process



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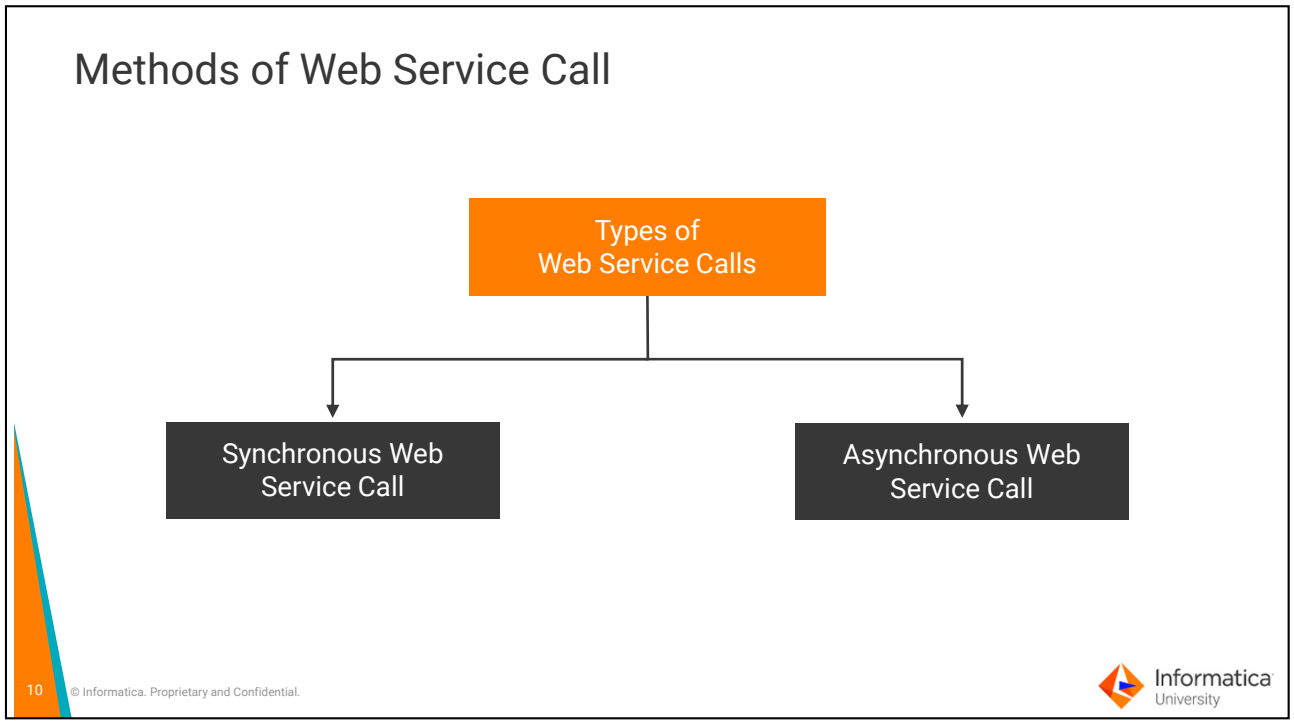
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Informatica allows its users to access REST/SOAP web services using service connectors. To add a web service in a Process, perform the following steps:

1. Create a Service Connector.
2. After you create a service connector, to use it in a Process, you must create a connection using that service connector.
3. Create a Process.
4. To access the service connector from Process, add a Service Step in the Process.
5. In the Service properties of the service step, select the **Service Type** as 'Connection'.
6. In the Connection field, select the connection you created using the service connector.
7. In the Action field, select the **action** you want the service to access.

Note: The action is configured in the service connector.

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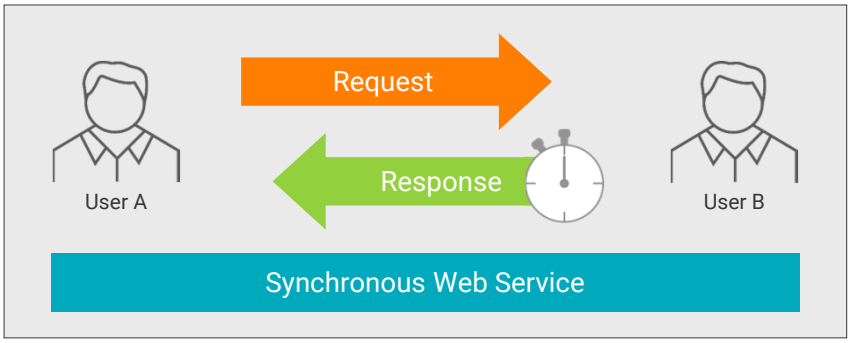
Informatica cloud supports two types of web service call:

1. Synchronous web service call
2. Asynchronous web service call

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Synchronous Web Service

- Synchronous web service provides an immediate response to a query



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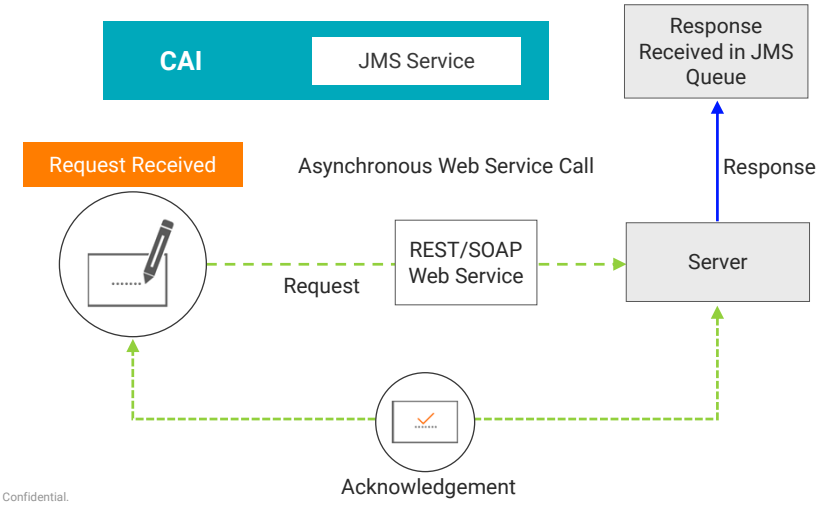
In a synchronous web service call, the calling service sends a request to another service and waits for the service to complete. Only when it receives the result of the service, the calling service continues processing. A timeout value can be defined so that if the service does not finish within the defined period, the call is assumed to have failed and the caller service continues processing. A synchronous service call returns an immediate response to a query.

Calling one process in another process using the Subprocess step, is an example of a Synchronous web service call.

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Asynchronous Web Service

- Client invokes the web service and resumes with other proceedings without waiting for a response



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In Asynchronous web service call, The client can invoke the web service and resume with other proceedings without waiting for a response. In an Asynchronous Process, a significant amount of time is required to process a request. A service call to Messaging service such as JMS is an example of an Asynchronous call.

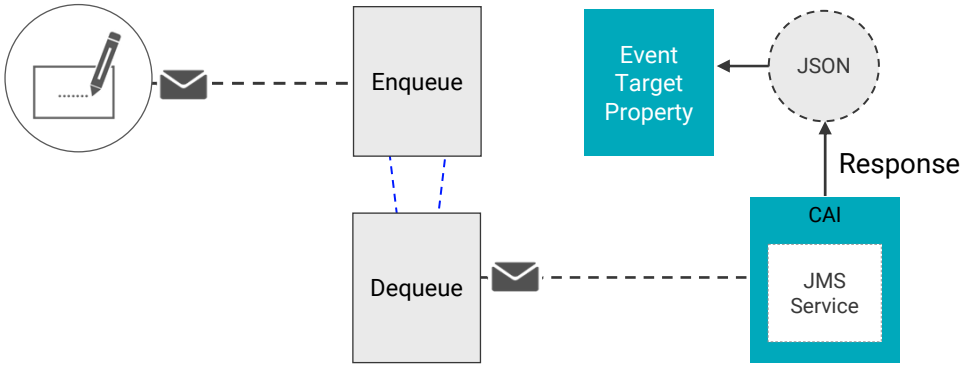
The following pointers explain how an Asynchronous Web Service works:

1. The client submits a request to an Asynchronous Web Service.
2. The Web Service receives the request and stores it in a request queue.
3. The Web Service also sends a receipt of confirmation to the client.
4. The listener associated with the provider, for example, JMS receives the message and initiates processing of the request.
5. After completion of the Process, the response is returned to the JMS queue. The queue returns a receipt confirmation message to the response service.
6. The response service then returns a confirmation message to the response queue to terminate the sequence.
7. Finally, the client can view the response stored in the JMS queue.

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Java Messaging Service

- Java Message Service (JMS) API is a Java message-oriented API
- Used to exchange messages between two or more clients



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The user creates and stores a request in the queue. The request message is dequeued and then processed by the JMS Service configured in CAI that generates a response.

In the next step, the file writer process is invoked to write the response to a file. Based on the response payload settings, a comma delimited file or a JSON file is generated, and a response is written to a location specified in the event target property field.

Enqueue: Process of storing an element in the queue.

Dequeue: Process of removing an element from the queue.

File Writer: In the file connection process, user specifies the event source and event target.

The event target is the location where you want to store the output of the process. The event target reads the data from a Process Object and writes it to an automatically generated comma delimited file.

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Topic

Implementing Synchronous Web Service Call

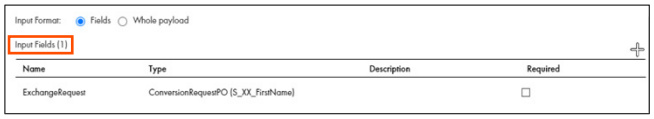
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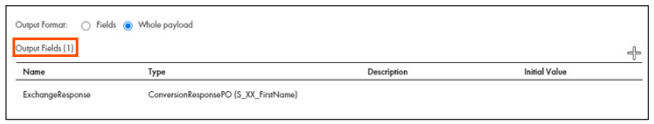
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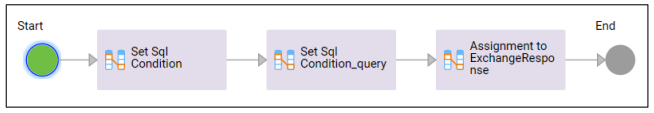
Implementing Synchronous Web Service Call

Step 1 Create an ExchangeRate Process and set Run on as Secure Agent as it will be called by another process

- Define Input, Output, and Temp fields
- Use Assignment step to assign values to fields
- Save and Publish Process







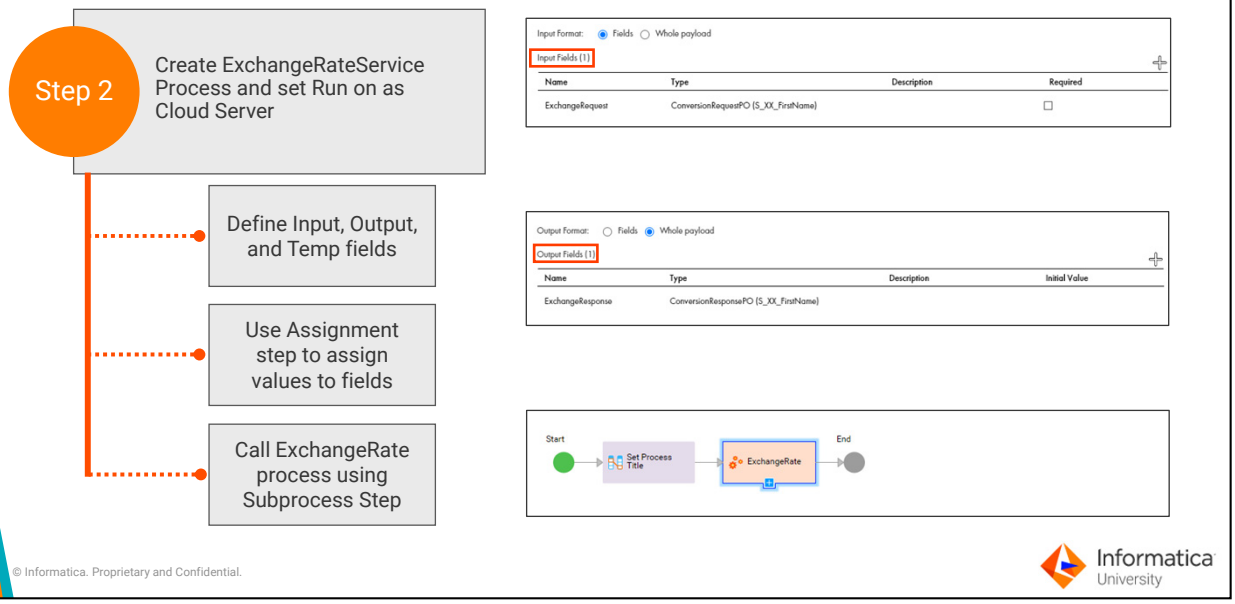
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In this example, you will create two processes – ExchangeRate process and ExchangeRateService process. One process runs on the selected secure agent and the other process executes on the cloud server. Using the Service URL generated by the process that executes on the cloud server, you will run the process using synchronous web service call.

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Implementing Synchronous Web Service Call (continued)



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In the ExchangeRateService process, perform the following steps:

1. Create an input field ‘ExchangeRequest’ of custom data type. Similarly, create an output field ‘ExchangeResponse’ of custom data type. Here, the custom data type is process objects.
2. To hold the temporary values during the process execution, define a temp field ‘Title’.
3. Use the Assignment step to fetch the data from input field, process the data using the given formula, and then save it in the temp field ‘Title’.
4. As you have already performed conversion of currency in ExchangeRate process, call the ExchangeRate process in this process using Subprocess step.
5. After that, you must save and publish the process to get the Service URL.
6. Run the process from the interface.
7. After a request is received by the process, Informatica Cloud gets the conversion rates from the MS SQL database and then performs arithmetic calculations and returns the converted value to the user.

Note: This is an example of synchronous web service call, as the client waits to get conversion rates from MS SQL database (server) before the client can proceed with the process execution.

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Lab Activity

10-1 Invoke a Synchronous Web Service Call

In this lab, you will perform the following:

- Create ExchangeRate process
- Create ExchangeRateService process
- Execute the ExchangeRateService process using a Rest Client

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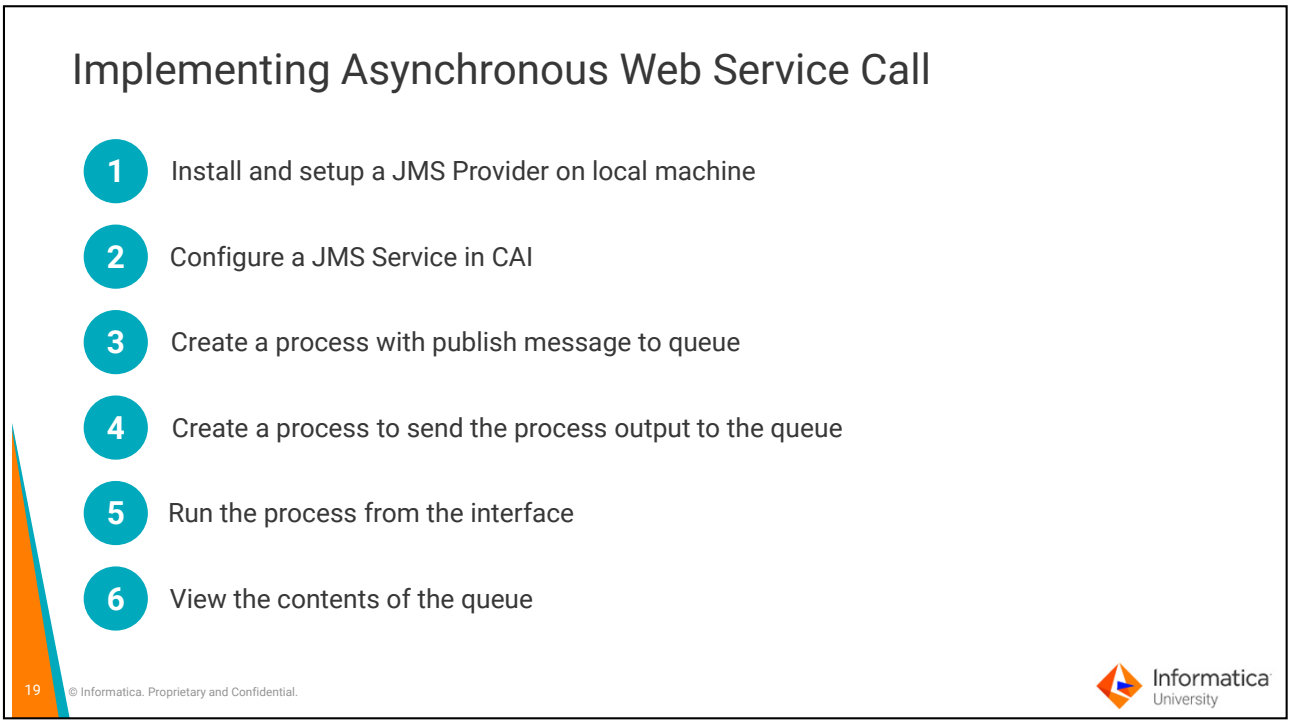
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Implementing Asynchronous Web Service Call

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To perform an asynchronous web service call, perform the following steps:

1. Install JMS provider ActiveMQ on the machine on which your Secure Agent is installed.
2. After you install the ActiveMQ, create a queue to read and write messages.
3. To configure the JMS messaging service in CAI, navigate to Application Integration Console service and select messaging service.
4. To configure an asynchronous web service call, in the messaging service you must set the Listener count to 0 so that the queue is not de-queued automatically.
5. Next, you must create **Publish_Message** process to publish the message to the queue created in ActiveMQ.
6. As you want to send the output of **ExchangeRate** service to the ActiveMQ queue, create **PublishExchangeRates** process and call **ExchangeRate** and **Publish_Message** as Subprocesses.
7. Save and publish the **PublishExchangeRates** process to get the Service URL.
8. Run the process from the interface.
9. To view the response, login to **ActiveMQ** and select the queue to view the message details.

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Implementing Asynchronous Web Service Call (continued)

Setup JMS Provider

- Create a queue named local-queue in ActiveMQ



Note The number of messages Enqueued and Dequeued indicates that the queue is empty.

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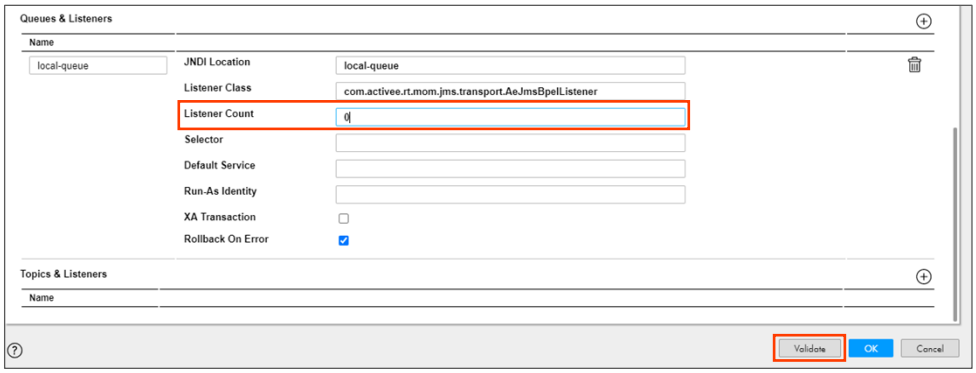
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Implementing Asynchronous Web Service Call (continued)

JMS Service Configuration

- Configure the service and set Listener Count as 0



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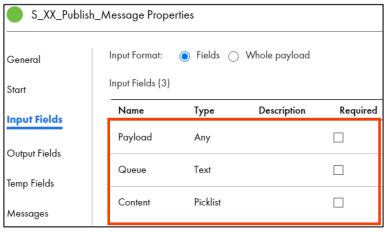
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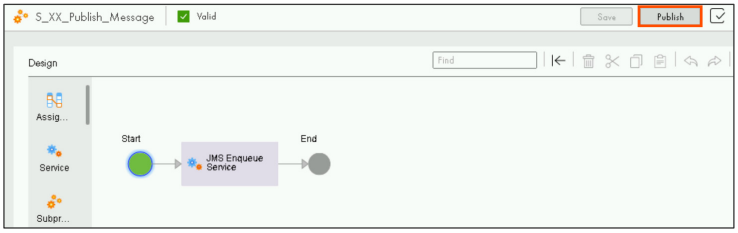
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Implementing Asynchronous Web Service Call (continued)

Create Publish_Message Process

- Define Input Fields and set content type as XML, JSON
- Call the JMS Enqueue service to send data in the queue





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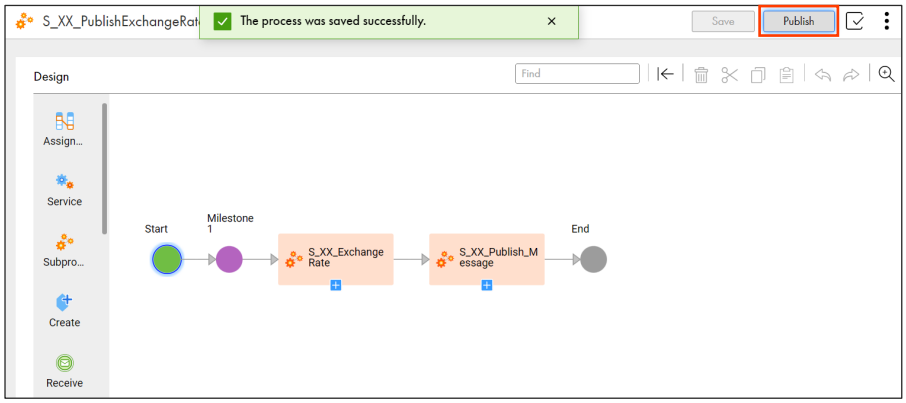
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Implementing Asynchronous Web Service Call (continued)

Create PublishExchangeRates Process

- Call ExchangeRate process to perform currency conversion
- Call the Publish_Message process to send the output of the ExchangeRate process to the queue



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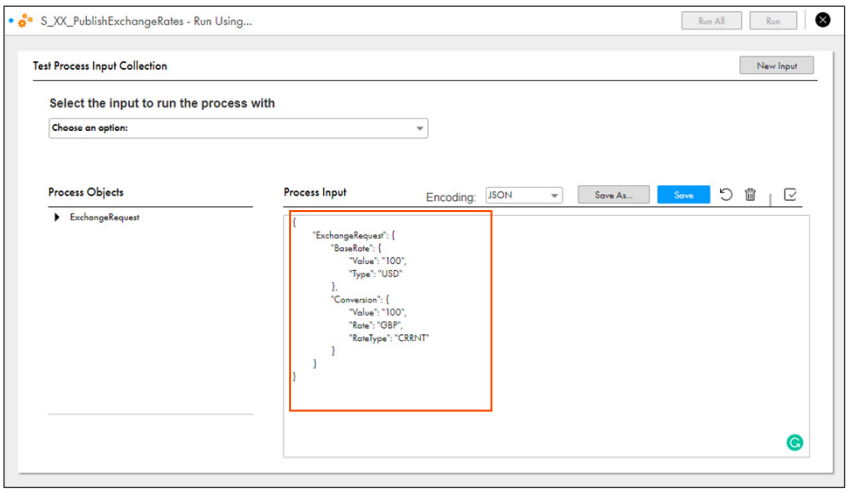
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Implementing Asynchronous Web Service Call (continued)

Process Input



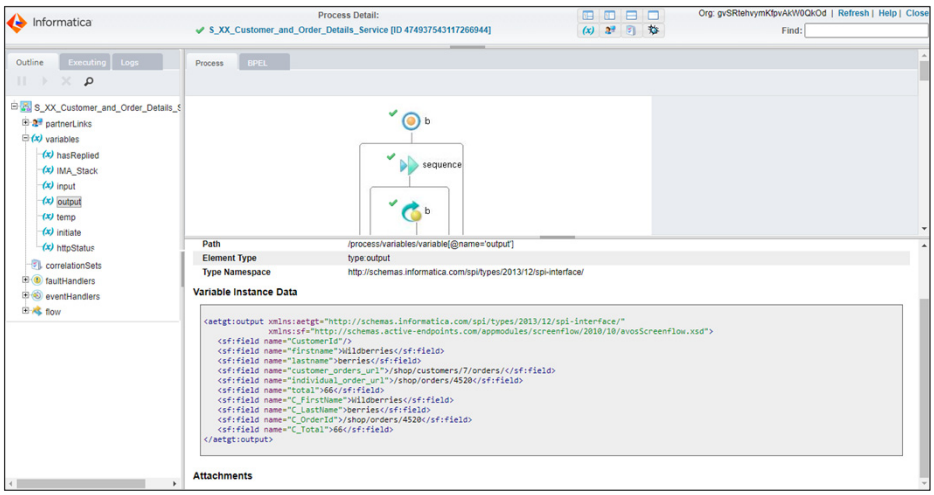
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Implementing Asynchronous Web Service Call (continued)

Process Output



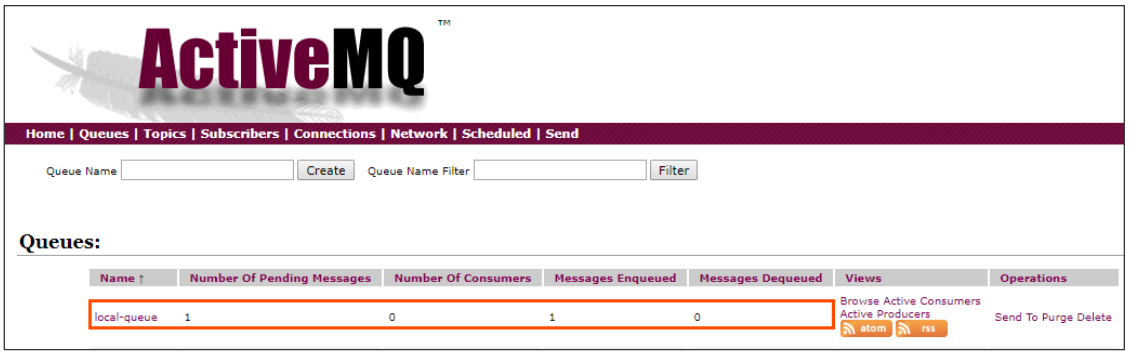
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Implementing Asynchronous Web Service Call (continued)

Process Output



Note The Message Enqueued in the local-queue is 1.

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Topic

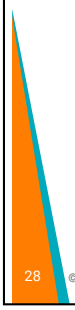
Knowledge Check

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In a _____, the calling service sends a request to another service and waits for the service to complete.

- Synchronous Web Service Call
- Asynchronous Web Service Call
- Java Messaging Service
- None of the above



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Answer: Synchronous web service call

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What does SOAP envelope do?

- It encapsulates all of the necessary details of the SOAP messages, which are exchanged between the web service and the client application.
- It includes authentication information and details about how to process the SOAP message.
- It contains the data that the client and web service provider pass between each other. SOAP messages are in the XML format.
- None of the above

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Answer: It encapsulates all of the necessary details of the SOAP messages, which are exchanged between the web service and the client application.

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Lab Activity

10-2 Invoke an Asynchronous Web Service Call

In this lab, you will perform the following:

- Install and setup a JMS Provider
- Configure a JMS Service in CAI
- Create a process with asynchronous web service call
- Run the process from a third-party application
- View the contents of the queue after message has been written

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Module Summary

This module showed you how to:

- List types of web services
- Add the Synchronous and Asynchronous web service calls to processes

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Module Objectives

After completing this module, you will be able to:

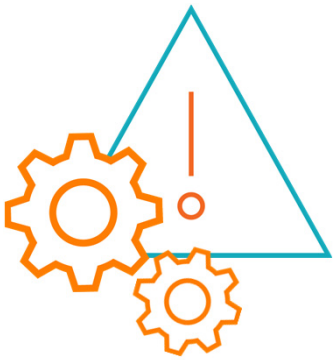
- Describe Fault Handling
- Explain the significance of Fault Handling
- Describe Fault Handling techniques

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Fault Handling Overview



Allows you to anticipate, detect, and resolve errors

Helps in normal flow of execution of an application

Plays a vital role in the application's user experience

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Fault handling is a process which allows you to anticipate, detect, and resolve errors that occur when you execute applications and processes. When an error occurs in an application, it prevents the normal flow of execution of the application. Fault handling is required for most of the applications, and it plays a vital role in providing a good user experience. In CAI, when you integrate applications with Process Designer, you can handle error conditions in the following two ways:

1. Catching a fault in a process using boundary events.
2. Returning faults from a process either with output fields or by adding a Throw step to the process to throw a fault.

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Need of Fault Handling

- Fault handling improves user experience
- Informs users that an error has occurred and what actions they must take

Fault Scenario	Resolution
User provides incorrect input	Display message to change the input and try again
Service is not available	Display message to access the service after some time
Feature is disabled in the application	Display message to contact Support
Application throws unexpected error	Display message to restart the application

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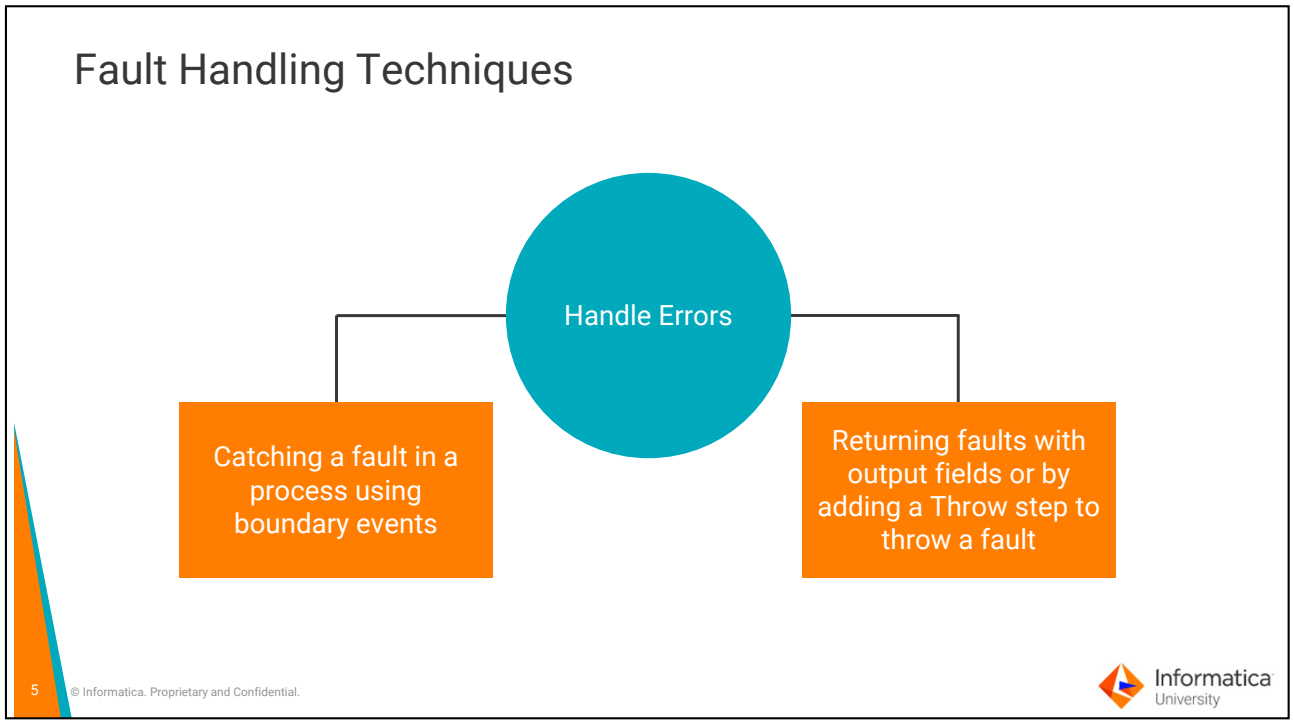
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Efficient fault handling improves the user experience. Fault handling informs users that something unexpected has happened in the application, and takes actions to resolve the error.

Faults can occur in following scenarios:

1. **User provides incorrect input**: If a user provides incorrect input, then a message is displayed stating that the user must change the input and try again.
2. **Service is not available**: If a user tries to access a service that is currently unavailable, then a message is displayed stating that the user must try to access the service after some time.
3. **Feature is disabled in the application**: If certain features are disabled in the users' application, then a message is displayed stating that the user must contact the Support team to get it enabled in the application.
4. **Application throws an unexpected error**: If the application throws an unexpected error, then a message is displayed stating that the user must restart the application.

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When you integrate applications with Process Designer, you can handle error conditions using the following two methods:

1. Catching a fault in a process using boundary events.
2. Returning faults from a process either with output fields or by adding a Throw step to the process to throw a fault.

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Fault Handling Using Boundary Events

A **Boundary Event** is used to catch errors that occur on the boundary of a particular step.

- The fault handler waits for a fault on the step.
- When the handler detects a fault, the step is interrupted, and the process follows the fault path.
- You can configure various fault handling procedures in the fault path.

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A boundary event is used to catch errors that occur on the boundary of a particular step, that is, within the scope of the step where it is defined. When you enable fault handling for a step in a process, you define a boundary event. When a process executes, the fault handler waits for a fault on the step. When the handler detects a fault, the step is interrupted, and the process follows the path you define for handling the fault. You can configure the following options in the fault path:

1. Send email notification and terminate the process
2. Retry after waiting for a specified time interval
3. Suspend the process
4. Complete the process using an alternative path

Note: A fault handler on a step is an interrupting boundary event. This means that when a fault occurs, the process follows the fault path and the main path from the step that is not executed.

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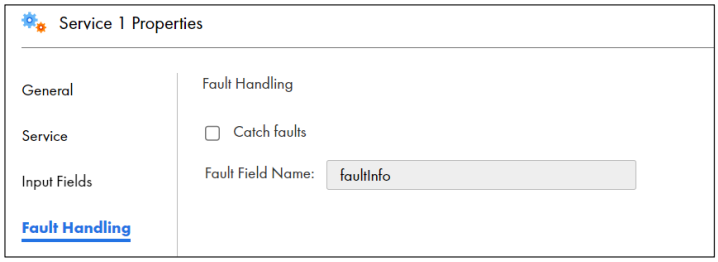
Fault Handling Using Boundary Events (continued)

Configuration

- You can enable fault handling on the following types of steps:

Service Step | Subprocess Step | Create Step

- On the Fault Handling tab for each of these Step types, you can set following properties:
  - Catch Faults
  - Fault Field Name



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In CAI, you can enable fault handling for the following step types:

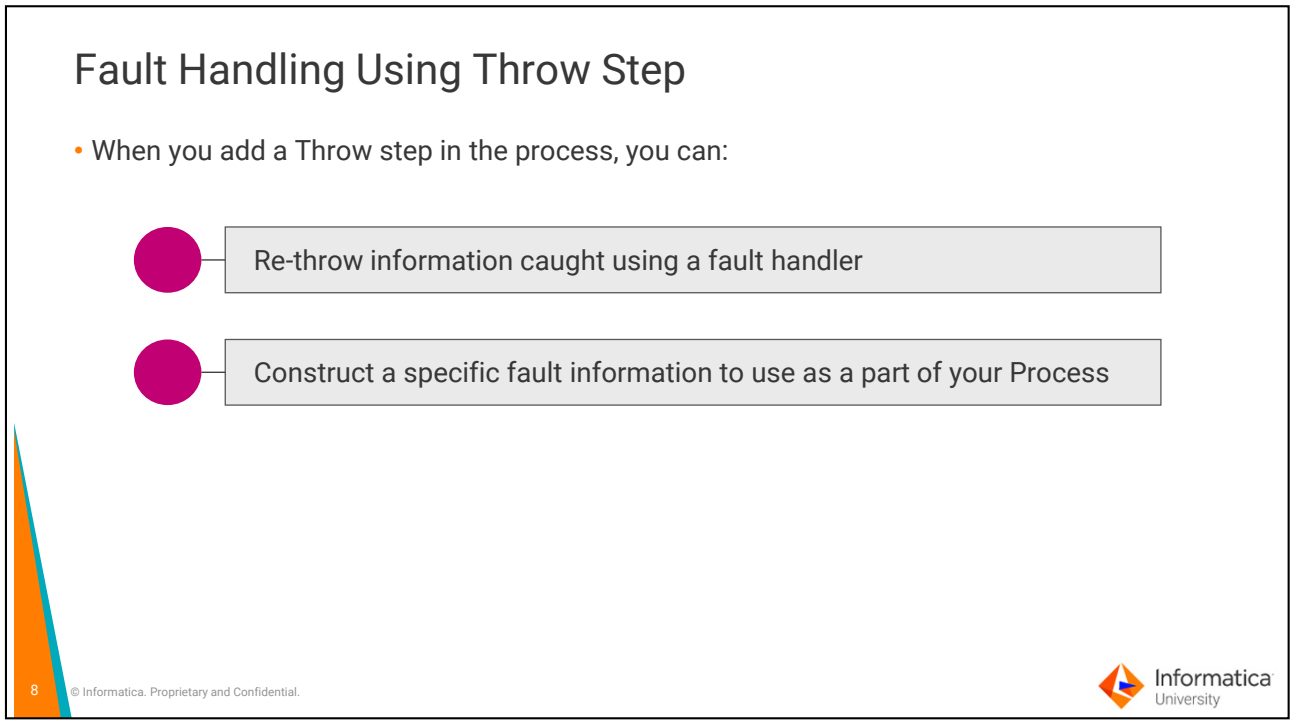
1. Service
2. Subprocess
3. Create

On the Fault Handling tab for each of these step types, you can set the following properties:

1. **Catch Faults**: Required. Check to enable fault handling for the step. By default, this option is disabled.
2. **Fault Field Name**: Required if Catch Faults is enabled. The fault name, which defaults to fault Info.

If you enable fault handling, the fault handler catches all faults on each individual step. Because it is interrupting, it creates a separate execution path in the process. You specify a fault name field, so you can get all the fault details in the process object.

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When you add a Throw step to a process, the error information is propagated by the Throw step. This allows you to catch the fault when you call it, for example, through a Subprocess step.

When you use the Throw step in a Process, the following point applies:

1. Throw is a terminating step, so it does not support any outbound links.
2. A Throw step can be used to re-throw information caught using a fault handler. In that case, the payload for the step is a faultInfo element that contains the parameters returned by the service or service connector.
3. You can use a Throw step to construct specific fault information that you want to use as a part of your process.

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Design Guidelines for Fault Handling

- Determine whether the boundary event occurs in a single step or in multiple steps.
- For a Parallel Path step, you must provide unique field names for each field that contains the fault information.
- You can handle faults in the following ways:
  - If there is an uncaught fault, suspend the process and debug it using the Application Integration Console.
  - Log the fault and continue the process.
  - Add a Wait step and then retry for a specific number of times before exiting.

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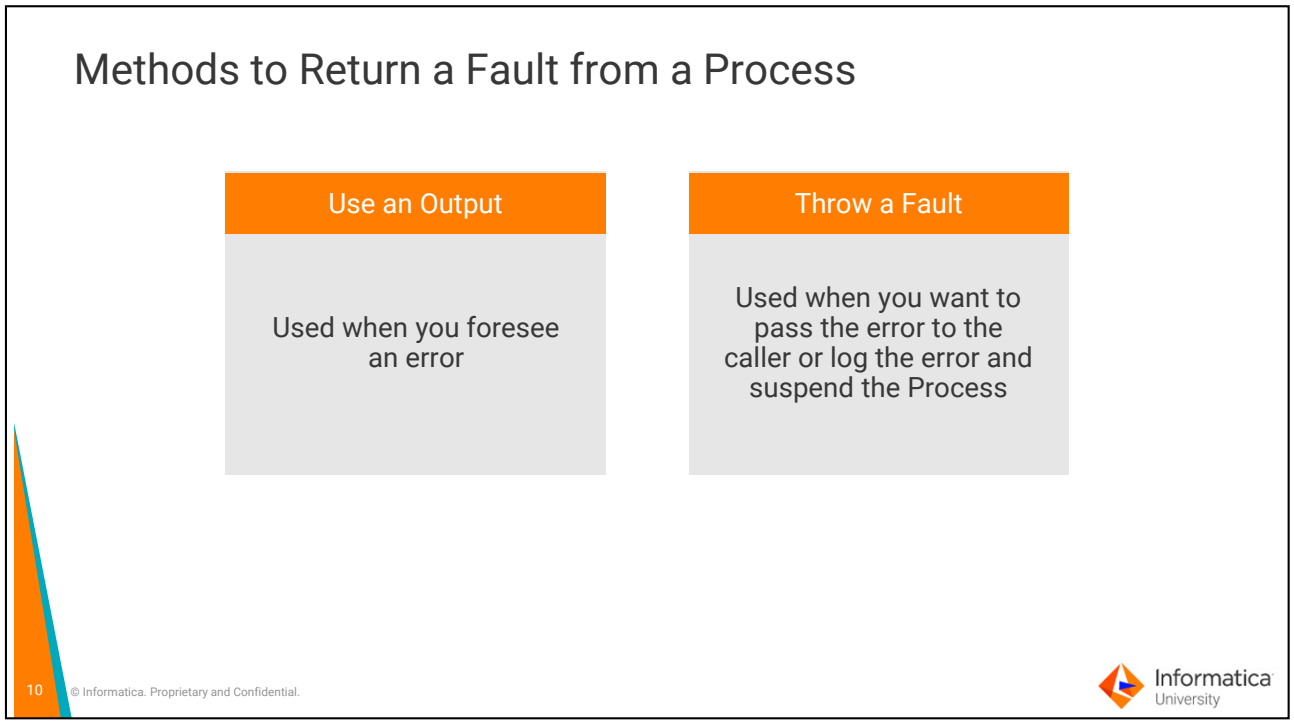
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When you define fault handlers, you must determine whether the boundary event occurs in a single step or in multiple steps. If the boundary event occurs in multiple steps, you can either add fault handling for each step individually or define the steps inside a wrapping process. If you define the steps inside a wrapping process, you can add the fault handler on a Sub-process step and invoke the wrapped process.

If you have a Parallel Path step for fault handling, you must provide unique field names for each field that contains the fault information. Since you define the fault information output field at the process level, there is a possibility that fault information from one path can overwrite the fault information of another path. To avoid this, you must set a unique field name for the fault information in each path of a Parallel Path step. You can handle faults in the following ways:

1. If there is an uncaught fault, suspend the process and debug it using the Application Integration Console. You must select 'Suspend on Uncaught Fault' in the Advanced process properties. You can also re-throw the fault after you catch it.
2. Log the fault, for example, by sending an email notification, and continue the process.
3. Add a Wait step to wait, then retry a specific number of times before exiting. This indicates that you must have a logic error with a counter and exit with a valid error message. If not, the process will go into an infinite loop.

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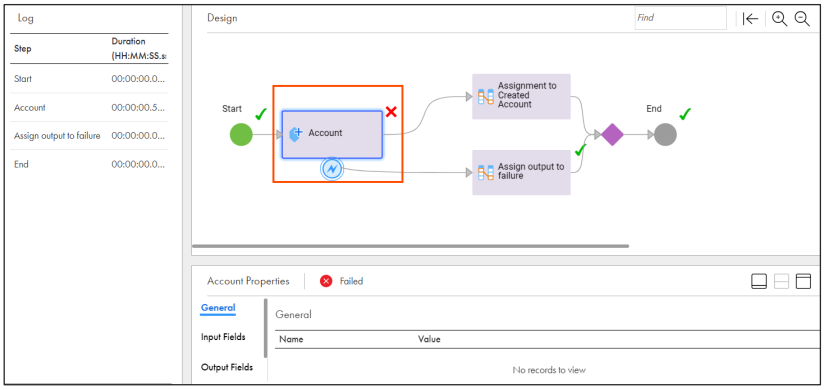
Use an output: This method is suitable when you foresee that the error may occur and you want to use a data decision step to branch the process based on the normal process flow.

Throw a fault: This method is suitable when the only appropriate response to the error is to pass the error to the caller, or log the error and suspend the process. For example, you attempt to access a web service that responds with the HTTP status code 403 Forbidden. This error indicates that the service understood the request but refused to take any action because the access was denied. Throwing a fault is usually the best option to handle this error because you are unlikely to resolve it in the normal process flow.

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View Process Detail

- View the process details in the **Processes** tab of the Application Integration Console.
- To see the Process View Detail page, select the **Process ID** for the process.
- This is an example of a faulted process in CAI:



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You can view the process details in the Processes tab of the Application Integration Console service. To access the Process View Detail page, click on the Process ID. Here you can see a chronological list of process activities, the process as it appears in the designer, and the detailed step information.

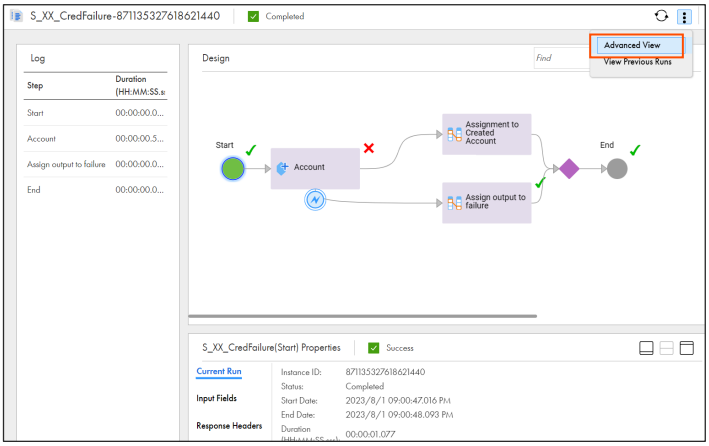
The image displayed on the slide shows an example of a faulted process in CAI.

The steps in the faulted process are indicated by an X symbol.

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Advanced View

- Allows you to see the Active Process Detail page of a process
- Active Process Detail page provides information about the process instance



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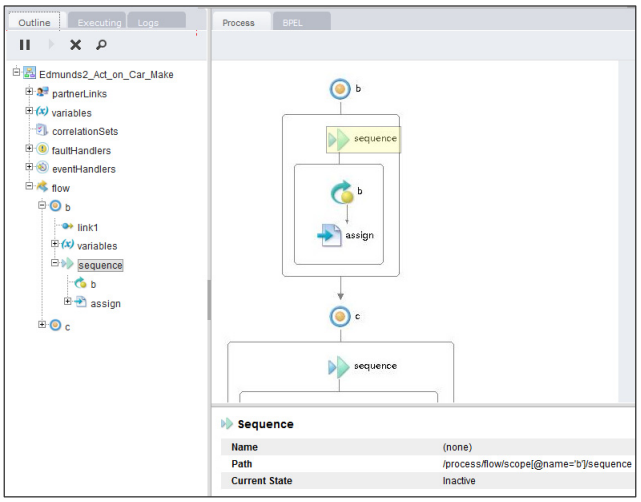
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The **Advanced View** option allows you to see the Active Process Detail page of a process. The Active Process Detail page provides information about the process instance. You can suspend the process and perform process exception management in the Active Process Detail page.

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Advanced View Components

- **Outline View:** Shows the structural elements of a BPEL process and the current process execution state of each activity
- **Properties View:** Shows process properties and current values for the selected element
- **Executing View:** Shows the current process execution state
- **Logs View:** Shows execution details
- **Process View:** Shows the main process flow on a canvas



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The following components are available in the Advance View:

Outline View: Shows the structural elements of a BPEL process and the current process execution state of each activity. You can select an element to view its properties and values.

Properties View: Shows process properties and current values for the element selected from the canvas.

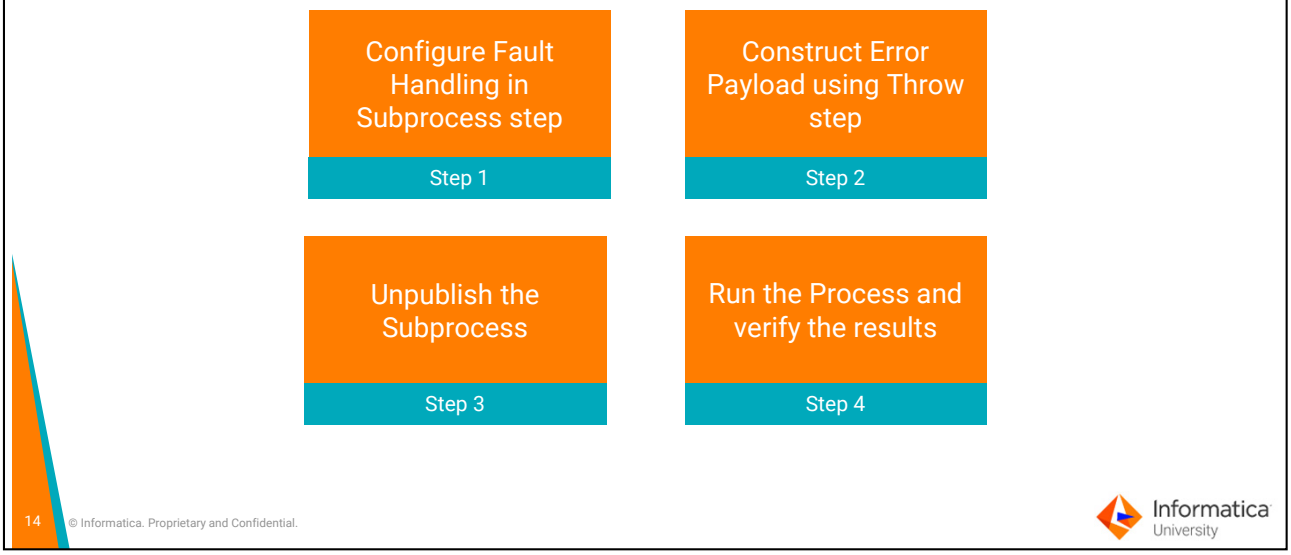
Executing View: Shows the current process execution state for a running process, stopping on the currently executing activity. If the process is completed, the execution view is blank.

Logs View: Shows execution details that you can use to analyse data changes, faults, and other details. You can also suspend a process and perform a process rewind.

Process View: Shows the main process flow on a canvas. If the process has event, fault and compensation handlers, you can view them by selecting the corresponding activity from the Outline.

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Example: Handle a Subprocess Failure



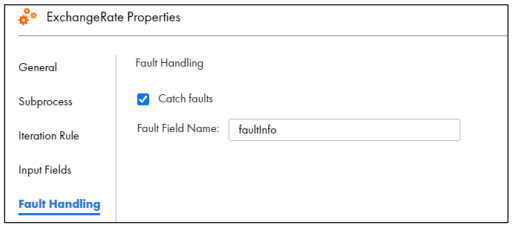
In the ExchangeRateService process, perform the following steps:

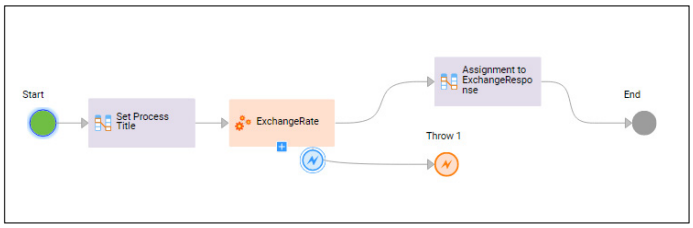
1. To configure fault handling in the subprocess step and check Catch faults options, IICS automatically adds two new paths to the process canvas.
2. To continue with the process execution, add an assignment step to one of the paths.
3. Configure the assignment step with the input ExchangeResponse field.
4. Now, to construct an error payload, you will add the Throw step to the other path.
5. Save the ExchangeRateService process and publish the process to get the Service URL.
6. To get an error while running the process, unpublish the ExchangeRate process that is called in the Subprocess step.
7. Run the process from the interface.
8. After the request is received by the process, IICS tries to call the ExchangeRate process to perform currency rate conversion.
9. As you have unpublished the ExchangeRate process, the process fails with Internal Server Error.
10. As the output of the process, you can see the custom message configured in the throw step of the process.

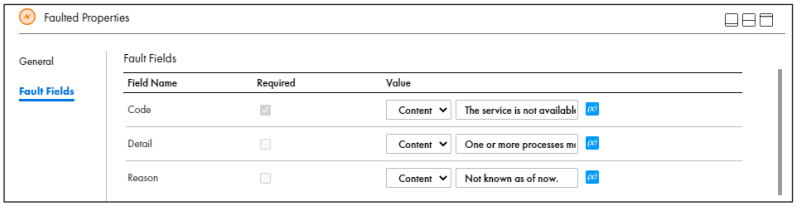
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Example: Handle a Subprocess Failure (continued)

Process Configuration







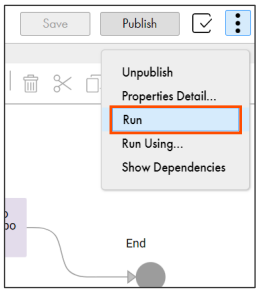
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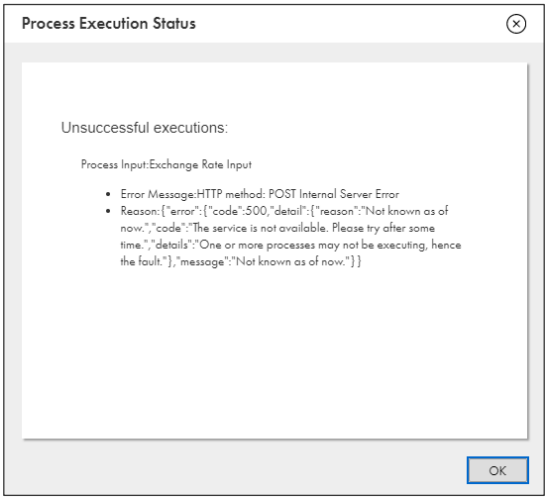
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Example: Handle a Subprocess Failure (continued)

Process Input and Output





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Topic

Knowledge Check

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In CAI, which of the following step types can trigger faults?

- Service Step
- Create Step
- Throw step defined in a process
- Host system interaction
- Other faults
- All of the above

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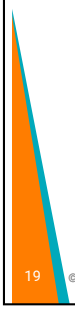
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Answer: All of the above

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A boundary event is used to catch errors that occur on the boundary of a particular step.

- True
- False



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Answer: True

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Lab Activity

11-1 Handle Sub-process Failures

In this lab, you will perform the following:

- Configure error handling for sub-process failure
- Run the process
- Check failed process instance

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Lab Activity

11-2 Handle Credential Failures

In this lab, you will perform the following:

- Change the Salesforce password to introduce an error
- Create a process
- Enable Fault Handling to simplify the error message

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Module Summary

This module showed you how to:

- Describe Fault Handling
- Explain the significance of Fault Handling
- Describe Fault Handling techniques

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Module 12

Introduction to Guides Designer

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Module Objectives

After completing this module, you will be able to:

- Describe Guides
- Identify the steps used in Guide Design

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Topic

Guides

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Guides Overview

Guides are a set of screens or steps that prompt users to review, enter, or confirm data

- Create a guide to automate flows that a user might need to go through often
- Run a guide within Application Integration or within the Salesforce Interface
- Provides iFrame URL that you can use for embedding guides within other web applications

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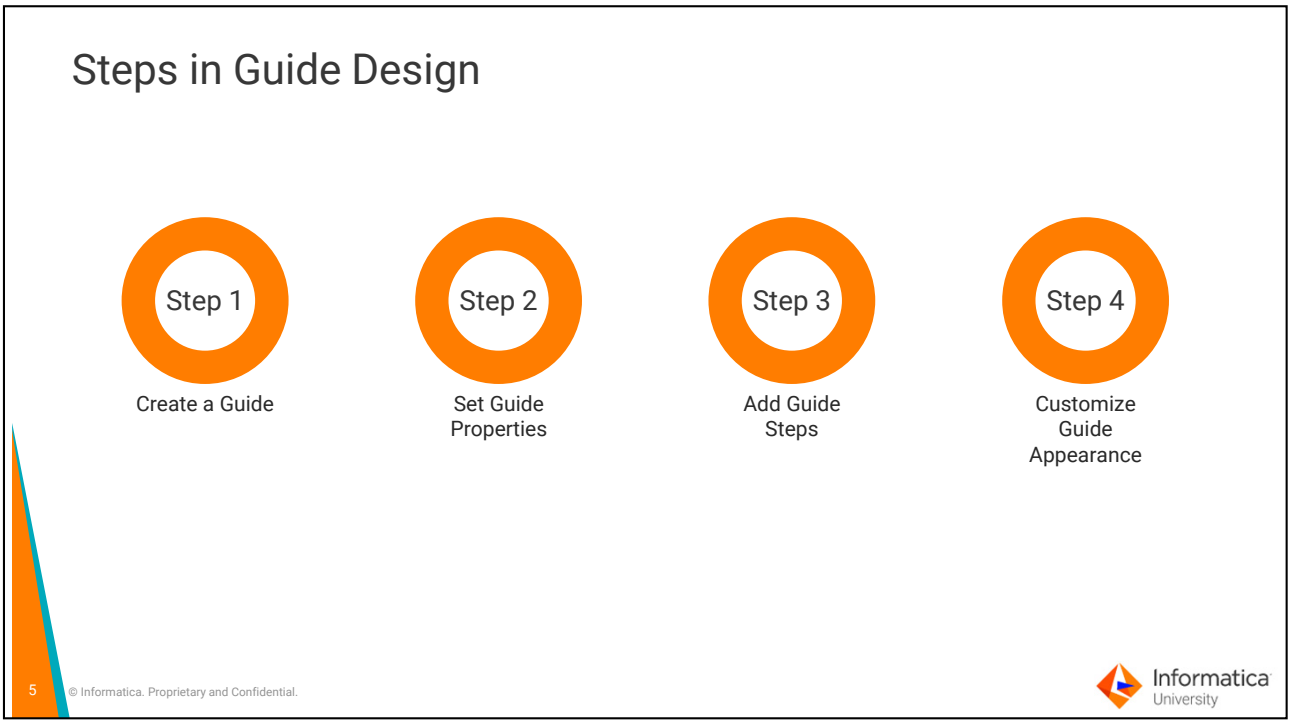
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A guide is a set of screens or steps that prompts users to review, enter, or confirm data. You can create a guide to automate flows that a user might need to go through often. You run a guide within Application Integration. If you have the Salesforce managed package, you can run certain guides within your Salesforce interface.

For example, you can create a guide when you want to automate the process that a sales manager follows to create a new account or if you want to update an existing account in Salesforce.

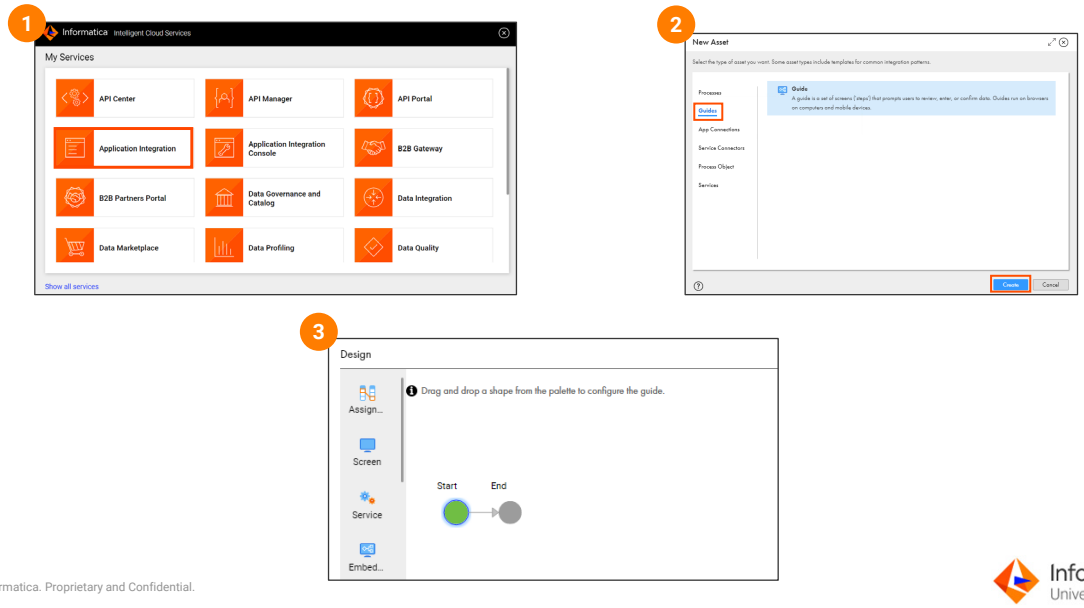
The guide designer generates an iFrame URL that you can use to embed guide within other web applications.

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Create a Guide



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To create a Guide, you must perform the following steps:

1. Login to your IICS Org and select the Application Integration Service.
2. From the Application Integration homepage, to open the New Asset dialogue box, click **New**.
3. From the list of Assets, select Guides and click **Create**.
4. A new Process Designer appears with Start and End steps that mark the start and end of Guide respectively.

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Guide Steps

Step Icon	Step Name	Function
	Start Step	Marks the starting of the Guide
	Assignment Step	Used to set value for a field
	Service Step	Used to call a service in a Guide
	Process Call Step	Used to invoke a Process from within a guide
	Create Step	Used to add new object instances
	Milestone Step	Indicates the end of one phase in a Guide's activities

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The above screen shows the type of steps supported in guides. As discussed about some of these steps in the Process Designer module, in this module, you will discuss the steps that are specific only to guides.

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Guide Steps (continued)

Step Icon	Step Name	Function
	Screen Step	Used to display, update, or enter content
	Embedded Guide Step	Used to better orchestrate and manage business processes
	Decision Step	Allows the Guide to take one of the defined paths
	Jump Step	Used to define the target a step can jump to
	End Step	Marks the ending of the Guide

8

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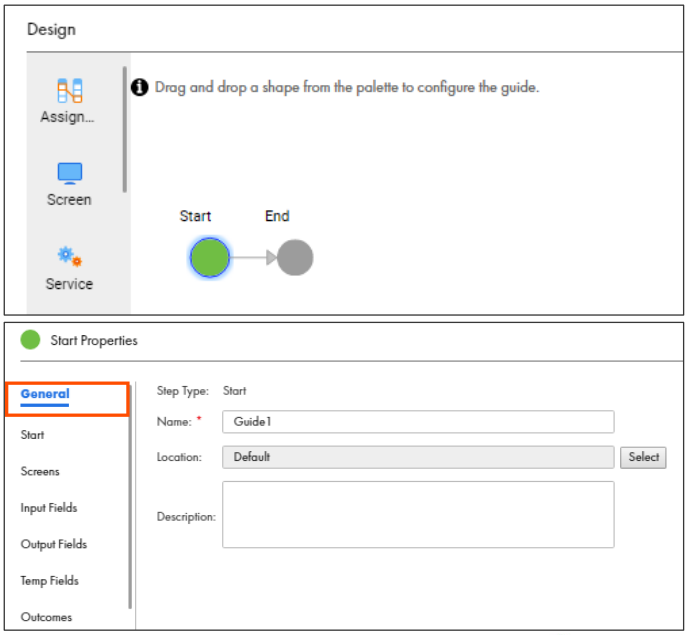
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Start Step

General Properties

- In the Start Step, you can view or edit guide properties.
- Specify the following **General** properties for Guides:
  - Name
  - Location
  - Description



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To view or edit guide properties, select the Start step from the canvas.

You can specify the following General properties for Guides:

Name: A descriptive name to identify the Guide. The name appears when you want to use it in other objects, such as a Process Call step.

Location: The location of the project or folder you want to save the Guide.

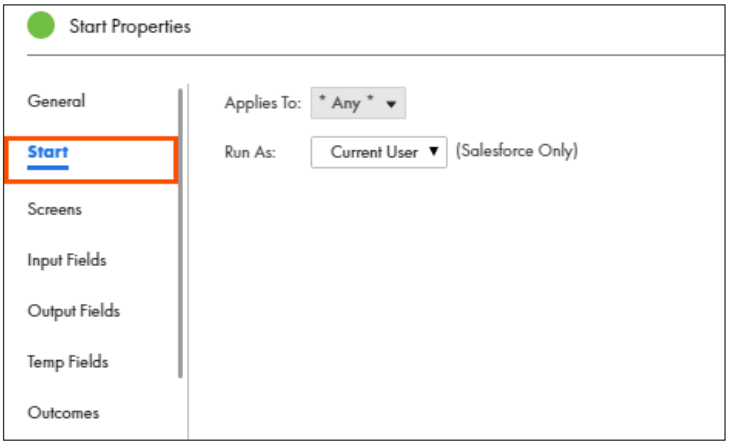
Description: This is an optional field. You can use this option to explain the purpose of the Guide.

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Start Step (continued)

Start Properties

- Specify the following Start properties for Guides:
  - Applies To
  - Run As



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1. **Applies To**: The type of object associated with the guide. The objects displayed in this list are the process objects you defined separately, or generated in a defined connection. Drill-down in the list to see process objects and connections, and then select one of these items. Select a process object only if the guide will be embedded in another guide. You must select 'Any' when you do not want to associate the guide with a specific object. When you select this option, the guide does not automatically have access to an object's fields. Use this object type when a guide creates a new object and does not access information within existing objects.

Note: If you run a guide, the **Applies To** setting must be either **Any** or a specific object in a service, but not a process object.

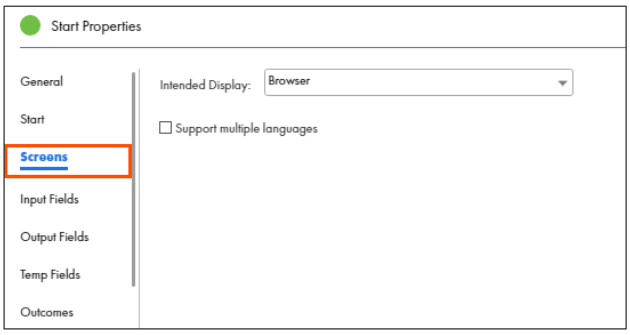
2. **Run As**: Used for Salesforce only. You can choose whether to run the guide as the **Current User** or **System**.

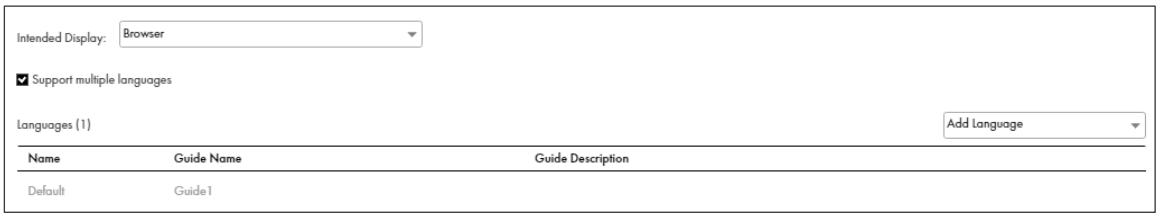
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Start Step (continued)

Screens Properties

- Configure the display and language settings for the Guide
- Specify the following screen properties for Guides:
  - Intended Display
  - Support multiple languages





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You can use the **Screens** tab to configure the location (Browser/ Mobile App) to display the guide and the languages that the guide supports.

Intended Display: The environment in which the guide will run.

Select Browser if you want the guide to run on a browser of the following devices:

Desktop, Laptop, and Tablet

Select Mobile App if you want the guide to run on the following mobile applications:

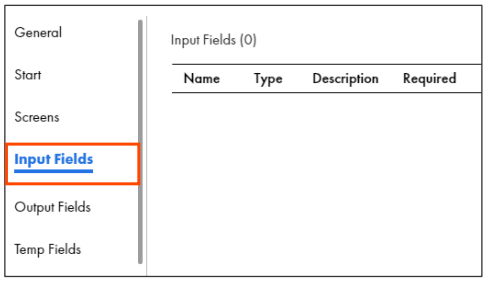
Informatica and Salesforce S1

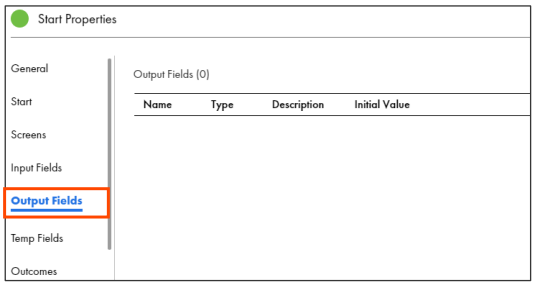
Support multiple languages: The list of languages supported by the guide for localization. Select a language from the Add Language list.

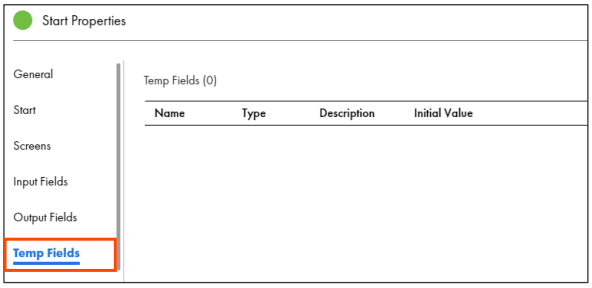
Start Step (continued)

Fields

- Fields present in the object to which the Guides apply
- In Guides, you can define 3 types of fields:
  - Input
  - Output
  - Temp







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Input Field: The Input field values are available in all steps of the process or guide.

An embedded guide may need to access information from the embedding process or guide. Creating input fields is a way to define this information so that it can be passed to the embedding guide.

If the field is required for the guide or process to execute, select Required.

Output Field: The Output field values are set when the process executes, and are then available in the steps that follow. You must define output fields only for use in an embedded process or guide. The output fields do not appear in lists until the values are returned from an embedded process or guide.

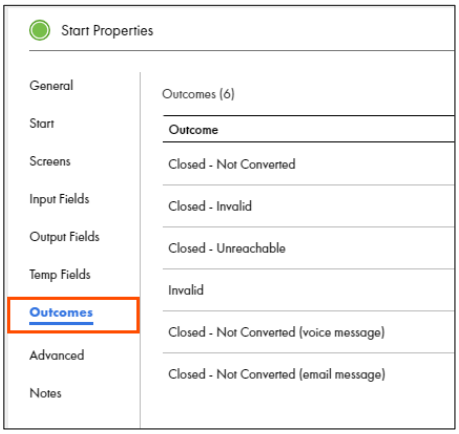
Temp Field: The Temporary field values can be used in all of the current guide's steps. However, a temporary field's value is not available to an embedded guide or process.

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Start Step (continued)

Outcomes Properties

- Configure how the Guide ended in the **Outcomes** tab



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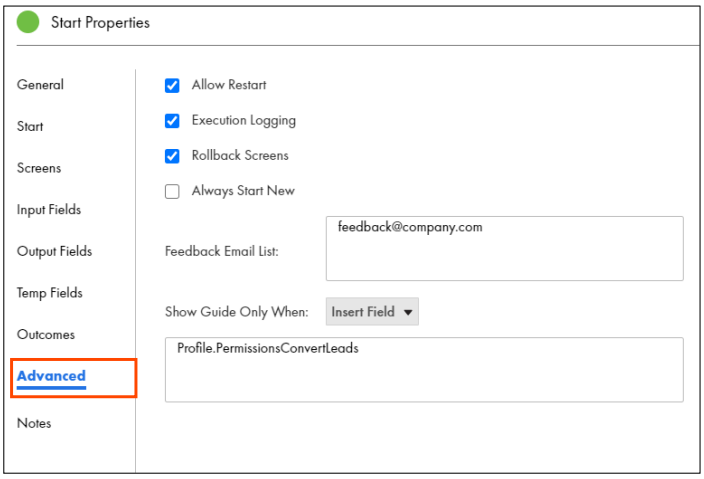
You can define various outcomes for a guide in the Outcomes tab. It allows you to define how the guide ended.

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Start Step (continued)

Advanced Properties

- Specify the following Advanced properties for Guides:
  - Allow Restart
  - Execution Logging
  - Rollback Screens
  - Always Start New
  - Feedback Email List
  - Show Guide Only When



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Allow Restart: Determines whether the users can restart this guide from the first step.

Execution Logging: If you select this option, Application Integration logs details about the guide steps and execution, so it is available for reports.

Rollback Screens: If selected, your application can undo changes made to the database when a user clicks Go Back One Step or clicks a step in the History panel. If an embedded guide enables rollback, then the undo option is applicable only to the scope of the embedded guide, unless the parent guide also enables rollback.

Always Start New: When selected, opening a guide always creates a new instance of the guide.

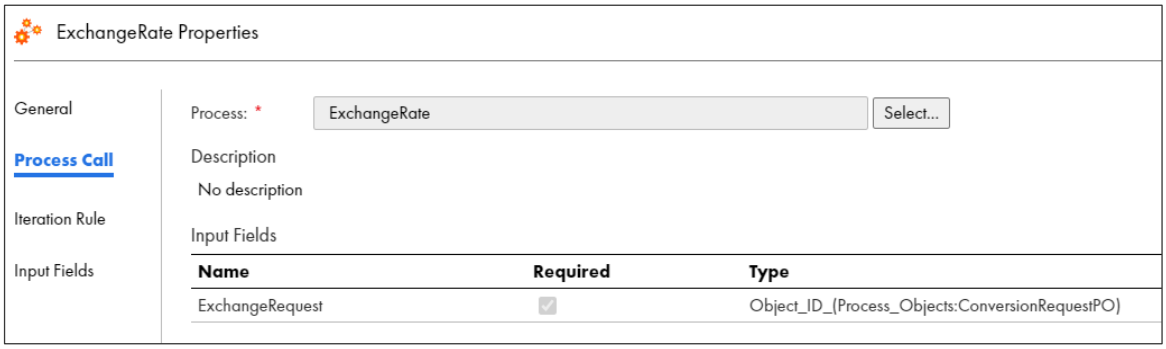
Feedback Email List: If you want users to send feedback, add one or more email addresses (comma-separated) to receive feedback.

Show Guide Only When: You can specify the criteria that determines when this guide displays in the Guides area on a user's Salesforce page.

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Process Call Step

- Process Call step embeds a Process within a Guide
- You can define the Input Fields for the called Process



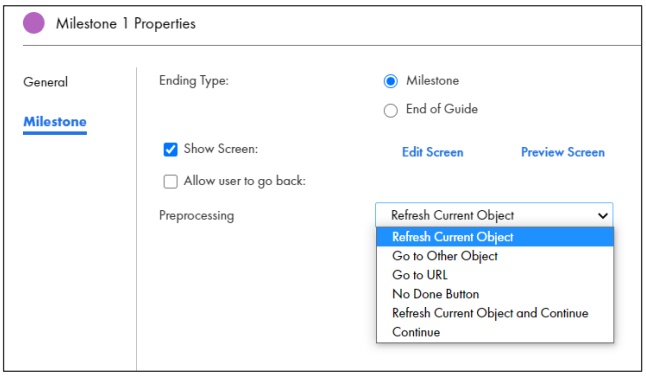
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A Process call step embeds a process within a guide. When the Process Call step executes, the called process executes instead of the Guide. You can also define input fields for the called process.

Milestone Step

- You can use Milestone step to:
  - Hand off work to someone else
  - Record an event that occurred
- Specify the following Milestone properties:
  - Show Screen
  - Allow user to go back
  - Preprocessing



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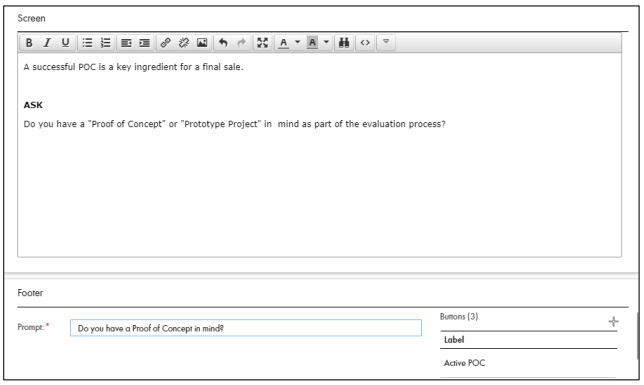
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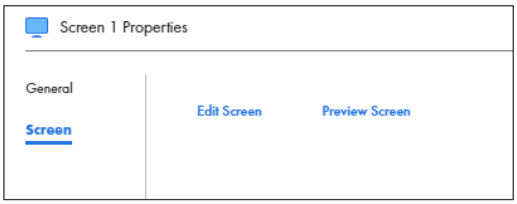
Milestones can be helpful when a user hands off work to someone else. When that user starts the guide, it starts at the step following the milestone. Milestones should also be used if the user normally stops executing the guide at a step to do something else. In some cases, you may simply be using a milestone step to record an event that occurred, and the guide will automatically continue to the next step.
Ending Type: Defines a way to end a guide. You can select the Milestone or End of guide options.
Show Screen: If selected, the user sees the milestone step. If not selected, the user does not see the milestone step. Instead, the action indicated in the On Done Button area immediately occurs after Milestone step executes.
Allow user to go back: If selected, user can go back to steps executed before the milestone. In either case, all the steps that executed within the guide are displayed in the guide's history. However, if the user cannot go back, the user cannot select one of the previously executed steps.
Preprocessing: Used to select an option from this list to configure the action that occurs after the Milestone step completes. You can select from the options -
Refresh Current Object, Go to Other Object, Go to URL, Continue, and Refresh Current Object, and Continue.

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Screen Step

- Use the Screen step to:
  - Show instructions about what is being displayed, what to enter, and so on
  - Update an object's data, based on user input





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You can use a Screen step to perform the following tasks:

1. Show instructions about what is being displayed, what to enter, and so on.
2. Update an object's data based on user input.

In the screen tab of screen step, you can select the Edit Screen option to place the fields and labels or prompts for the user.

You can insert the following field types when you edit a Screen step:

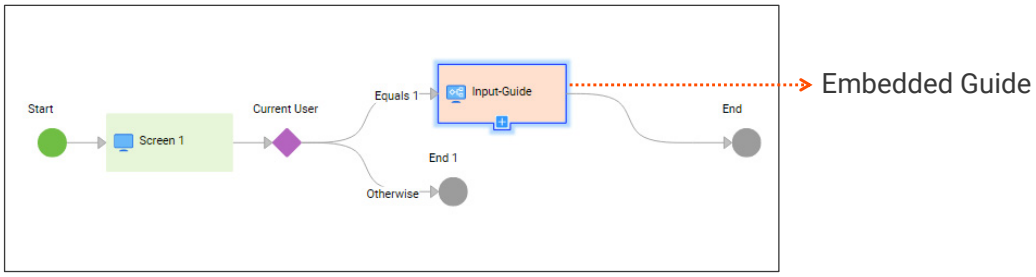
1. **Editable Fields**: Fields that the user can edit. You can proceed to the next step even if you do not enter values in these fields.
2. **Read-only Fields**: Fields that the end user cannot edit.

To preview the screen content, select the Preview screen option.

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Embedded Guide Step

- Used to better orchestrate and manage business processes
- Embeds a guide within another guide
- Works similar to a Subprocess step



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Embedded Guide step is used to better orchestrate and manage business processes. It allows users to embed a guide within another guide. This step works similar to the Subprocess step.
You must define the following properties for the Embedded Guide step:

General: The name of the Embedded Guide step

Guide: Select the guide that you want to embed within the current guide. The current guide is called the embedding guide. The guide that the embedding guide embeds is called the embedded guide.

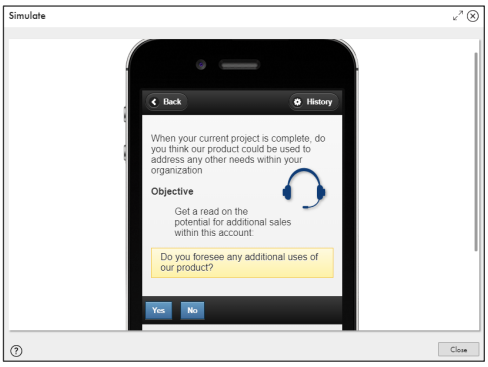
Iteration Rule: Configure whether you want the embedded guide to run on a single object or on all the objects.

Input Fields: Displays the input fields available for the selected object. If the embedded guide receives one of these fields as input, the value of that field may be updated while the embedded guide executes. If so, the value is updated in the parent guide after this embedded guide completes the execution.

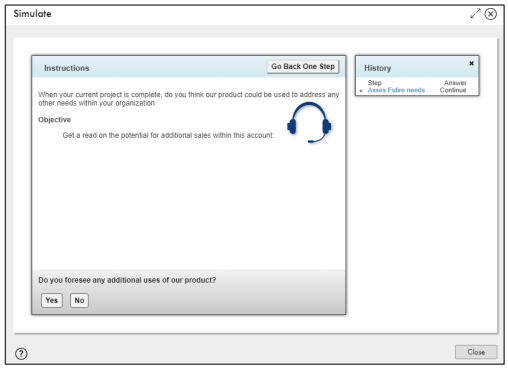
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Guide Simulation

- To view guide appearance, use the **Simulate** option.
- You can simulate a guide from the beginning or from the selected step.
- The type of simulation you see depends on configured Intended Display.



Mobile Display



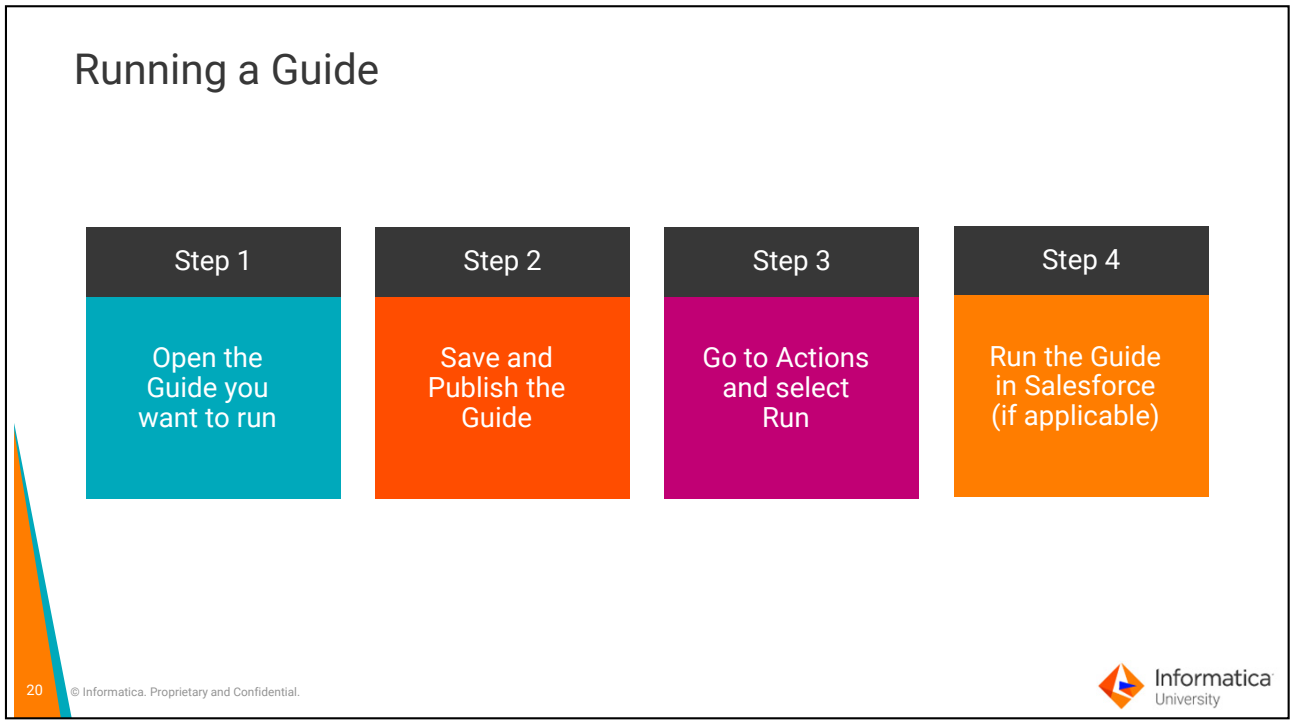
Browser Display

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In CAI, you can view the guide appearance, using the Simulate option. You can simulate a guide from the beginning or from the selected step. The type of simulation depends on the configured Intended Display in the Start step of the Guide.

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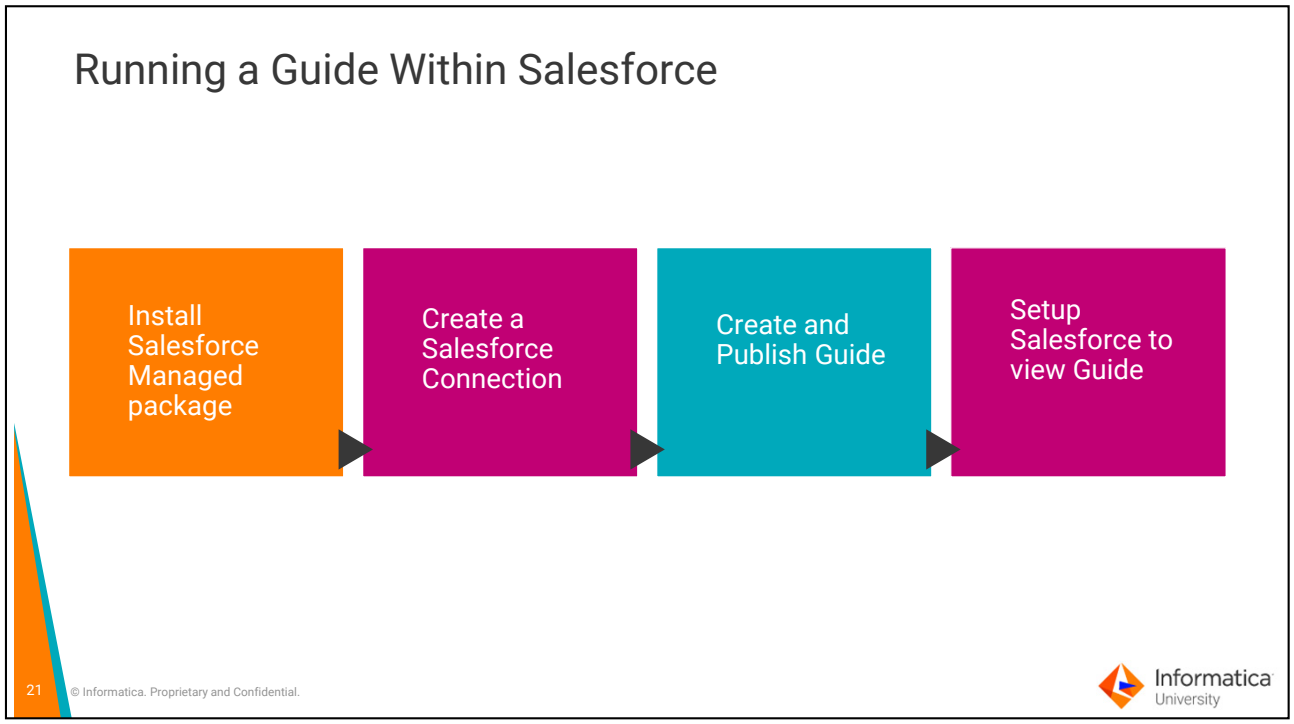
You must perform the following steps to run a guide:

1. Open the guide you want to run.
2. To run a guide, you must first publish the guide.
3. After you publish the guide, select Actions -> Run.

Note: To run a guide in Application Integration, the guide must be published and the Applies to option must be set to Any.

4. For the Guides that contains Salesforce Object, run the guide in Salesforce.

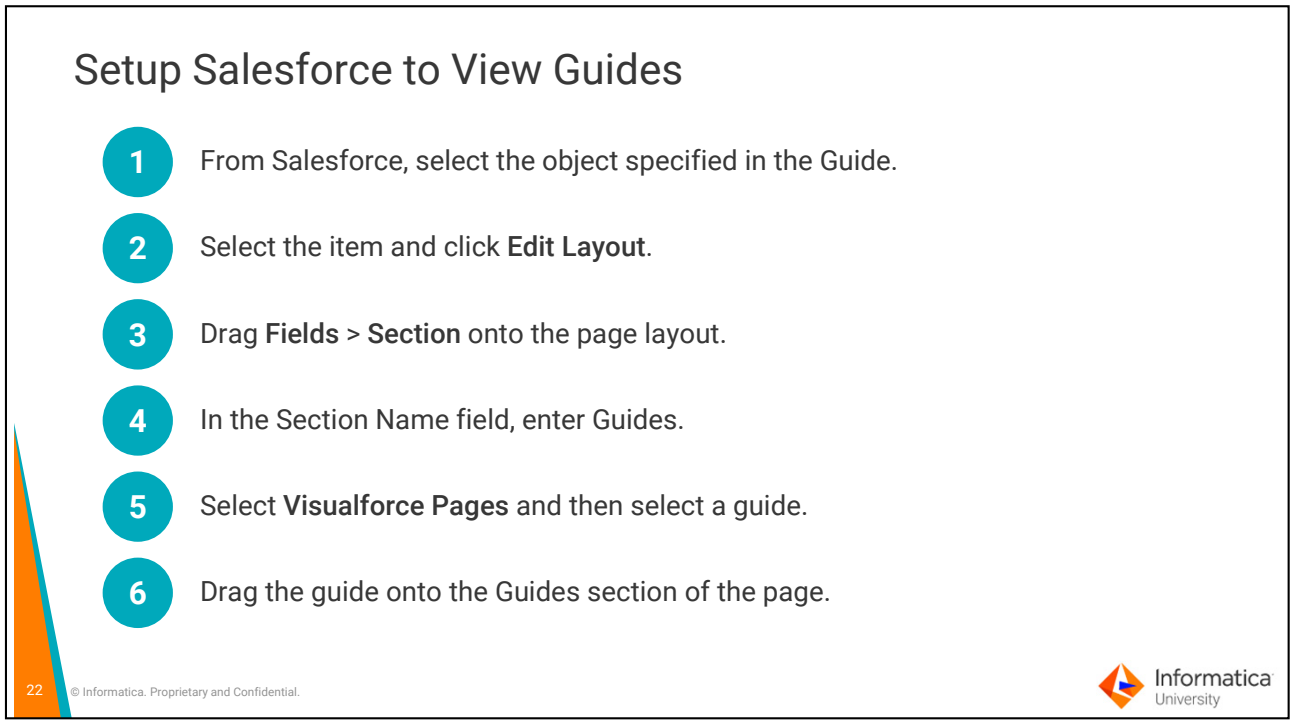
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To run a guide within Salesforce, perform the following steps:

1. You must have the salesforce managed package installed in your salesforce Org.
2. To use the salesforce objects in CAI, you must create a salesforce connection.
3. If you want to run a guide from Salesforce, the guide must be associated with a Salesforce object.
4. In the properties section of the Start step, you must set the Start > Applies to property to a Salesforce object. Configure and publish the guide.
   Note: If you set the property to Home, the guide appears on your Salesforce Home page.
5. To run the guide on Salesforce, you must setup the salesforce Org.

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To run the guides that contain Salesforce object, you need to setup your Salesforce account to view the guides. To setup salesforce account, you must perform the following steps:

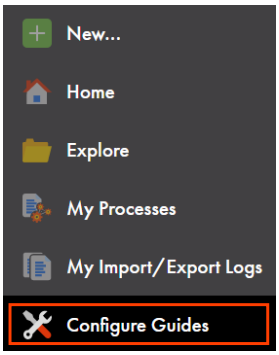
1. Login to your Salesforce account and select the Object used in the Applies To field of the Start step of the guide.
2. From the Object, select any item for which you want to see the Guide and click Edit Layout.
3. To add another section to the layout page, from the Layout window, drag the Fields section and drop it into the page layout area.
4. Name the newly added section as Guides.
5. After that, from the Visualforce page section, select the guide and drag the guide on the Guides section.
   Note: A Visualforce page is a Salesforce webpage that is created using a unique tag-based mark-up language. Each tag in Visualforce language corresponds to some user interface component like section of a page, a list view, or a field of an object.
6. Click the link to run the guide.

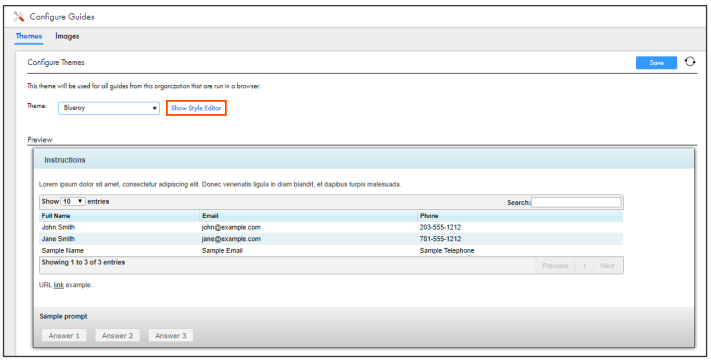
Note: All published guides that apply to this Salesforce object appear as links here.

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Guide Theme

- Themes allow you to customize the appearance of a guide
- Use the Configure Guides page to specify guide themes





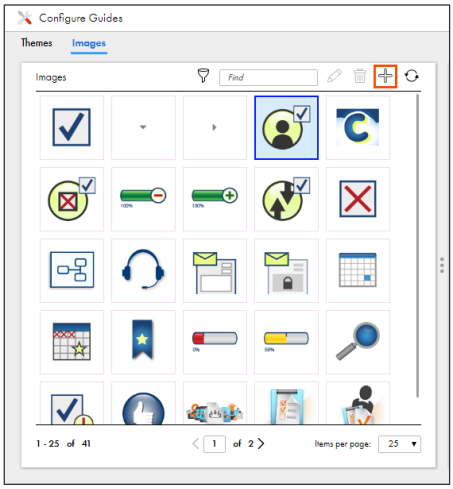
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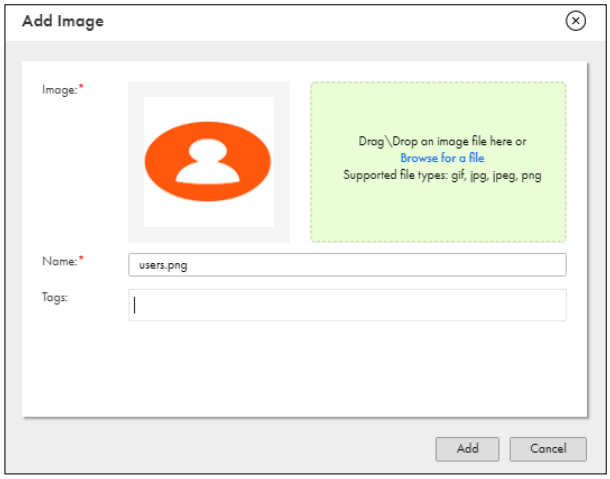
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You can customize the appearance of a guide in the Configure Guides page. You can select from a list of built-in themes to determine what a guide will look like when a user runs the guide on a desktop browser. In the Configure Guides page, use the Theme tab to select the theme from Guide. Click **Show Style Editor** to view and edit the CSS for a guide step, paragraph, or character in the theme. When you select, configure, and save a theme, it applies to all the guides in your organization.

Guide Images

- **Images** tab allows you to view and edit icons you want to use in guides





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You can use the Images tab to view and edit icons that you want to use in guides. You can use the Add Image option to upload .gif, .jpg, .jpeg, or .png images from your system to the Application Integration image library.

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Demonstration

Create and Run a Simple Guide

In this demonstration, the instructor will demonstrate the following:

- The steps to create and run simple Guide

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Topic

Knowledge Check

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Can you customize the appearance of a guide in CAI?

- Yes
- No

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Answer: Yes

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Which of the following steps is used to invoke a process from within a guide?

- Start Step
- Assignment Step
- Process Call Step
- Milestone Step

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Answer: Process Call Step

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Lab Activity

12-1 Create a Simple Guide

In this lab, you will perform the following:

- Create a Guide
- Run it through the interface

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Lab Activity

12-2 Create a Guide to Invoke the Exchange Rate Process

In this lab, you will perform the following:

- Create a Process Object
- Create a Guide using the exchange rate process

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Module Summary

This module showed you how to:

- Describe Guides
- Identify the steps used in Guide Design

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Module 13
Monitoring

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Module Objectives

After completing this module, you will be able to:

- Monitor and manage your processes using Application Integration Console

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Topic

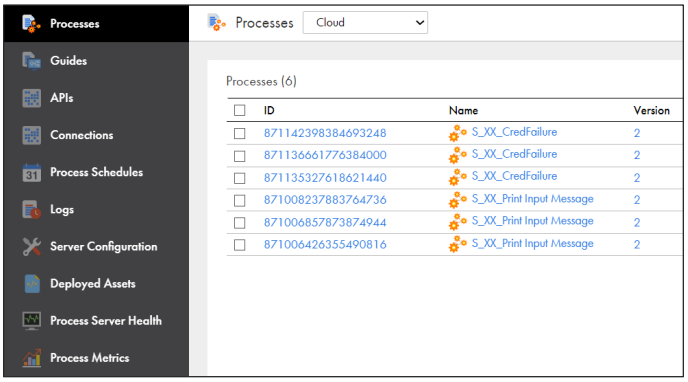
The Application Integration Console

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The Application Integration Console

- The Application Integration Console provides options to monitor and manage your processes, whether they are running on the Informatica Cloud Server or on a Secure Agent
- The Application Integration Console includes the following tabs:
  - Processes
  - Guides
  - APIs
  - Connections
  - Process Schedules
  - Logs
  - Server Configuration
  - Deployed Assets
  - Process Server Health
  - Process Metrics



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Topic

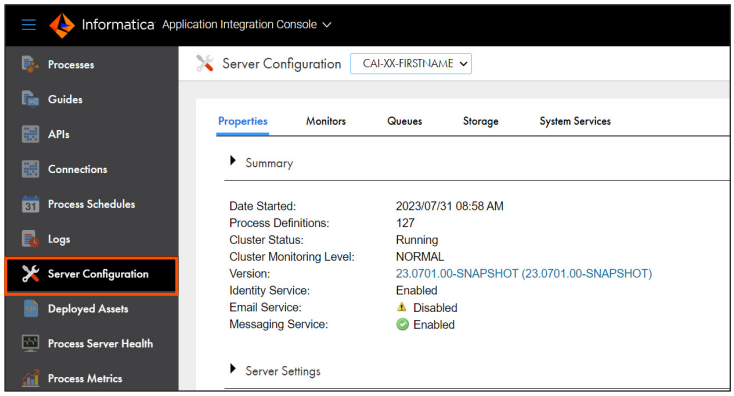
Server Configuration

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Server Configuration

- The Server Configuration page displays the following details:
  - Tenant or Process Server Properties
  - Monitors settings
  - Queues settings
  - Storage and scheduled maintenance settings
  - System Services settings



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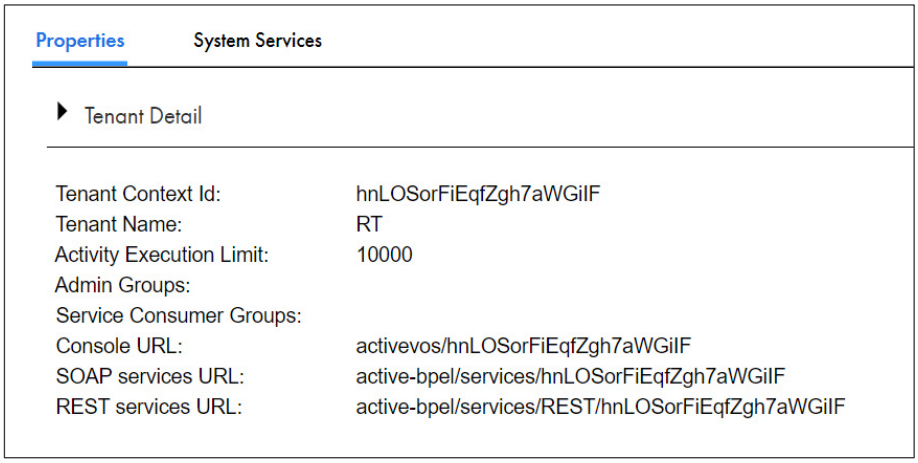
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Server Configuration (continued)

Tenant Detail

- The Tenant Detail section displays information about a tenant



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The Tenant Detail section displays the following information about a tenant:

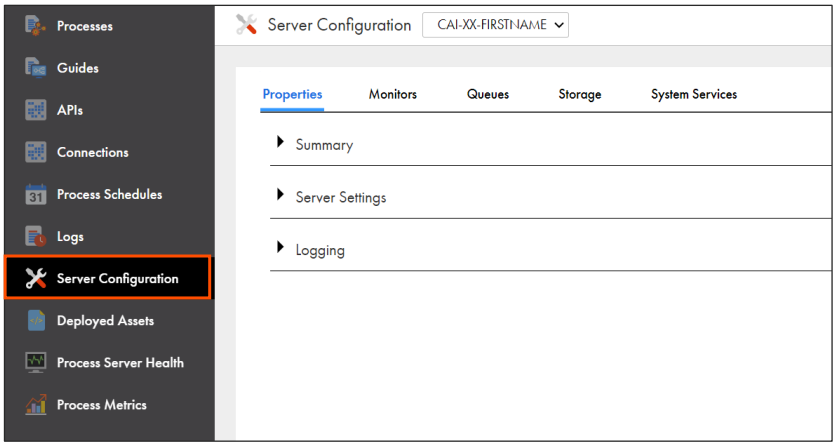
- **Tenant Context ID:** The organization ID
- **Tenant Name:** A descriptive name
- **Activity Execution Limit:** Use this option to suspend or terminate processes that use infinite loops and a high number of resources. The default value is 10,000.
- **Admin Groups:** The names of admin groups defined for the tenant
- Service Consumer Groups: The names of service consumer groups defined for the tenant
- **Console URL:** The URL that is used to invoke the tenant's Application Integration Console
- **SOAP Services URL:** The URL to which a SOAP request can be sent
- **REST Services URL:** The URL to which a REST request can be sent

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Server Configuration (continued)

Properties

- The Properties tab includes the following sections:
  - Summary
  - Server Settings
  - Logging



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The Properties tab includes the following sections:

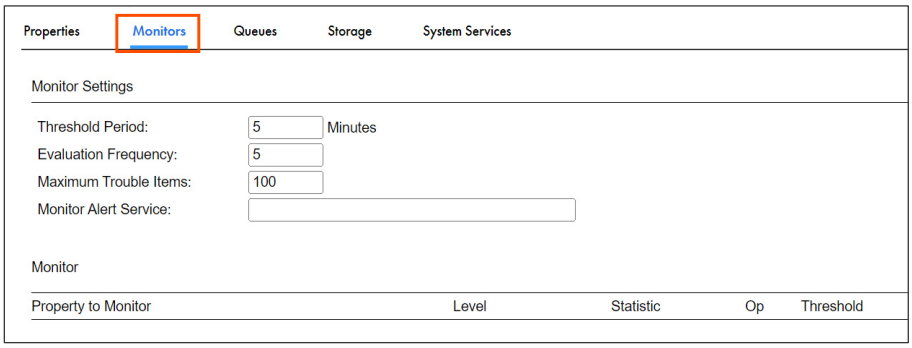
- **Summary:** Provides an overview of the engine that executes BPEL processes.
- **Server Settings:** Allows you to make configuration changes without stopping and restarting the engine.
- **Logging:** Allows you to select logging levels for the Process Server and for processes.

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Server Configuration (continued)

Monitors

- Select engine properties to monitor on the Monitors tab
- For each property, you can provide a statistic and threshold that alerts you to a warning or error condition



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You can decide the frequency and interval of monitoring periods, as described below:

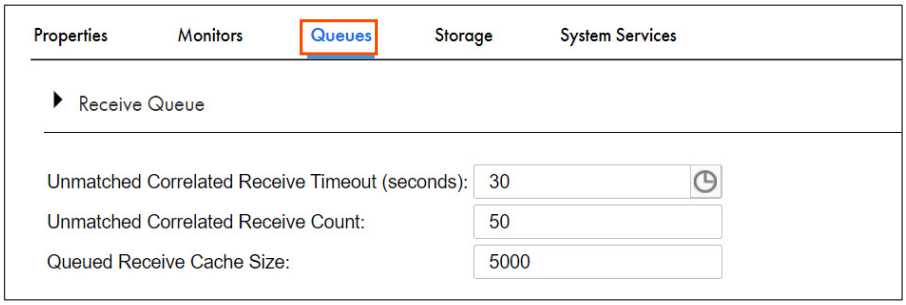
- **Threshold Period:** Period for collecting and aggregating statistics. The default is five minutes.
- **Evaluation Frequency:** Number of times during the threshold interval the statistics are evaluated.
- **Maximum Trouble Items:** Number of error/warning items per engine to display on the Monitoring page of the Application Integration Console.
- **Monitor Alert Service:** Name of the service that will run when errors and warnings occur for monitored properties.

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Server Configuration (continued)

Queues

- Use the Queues tab to set the size for how many queued receives to hold in memory to reduce the number of queries to the database
- Sizing depends on server memory constraints



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The information on the Queues page is as follows:

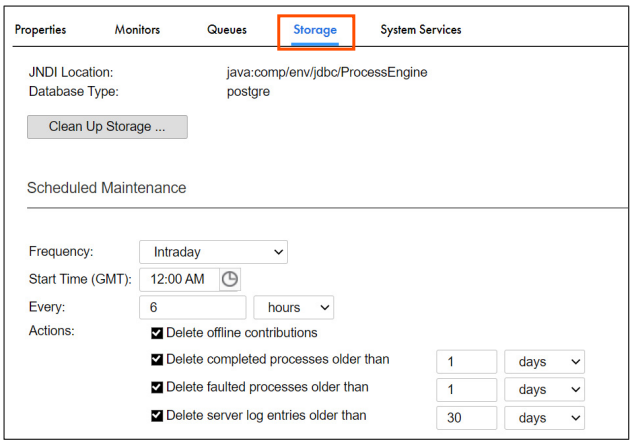
- **Unmatched Correlated Receive Timeout:** The amount of time (in seconds) that the engine waits for a correlated message to be matched to a receive activity.
- **Unmatched Correlated Receive Count:** The maximum number of unmatched correlated receives the engine will hold in memory at a time.
- **Queued Receive Cache Size:** The size of the cache memory that holds activity information when the system is busy with other requests.

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Server Configuration (continued)

Storage

- The Storage page displays database configuration properties and allows you to maintain the database



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The Storage tab displays the following information:

- **JNDI Location:** The Java Naming and Directory Interface (JNDI) context that specifies where to look for the database. For example,jdbc/ActiveVOS.
- **Database Type:** The type such as MySQL.

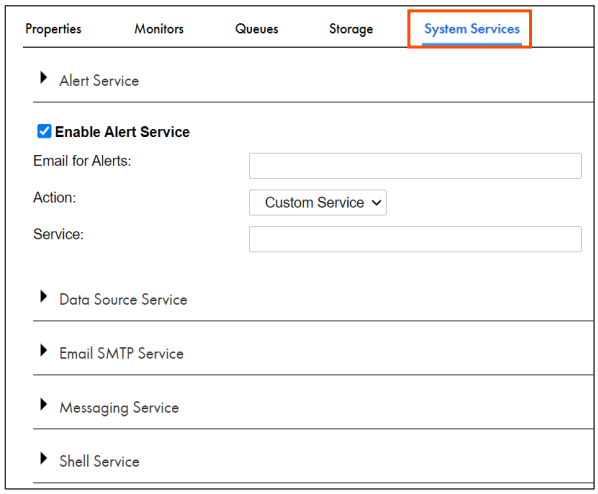
Note: You only see the Storage tab if you have selected a secure agent.

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Server Configuration (continued)

System Services

- The System Services tab includes the following sections:
  - Alert Service
  - Email Service
  - Shell Service
  - Messaging Service
  - Data Source Service



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The System Services tab includes the following sections:

- **Alert Service:** You can configure an email or a custom service to run when a process stops because of a fault.
- **Email Service:** The Email Service settings enable you to send an email to a specified administrator.
- **Shell Service:** The Shell Service option enables you to define processes that include a shell command in a Service Call step, for processes that run on a Secure Agent.
- **Messaging Service:** The Messaging Service allows you to configure parameters for one or more JMS Messaging Services so that Process Server can establish connections to the JMS provider.
- **Datasource Service:** The Data Source Service lets you create a "pool" of open connections between all of the application's current users.

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Topic

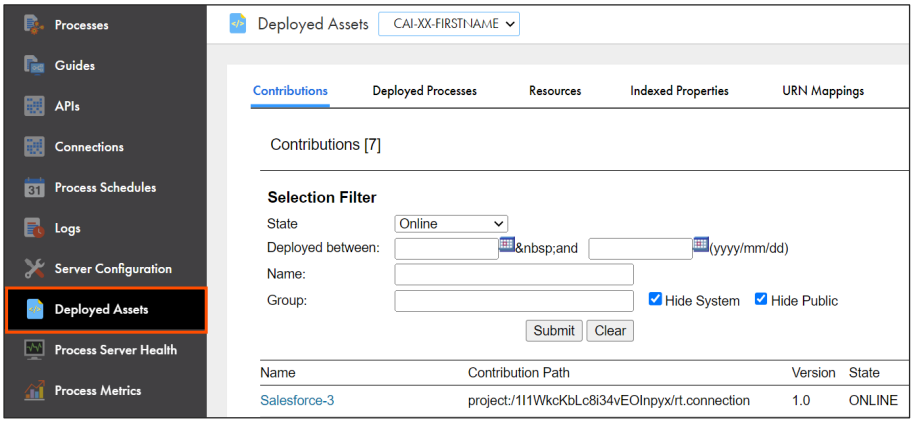
Deployed Assets

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Deployed Assets

- Use the Deployed Assets page to view and manage the following assets:
  - Contributions
  - Deployed Processes
  - Resources
  - Indexed Properties
  - URN Mappings



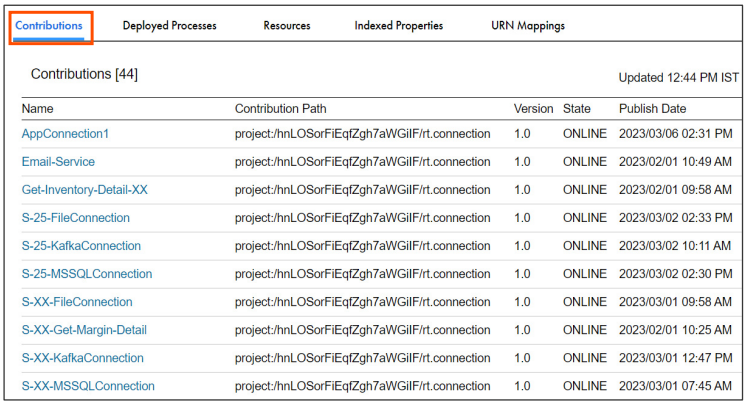
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Deployed Assets (continued)

Contributions

- The Contributions tab lists all Business Process Archive (BPR) deployments
- Their names are based upon the Process Developer project from which they were deployed



Name	Contribution Path	Version	State	Publish Date
AppConnection1	project:/hnLOSorFiEqfZgh7aWGilF/rt.connection	1.0	ONLINE	2023/03/06 02:31 PM
Email-Service	project:/hnLOSorFiEqfZgh7aWGilF/rt.connection	1.0	ONLINE	2023/02/01 10:49 AM
Get-Inventory-Detail-XX	project:/hnLOSorFiEqfZgh7aWGilF/rt.connection	1.0	ONLINE	2023/02/01 09:58 AM
S-25-FileConnection	project:/hnLOSorFiEqfZgh7aWGilF/rt.connection	1.0	ONLINE	2023/03/02 02:33 PM
S-25-KafkaConnection	project:/hnLOSorFiEqfZgh7aWGilF/rt.connection	1.0	ONLINE	2023/03/02 10:11 AM
S-25-MSSQLConnection	project:/hnLOSorFiEqfZgh7aWGilF/rt.connection	1.0	ONLINE	2023/03/02 02:30 PM
S-XX-FileConnection	project:/hnLOSorFiEqfZgh7aWGilF/rt.connection	1.0	ONLINE	2023/03/01 09:58 AM
S-XX-Get-Margin-Detail	project:/hnLOSorFiEqfZgh7aWGilF/rt.connection	1.0	ONLINE	2023/02/01 10:25 AM
S-XX-KafkaConnection	project:/hnLOSorFiEqfZgh7aWGilF/rt.connection	1.0	ONLINE	2023/03/01 12:47 PM
S-XX-MSSQLConnection	project:/hnLOSorFiEqfZgh7aWGilF/rt.connection	1.0	ONLINE	2023/03/01 07:45 AM

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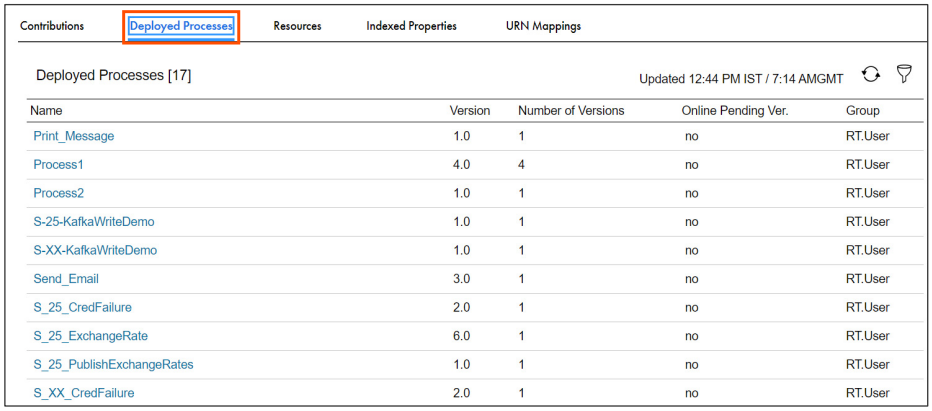
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Deployed Assets (continued)

Deployed Processes

- The Deployed Processes tab shows a list of processes that have been deployed to the cloud or to a Secure Agent



Name	Version	Number of Versions	Online Pending Ver.	Group
Print_Message	1.0	1	no	RT.User
Process1	4.0	4	no	RT.User
Process2	1.0	1	no	RT.User
S-25-KafkaWriteDemo	1.0	1	no	RT.User
S-XX-KafkaWriteDemo	1.0	1	no	RT.User
Send_Email	3.0	1	no	RT.User
S_25_CredFailure	2.0	1	no	RT.User
S_25_ExchangeRate	6.0	1	no	RT.User
S_25_PublishExchangeRates	1.0	1	no	RT.User
S_XX_CredFailure	2.0	1	no	RT.User

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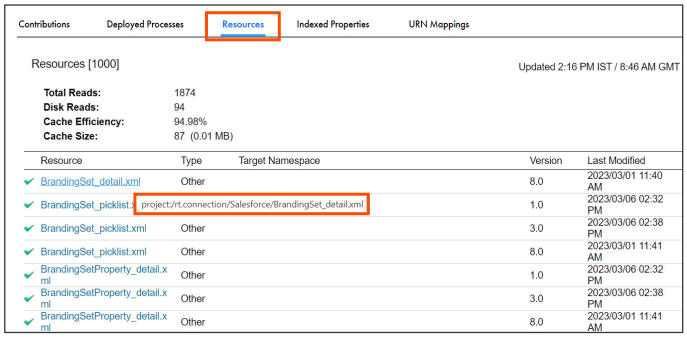
Note: Unlike the Processes page, where you see an entry for each process instance, you see an entry for each process created on the Deployed Processes tab.

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Deployed Assets (continued)

Resources

- The Resources tab shows a list of retrieved resources during process execution
- This page shows the Name, Type, Target Namespace, Version, and the Last Modified date for each resource
- Rest your mouse on the Resource name to view the physical location where the resource was loaded from



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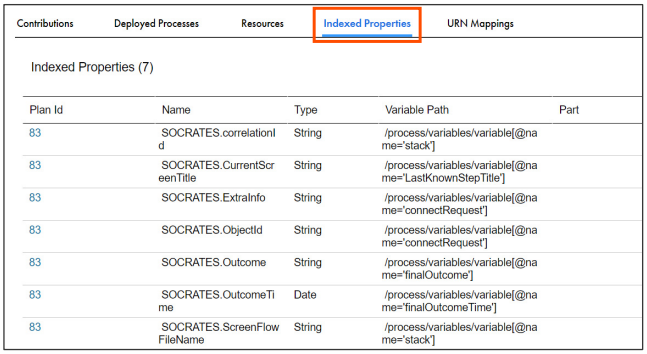
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Deployed Assets (continued)

Indexed Properties

- An indexed property is a variable property that serves as a selection filter for active processes
- It can also be used in defining events for business event processing
- Using an indexed property in a selection query provides a fast way to filter processes based on important data items



Contributions | Deployed Processes | Resources | Indexed Properties | URN Mappings

Indexed Properties (7)

Plan Id	Name	Type	Variable Path	Part
83	SOCRATES.correlationId	String	/process/variables/variable[@name='stack']	
83	SOCRATES.CurrentScreenTitle	String	/process/variables/variable[@name='LastKnownStepTitle']	
83	SOCRATES.ExtraInfo	String	/process/variables/variable[@name='connectRequest']	
83	SOCRATES.ObjectId	String	/process/variables/variable[@name='connectRequest']	
83	SOCRATES.Outcome	String	/process/variables/variable[@name='finalOutcome']	
83	SOCRATES.OutcomeTime	Date	/process/variables/variable[@name='finalOutcomeTime']	
83	SOCRATES.ScreenFlowFileName	String	/process/variables/variable[@name='stack']	

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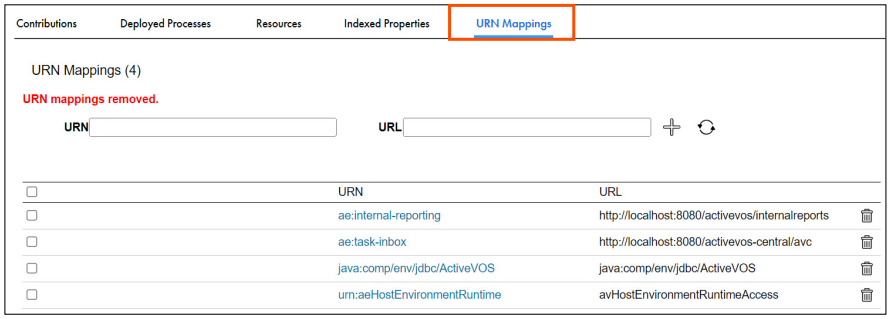
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Deployed Assets (continued)

URN Mappings

- On the Universal Resource Name (URN) Mappings tab, there are system mappings and user-defined mappings:
  - **System Mappings**: Process Server provides the default URN Mappings for system services. You may need to manually add a mapping if it wasn't part of your initial configuration.
  - **User-Defined Mappings**: You can assign a physical address to a URN. The URN is a logical address of a partner link, specified in a deployment resource.



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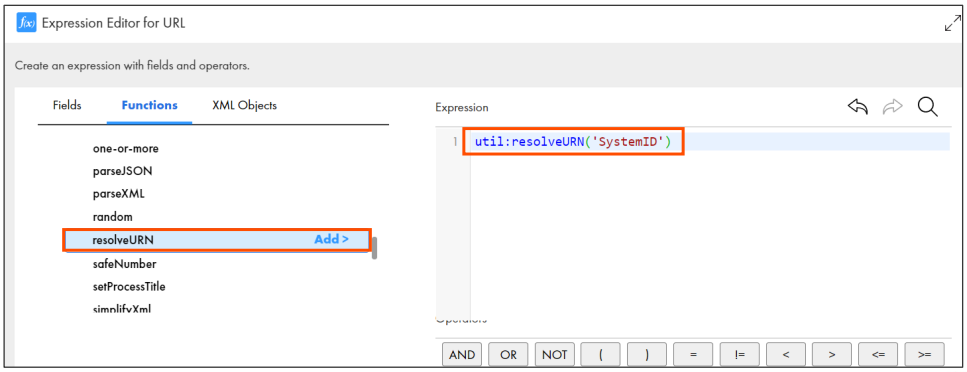
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Deployed Assets (continued)

URN Mappings - util:resolveURN(URN)

- The util:resolveURN(urn) is a built-in function which retrieves the URN \URL mapping specified in the application integration console in an IICS org
- It is available for use in the CAI process via an assignment step under the Formula option



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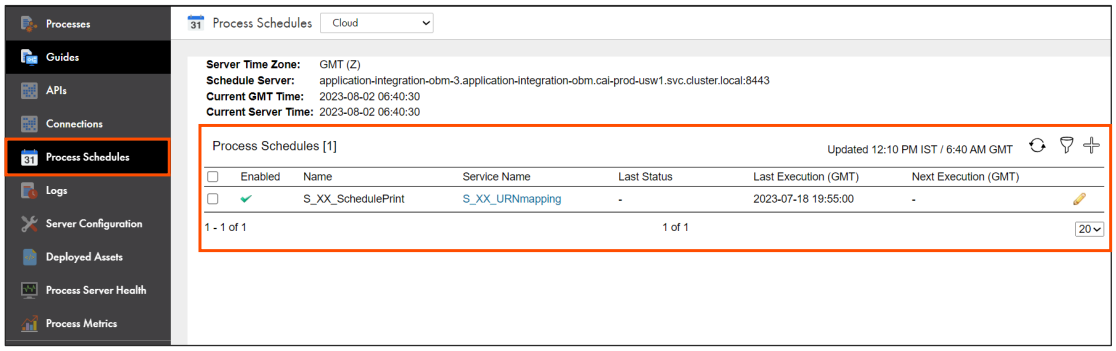
Topic

Process Schedules

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Process Schedules

- A BPEL process may be created to perform routine maintenance or another recurring task
- You can deploy the process and then set a schedule to run it, such as weekly or on a specific day of the month



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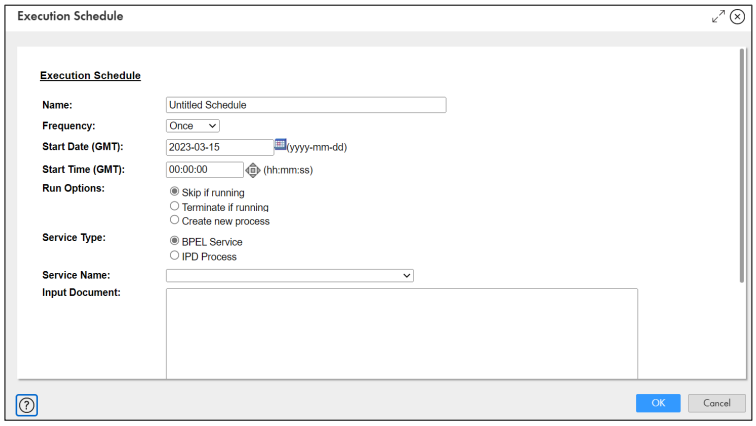
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Process Schedules (continued)

Creating a New Process Schedule

- To create a scheduled process, click the '+' icon at the top of the Process Schedule page
- A new window **Execution Schedule** opens



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In the Input Document/POST Body field, use the following details:

- If the process is implemented using Process Developer and of BPEL type, follow either of the below steps:
  - If the process implements a SOAP interface, enter the SOAP input message data contained within the SOAP Body Element, that is, the XML input message used to invoke the process. The data must conform to the WSDL interface of the process.
  - If the process implements a REST interface, you first need to create a SOAP wrapper for the REST service. Then, follow the instructions above.
- If you use Process Designer to implement the process and the process is of the Informatica Process Definition (IPD) type, enter the JSON body of the POST message used to invoke the process.

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Topic

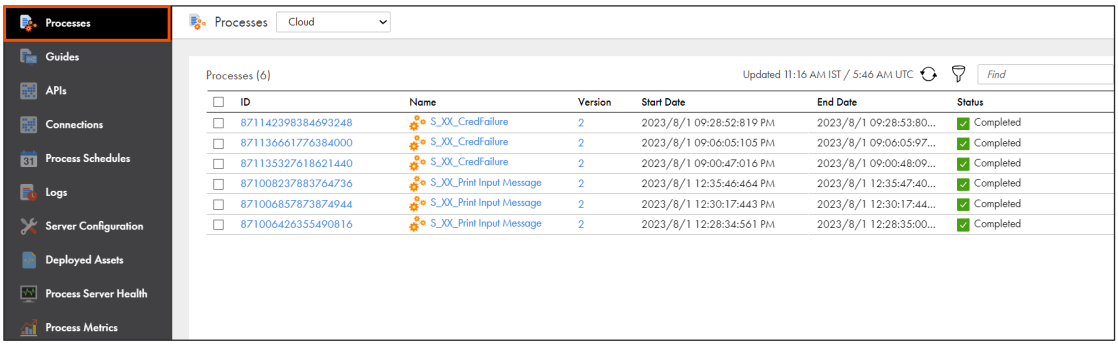
Processes

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Processes

- The Processes page shows a list of process instances that are running or have been running in the organization, along with the process version



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Processes (continued)

Process States

- The following table shows the different process states:

Process State	Meaning
Compensatable	A sub-process is complete and eligible for compensation.
Completed	Normal completions.
Faulted	Processes completed with a fault
Running	Normally running processes.
Suspended	A process suspended for any reason.

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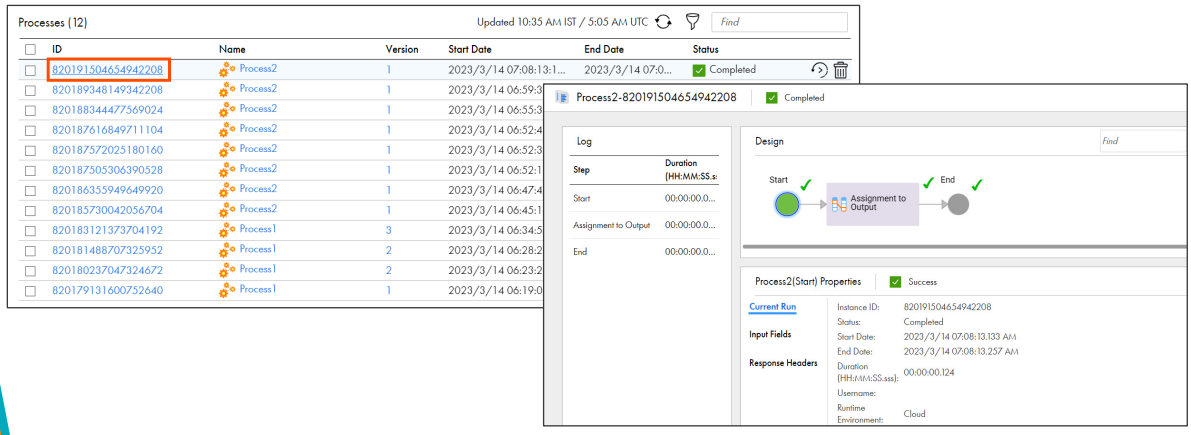
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Processes (continued)

Process View Detail Page

- Click a process ID on the Processes page to see the Process View Detail page.



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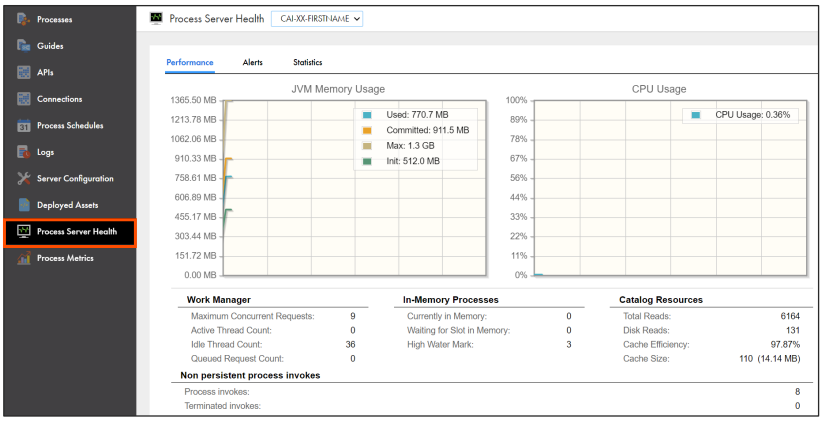
Process Server Health

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Process Server Health

- Process Server Health displays the following tabs:
  - Performance
  - Alerts
  - Statistics



Note: You must select a Secure Agent to see Process Server health details.

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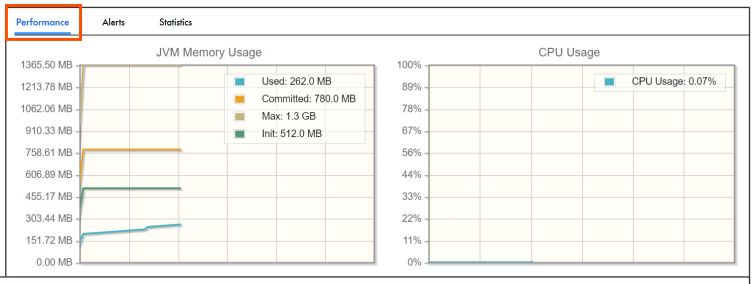
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Process Server Health (continued)

Performance

- On the Performance tab, you can see the following details:
  - Overall performance data
  - Performance and throughput of service requests
  - Resource and service bottlenecks



Work Manager		In-Memory Processes		Catalog Resources	
Maximum Concurrent Requests:	7	Currently in Memory:	0	Total Reads:	4237
Active Thread Count:	0	Waiting for Slot in Memory:	0	Disk Reads:	117
Idle Thread Count:	40	High Water Mark:	2	Cache Efficiency:	97.24%
Queued Request Count:	0			Cache Size:	110 (6.57 MB)

Non persistent process invokes	
Process invokes:	26
Terminated invokes:	0

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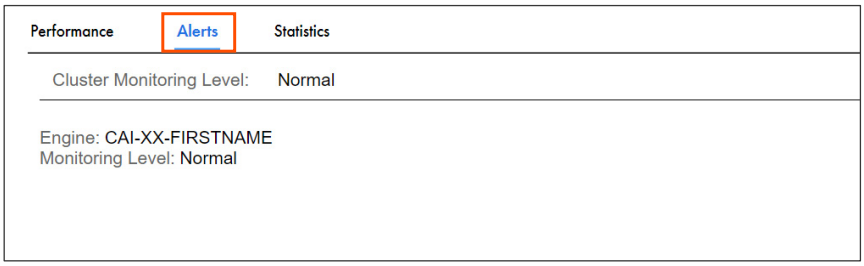
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Process Server Health (continued)

Alerts

- On the Alerts tab, you see alerts triggered by server statistics monitoring thresholds
- During each threshold interval, if an error or warning occurs, it is reported at the end of an evaluation period
- Each time you refresh your browser or open the Monitoring page, monitoring statistics are updated



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Process Server Health (continued)

Statistics

- Use the Statistics tab to view the monitoring properties you configured, or which are the default
- Process Server collects engine statistics and then aggregates them by intervals

Performance Alerts Statistics

Server Statistics

Updated 12:33 PM IST / 7:03 AM GMT

Property Name	Monitored	Current Interval (5 min.) Samples	Min	Max	Avg	Since Start or Reset Samples	Min	Max	Avg
Faulted/Faulting processes (count)		0	---	---	---	0	---	---	---
Process cache efficiency (percent)		0	---	---	---	110	0%	100%	100%
Time to obtain process (ms)		0	---	---	---	110	0	18	0
Work manager work start delay (ms)		425	0	5	0	1572460	0	1063	0
Database connection acquisition time (ms)		144	0	2	0	74731	0	675	0
Discarded unmatched correlated receives (count)		0	---	---	---	0	---	---	---
Time to validate messages (ms)		0	---	---	---	111	0	104	14
Critical storage exceptions (count)		0	---	---	---	0	---	---	---
Deadlock retry attempts (count)		0	---	---	---	0	---	---	---

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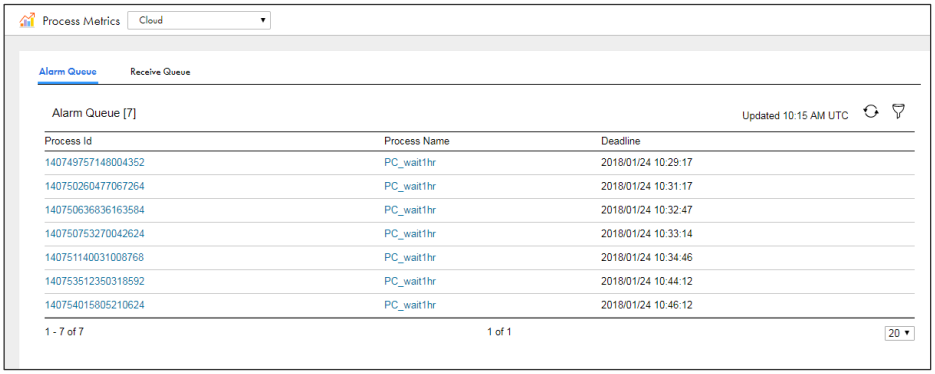
Process Metrics



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Process Metrics

- Use Process Metrics to view Alarm Queue and Receive Queue:
  - **Alarm Queue Metrics**: Lists the alarms in the alert queue, organized by process
  - **Receive Queue Metrics**: You need to select one or more options from the filter option list to view a selection of active receives



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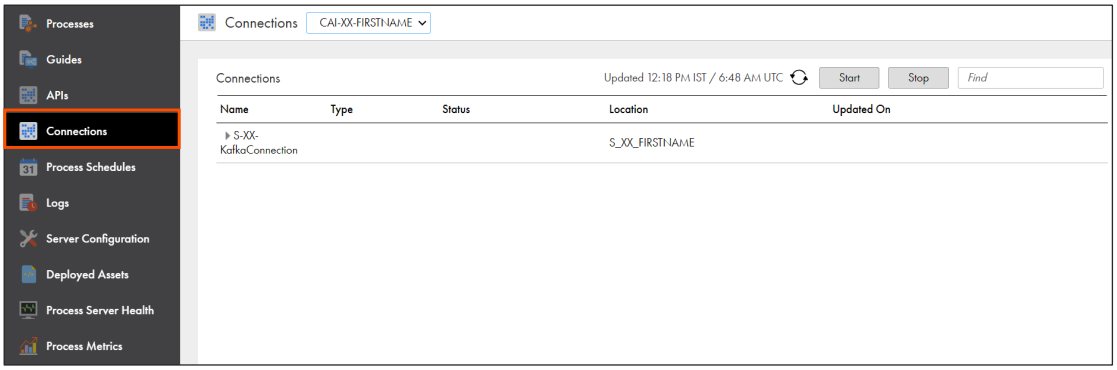
Connections

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Connections

- The Connections page shows a list of event sources in listener-based connections that have been published on a selected Secure Agent, along with the connection type and status.
- Use the Connections page to view, start, or stop event sources in listener-based connections.



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Topic Guides

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Guides

- Use the Guides page to view a list of guides that have been running
- The list includes guides that were completed successfully, completed with a fault, and guides that are still running
- Guides remain on the page for the following durations:

Status	Duration
Successful guides	7 days
Faulted or prematurely terminated guides	18 days
Running guides	As long as there is a running instance of the guide. After all instances are complete, it depends on whether there were successful or faulted instances.

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Topic

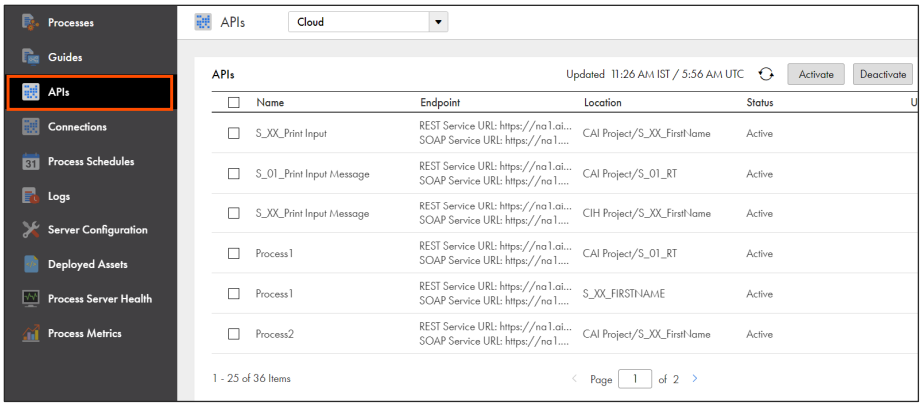
APIs

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APIs

- The APIs page shows a list of APIs of published processes that use the REST/SOAP binding, along with the REST and SOAP endpoints and their status.
- Use the APIs page to activate or deactivate the APIs of published processes.



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Topic

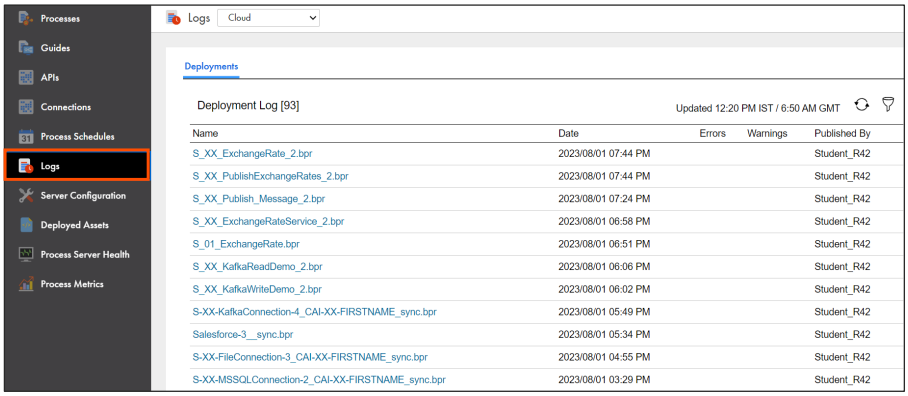
Logs

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Logs

- Use the Logs page to see server logs and deployment logs:
  - The **Deployments** tab shows a list of logs generated when new and modified business process archive (.bpr) files are deployed
  - The **Server** tab captures events occurring on the server so that you can view and troubleshoot problems



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Topic

Knowledge Check

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What does the Server Configuration page display?

- Monitor settings
- Queue settings
- Storage and scheduled maintenance settings
- System services settings
- All of the above

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Answer: All of the above

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Which of the following tabs is used to view and manage the URN Mappings?

- Server Configuration
- Deployed Assets
- Process Server Health
- Process Metrics

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Answer: Deployed Assets

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Lab Activity

13-1 Create and Use URN Mapping

In this lab, you will perform the following:

- Create a URN to URL Mapping
- Use the URN Mapping in a process

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Lab Activity

13-2 Create a New Process Schedule

In this lab, you will perform the following:

- Create a Process Schedule for an existing process

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Lab Activity

13-3 Start and Stop an Event Source in a Listener-based Connection

In this lab, you will perform the following:

- Stop the Event Source in the Kafka connection
- Start the Event Source in the Kafka connection

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Module Summary

This module showed you how to:

- Monitor and manage your processes using Application Integration Console

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Module 14
API Management

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Module Objectives

After completing this module, you will be able to:

- Discuss API Manager
- Create Managed API
- Discuss Authentication for Managed APIs
- Describe API Portal

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Topic

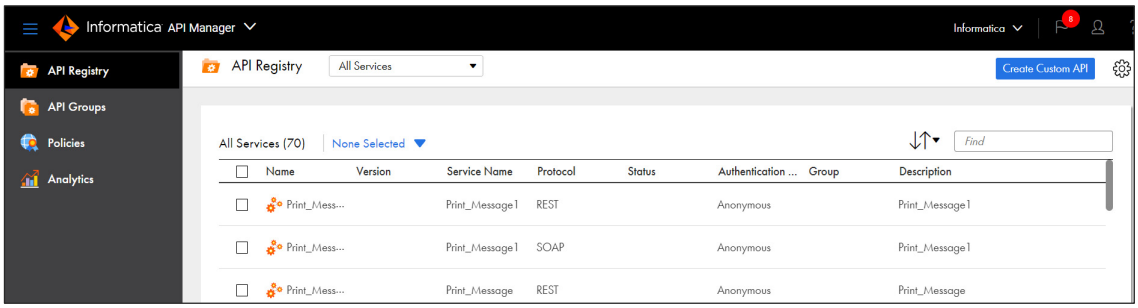
API Manager

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API Manager

- API Manager is a cloud-based service that:
  - Manages APIs for enterprise services and processes built in CAI
  - Enables the organization to deploy, manage, and control the use of APIs
  - Ensures that internal or external users use organization APIs safely and securely
  - Enables administrators to set policies and access authorization rights



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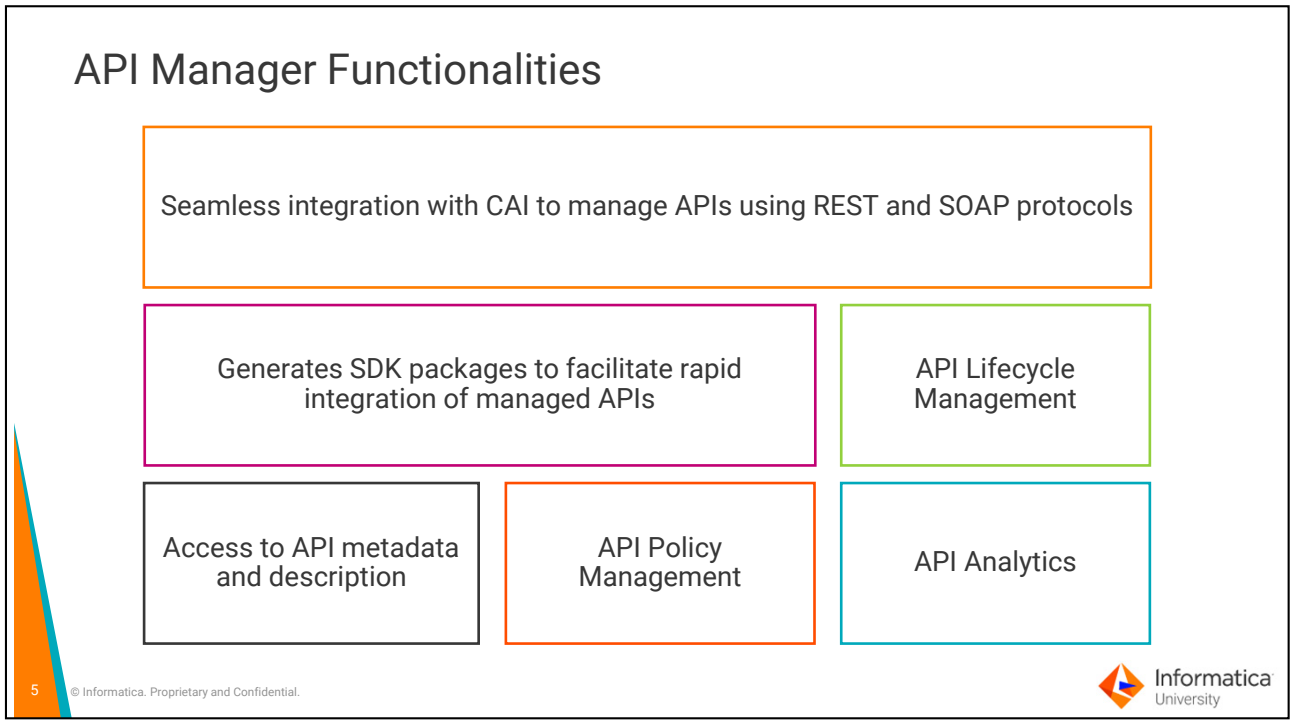
The API Manager is a cloud-based service that manages APIs for enterprise services and processes built in Informatica Cloud Application Integration.

API Manager enables the organization to deploy, manage, and control the use of APIs.

API Manager ensures that internal or external users can use the organization APIs safely and securely.

Administrators use API Manager to create managed APIs, set policies, and access authorization to control the use of managed APIs.

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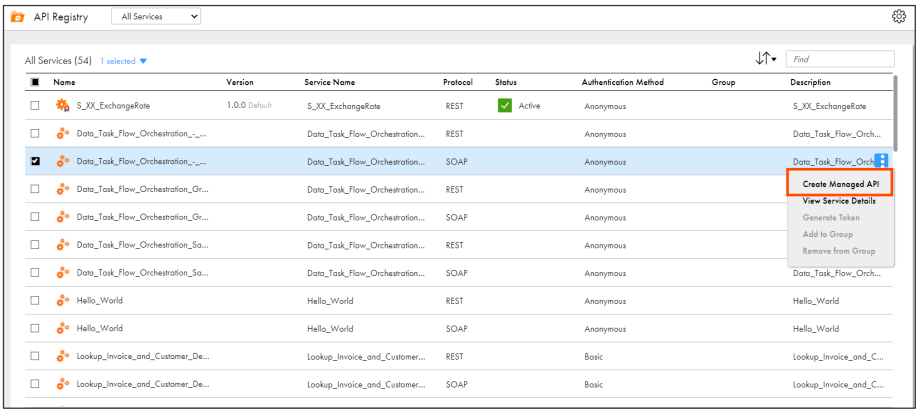
API Manager provides the following functionalities:

1. Seamless integration with CAI to manage APIs using REST and SOAP protocols.
2. Generates Software Development Kit (SDK) packages to facilitate rapid integration of managed APIs into organization applications.
3. API Lifecycle Management includes activating, deactivating, and deleting APIs.
4. Access to API metadata and description.
5. API Policy Management.
6. API Analytics include a dashboard, a security exception log, and an activity log.

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Create Managed API

- You can create Managed APIs in the API Registry tab.
- From the Actions menu, select **Create Managed API**.



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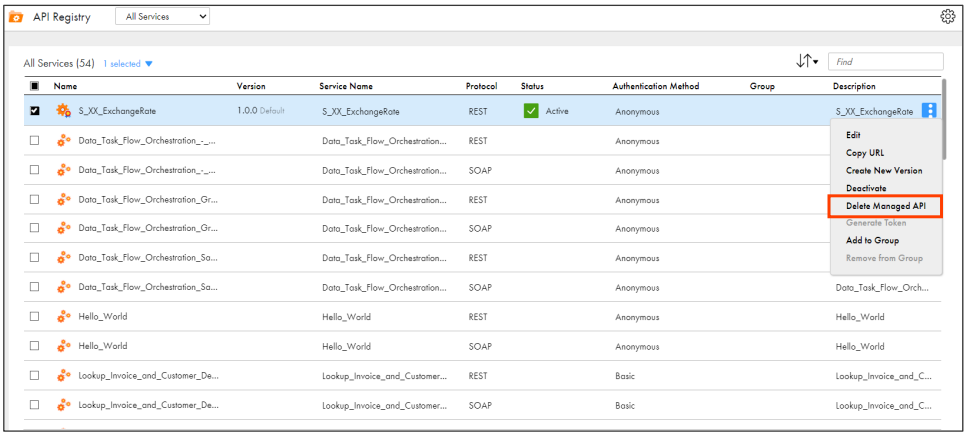
To create and use a managed API, perform the following steps:

1. Select the API Manager service.
2. Select the service for which you want to create a managed API.
3. To create the managed API, select **Actions** > **Create Managed** API.
4. When API Manager creates a managed API, it activates the managed API.
5. You can optionally define privacy policies, IP filtering policies, rate limit policy, and so on for the selected API under **Actions** > **View Details**.
6. After the Managed API URL is created, you can copy the managed API URL and provide the URL to API consumers.

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Delete Managed API

- You can delete Managed API in the API Registry tab.
- From the Actions menu, select **Delete Managed API**.



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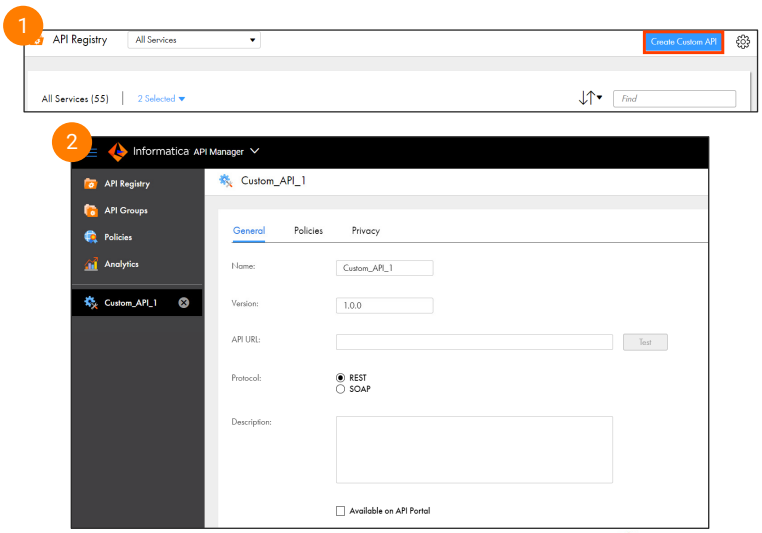
To delete a managed API, in the API Registry page, select the API. From the Actions menu, select Delete Managed API.

Note: The services on which the API was based is not affected.

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Customize Managed API URL

- You can customize the URLs of managed APIs so that the API domain name replaces the Informatica domain name as the base URL.
- To create a custom API:
  - From the API Registry page, click **Create Custom API**.
  - In the Custom API page, enter the Name, Version, URL, and Protocol for the API.



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You can customize the URLs of managed APIs so that the API domain name replaces the Informatica domain name as the base URL.
Example: Your organization's API domain is org-subdomain.com, and the name of the managed API is Customers. When you create a managed API, API Manager creates one of the following API URLs, based on whether or not you configured them:

1. If you don't customize the API URLs, API Manager creates the following URL:
   https://apigw-pod1.dm-us.informaticacloud.com/t/org-subdomain.com/Customers

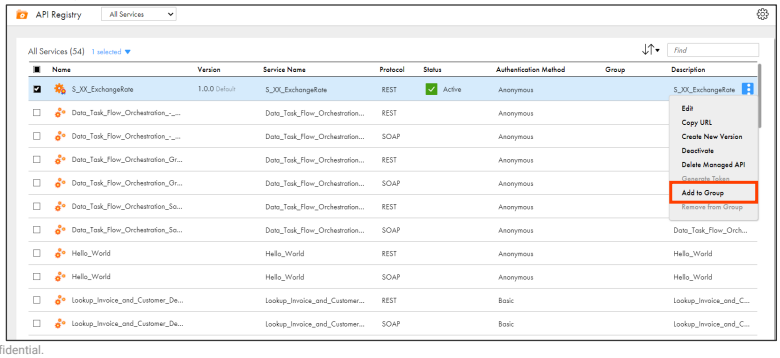
2. If you customize the API URLs, API Manager creates the following URL:
   https://org-subdomain.com/Customers

Note: Before you configure API URL customization, you must contact Informatica Global Support and provide Informatica with a certificate for the organization API domain.

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API Group

- API Group page is used to:
  - View and edit managed API groups
  - Generate a JSON web token for the group
  - Delete the group
- Use the API Registry page to add or remove APIs from the group



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The API Group page is used to:
1. View and edit managed API groups.
2. Generate a JSON web token for the group.
3. Delete the group.

To add an API to a group, you must use the 'Add to Group' option in the API Registry page.

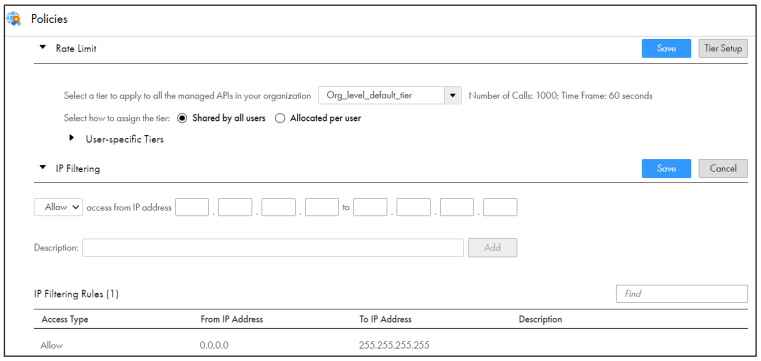
After you create the group, use the 'Generate Token' option to generate JSON web Token for the group.

To delete an API from the group, you can use the ‘Remove from Group’ option in the API Registry page or select the group in the API group page and from the Actions menu, click **Delete**.

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API Policies

- Policies page is used to define:
  - Rate Limit Policy
  - IP Filtering Policy
  - Rate limit policy controls the number of times a managed API can be invoked
  - IP filtering policy defines the range of IP addresses that are allowed or denied permission to invoke managed APIs



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The Policies page of API Manager is used to define: Rate Limit Policies and IP Filtering Policies.

The rate limit policy controls the number of times a managed API can be invoked during a designated time period for all the organizations' managed APIs.

Note: The rate limit rule cannot be deleted, but can be updated.

You can create IP filtering policies that apply to all the managed APIs in the organization. The IP filtering policy defines the range of computer IP addresses that are allowed or denied permission to invoke managed APIs. The order of the policy rules determines the precedence.

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API Analytics

- API Analytics provides a graphical overview of activities and API usage, and the ability to drill down to specific activities and events.



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API analytics provides a graphical overview of an activity and API usage, and the ability to drill down to specific activities and security events.

Overview tab: The Overview dashboard offers a collection of panels that contains reports about managed APIs. You can obtain visual summary information about APIs, such as trends in usage over time, APIs with the most invocations, and the most frequent users.

Activity Log tab: When users invoke API calls, the organization tracks general API usage activity for API access instances and access exceptions. API Manager creates an activity log for all the API access instances, and a security log to track any access exceptions that users create while invoking managed APIs.

Event Log tab: The organization creates IP filtering, rate limiting, and basic authentication policies. If a user breaches these policies, API Manager logs exceptions to the security log.

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Topic

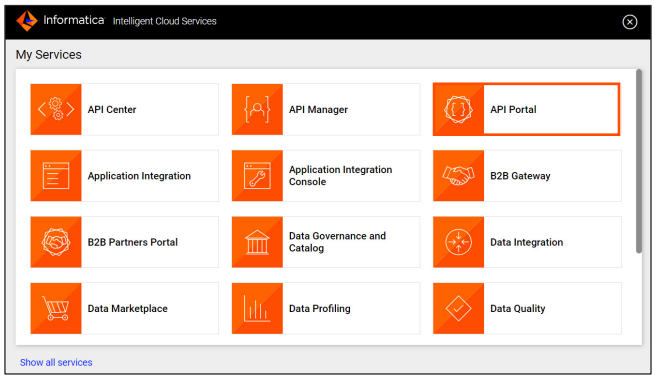
API Portal

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API Portal

- API Portal provides API consumers with secure access to managed APIs along with the details such as status, authentication type, and access control policies.
- API consumers also use the API Portal to:
  - Generate SDK packages
  - Interact with managed APIs
- To access API Portal, the organization administrator must assign you the Service Consumer privilege, and the View API Portal privilege in API Manager.



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API Portal provides API consumers with secure access to managed APIs along with the details of a managed API. The details include status, authentication type, and access control policy. You can also drill down to further details with Swagger or WSDL.

API consumers can use API Portal to generate SDK packages to facilitate rapid integration of managed APIs into applications. Consumers can generate SDK packages that provide a set of resources to enable integration for Java, Android, JavaScript, Node.js, Python, Ruby-on-Rails, C#.NET, ASP.NET5, or C# applications.

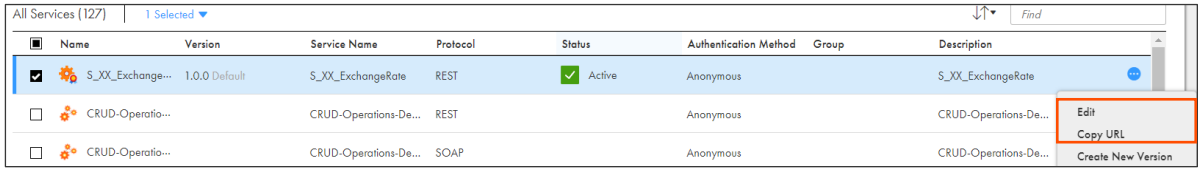
API consumers can also use API Portal to interact with managed APIs and view API usage analytics.

To access API Portal, the organization administrator must assign you the Service Consumer privilege and the View API Portal privilege in the API Manager.

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API Registry

- API Registry page enables you to:
  - Securely edit the details of the managed APIs and copy the URL of a managed API to use in an application
  - Access a Swagger Interface for REST managed APIs or a WSDL interface for SOAP managed APIs



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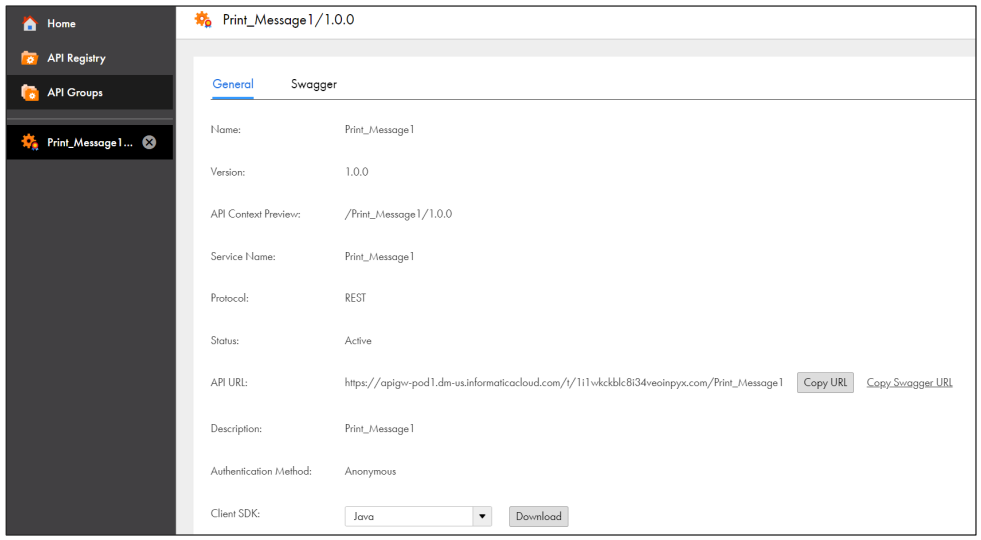
API Registry page enables you to securely view details of the managed APIs and copy the URL of a managed API to use in an application.

You can also access a Swagger interface for REST managed APIs or a WSDL interface for SOAP managed APIs.

You can use the Swagger or WSDL interfaces to test and debug API execution.

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API Registry Properties



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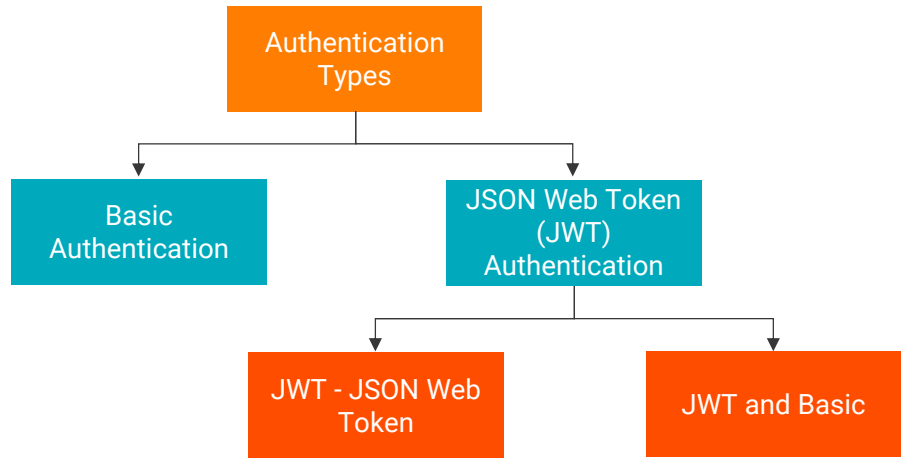
To view the managed APIs, you must use the API Registry page of API Manager. The API Registry page shows managed APIs in alphabetical order. To sort the managed APIs, click the column title. If a managed API exists for a service that is unavailable or deleted, the API Registry greys it out in the display.

The API Registry page properties are:

1. **Name**: Name of the managed API.
2. **Version:** Version of the API.
3. **Protocol**: REST or SOAP protocol.
4. **Status**: Active, Inactive, or Service not available.
5. **API URL:** URL of the managed API.
6. **Group:** The group that the API belongs to.
7. **Description**: Description provided by Informatica CAI developer for the service.
8. **Authentication**: Basic Authentication and Anonymous.
   i. **Basic Authentication** – API Consumer must provide IICS username and password.
   ii. **OAuth 2.0.** – API consumer provides an OAuth 2.0 access token to invoke the managed API.
   iii. **JWT - JSON Web Token –** API consumer passes the JWT token as a bearer.

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Authentication for Managed APIs



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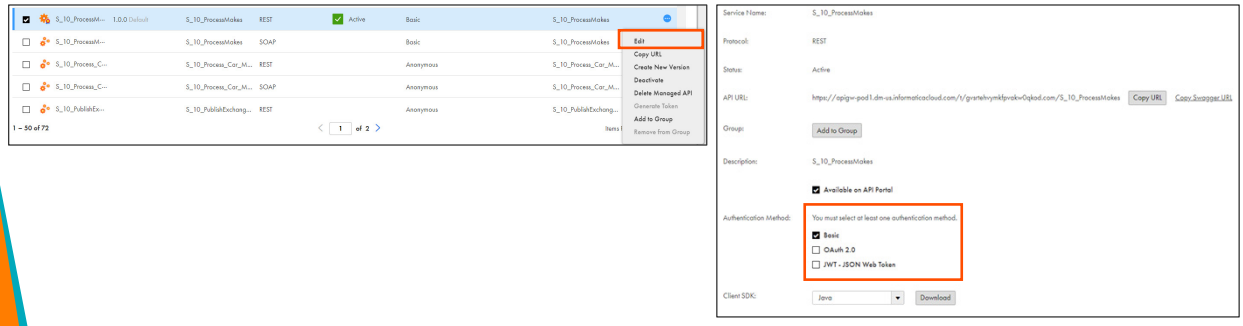
In CAI, you can configure basic authentication and JSON Web Token (JWT) authentication for managed Informatica Cloud Application Integration APIs.

There are two methods to configure JWT authentication in a managed API
1. JWT − JSON Web Token
2. JWT and Basic

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Set Authentication Type for Managed API

- To set authentication type:
  - From the **API Registry** page, select the **Managed API** of interest.
  - From Actions menu, select 'Edit'.
  - In the General tab, select the desired Authentication Method.



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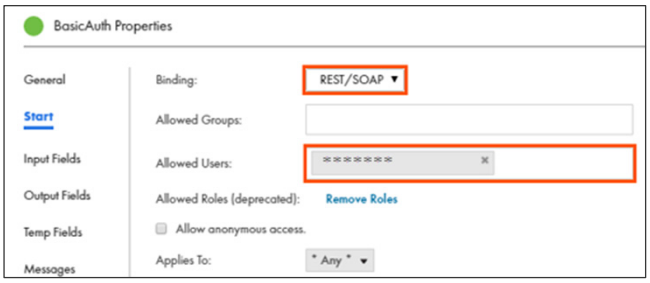
You can set the authentication type for a managed API in the API Registry page of the API Manager service.

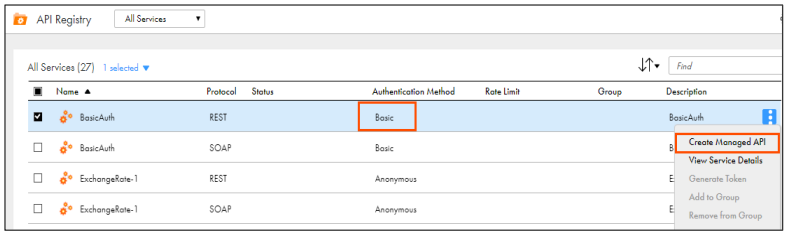
To set authentication type, perform the following steps:

1. From the API Registry page, select the managed API for which you want to set the authentication type.
2. Select the managed API and from Actions menu, select View Details.
3. In the General tab of managed API, from the Authentication Method drop-down, select the desired authentication method.
4. To apply the authentication, save the changes performed in the Managed API.

Basic Authentication

- Requirements to configure Basic Authentication:
  - The associated process uses REST/SOAP binding.
  - The associated process defines the user groups and users who can access the process service URL at runtime.
  - The associated process is published and exposed as a service.





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To configure basic authentication for a managed API, the associated process must meet all of the following requirements:

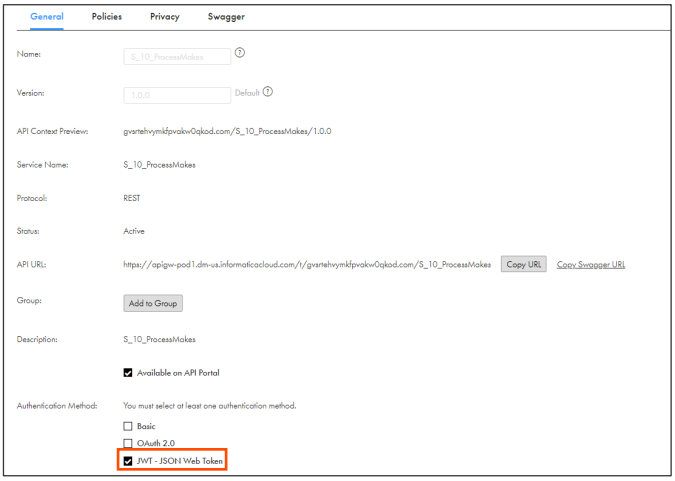
1. The associated process uses HTTP/SOAP binding.
2. The associated process defines the user groups and users who can access the process service URL at run time.
3. The associated process is published and exposed as a service.

When you configure basic authentication for a managed API, the user groups and users that are provided access to the API in Informatica Cloud Application Integration can invoke the API.

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JSON Web Token (JWT) Authentication

- Used to configure JWT authentication with tokens
- Generate a token using API Manager or API Portal, and set an expiration date for the token



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JWT is an open standard that helps in the secure transmission of information between API consumers and REST web services such as Informatica Cloud Application Integration service APIs.

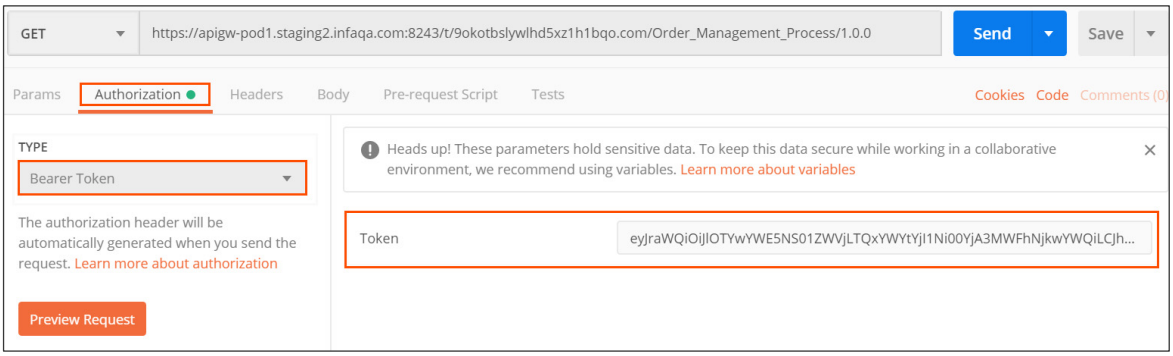
This authentication method is used to configure JWT authentication with tokens. You can generate a token using API Manager or API Portal. After you set the authentication method as JWT, you can set an expiration date for the token and generate token for the API using the 'Generate' option.
API Manager uses the Coordinated Universal Time (UTC) time zone for the JWT token expiration and uses the current time on your computer as the baseline time for the token expiration. The token expires on the expiration date you configure and a minute earlier than the time at which you generated the token. After the token expires, you cannot refresh it. You must generate a new token.

Note:

1. You can configure JSON web token authentication for a managed API only if the associated process uses basic authentication in Informatica Cloud Application Integration. You cannot configure JWT authentication if the associated process allows anonymous access.
2. If you want to generate the token from API Portal, you must select 'Available on API Portal' option for managed API.

Invoking API with JWT Authentication

- You can invoke the API by passing the token as a bearer token in the Authorization header



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You can invoke an API based on the authentication method that it uses.

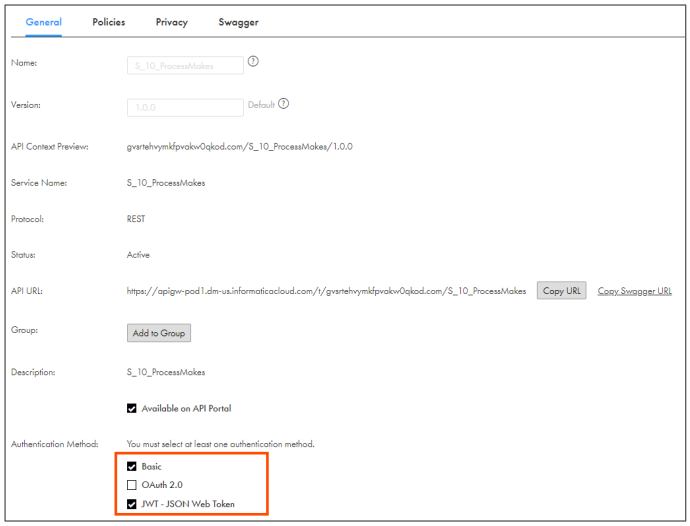
To invoke a managed API with JWT - JSON Web Token authentication method, you can invoke the API by passing the token as a bearer token in the HTTP Authorization header.

The image shows an API invoked through Postman with the authorization type set to Bearer Token and the token specified.

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JWT and Basic Authentication

- Used to configure JWT authentication with tokens and allow basic authentication for the managed API



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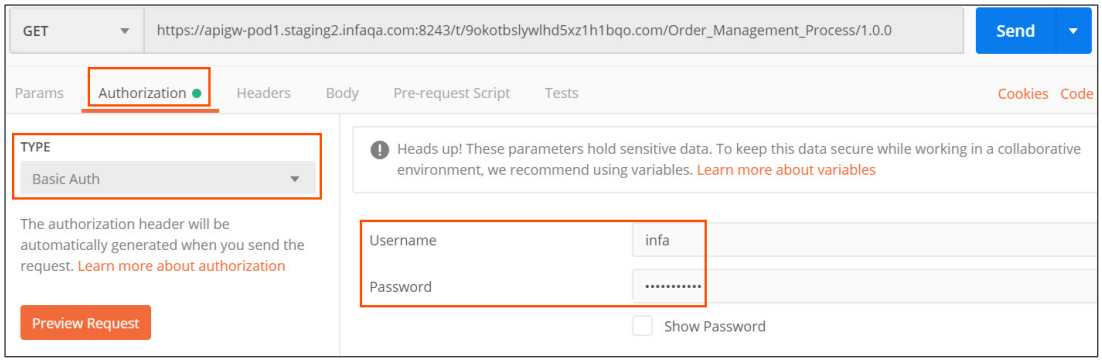
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This authentication method is used to configure JWT authentication with tokens and allow basic authentication for the managed API.

When you use this type of authentication, you can generate a token and set an expiration date for it. You can also allow the API to be accessed by the allowed user groups and allowed users defined for the API in Informatica Cloud Application Integration.

Invoking API with JWT and Basic Authentication

- You can invoke the API by one of the following methods:
  - Passing the token as a bearer token in the HTTP Authorization header
  - Providing the login credentials of the allowed user groups or users



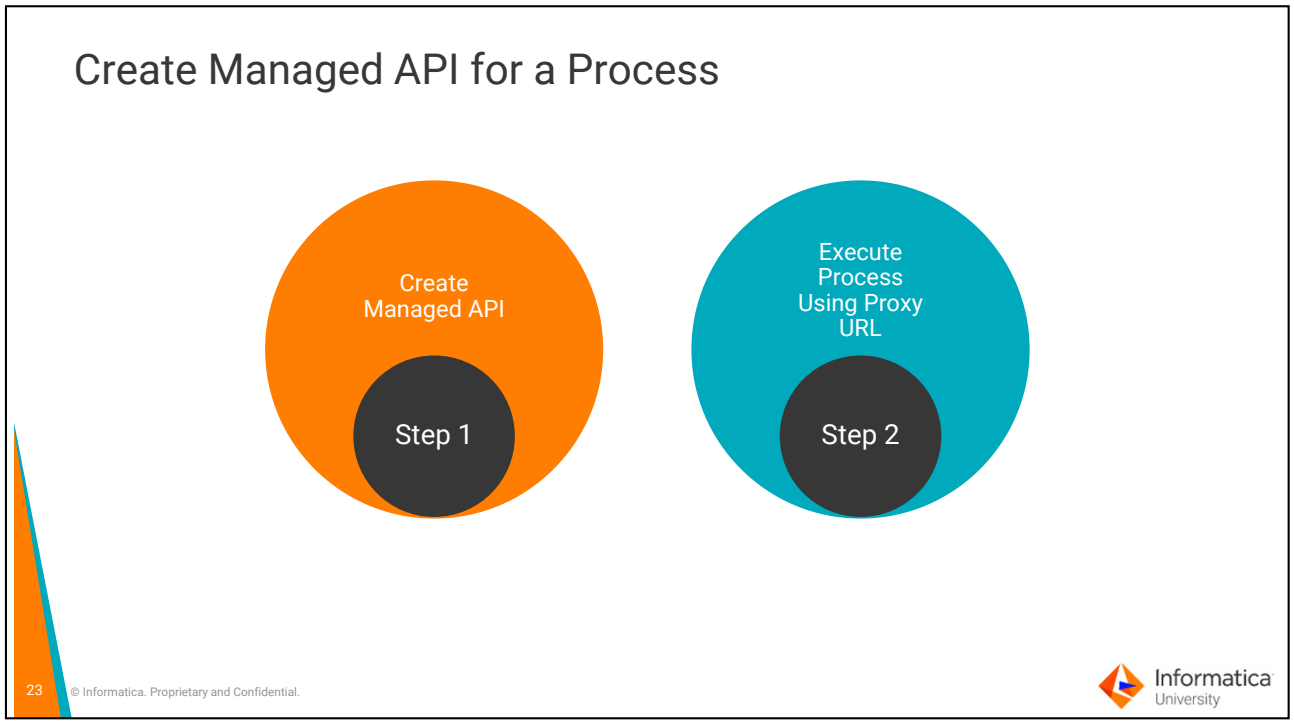
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To invoke a managed API with JWT and basic authentication method, you can invoke the API by following two methods:

1. Passing the token as a bearer token in the HTTP Authorization header.
2. Providing the login credentials of the allowed user groups or users.

The image shows an API invoked through Postman with the authorization type set to Basic Auth, and the user name and password specified.



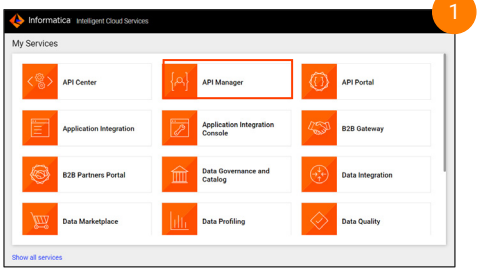
To create the managed API for ExchangeRateService process, perform the following steps:

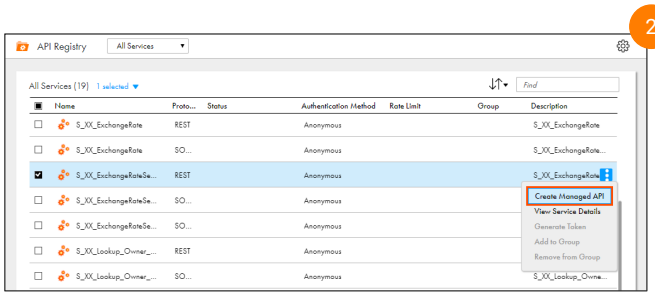
1. From the My Services window, select **API Manager service**.
2. To create a Managed API, select **API Registry** tab.
3. Select the ExchangeRateService process and from the Actions menu, select 'Create Managed API'.
4. To find the managed API URL generated by the system, select **Actions** > **View Details**.
5. Copy the API URL and execute the process using a third-party tool like Postman.

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Create Managed API for a Process (continued)

Process Configuration







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Topic

Knowledge Check

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Can we apply IP filtering policy to API Groups?

- Yes
- No

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Answer: Yes

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Why do we use the API Registry page?

- To securely view details of the managed APIs and copy URL of a managed API to use in an application
- To access a Swagger interface for REST managed APIs
- To access WSDL interface for SOAP managed APIs
- All of the above

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Answer: All of the above

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Lab Activity

14-1 API Manager

In this lab, you will perform the following:

- Create a Managed API for the ExchangeRate process

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Module Summary

This module showed you how to:

- Discuss API Manager
- Create Managed API
- Discuss Authentication for Managed APIs
- Describe API Portal

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Module 15
CAI and CDI Integration

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Module Objectives

After completing this module, you will be able to:

- Define Cloud Data Integration
- Create a Connection
- Create a Mapping
- Integrate CAI and CDI

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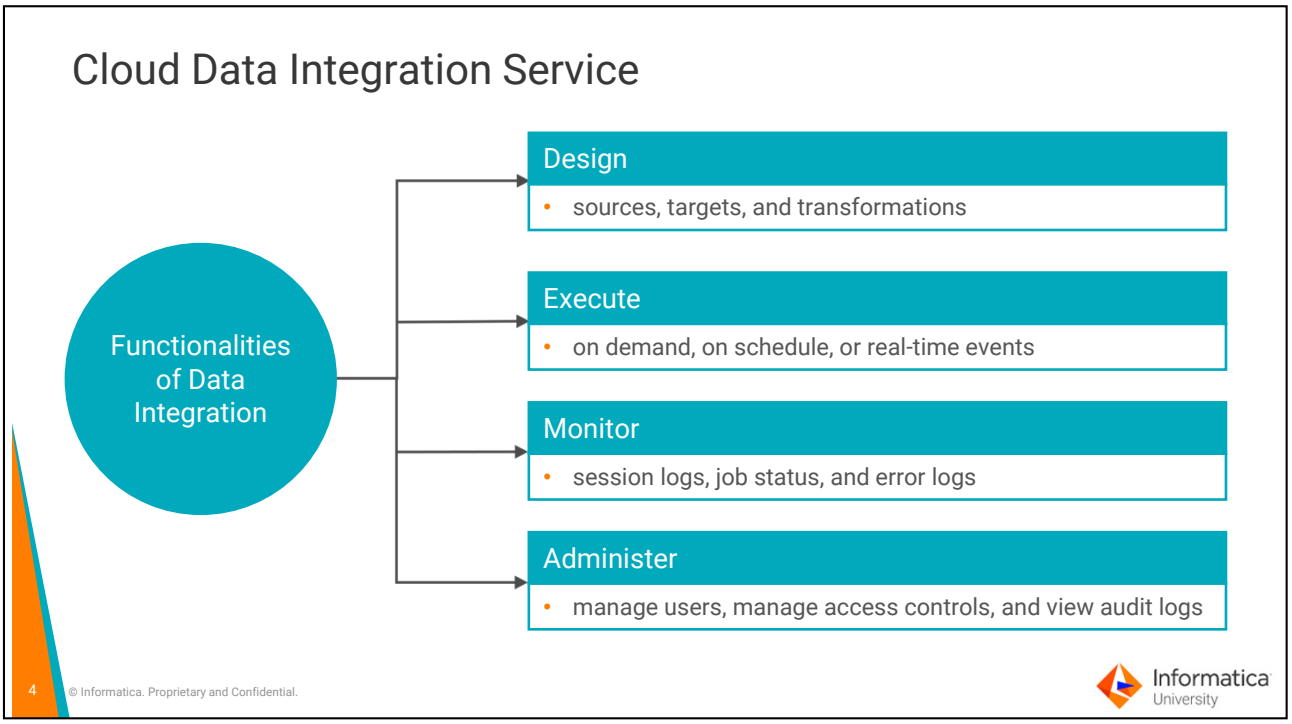
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Topic

CDI Overview

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You can use the IICS Data Integration service to design, execute, monitor, and administer tasks.

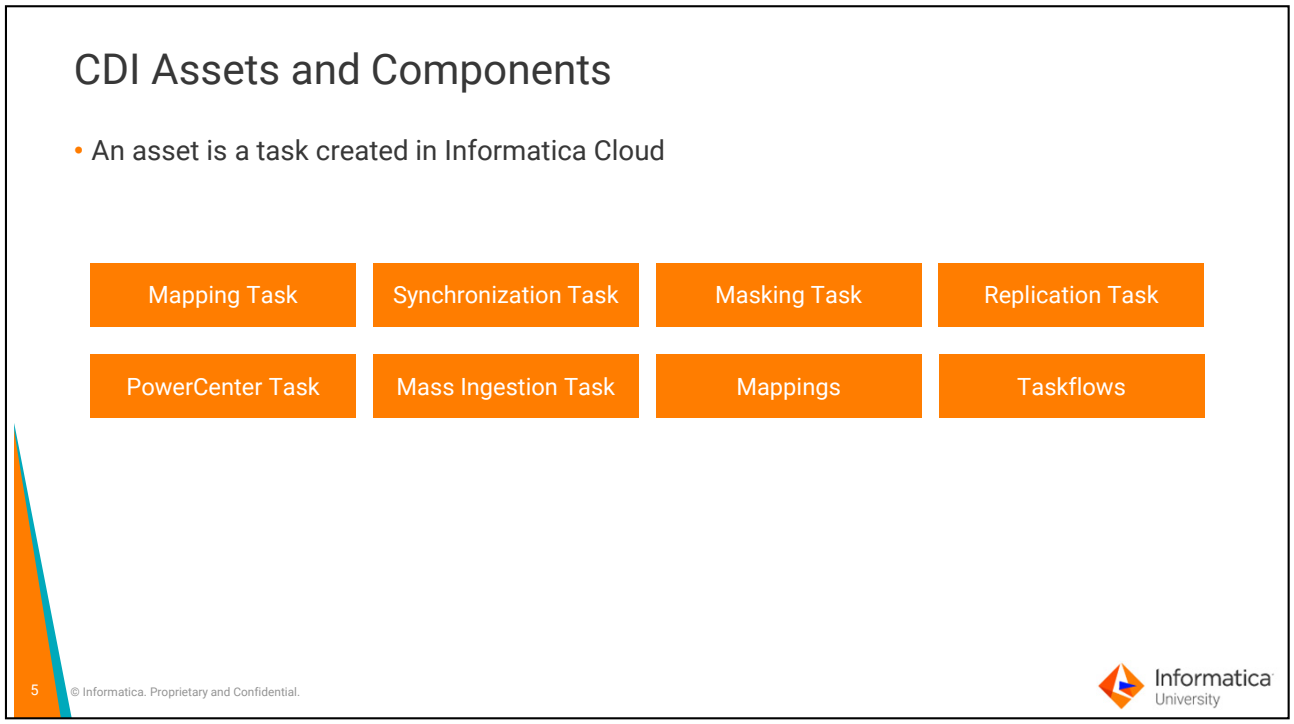
While designing a task, you can select from a variety of sources, targets, and transformations to transform and map data.

You can execute a task on demand or based on a schedule, or real-time events such as an outbound message triggered in Salesforce.

While monitoring a task, you can view session logs, job status, and error logs in the jobs section of the Data Integration Service.

As a part of the administrative tasks, you can manage users, manage their access rights and privileges, and view audit logs.

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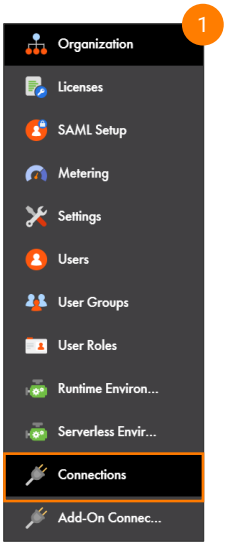
An asset is a task created in Informatica Cloud. Using the Data Integration service, you can create a Mapping task, Synchronization task, Masking task, Replication task, PowerCenter task, Mass Ingestion task, Mappings, and Taskflows.

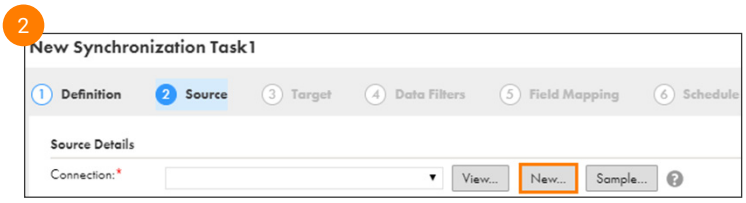
A component is used within an asset. For example, to use a saved query in a synchronization task, you need to first create the saved query component, and then refer to it in the synchronization task asset.

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Creating a Connection in CDI

- Connections are available to all the users in the Org.
- Before creating any connection, you must install a Secure Agent.
- Create Connection:
  - From **Administrator** service
  - On-the-fly via the wizard when configuring a task





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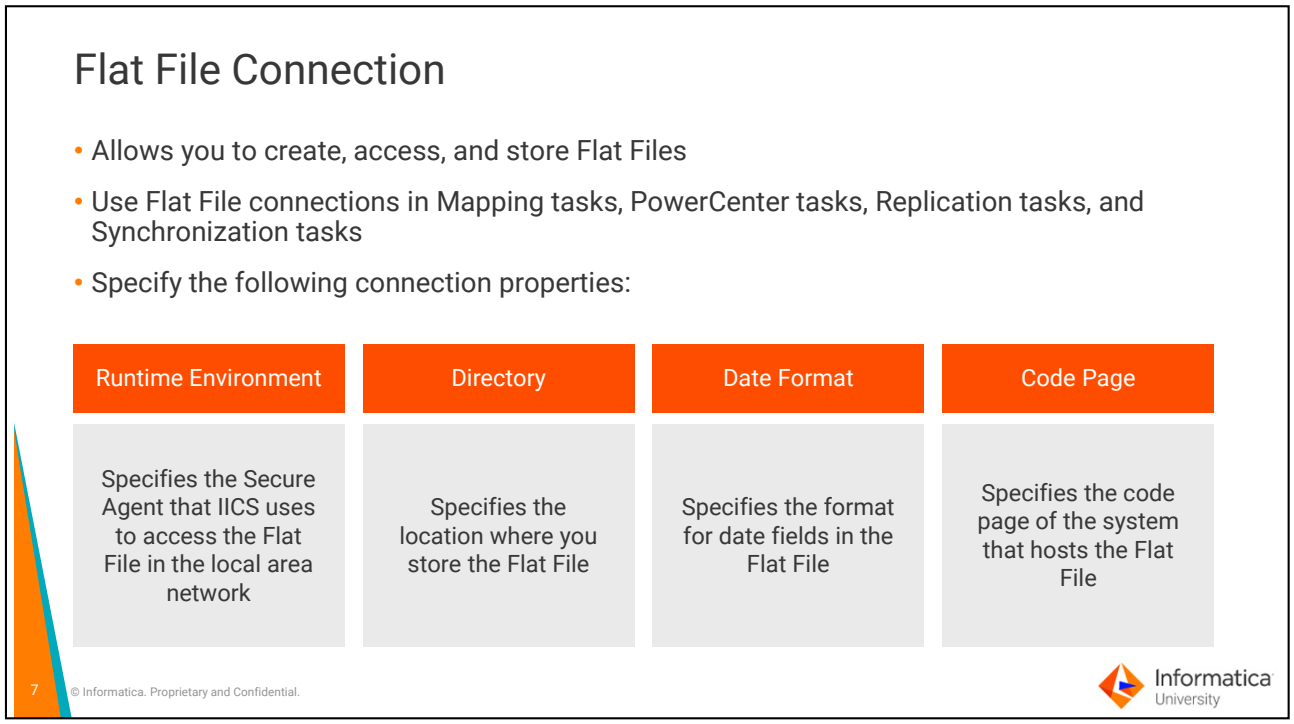
You can create connections in IICS by selecting 'Connections' from the Administrator service.

You can also create connections on-the-fly using the wizard when you configure a task. For example, you can create connections using the Synchronization task wizard.

When you create a connection, the connection becomes available to all the users in the Org.

Except for Software-as-a-Service (SaaS) applications, all connections depend on the Secure Agent. Therefore, you must first install the Secure Agent in your IICS Org before creating any connection.

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Flat File connections allow you to create, access, and store flat files. You can use Flat File connections in mapping tasks, PowerCenter tasks, Replication tasks, and Synchronization tasks. When you select a Flat File connection, you must choose the formatting options for the Flat File. When you choose the formatting options in a Source, Lookup, or Target transformation, you must specify whether the Flat File is a delimited Flat File or a fixed-width Flat File.

When you create a Flat File connection, you must specify some connection properties:

- **Runtime Environment**: Specifies the Secure Agent that IICS uses to access the Flat File on the local area network.
- **Directory**: This is the location where you store the Flat File. You must enter the full path to the directory, or you can click the Browse button to locate and select the directory.
- **Date Format**: This is the format for date fields in the Flat File.
- **Code Page**: This is the code page of the system that hosts the Flat File.

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Lab Activity

15-1 Create a Flat File Connection

In this lab, you will perform the following:

- Create a Flat File connection

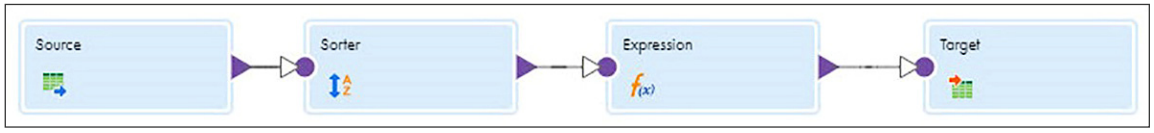
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Mapping

- Defines the flow of data from the source to the target
- Add transformations to perform data transformation tasks
- Define rules for fields that are part of transformations
- Links visually represent the flow of data in a mapping



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A mapping is an object that you create in the Cloud Mapping Designer. It defines the flow of data from the source to the target.

To create a mapping, you must add transformation shapes to the mapping flow. The transformations perform various data transformation tasks.

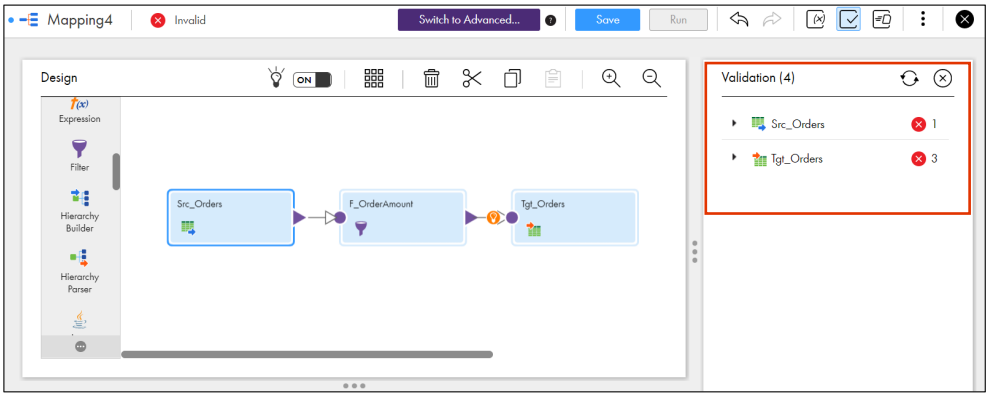
You can also define rules for fields that come into the transformations. The rules are based on certain criteria and allow you to create flexible mappings. For example, you can choose to include or exclude fields of a certain data type. This allows the mapping to remain valid even when you add new fields to the source.

The links in the mapping visually represent the flow of data in the mapping.

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Validate Mapping

- When you save a mapping, the Mapping Designer validates the mapping
- The status of the mapping can be valid or invalid
- Use the Validation panel to view the location and details of mapping errors



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When you save the mapping, the Mapping Designer automatically validates the mapping. The status of the mapping is displayed in the header of the Mapping Designer. The status can be either valid or invalid.

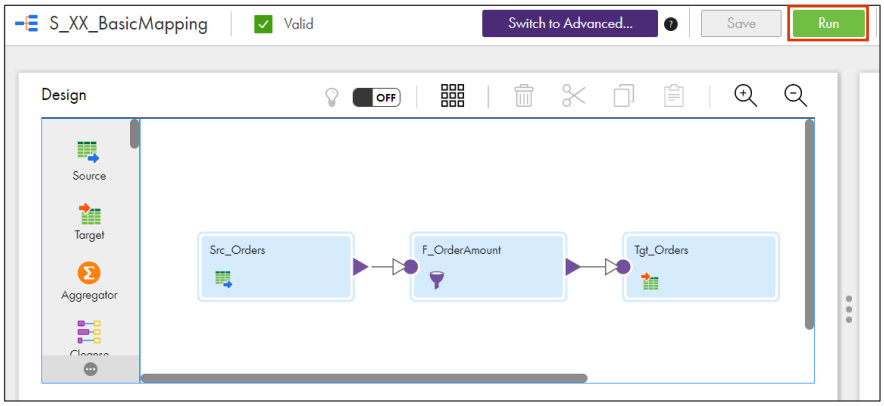
If the mapping is invalid, you can use the Validation panel to view the location and details of the errors. The Validation panel displays a list of transformations in the mapping. An error icon is displayed next to the transformation that includes errors.

As you can see in the image, the sample mapping has four errors in the Target Transformation.

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Run a Mapping

- When the mapping is valid, you can perform a test run to verify the results of the mapping
- Test run creates a temporary mapping task
- Data Integration service deletes the temporary mapping task after the test run is complete



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After you check the validity of a mapping, and ensure that it is free of errors, you can perform a test run to verify the results of the mapping.

In the test run, you run a temporary mapping task. The task reads data from the source, writes data to the target, and performs all transformation logics in the data flow. The Data Integration service deletes the temporary task after the test run is complete.

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Lab Activity

15-2 Create a Basic Mapping and a Mapping Task

In this lab, you will perform the following:

- Use the Mapping Designer to create a mapping
- Create a Mapping Task

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Invoke a Mapping in CAI

- A Service step can be used to interact with Data Integration applications
- Perform the following steps to create a Run Cloud Task Service step:
  - Insert a Service step
  - In the Service properties, select the Service Type as **System Service** and the Action as **Run Cloud Task**
  - Configure the following required properties:
    - **Task Name**: The name of the cloud task that you want Process Designer to start.
    - **Wait for Task to Complete**: If you select this, Process Designer waits for the task to complete before resuming execution.
    - **Max Wait**: The maximum number of seconds that Process Designer waits for a task to complete. The default value is 300 seconds, or 5 minutes.
    - **Fail on Cloud Task Errors**: Faults when the synchronization or mapping fails.



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Perform the following steps to create a Run Cloud Task Service step:

1. Insert a Service step.
2. In the Service properties, select the action Run Cloud Task.
3. Configure the following required properties:

- **Task Name**: The name of the cloud task that you want Process Designer to start.
- **Wait for Task to Complete**: If you select this, Process Designer waits for the task to complete before resuming execution.
- **Max Wait**: The maximum number of seconds that Process Designer waits for a task to complete. The default value is 300 seconds, or 5 minutes.

Important: You must enter a Max Wait value of less than 604800 seconds, that is, seven days or fewer. Process Designer waits for a maximum of seven days for a task to complete.

- **Fail on Cloud Task Errors**: Faults when the synchronization or mapping fails.

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Topic

Knowledge Check

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Using the Data Integration service, you can _____.

- Create Mapping task, Synchronization task, and Masking task
- Create Replication task, PowerCenter task, and Mass Ingestion task
- Create Mapping and Taskflows
- Create a Mapping task, Synchronization task, Masking task, Replication task, PowerCenter task, Mass Ingestion task, Mappings, and Taskflows.

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Answer: Create a Mapping task, Synchronization task, Masking task, Replication task, PowerCenter task, Mass Ingestion task, Mappings, and Taskflows.

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Which step can you use to invoke a CDI Mapping in CAI?

- Process
- Service
- Assignment
- Subprocess

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Answer: Service

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Lab Activity

15-3 Invoke a Mapping in CAI

In this lab, you will perform the following:

- Create a Process
- Use a Service step and add a Cloud Task
- Execute the Process

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Module Summary

This module showed you how to:

- Define Cloud Data Integration
- Create a Connection
- Create a Mapping
- Integrate CAI and CDI

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Module 16

Troubleshooting, Tips and Tricks, Best Practices

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Module Objectives

After completing this module, you will be able to:

- View Asset Dependencies
- List Best Practices for CAI Assets
- Define Logging Types in CAI
- Discuss Frequently Faced Issues and Workarounds

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Topic

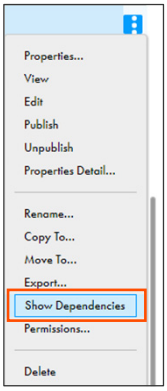
Managing Assets and Processes

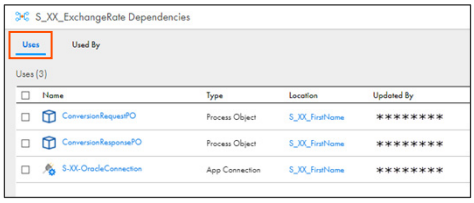
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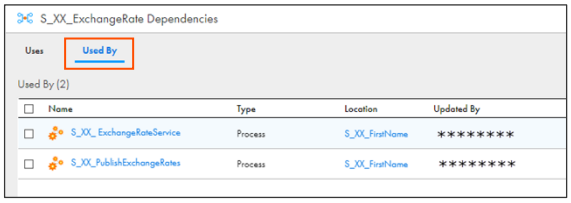
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Asset Dependencies

- Asset Dependencies allow you to view object dependencies before performing certain operations on an asset.
- You can view object dependencies for an asset on the **Explore** page.
- To view object dependencies for an asset, in the row that contains the asset, click **Actions**, and select **Show Dependencies**.
- The Dependencies page opens the Uses tab by default.







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Asset Dependencies allow you to view the object dependencies before performing certain operations on an asset.

Assets cannot be deleted under the following conditions:

1. If another object depends on the asset.
2. To delete an asset, first delete the dependent objects and then delete the asset.

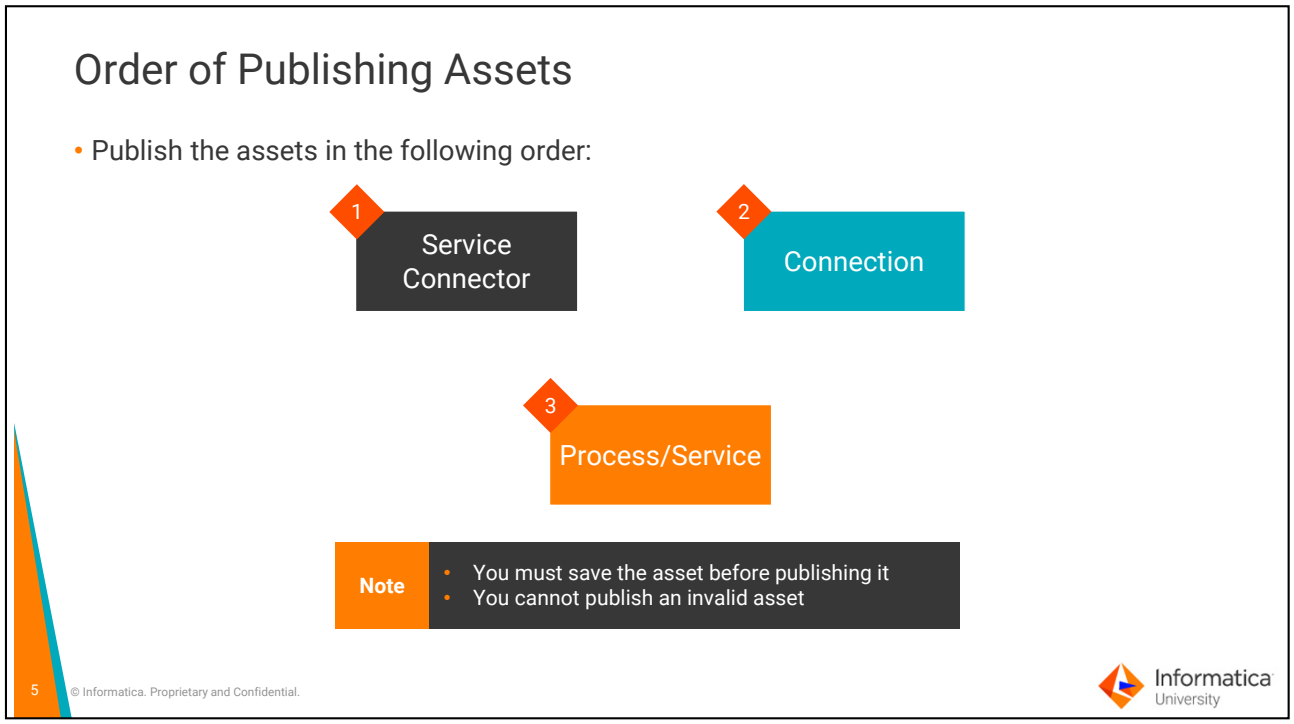
You can find the dependent objects by viewing the asset dependencies.

You can view object dependencies for an asset on the Explore page.

To view object dependencies for an asset:

1. In the row that contains the asset, click Actions.
2. Select Show Dependencies.
3. The Dependencies page opens the Uses tab by default.

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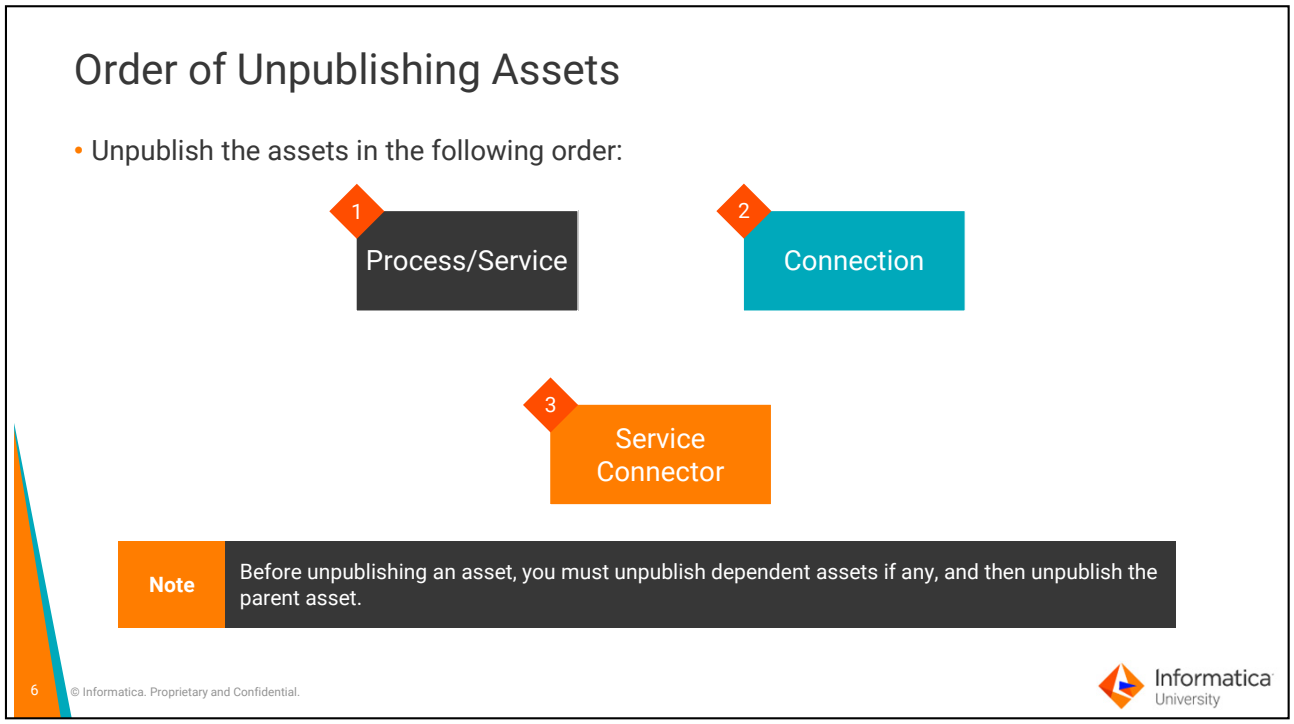


When you publish any asset in CAI, you must publish the assets in the following order:

1. Service Connector
2. Connection
3. Process/Service

You must follow this order when you want to connect to a third-party service that uses a connector and the respective connection in a Process. In this case, you will first need to create a service connector and create a connection using the service connector. Then you can use the created connection in a Process.

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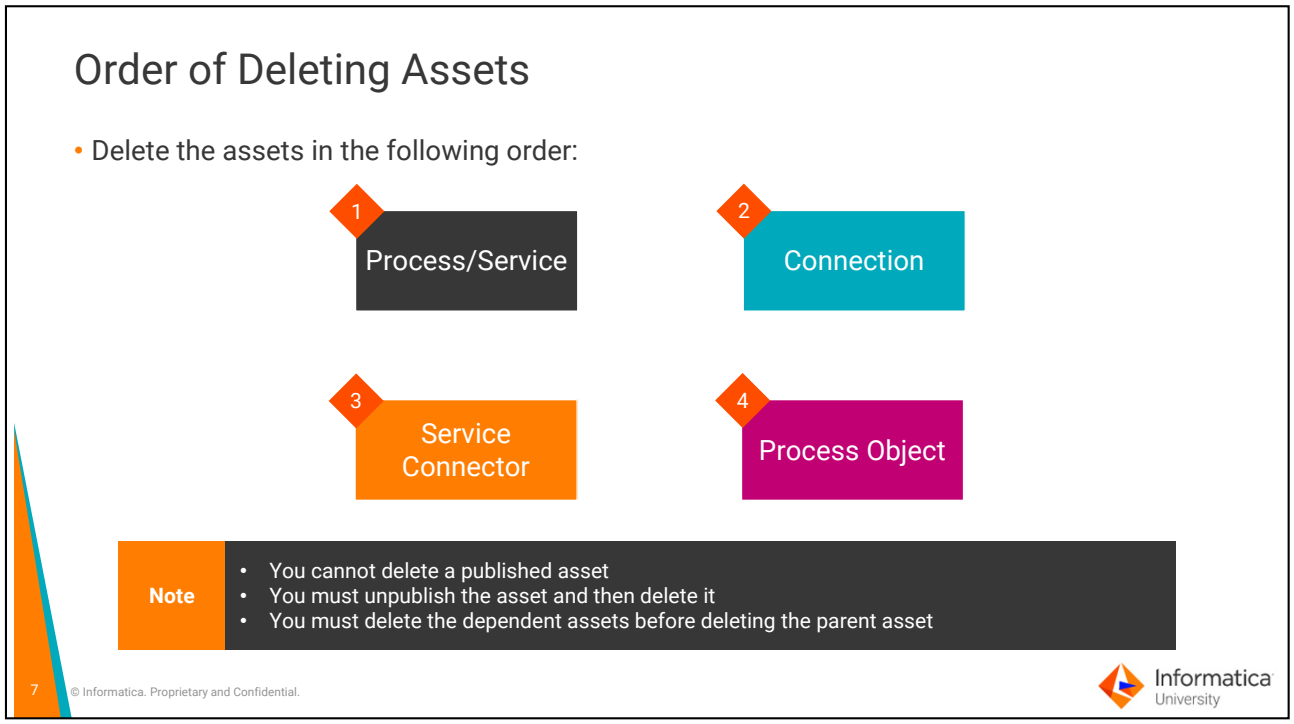


When you unpublish any asset in CAI, you must unpublish the assets in the following order:

1. Process/Service
2. Connection
3. Service Connector

In CAI, it is not allowed to unpublish a parent asset that contains another asset. You must unpublish the dependent asset first. As the Process is a parent asset to a Connection or a Service Connector, you must follow the above mentioned order while unpublishing an asset.

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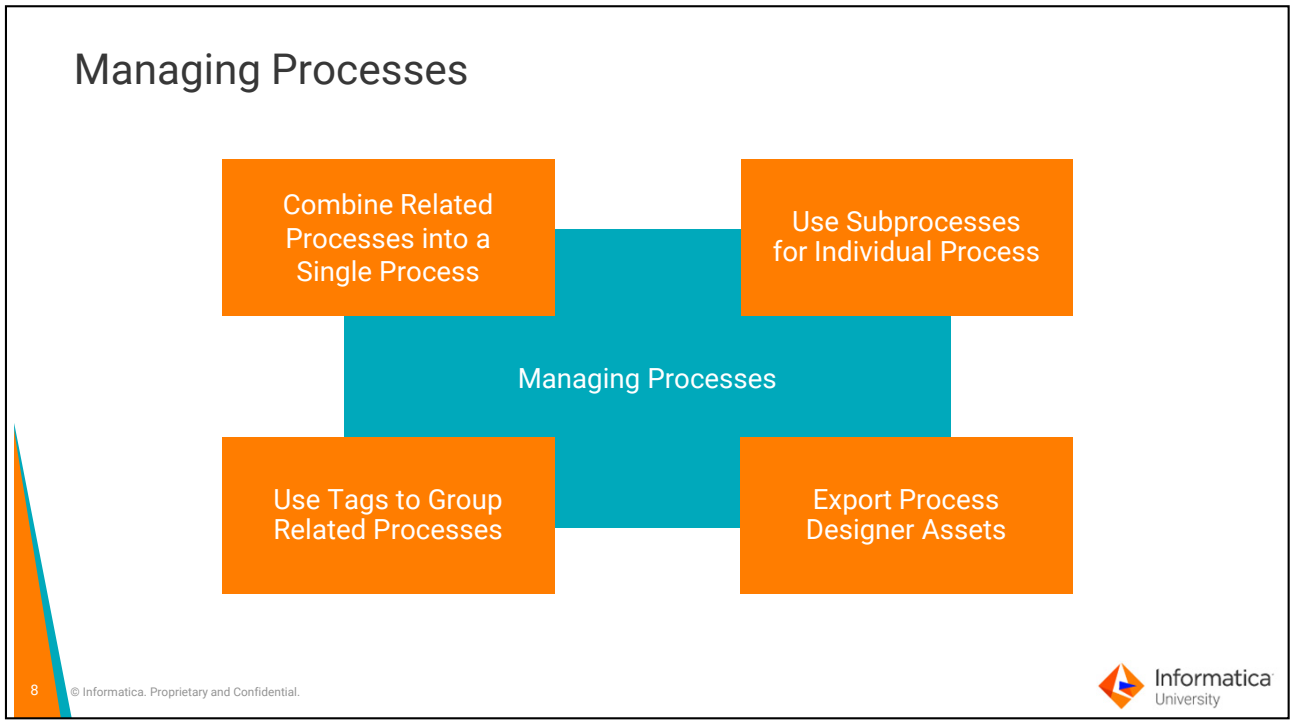
When you delete any asset in CAI, you must delete the assets in the following order:

1. Process/Service
2. Connection
3. Service Connector
4. Process Object

When you delete an asset, note the following points:

1. You cannot delete a published asset. You must unpublish the asset before deleting.
2. You must delete the dependent assets first. You can check an asset's dependency by selecting the asset and checking the Show Dependencies tab.

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Combining Related Processes into a Single Process: To reduce the overall number of Processes, it is recommended that you combine related processes into a single process.
Use Subprocesses for Individual Process: Subprocesses allows you to maintain the logic of individual process in one artifact, while grouping related processes together in the parent process.

Use Tags to Group Related Processes: You can use the 'Tag' feature in the Process Designer to group related processes in the same folder. When you tag similar processes, the folder structure of the Process Designer makes it easier to navigate to the relevant process when modifications are required.

Use Description and Notes: You must use the Description field to provide meaningful description to the process. You must also use the Notes section to record all the documentation associated with the process. By using Description and Notes sections, developers can easily access the documentation for the process.

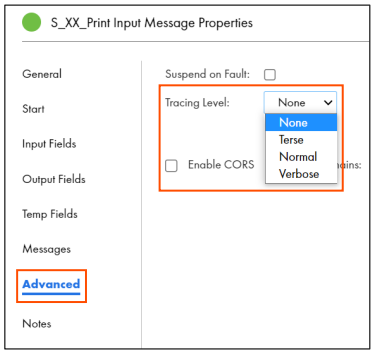
Exporting Process Designer Assets: To ensure that processes can be restored to a previous state if any issues arise, it is recommended that you periodically export all assets during the development and modification phases.

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Set Logging Type

- Tracing levels define the amount of data written in the log file when you execute a Process
- You can define logging level in the **Advanced** tab of Start properties
- Tracing Level determines the persistence and logging level for the Process

Tracing Level	Persistence Level	Logging Level
None	Brief	None
Terse	Brief	Fault
Normal	Brief	Execution with Service Data
Verbose	Final	Execution with Data



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Tracing levels define the amount of data written in the log file when you execute the Process. The log files can help in analyzing errors in case the Process fails. For each process, you can set the tracing level in the Advanced tab of Start properties. CAI supports the following 4 tracing levels:

1. None
2. Terse
3. Normal
4. Verbose

The tracing level also determines the Logging level and Persistence Level that are handled separately by Process Developer as shown in the above table.

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Set Logging Type (continued)

Persistence Level

- Persistence Level refers to the storage of active processes.
- When a process runs on the server, all the state and variable data is stored in the Process Server database.
- You can set the following Persistence levels for a Process:
  - Full
  - Persist
  - Final
  - Brief
  - None

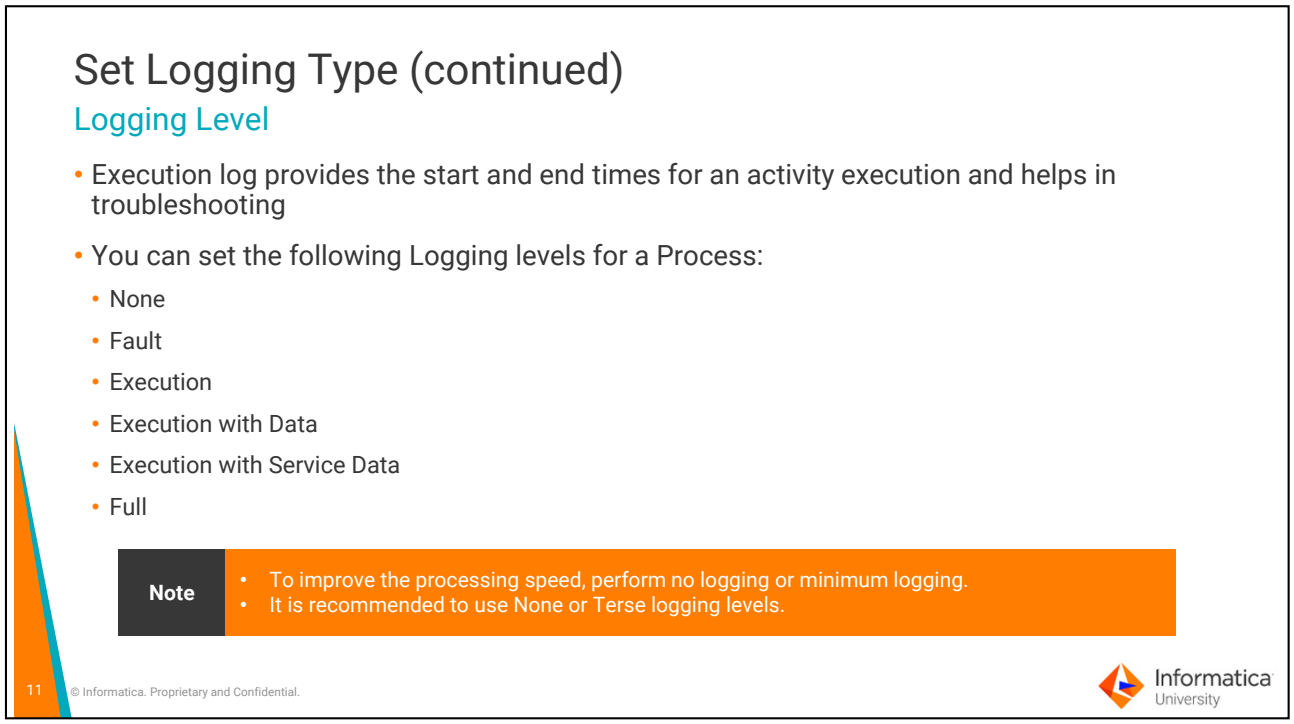
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Persistence refers to the storage of active processes. When a process runs on the server, all the state and variable data is stored in the process server database.
You can set the following Persistence levels for a process:

Full: The is the default persistence level. In this, all the process instance information, is stored. Thus, in the event of a server failure, a running process can be fully recovered as the system maintains a journal of changes intended for the database.
Persist: This level does not support journaling. If a process is running, and the server fails, then the process is suspended.
Final: Stores only the final state of the process (completed or faulted) and process variables.
Brief: This is the minimum level for process logging. Process Server stores only the start and completion times and the final state of the Process. It stores state and process variables only if the process faults.
None: No process information is stored in the server database when a process terminates. The process instance is not listed in the Processes page.

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Execution log provides the start and end times for activity execution and helps in troubleshooting.

You can set the following Logging levels for a process:

None: The Process Server does not log any information. Use this option to enhance performance.
Fault: The Process Server logs only fault information. If no faults occur in a process, the Process Logging level defaults to None. You must use this information to reduce the log file size.
Execution: This is the default option. The Process Server logs all execution statements except for the Will Not Execute statements.
Execution with Data: The Process Server logs all execution statements except for the Will Not Execute statements, but includes variable, expression, and partner link data.
Execution with Service Data: The Process Server logs all the execution and fault information.
Full: The Process Server logs all the execution statements as well as Will Not Execute statements. For example, the Process Server logs all fault handling statements that are not executed.

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Topic

Common Issues and Troubleshooting

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Salesforce Connection

Issue: OBM URL not visible	Workaround
• When you use a Salesforce connection in a process and you do not see the Salesforce OBM URL	• Suffix all the Salesforce connection names with the term 'Salesforce‘

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Salesforce Connection (continued)

Issue: 403 Forbidden error	Workaround
• All Salesforce outbound messages to CAI process throw 403 forbidden error. • The same process could be initiated from the SOAP UI without any issue. • This issue occurs when the Salesforce ICRT managed package is not correctly configured to point to the appropriate IICS Org.	• Set the correct IICS Org URL in Salesforce. • You can set the Org URL under Setup > Guide Setup > Advanced Settings.

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Secure Agent

Issue: Service URL does not work	Workaround
• The service URL of a process deployed to the Secure Agent does not work if you change the server host name or the HTTPS port number.	• If you change the server host name or HTTPS port number, restart the Secure Agent before you invoke a process service URL.

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Exposing Process as a WSDL

Issue: Import issue	Workaround
• When you define a process exposed as a WSDL, an error occurs while importing the WSDL in the SOAP client. • This error occurs if your process includes a space in a field or event name.	• Remove any spaces from field or event names and republish the process.

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Process Object

Issue: Unknown Process Object	Workaround
• When you export a process that references a process object and import it into another organization, the process no longer references the process object. • When you validate the process, you see the following Start step error: • Error: Field "<FieldName>" references to unknown Process Object: "<ProcessObjectName>"	• Save the process and log out of Informatica Intelligent Cloud Services. • Log in to IICS and select the Application Integration service. • Open and validate the process. • You must see that the validation error no longer appears.

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Guides

Issue: Incorrect path	Workaround
• In Guide, when you use a process object field in a Decision step, the guide does not always follow the correct path.	• Use an Assignment step to assign the process object field to a temporary field. • Use the temporary field in the Decision step.

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Guides (continued)

Issue: Publish fails	Workaround
• When you publish a guide, you see the following error: • Error: AccesstoAll was not published due to the following error: HTTP method: POST Bad Request Status Code: 400 • This issue occurs when user is logged in to CAI and Salesforce at the same time.	• Logout from the Salesforce account. OR • Close the browser. • Delete cache/cookies. • Go to the Incognito mode of Chrome and then try to publish the guides.

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Process

Issue: Internal Application Provider Error	Workaround
• When you import a process, following error is seen in a log file: • Error: Internal application provider error Request to external service [frs] failed. cause [httpStatus: 400 message: DocRef validation failed • This issue occurs when the exported process has a dependency on subprocess which has not been exported in the archive	• Login to Source Org • Re-save the process by changing the description • Export the process again • Use this exported .zip to import

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Service Connector from Swagger File

Issue: After import UI is stuck

- When you create a service connector using a swagger file, the screen freezes after importing the swagger on the UI
- Unable to proceed further after clicking the Next button
- This issue occurs if the OperationId in swagger has spaces in between

Workaround

- Open the Swagger file
- Edit all the operationIds and remove spaces from them

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JDBC Connection

Issue: Publish fails	Workaround
• When you create a JDBC connection and the test for connection is successful but the publish fails • You can see the following error in Catalina logs: • Error:javax.xml.ws.soap. SOAPFaultException: The host did not accept the connection within timeout of 60000 ms	• Whitelist the CAI DNS on your proxy • Turn ON the async URN mapping in CAI • Navigate to Application Integration Console > Deployed Assets > URN Mapping • Add the following URN and URL: • URN: ae:agent-response-mechanism • URL: async

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Module Summary

This module showed you how to:

- View Asset Dependencies
- List Best Practices for CAI Assets
- Define Logging Types in CAI
- Discuss Frequently Faced Issues and Workarounds

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Module 17

CAI Live Project Walk-Through

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Module Objectives

After completing this module, you will be able to:

- Create and invoke a process that uses Inventory Details
- Use the API Manager service to create a proxy API

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Scenario

- In the Order Management process, you use Application Integration to create a process that fetches inventory details and helps you to obtain the margin and commission details. You will also create a managed API using the API Manager Service.
- To achieve this, you must create the following CAI assets and bind them to create the process:
  - Import a service connector and a connection and use them in a process
  - Import a Swagger JSON URL to create a service connector
  - Create a process that uses steps such as Assignment, Service, Parallel Paths, and Subprocess
  - Connect the process to an email service
  - Invoke the process with and without errors
  - Update the process to enable fault handling
  - Create a managed API using the IICS API Manager service

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Steps Overview

Step 1	Step 2	Step 3	Step 4
Import the Service Connector and Connection	Publish the connector and connection	Create and invoke Order Management process	Create a Service Connector for the margin service
Step 5	**Step 6**	**Step 7**	**Step 8**
Create a connection that points to the margin service	Modify the process to add the margin details	Connect to an Email Service	Create a Subprocess
Step 9	**Step 10**	**Step 11**	**Step 12**
Add the Subprocess to the Main Process	Invoke the modified Order Management Process	Enable Fault Handling	Create a Managed API

4

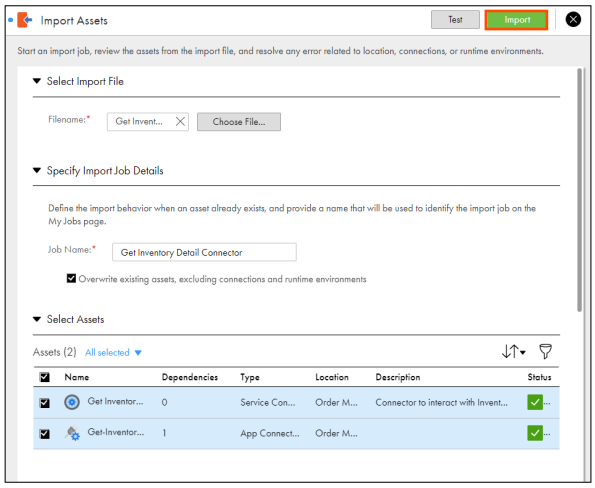
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Import a Service Connector and Connection

- Go to **https://network.informatica.com/docs/DOC-17733** and download the **Get Inventory Detail Connector.zip** file.
- In Application Integration, click **Explore** > **Import**.
- Select **Get Inventory Detail** and **Get-Inventory-Detail** and copy them to **S_XX_Order_Management**.
- Publish the service connector.
- Open the **Get-Inventory-Detail** connection and set its Type to **S_XX_Order_Management** > **Get Inventory**.



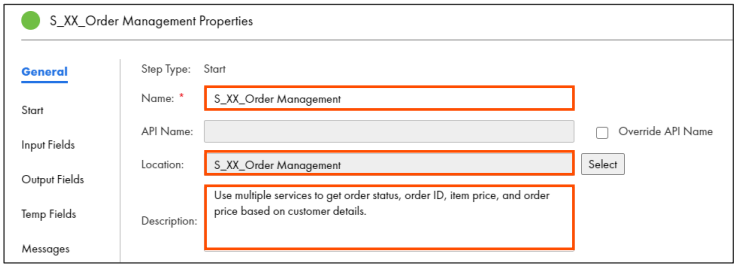
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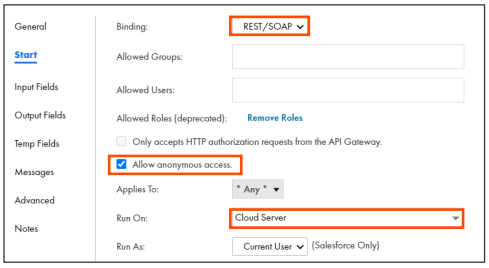
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Create a New Process

- Create a Process.
- Specify the following **General** properties for the Process:
  - Name
  - Location
  - Description
- Specify the following **Start** properties for the Process:
  - Binding
  - Allowed Roles
  - Run On





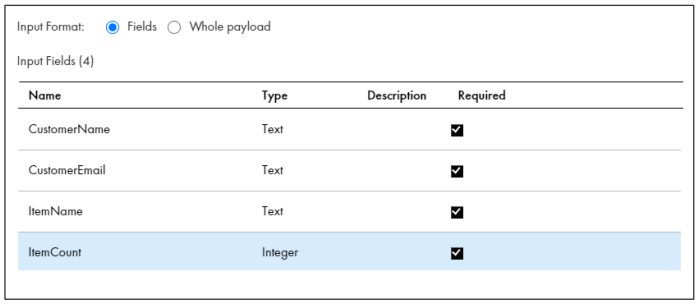
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Create a New Process (continued)

- Specify the following Input properties for the Process:

Name	Type	Required
CustomerName	Text	Yes
CustomerEmail	Text	Yes
ItemName	Text	Yes
ItemCount	Integer	Yes



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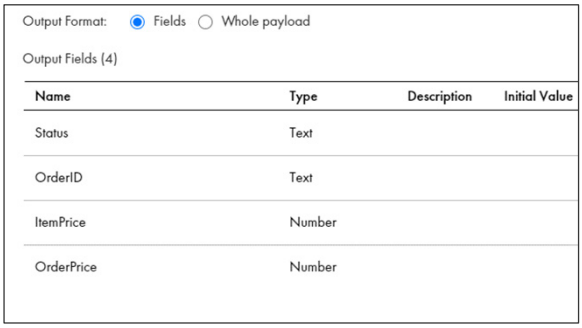
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Create a New Process (continued)

- Specify the following Output properties for the Process:

Name	Type
Status	Text
OrderID	Text
ItemPrice	Number
OrderPrice	Number



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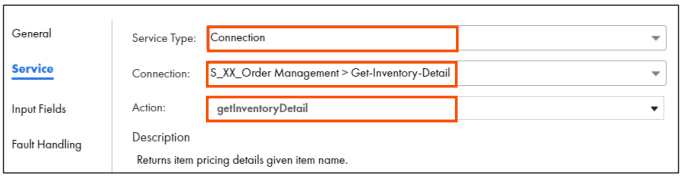
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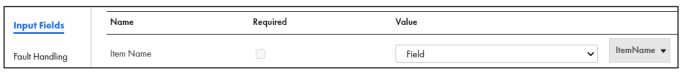
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Get Inventory Details

- Drag and drop a Service step from the palette onto the canvas.
- Specify the following General properties:
  - Name
- Specify the following Service properties:
  - Service Type
  - Connection
  - Action
- Specify the following Input properties:

Name	Value
Item Name	Field





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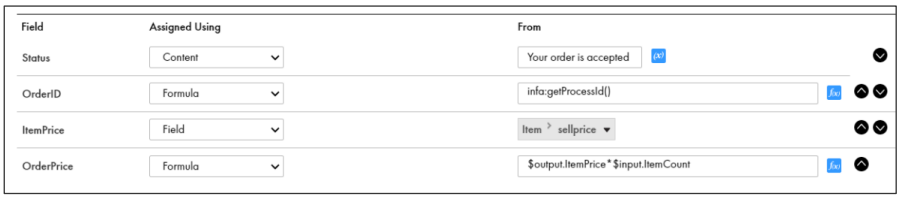
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Configure Output

- Drag and drop an Assignment step from the palette onto the canvas.
- Specify the following General properties:
  - Name
- Specify the following Assignment properties and save it:

Target	Value Type	Value
Status	Content	Your order is accepted.
OrderID	Formula	infa:getProcessId()
ItemPrice	Field	item>sellprice
OrderPrice	Formula	$output.ItemPrice*$input.ItemCount



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Invoke the Process

- Copy the Service URL.

Open a text editor and add the following input parameters to the service URL:

```
<Service
URL>?CustomerName=TestConsumer&CustomerEmail=testconsumer@m
ailinator.com&ItemName=item1&ItemCount=2
```

- Open a browser and paste the service URL with the input parameters.

{"Status":"Your order is accepted","OrderID":"54476385406892851 2","ItemPrice":50.0,"OrderPrice":100.0}

Note Enter your Informatica Intelligent Cloud Services credentials, if prompted.

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Create a Service Connector to the Margin Service

- Create a Service Connector.
- Specify the following properties in the Swagger Source tab:

Property Name	Property Content
Name	S_XX_Get Margin Detail
Location	S_XX_Order Management
Description	Access the margin service at **https://na1.ai.dm-us.informaticacloud.com/active-bpel/rt/Calculate_Margin_Service?Swagger**
Swagger Source	URL
Swagger URL	https://na1.ai.dm-us.informaticacloud.com/active-bpel/rt/Calculate_Margin_Service?Swagger
User Name	testuser2
Password	password2#

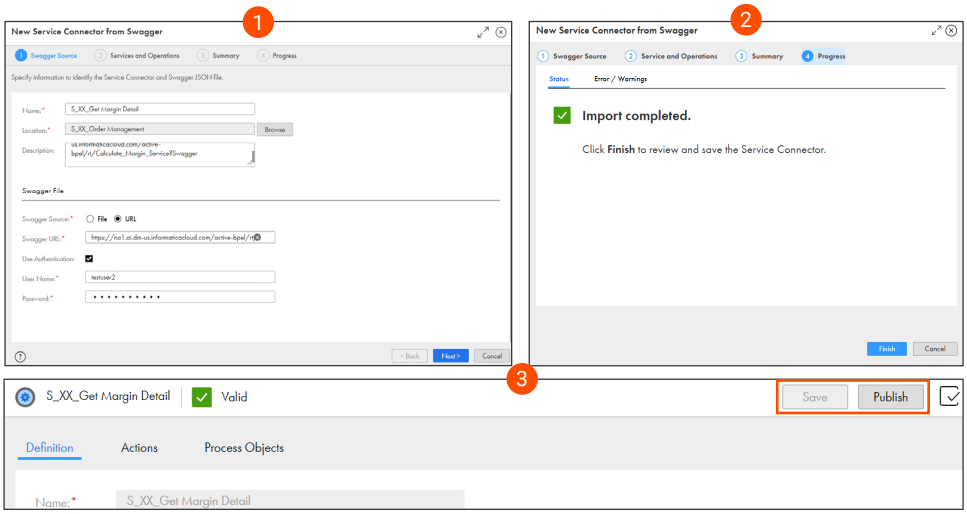
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Create a Service Connector to the Margin Service (continued)

- Save and publish the changes.



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Create a Connection

- Create a new Connection.
- Specify the following information in the Properties tab:

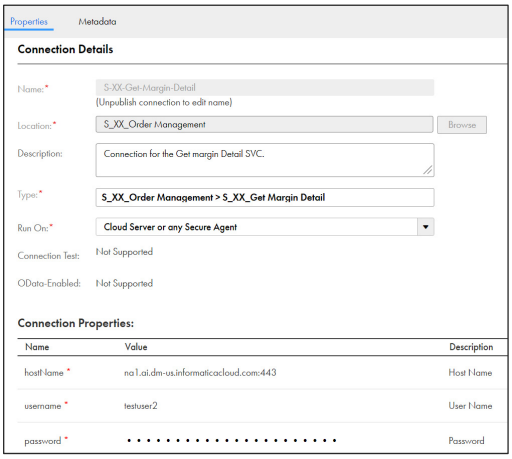
Property Name	Property Content
Name	S-XX-Get-Margin-Detail
Location	S_XX_Order Management
Description	Connection for the Get Margin Detail SVC.
Type	S_XX_Order Management>S_XX_Get Margin Detail
Run On	Cloud Server or any Secure Agent
Hostname	na1.ai.dm-us.informaticacloud.com:443
Username	testuser2
Password	password2#

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Create a Connection (continued)

- Save and publish the Connection.





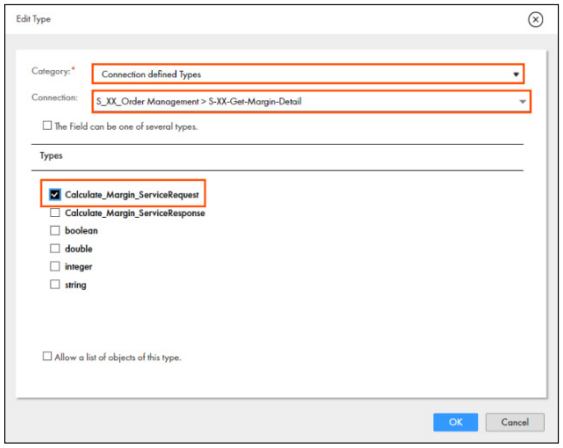
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Add a Temporary Field

- In the **S_XX_Order Management** process, select the **Start** step.
- Select the **Temporary Fields** tab and create a field with the following properties: (This step serves as the input to the Get-Margin-Detail connection.)
  - Name: **InventoryDetails**
  - Type: **More Types** > **Connection Defined Types** > **Order Management** > **Get-Margin-Detail** > **Calculate_Margin_ServiceRequest**



Note

The four fields contained within **Calculate_Margin_ServiceRequest** appear as options in the Assignment step.

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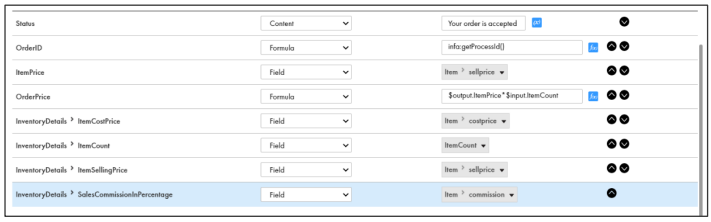
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Assign Values

- Select the **Assign Order Details** step and in the **Assignments** tab, assign values to the following fields:
  This step assigns values to the **Calculate_Margin_ServiceRequest** fields.

Target	Value Type	Value
InventoryDetails > ItemCostPrice	Field	Item>costprice
InventoryDetails > ItemCount	Field	ItemCount
InventoryDetails > ItemSellingPrice	Field	Item>sellprice
InventoryDetails > SalesCommissionInPercentage	Field	Item>commission



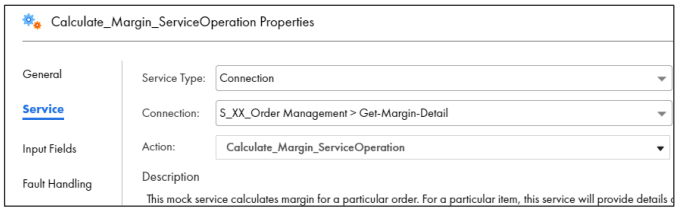
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Configure the Process to Get the Margin Details

- Drag and drop a Service step from the palette onto the canvas.
- Select the Service tab and perform the following tasks:
  - Select the General tab and enter **Get Margin Detail** in the Name field.
  - Set the Service Type to **Connection**.
  - Set the Connection to **S_XX_Order Management** > **Get-Margin-Detail**.
  - Set the Action to **Calculate_Margin_ServiceOperation**.
  - Select the Input Fields tab and assign the value as **Field** > **InventoryDetails**.



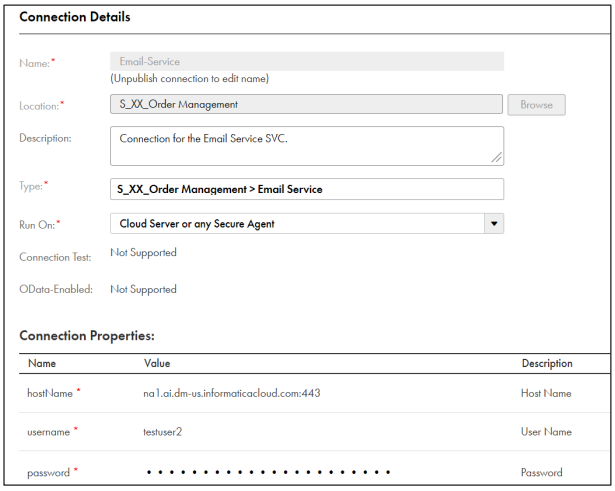
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Connect to an Email Service

- Create an email service connector and a connection
- You later use these connections in the process to send emails to consumers and service providers
- Use a Swagger URL to connect to the following service: https://na1.ai.dm-us.informaticacloud.com/active-bpel/rt/Email_Service?swagger
- Finally, create a connection for the email service connector



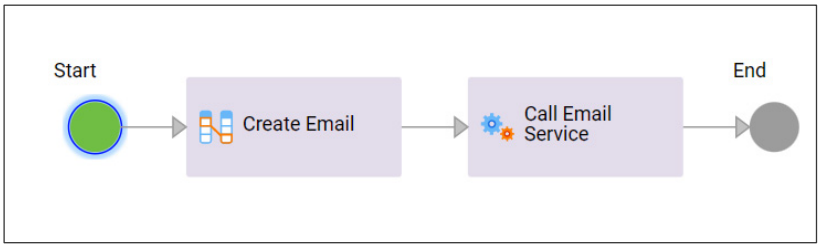
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Create a Subprocess

- Create a process that uses the email connection to send emails to consumers and vendors.
- When you add this process to the Order Management process, it becomes a subprocess.



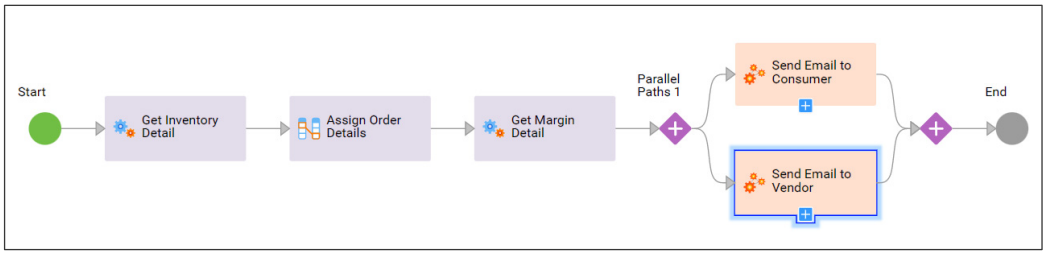
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Add the Subprocess to the Main Process

- Add a Parallel Paths step with two branches and add a service step on each branch.
- The first branch sends an email to the consumer and the second branch sends an email to the vendor.



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Invoke the Process Successfully

- Invoke the modified process with two sets of input parameters.
- To copy the Service URL, from the Actions menu, select Properties Details.

In a browser, paste the following parameterized service URL:

```
<serviceURL>?CustomerName=TestConsumer&CustomerEmail=testco
nsumer@mailinator.com&ItemName=item1&ItemCount=2
```



Note You will see a successful response.

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Invoke Process to Show a Fault

- To copy the Service URL, from the Actions menu, select Properties Details.

In a browser, paste the following parameterized service URL:

```
<serviceURL>?CustomerName=TestConsumer&CustomerEmail=testco
nsumer@mailinator.com&ItemName=Newitem1&ItemCount=2
```



Note You will see an HTTPS 400 error response. This is due to change of item name to Newitem1, which is an invalid value.

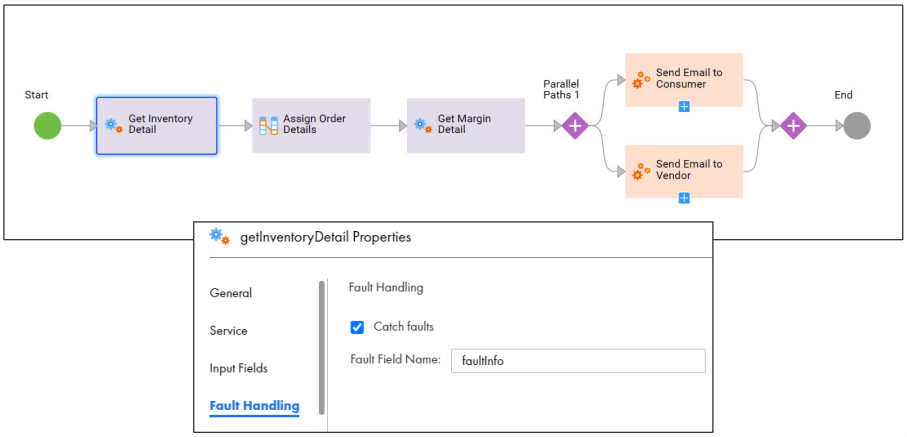
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Enable Fault Handling

- Open the S_XX_Order_Management process and select the **Get Inventory Detail** step.
- Select the **Fault Handling** tab and enable **Catch faults**.



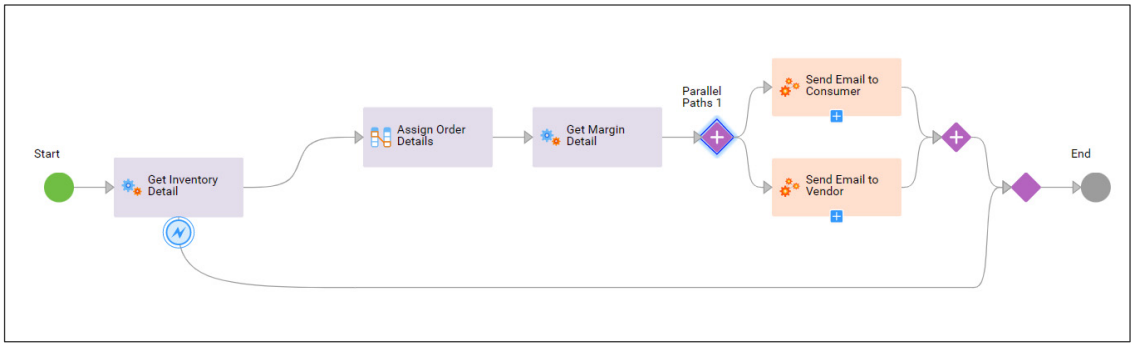
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Configure the "No Fault" Branch

- Drag and drop all the steps to the “no fault” branch except the End step.



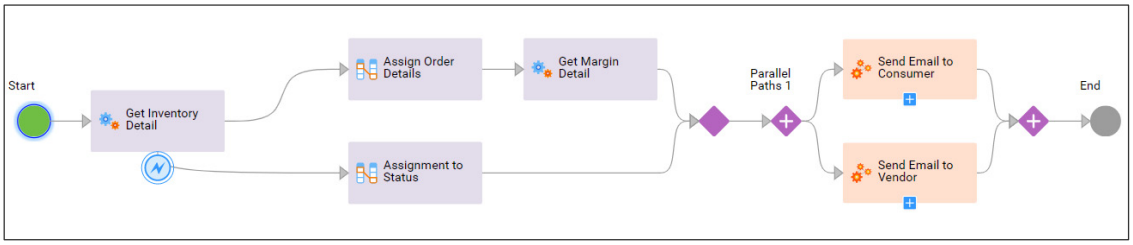
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Configure the Fault Branch

- Drag and drop an Assignment step on to the "fault" branch of the Get Inventory Detail step.
- Select the General properties tab and enter **Assignment to Status** in the name field.
- Select the Assignments tab and add the Status field with the Value as **Content** > **Item not in stock**. **Apologies for the inconvenience**.
- Save and publish the process.



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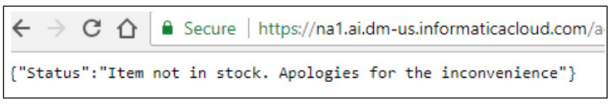
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Invoke the Process

- To copy the Service URL, from the Actions menu, select Properties Details.

In a browser, paste the following parameterized URL:

```
<serviceURL>?CustomerName=TestConsumer&CustomerEmail=testco
nsumer@mailinator.com&ItemName=Newitem1&ItemCount=2
```



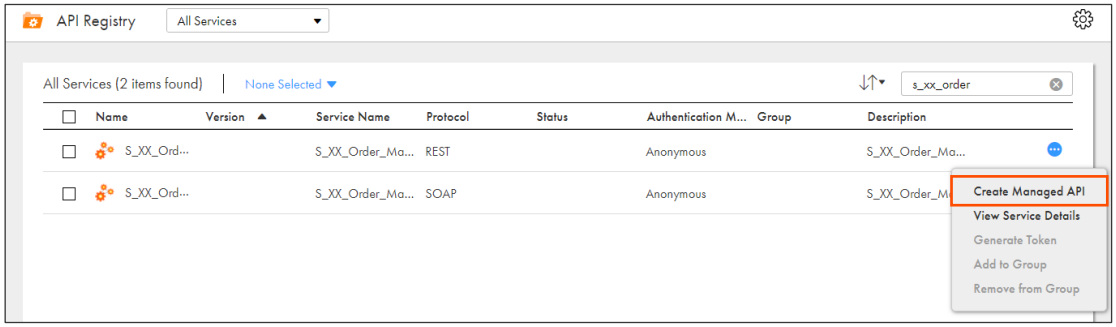
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Create a Managed API

- In IICS, click **My Services** and select the **API Manager** service.
- On the API Registry page, click **Actions** > **Create Managed API**



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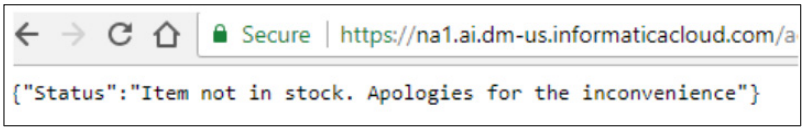
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Invoke the Managed API

- Next to S_XX_Order_Management, click **Actions** > **Copy URL**.

In a browser, paste the following parameterized URL:

```
<URL>?CustomerName=TestConsumer&CustomerEmail=testconsumer@
mailinator.com&ItemName= item1&ItemCount=2
```



Note You see the same result when you invoke the process with the Application Integration service URL.

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Lab Activity

17-1 CAI Live Project- Order Management

In this lab, you will perform the following:

- Create and invoke the Order Management Process
- Connect to a Margin Service
- Modify the Process to add the Margin Details Service
- Create a Service Connector to the Email Service
- Create an Email Connection
- Create a process that uses the email connection to send emails
- Invoke the Order Management Process
- Enable Fault Handling
- Create a Managed API

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Module Summary

This module showed you how to:

- Create and invoke a process that uses Inventory Details
- Use the API Manager service to create a proxy API

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